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INVITED REVIEW

Fairness of artificial intelligence in healthcare: review and recommendations

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Abstract

In this review, we address the issue of fairness in the clinical integration of artificial intelligence (AI) in the medical field. As the clinical adoption of deep learning algorithms, a subfield of AI, progresses, concerns have arisen regarding the impact of AI biases and discrimination on patient health. This review aims to provide a comprehensive overview of concerns associated with AI fairness; discuss strategies to mitigate AI biases; and emphasize the need for cooperation among physicians, AI researchers, AI developers, policymakers, and patients to ensure equitable AI integration. First, we define and introduce the concept of fairness in AI applications in healthcare and radiology, emphasizing the benefits and challenges of incorporating AI into clinical practice. Next, we delve into concerns regarding fairness in healthcare, addressing the various causes of biases in AI and potential concerns such as misdiagnosis, unequal access to treatment, and ethical considerations. We then outline strategies for addressing fairness, such as the importance of diverse and representative data and algorithm audits. Additionally, we discuss ethical and legal considerations such as data privacy, responsibility, accountability, transparency, and explainability in AI. Finally, we present the Fairness of Artificial Intelligence Recommendations in healthcare (FAIR) statement to offer best practices. Through these efforts, we aim to provide a foundation for discussing the responsible and equitable implementation and deployment of AI in healthcare.

Keywords Fairness · Bias · Artificial intelligence · Healthcare · Medicine · Review

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Introduction

Fairness is one of the core principles of artificial intelligence (AI) ethics [1–3], and in recent years, there has been an increase in efforts focusing on fairness in AI, with a growing number of publications highlighting the need for improvement [4–9]. Various biases are involved in developing and applying AI, and these biases can affect fairness by erroneously skewing AI results [10]. In the medical field, bias and discrimination in AI have been studied in various domains [11, 12]. The World Medical Association's Geneva Declaration cites factors such as "age, disease or disability, creed, ethnic origin, gender, nationality, political affiliation, race, sexual orientation, social standing or any other factor" as examples that should not influence a physician's obligation to their patients [13]. Therefore, fairness concerns arise if AI does not perform adequately for specific patients.

AI research in radiology is an active field in healthcare owing to its affinity for imaging [14], with the number of AI-related publications and medical device certifications increasing annually [15–17]. One important reason for this is the global shortage of radiologists [18–21]. In particular, Japan has numerous publications on AI use in the field of radiology, including X-ray [22–25], mammography [26–30], US [31], CT [32–40], MRI [41–50], and PET [51, 52]. This surge in Radiological AI publications in Japan could be related to Japan having both the lowest number of radiologists per capita and the highest number of CT and MRI machines per capita among the Organization for Economic Co-operation and Development (OECD) countries [53]. Furthermore, owing to the coronavirus disease 2019 (COVID-19) pandemic, the number of COVID-19-related studies from Japan has increased [54–61], with a significant increase in such research focusing on AI [62–64].

As physicians in this era of AI clinical practice, we must be mindful of fairness concerns arising from AI bias in healthcare to provide better care to all patients. This review aims to provide a comprehensive overview of concerns related to AI fairness, discuss strategies to mitigate AI biases, and emphasize the need for collaboration among stakeholders to ensure equitable AI integration. In doing so, it lays the foundation for discussing the responsible and equitable implementation and deployment of AI in healthcare.

First, fairness in healthcare is discussed. We then discuss the issue of bias in AI systems used in healthcare. Next, we suggest strategies to reduce bias such as using diverse data, validating algorithms, and educating clinicians and patients regarding AI. We then discuss ethical and legal issues such as patient consent, data privacy, accountability, and the need for transparency in AI systems. Collaboration is key in this context; therefore, we explore the roles of various stakeholders, including physicians, AI researchers, policymakers, regulatory authorities, patients, advocacy groups, and professional associations. We include the best practices recommended for fairness in AI and the areas where more research is needed. Finally, we conclude the paper with a summary of our main findings.

Fairness concerns in healthcare

Defining fairness in healthcare

Fairness in healthcare is a multidimensional concept that includes the equitable distribution of resources, opportunities, and outcomes among diverse patient populations [65]. The concept of fairness is based on the fundamental ethical principles of justice, beneficence, and non-maleficence. Healthcare systems must provide access to high-quality care for all individuals without discrimination. In the context of radiology, fairness in AI refers to the development and deployment of unbiased AI that provides accurate diagnoses and treatments for all patients regardless of their social status or ethnic differences. Achieving this fairness requires a comprehensive understanding of the potential causes of bias in AI and development of strategies to mitigate these biases [66].

Biases of AI in healthcare

Generally, biases in AI can emerge from two main sources: the data used for algorithm training (data bias) and inherent design or learning mechanisms of the algorithm itself (algorithmic bias). However, in the healthcare context, additional biases may arise because of the complex nature of human interactions and decision-making processes. These additional biases can be classified into two types: those that originate from AI–clinician interactions and those that originate from AI–patient interactions [11]. An overview of these biases is shown in Fig. 1.

Data biases

Data bias refers to problems arising from the collection and organization of data used in AI training that can potentially have harmful effects on fairness and accuracy [67]. The types of data biases include minority bias, missing data bias, informativeness bias, and training–serving skew [11]. Minority bias occurs when the number of protected group members in the dataset is insufficient for AI to learn accurate statistical patterns. This can lead to decreased

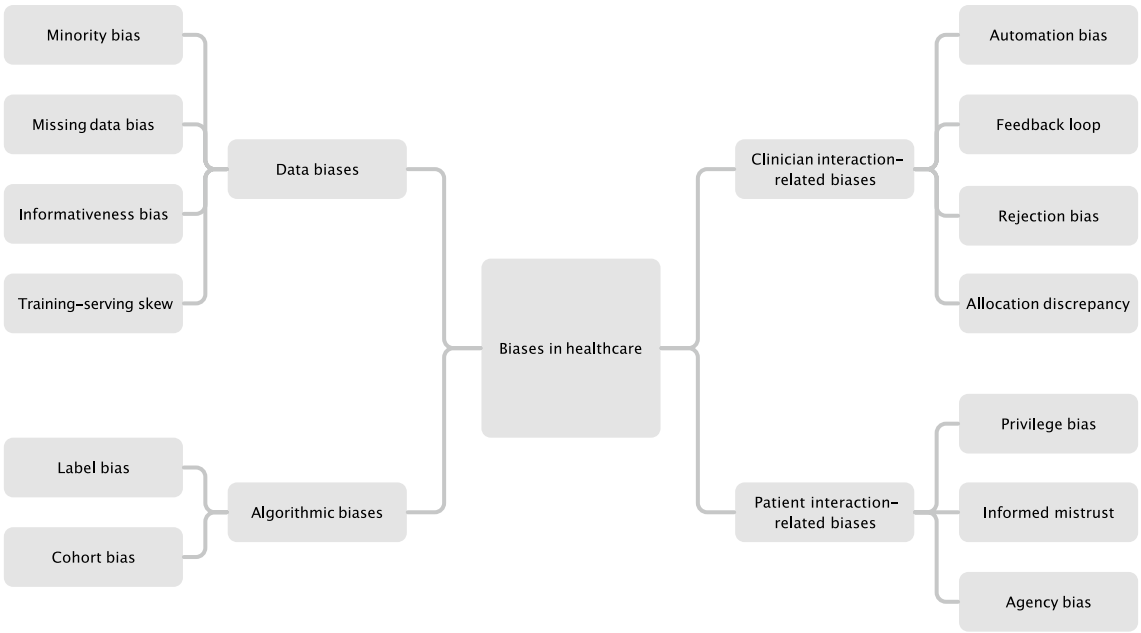


Fig. 1 Biases in healthcare of AI

performance and biased results when the algorithm is applied to these underrepresented groups. For example, many cardiovascular risk prediction algorithms have a history of being trained primarily on male patient data [68, 69]. This has led to an inaccurate risk assessment in female patients with different symptoms and risk factors. Missing data bias occurs when data from protected groups are missing nonrandomly, making it difficult for AI to generate accurate predictions. For example, if patients in contact isolation have fewer vital sign records than other patients, the algorithm may struggle to identify clinical deterioration. Informativeness bias occurs when the features used for detection are not as apparent for certain protected groups, lowering their informativeness when predictions are made. For example, identifying melanoma from images of patients with dark skin is more challenging than those with light skin [70, 71]. Training–serving skew refers to the mismatch between the data used for AI training and those used during deployment. This can arise from nonrepresentative training data due to selection bias or from the deployment of the model on patients with a population prevalence different from that of the training data. In a study training AI to diagnose pneumonia from chest X-rays, the performance on unseen data from the institution where the training data were collected was significantly higher than its performance on data collected from external hospitals [72]. This common scenario means that estimations of AI performance based on internal test data may overestimate its real-world performance on external data [73–75].

Algorithmic biases

Algorithmic bias refers to problems arising from the development and implementation of AI, which can negatively affect fairness and effectiveness. Even with representative data without data bias, AI can exhibit bias because of its inherent design or learning mechanisms. Algorithmic biases include label and cohort bias [11]. Label bias is a broad concept that includes test referral and interpretation bias. This occurs when AI training uses inconsistent labels, which may be influenced by healthcare disparities rather than universally accepted truths. This can lead to biased decision-making based on inaccurate or inconsistent information in the AI algorithms. For example, significant racial bias has been observed in commercially available algorithms used to predict patients' healthcare needs. Although several biases were affected, a major contributing factor to this algorithm's bias was its design, which used cost as a proxy for healthcare needs, leading to an underestimation of the needs of Black patients compared with those who are White with similar conditions [76]. Cohort bias occurs when AI is developed based on traditional or easily measurable groups without considering other potentially protected groups or varying levels of granularity. For example, mental health disorders have been underdiagnosed or misdiagnosed within lesbian, gay, bisexual, transgender, queer or questioning, intersex,

asexual, and other (LGBTQ +) populations [77]. One reason for this is that algorithms often do not take the granularity of the LGBTQ + population into account and rely only on information about biological males and females. AI trained on such data may continue to overlook or misdiagnose mental health issues in these populations, potentially perpetuating existing disparities in mental healthcare.

Clinician interaction-related biases

When healthcare professionals interact with AI, biases related to interactions can occur, potentially affecting the algorithm's performance, fairness, and adoption [11]. One such bias is automation bias, which refers to the tendency to overly rely on AI when tasks are transferred from healthcare professionals to AI programs [78]. Overconfidence in algorithms can result in inappropriate actions based on inaccurate predictions. One study found that incorrect AI advice negatively affected radiologists' mammogram reading performance across all expertise levels. Inexperienced radiologists are more likely to follow incorrect AI suggestions [79]. Another bias related to interactions is the feedback loop [80]. This occurs when clinicians accept AI recommendations even if they are incorrect, leading the algorithm to relearn and perpetuate the same mistakes. Rejection bias refers to the conscious or unconscious desensitization to excessive alerts. Alert fatigue is a manifestation of this bias, as clinicians may ignore important alerts owing to an overwhelming number of false alarms [81, 82]. Finally, an allocation discrepancy occurs when the positive predictive values for protected groups are disproportionately low, leading the AI to withhold necessary resources, such as clinical attention or social services. Such resource allocation discrepancies can exacerbate disparities in care and outcomes among the affected groups.

Patient interaction-related biases

Biases related to interactions between patients and AI or the systems that incorporate them include privilege bias, informed mistrust, and agency bias [11]. Privilege bias occurs when certain populations cannot access AI in care settings or when these algorithms require technology or sensors that are not available to all populations [83]. This can lead to an unequal distribution of AI-driven healthcare benefits, potentially exacerbating existing healthcare disparities. Informed mistrust refers to the skepticism protected groups may have toward AI owing to historical exploitation and unethical practices in healthcare [84, 85]. This mistrust may lead these patients to avoid care or intentionally conceal information from clinicians or systems using AI. Agency bias arises when protected groups lack a voice in the development, use, and evaluation of AI [86]. These groups may lack the access, resources, education, or political influence necessary to detect AI biases, voice concerns, and affect change. This lack of agency can result in AI inadequate at considering the needs and perspectives of protected groups, potentially leading to biases and disparities in healthcare outcomes.

Strategies to mitigate bias

Diverse and representative data

One of the most effective methods of mitigating AI biases is to ensure the use of diverse and representative datasets during AI development and training [67, 87]. This process entails carefully collecting and incorporating data from a wide range of sources to accurately reflect the demographics, characteristics, healthcare needs, and potential disparities in the target population. This diversity is not only critical for developing AI systems capable of catering to a multitude of patient requirements but also for fostering trust and confidence in AI-driven healthcare solutions. By incorporating data from various patient populations, age groups, disease stages, cultural and socioeconomic backgrounds, and healthcare settings, AI can learn to recognize, diagnose, and treat a broad spectrum of patient conditions with greater precision and contextual understanding. This comprehensive approach to data collection and curation prevents potential biases from occurring in AI systems, resulting in a reduction of disparities and promotion of equity in healthcare outcomes [76]. Furthermore, a diverse and representative dataset ensures that AI algorithms are rigorously tested across different scenarios, thereby enhancing their overall performance and utility. This enables healthcare providers to rely on AI-driven diagnostics and treatment recommendations, leading to improved patient care and reduced clinician workload.

Algorithm auditing and validation

Regular audits and AI validation play crucial roles in identifying and addressing potential biases and ensuring that AI systems remain fair, accurate, and effective in diverse healthcare settings. Independent audits by external experts or organizations can be conducted to evaluate the fairness, accuracy, and performance of AI, with adjustments made to the algorithms to correct identified biases [88]. The healthcare landscape is constantly changing; therefore, there is no guarantee that an AI algorithm with high performance will maintain its high performance in the future [89]. Validation studies are essential for verifying the effectiveness of AI in different patient populations and conditions [72]. The establishment of a dedicated department within hospitals for algorithm quality control has been advocated [90]. This department should be

responsible for continuously monitoring AI performance, identifying potential biases, and making the necessary updates to algorithms. This proactive approach to quality control would ensure that AI systems are held accountable and maintain their effectiveness in providing accurate and equitable care for all patients. Considering the growing prevalence of medical AI, practitioners must remain vigilant and evaluate key indicators, such as underdiagnosis rates and other health disparities, during the algorithm development process and after deployment. This ongoing evaluation will help identify and rectify emerging issues, ensuring that AI systems continue to serve patients effectively and equitably.

Education to both clinicians and patients

Educating clinicians and patients on the biases inherent in AI is crucial for fostering a shared understanding and promoting fairness in healthcare [1]. This educational process involves raising awareness of potential biases, sharing best practices to address them, and encouraging open discussions on the implications of AI in healthcare decision-making. Clinicians aware of AI biases can avoid overreliance on AI-generated results and make decisions based on more accurate information [91]. This increased awareness enables healthcare professionals to critically evaluate AI recommendations, weigh potential risks and benefits, and consider alternative sources of information when making patient care decisions. Additionally, clinicians can advocate for, and participate in, the development and evaluation of AI systems to ensure that their expertise and experience are incorporated into the models, further enhancing their accuracy and reliability. Patients who understand AI biases can make more informed and satisfactory decisions [92, 93]. By being aware of the potential limitations and biases of AI-generated recommendations, patients can engage in more meaningful conversations with their healthcare providers regarding treatment options and play a more active role in their care. This empowerment promotes patient-centered care and ensures that individual preferences, values, and circumstances are considered when making healthcare decisions. To foster a culture of continuous learning and improvement of AI, creating channels for feedback and collaboration among healthcare professionals and patients is essential. This can be achieved through workshops, conferences, online forums, or interdisciplinary collaborations that bring together diverse perspectives and experiences. By sharing knowledge, insights, and best practices, they can work together to identify and address biases and continuously refine AI systems to better serve the needs of all patients.

Ethical and legal considerations

Data privacy and security

Ensuring data privacy is an important ethical and legal consideration for AI fairness, as it has a significant impact on patient autonomy, trust in AI, and compliance with legal frameworks. Respecting patient autonomy and protecting confidential medical information is the foundation of ethical AI implementation, which can only be achieved by addressing important issues related to data privacy [94, 95]. One such issue is obtaining informed consent for data use [96, 97]. Patients must fully understand how their data are used, shared, and stored by AI. To achieve this, transparent communication regarding the purpose, risks, and benefits of data sharing is required to enable patients to make informed decisions regarding participating in AI-driven healthcare initiatives. Protecting data storage and transmission is an important aspect of data privacy [98]. Robust security measures, such as encryption and anonymization techniques, are required to protect patient data from unauthorized access, data breaches, and other cybersecurity threats. Moreover, strict access controls and audit mechanisms must be implemented to monitor and track data use, ensure accountability, and prevent data misuse. Compliance with privacy regulations, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union, is essential for legally and ethically sound AI practice [99]. These regulations provide strict guidelines for the collection, storage, and processing of personal health information, and AI researchers and healthcare professionals must adhere to standardized data protection protocols. By addressing these challenges and ensuring data privacy, AI developers and healthcare professionals can foster trust in AI, maintain patient autonomy, and adhere to ethical and legal standards. This promotes the development and implementation of fair and equitable AI-driven healthcare solutions that respect the privacy and dignity of all patients.

Liability and accountability

To address the potential errors, harmful outcomes, and biases in the predictions generated by AI, clear guidelines for responsibility and accountability in healthcare AI must be established. This includes determining the roles and responsibilities of various stakeholders, such as physicians, AI developers, and healthcare institutions, in cases where misdiagnoses or other patient harm occur [100, 101]. Physicians should be responsible for verifying AI-generated diagnoses and integrating them into the clinical decision-making process. This may involve critically evaluating the AI outputs, considering them along

with other relevant clinical information, and making informed decisions regarding patient care. Conversely, AI developers have a responsibility to ensure the accuracy, reliability, and fairness of their algorithms. This includes addressing biases and continuously improving the algorithms based on feedback from the clinical community. Developers also need to provide clear guidance on the intended use and limitations of AI solutions, enabling physicians to make informed judgments regarding their application in patient care. Healthcare institutions play a crucial role in overseeing the integration of AI solutions into clinical workflows [90]. They must ensure that the necessary infrastructure, training, and support are available for the safe and effective use of AI. This includes developing policies and procedures for managing potential risks and harmful outcomes as well as monitoring and evaluating AI performance to ensure continuous quality improvement. A robust framework for accountability and responsibility enables AI stakeholders to address potential ethical and legal issues more effectively. As a result, trust in AI-driven healthcare solutions can increase, fostering responsible use and improving patient outcomes and overall care quality.

Transparency and explainability

Transparency and accountability are essential elements of ethical AI as they enable healthcare professionals and patients to understand the basis of AI-generated predictions and foster trust in AI [102]. To enhance the accountability of AI, developing interpretable algorithms, visualizing decision-making processes, and providing comprehensible explanations for AI predictions are important [103]. By improving transparency and accountability in AI in healthcare, both healthcare professionals and patients can be supported in making informed decisions, and ethical and legal concerns associated with the use of AI in healthcare can be addressed [104]. However, recognizing the limitations of explainability is important [104–107]. Even with saliency maps that visualize the areas of an image contributing to AI judgment, humans must decipher the meaning behind the explanation. When people favor meanings that confirm their beliefs or hypotheses, this is called confirmation bias. In other words, humans tend to interpret explanations positively even if the AI is not accurate or trustworthy. Recognizing the limitations of explainable AI is important for maintaining a realistic perspective of its potential benefits and drawbacks. By striking a balance among transparency, accountability, and understanding the limitations of explainable AI, healthcare professionals can address the ethical and legal concerns associated with the use of AI in healthcare.

Collaboration among stakeholders

Physicians, AI researchers, and AI developers

Collaboration among physicians, AI researchers, and AI developers is essential to address fairness concerns in AI [108, 110]. Physician participation can provide valuable domain expertise and insights for AI researchers. Recent AI, developed to utilize images [14, 111], and radiologists are particularly well matched. A cycle of improvement can be achieved through communicating expertise in the field with physicians and sharing their experience in using AI in actual medical practice. By working together, stakeholders can identify potential biases and develop effective strategies to mitigate them, ensuring that AI is fair, equitable, and effective. Additionally, empirical research on AI biases is often difficult for independent researchers to analyze as large-scale deployed algorithms are generally proprietary [112]. This makes it difficult to adequately assess bias, and active collaboration of not only AI researchers but also AI developers from companies is essential.

Policymakers and regulatory authorities

Policymakers and regulatory authorities play a crucial role in ensuring AI fairness through establishing comprehensive guidelines, standards, and regulations that govern the development and deployment of AI in healthcare [95, 113–115]. Through proactively shaping policies, they can promote the development of frameworks for AI design, training, and validation, ensuring that AI is built with fairness and inclusivity in mind. Fostering transparency and accountability in AI is also an important aspect of their responsibilities [5, 116]. Policymakers and regulatory authorities can implement requirements for AI developers to disclose their methodologies, data sources, and performance metrics, allowing for a better evaluation and comparison of AI. Furthermore, policymakers and regulatory authorities can allocate resources and funding towards AI innovation and research as well as towards addressing issues on fairness and equity in AI-driven healthcare. Through formulating policies that encourage the development of AI technologies focused on health equity, policymakers and regulators can minimize bias and ensure that all patients benefit from AI, regardless of their background or circumstances. This will contribute to a more equitable healthcare system, in which AI-driven solutions can improve patient outcomes and reduce disparities in access to high-quality care.

Patients and advocacy groups

Patients and advocacy groups serve a crucial function in advancing AI fairness as they contribute valuable insights

and firsthand experiences to the conversation. They can provide valuable insights into the needs and preferences of diverse patient populations, ensuring that AI addresses the specific challenges faced by various communities [86, 117, 118]. As patients directly affected by the AI output, they have a vested interest in identifying areas in which AI may be subject to potential biases and disparities in healthcare outcomes. By collaborating with patients and advocacy groups, physicians, AI researchers, and AI developers can gain a deeper understanding of the unique challenges and concerns faced by various patient populations and promote the development of more equitable and effective AI solutions tailored to individual needs [119]. This can also help build trust in AI-driven healthcare [84, 85]. By giving them a voice in the design, implementation, and evaluation of AI, organizations can demonstrate their commitment to patients by addressing their concerns and enhancing transparency.

Professional associations

Professional associations are pivotal in steering the development and implementation of AI-driven healthcare solutions, addressing ethical challenges, and promoting best practices. Establishing guidelines, standards, and ethical frameworks, fostering interdisciplinary collaborations, and facilitating open dialogue among all stakeholders will bridge this gap. Their unique position allows them to contribute to the development of fair and transparent policies and practices while ensuring that AI technologies are developed and deployed responsibly, equitably, and in the best interests of patients.

Recommendations and future directions

Best practices in healthcare for fairness of AI

To promote AI equity in healthcare and ensure fair and accurate care for all patients, developing a comprehensive strategy that addresses biases at multiple levels as well as ethical, legal, and practical concerns is essential. This approach should foster collaboration among key stakeholders to achieve equitable AI-driven healthcare solutions. We present the following recommendations, called the FAIR (Fairness of Artificial Intelligence Recommendations in healthcare) principles, which aim to ensure fair and equitable AI-driven healthcare solutions (see Table 1):

1. Ensuring diverse and representative data in AI development

Utilize diverse and representative data during AI development and training. This ensures that AI systems can better recognize, diagnose, and treat a wide range of patient conditions, reduce disparities, and promote equity in healthcare outcomes.

2. Independent audits and validation of AI algorithms

Implement regular audits and validate AI algorithms by independent experts or organizations. This ensures objectivity and transparency in the evaluation process and helps identify potential biases, leading to necessary adjustments in the algorithms. Establish a dedicated system within hospitals for algorithm quality control to monitor AI performance continuously, identify potential biases, and update algorithms accordingly.

3. Education on AI biases for clinicians and patients

Educate clinicians and patients on the biases inherent in AI with ongoing education as needed. This will promote a shared understanding and encourage open discussions on the implications of AI in healthcare decision-making by creating channels for feedback and collaboration among healthcare professionals and patients. This can be achieved through workshops, conferences, online forums, and interdisciplinary collaborations.

4. Strengthening data privacy and security measures

Strengthen data privacy and security measures, ensuring compliance with existing legal frameworks such as HIPAA and GDPR. Develop transparent communication protocols to educate patients regarding data usage, storage, and sharing, allowing them to make informed decisions regarding participating in AI-driven healthcare initiatives.

5. Establishing liability and accountability frameworks

A robust framework for liability and accountability should be established, clearly defining the roles and responsibilities of physicians, AI developers, and healthcare institutions. Encourage continuous feedback and improvement of AI algorithms while maintaining transparency and providing guidance on the intended use and limitations of AI solutions.

6. Enhancing AI transparency and explainability

Enhance transparency and explainability in AI by developing interpretable algorithms, visualizing the model's decision-making processes, and providing explanations for AI predictions. Recognize the limitations of explainable AI and address potential biases to prevent overreliance on AI-generated outputs.

Table 1 FAIR (Fairness of Artificial Intelligence Recommendations in healthcare) statement

1. Ensuring diverse and representative data in AI development
Utilize diverse and representative data during AI development and training. This will ensure that AI systems can better recognize, diagnose, and treat a wide range of patient conditions, reducing disparities and promoting equity in healthcare outcomes
2. Independent audits and validation of AI algorithms
Implement regular audits and validation of AI algorithms by independent experts or organizations. This will ensure objectivity and transparency in the evaluation process and help identify potential biases, leading to necessary adjustments in the algorithms. Establish a dedicated system within hospitals for algorithm quality control to continuously monitor AI performance, identify potential biases, and update algorithms accordingly
3. Education on AI biases for clinicians and patients
Educate clinicians and patients on the biases inherent in AI with ongoing education as needed. This will promote a shared understanding and encourage open discussions on the implications of AI in healthcare decision-making by creating channels for feedback and collaboration among healthcare professionals and patients. This can be achieved through workshops, conferences, online forums, and interdisciplinary collaborations
4. Strengthening data privacy and security measures
Strengthen data privacy and security measures, ensuring compliance with existing legal frameworks such as HIPAA and GDPR. Develop transparent communication protocols to educate patients regarding data usage, storage, and sharing, allowing them to make informed decisions regarding participating in AI-driven healthcare initiatives
5. Establishing liability and accountability frameworks
Establish a robust framework for liability and accountability, clearly defining the roles and responsibilities of physicians, AI developers, and healthcare institutions. Encourage continuous feedback and improvement of AI algorithms, while maintaining transparency and providing guidance on AI solutions' intended use and limitations
6. Enhancing AI transparency and explainability
Enhance transparency and explainability in AI by developing interpretable algorithms, visualizing decision-making processes, and providing understandable explanations for AI predictions. Recognize the limitations of explainable AI and address potential biases to prevent overreliance on AI-generated outputs
7. Collaboration between physicians, AI researchers, and developers
Foster collaboration between physicians, AI researchers, and developers to share expertise, identify potential biases, and develop strategies to mitigate them. Encourage active participation of AI companies to support independent research on AI biases and improve algorithm fairness
8. Policymaker and regulatory authority involvement
Engage policymakers and regulatory authorities in developing comprehensive guidelines, standards, and regulations to ensure AI fairness, promote transparency and accountability, and allocate resources to support research and innovation in AI-driven healthcare
9. Patient and advocacy group participation in AI development and evaluation
Involve patients and advocacy groups in the design, implementation, and evaluation of AI solutions, giving them a voice in the decision-making process. Leverage their insights and experiences to address unique challenges and promote the development of equitable AI solutions tailored to individual needs
10. Professional association support
Professional associations help establish guidelines, standards, and ethical frameworks, and promote interdisciplinary collaborations and open discussions among all stakeholders. Their unique position enables them to aid in creating fair and transparent policies and practices

AI artificial intelligence, *HIPAA* the health insurance portability and accountability act, *GDPR* the general data protection regulation

7. Collaboration between physicians, AI researchers, and developers

Foster collaboration among physicians, AI researchers, and developers to share expertise, identify potential biases, and develop strategies to mitigate them. Active participation of AI companies should be encouraged to support independent research on AI biases and improve algorithm fairness.

8. Policymaker and regulatory authority involvement

Engage policymakers and regulatory authorities in developing comprehensive guidelines, standards, and regulations to ensure AI fairness; promote transparency and accountability; and allocate resources to support research and innovation in AI-driven healthcare.

9. Patient and advocacy group participation in AI development and evaluation

Involve patients and advocacy groups in the design, implementation, and evaluation of AI solutions, giving them a voice in the decision-making process. Leverage their insights and experiences to address unique challenges and promote the development of equitable AI solutions tailored to individual needs.

10. Professional association support

Professional associations help establish guidelines, standards, and ethical frameworks, and promote interdisciplinary collaborations and open discussions among all stakeholders. Their unique position enables them to create fair and transparent policies and practices.

By implementing these recommendations and addressing biases in data and algorithms, stakeholders in the AI-driven healthcare sector can foster trust, transparency, and inclusivity. This will ensure that AI technologies are developed and deployed ethically, responsibly, and equitably for the benefit of all patients regardless of their differences. Ultimately, this approach contributes to a more equitable healthcare system and improves patient outcomes.

Research gaps and future work

Several research gaps and opportunities for future research to address concerns regarding AI bias and fairness exist. Randomized controlled trials should be conducted to explore the potential of AI in improving patient care and outcomes. These trials should include diverse populations and AI that are tailored to the specific needs of different demographic groups. The long-term impact of AI adoption in healthcare on patient treatment, outcomes, and physician workload should be investigated. The models should be monitored regularly to address biases that may emerge over time [89]. Developing new technologies for explainability and transparency is necessary to enable healthcare professionals and patients to better understand AI-generated predictions, foster trust in AI, and ensure its ethical deployment.

Conclusion

In this review, we first defined fairness in AI in the healthcare domain, introduced various biases with examples and potential countermeasures, and emphasized the importance of collaborating with stakeholders. Subsequently, we discussed important ethical and legal issues. As a result, we summarized the best practices into the FAIR statement. This includes preparing diverse and representative data, continuously validating AI, educating physicians and patients, and emphasizing the importance of interdisciplinary collaboration. Although implementing each best practice is difficult, these efforts have become increasingly important as AI integration advances in the medical field. Furthermore, AI technology is still evolving, and the situation is constantly changing, with new challenges emerging one after another. The emergence of tools like Chat Generative Pre-trained Transformer (ChatGPT) is expected to greatly change white-collar jobs, and physicians are no exception [120–122]. We are currently in an era in which physicians require flexible thinking and the ability to respond quickly to new technologies.

Since its inception, AI has influenced several aspects of modern society, leading to notable advancements. The medical field has not remained untouched by this wave of change, with radiology particularly poised to harness the power of AI [123]. Given this unique position, the radiology community has a vital responsibility to share its experience in actively integrating AI into medicine [124–126], providing invaluable guidance and insight for other medical specialties. As pioneers in the implementation of AI, radiologists should champion AI equity in healthcare. Our early experience navigating the complex landscape of AI adoption and overcoming the challenges associated with its deployment can serve as a roadmap for other medical professionals. By doing so, we can ensure that AI benefits all patients, regardless of their backgrounds, and contributes to the greater good of society.

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Declarations

Conflict of interest There are no conflicts of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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References

1. Mittelstadt BD, Allo P, Taddeo M, Wachter S, Floridi L. The ethics of algorithms: Mapping the debate. Big Data Soc. 2016;3:2053951716679679. SAGE Publications Ltd. https://doi.org/10.1177/2053951716679679
2. Jobin A, Ienca M, Vayena E. Artificial intelligence: the global landscape of ethics guidelines. Nat Mach Intell. 2019;1:389–99. https://doi.org/10.1038/s42256-019-0088-2.
3. Tsamados A, Aggarwal N, Cowls J, Morley J, Roberts H, Taddeo M, et al. The ethics of algorithms: key problems and solutions. AI & Soc. 2022;37:215–30. https://doi.org/10.1007/s00146-021-01154-8.

4. Kleinberg J, Lakkaraju H, Leskovec J, Ludwig J, Mullainathan S. Human decisions and machine predictions. Q J Econ. 2018;133:237–93. https://doi.org/10.1093/qje/qjx032.
5. Edwards V. Slave to the algorithm: Why a right to an explanation is probably not the remedy you are looking for. Duke Law Technol Rev. https://heinonline.org/hol-cgi-bin/get_pdf.cgi?handle=hein.journals/dltr16&section=3.
6. Binns R. Fairness in machine learning: Lessons from political philosophy. In: Friedler SA, Wilson C, editors. Proceedings of the 1st conference on fairness, accountability and transparency. PMLR; 2018. p. 149–59
7. Selbst AD, Boyd D, Friedler SA, Venkatasubramanian S, Vertesi J. Fairness and abstraction in sociotechnical systems. In: Proceedings of the conference on fairness, accountability, and transparency. New York, USA. Association for Computing Machinery; 2019. https://doi.org/10.1145/3287560.3287598
8. Wong P-H. Democratizing algorithmic fairness. Philos Technol. 2020;33:225–44. https://doi.org/10.1007/s13347-019-00355-w.
9. Abebe R, Barocas S, Kleinberg J, Levy K, Raghavan M, Robinson DG. Roles for computing in social change. In: Proceedings of the 2020 conference on fairness, accountability, and transparency. New York, USA. Association for Computing Machinery; 2020. https://doi.org/10.1145/3351095.3372871
10. Bærøe K, Gundersen T, Henden E, Rommetveit K. Can medical algorithms be fair? Three ethical quandaries and one dilemma. BMJ Health Care Inform. 2022. https://doi.org/10.1136/bmjhci-2021-100445.
11. Rajkomar A, Hardt M, Howell MD, Corrado G, Chin MH. Ensuring fairness in machine learning to advance health equity. Ann Intern Med. 2018;169:866–72. https://doi.org/10.7326/M18-1990.
12. Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A. A survey on bias and fairness in machine learning. ACM Comput Surv. 2021;54:1–35. https://doi.org/10.1145/3457607.
13. World Medical Association. Declaration of Geneva. World Medical Association; 1983
14. Ueda D, Shimazaki A, Miki Y. Technical and clinical overview of deep learning in radiology. Jpn J Radiol. 2019;37:15–33. https://doi.org/10.1007/s11604-018-0795-3.
15. Yuba M, Iwasaki K. Systematic analysis of the test design and performance of AI/ML-based medical devices approved for triage/detection/diagnosis in the USA and Japan. Sci Rep. 2022;12:16874. https://doi.org/10.1038/s41598-022-21426-7.
16. Zhu S, Gilbert M, Chetty I, The SF. landscape of FDA-approved artificial intelligence/machine learning-enabled medical devices: an analysis of the characteristics and intended use. Int J Med Inform. 2021. https://doi.org/10.1016/j.ijmedinf.2022.104828.
17. Kelly BS, Judge C, Bollard SM, Clifford SM, Healy GM, Aziz A, et al. Radiology artificial intelligence: a systematic review and evaluation of methods (RAISE). Eur Radiol. 2022;32:7998–8007. https://doi.org/10.1007/s00330-022-08784-6.
18. Rimmer A. Radiologist shortage leaves patient care at risk, warns royal college. BMJ. 2017. https://doi.org/10.1136/bmj.j4683.
19. Nakajima Y, Yamada K, Imamura K, Kobayashi K. Radiologist supply and workload: international comparison–working group of Japanese college of radiology. Radiat Med. 2008;26:455–65. https://doi.org/10.1007/s11604-008-0259-2.
20. Gourd EE. UK radiologist staffing crisis reaches critical levels. Lancet Oncol. 2017. https://doi.org/10.1016/S1470-2045(17)30806-9.
21. Mollura DJ, Culp MP, Pollack E, Battino G, Scheel JR, Mango VL, et al. Artificial intelligence in low- and middle-income countries: innovating global health radiology. Radiology. 2020;297:513–20. https://doi.org/10.1148/radiol.2020201434.
22. Toda N, Hashimoto M, Iwabuchi Y, Nagasaka M, Takeshita R, Yamada M, et al. Validation of deep learning-based computer-aided detection software use for interpretation of pulmonary abnormalities on chest radiographs and examination of factors that influence readers' performance and final diagnosis. Jpn J Radiol. 2023;41:38–44. https://doi.org/10.1007/s11604-022-01330-w.
23. Ueda D, Ehara S, Yamamoto A, Iwata S, Abo K, Walston SL, et al. Development and validation of artificial intelligence-based method for diagnosis of mitral regurgitation from chest radiographs. Radiol Artif Intell. 2022. https://doi.org/10.1148/ryai.210221.
24. Ueda D, Yamamoto A, Ehara S, Iwata S, Abo K, Walston SL, et al. Artificial intelligence-based detection of aortic stenosis from chest radiographs. Eur Heart J Digit Health. 2022;3:20–8. https://doi.org/10.1093/ehjdh/ztab102.
25. Matsumoto T, Ehara S, Walston SL, Mitsuyama Y, Miki Y, Ueda D. Artificial intelligence-based detection of atrial fibrillation from chest radiographs. Eur Radiol. 2022;32:5890–7. https://doi.org/10.1007/s00330-022-08752-0.
26. Ueda D, Yamamoto A, Takashima T, Onoda N, Noda S, Kashiwagi S, et al. Visualizing "featureless" regions on mammograms classified as invasive ductal carcinomas by a deep learning algorithm: the promise of AI support in radiology. Jpn J Radiol. 2021;39:333–40. https://doi.org/10.1007/s11604-020-01070-9.
27. Uematsu T, Nakashima K, Harada TL, Nasu H, Igarashi T. Comparisons between artificial intelligence computer-aided detection synthesized mammograms and digital mammograms when used alone and in combination with tomosynthesis images in a virtual screening setting. Jpn J Radiol. 2023;41:63–70. https://doi.org/10.1007/s11604-022-01327-5.
28. Ueda D, Yamamoto A, Onoda N, Takashima T, Noda S, Kashiwagi S, et al. Development and validation of a deep learning model for detection of breast cancers in mammography from multi-institutional datasets. PLoS ONE. 2022. https://doi.org/10.1371/journal.pone.0265751.
29. Honjo T, Ueda D, Katayama Y, Shimazaki A, Jogo A, Kageyama K, et al. Visual and quantitative evaluation of microcalcifications in mammograms with deep learning-based super-resolution. Eur J Radiol. 2022. https://doi.org/10.1016/j.ejrad.2022.110433.
30. Ueda D, Yamamoto A, Takashima T, Onoda N, Noda S, Kashiwagi S, et al. Training, validation, and test of deep learning models for classification of receptor expressions in breast cancers from mammograms. JCO Precis Oncol. 2021;5:543–51. https://doi.org/10.1200/PO.20.00176.
31. Ozaki J, Fujioka T, Yamaga E, Hayashi A, Kujiraoka Y, Imokawa T, et al. Deep learning method with a convolutional neural network for image classification of normal and metastatic axillary lymph nodes on breast ultrasonography. Jpn J Radiol. 2022;40:814–22. https://doi.org/10.1007/s11604-022-01261-6.
32. Ichikawa Y, Kanii Y, Yamazaki A, Nagasawa N, Nagata M, Ishida M, et al. Deep learning image reconstruction for improvement of image quality of abdominal computed tomography: comparison with hybrid iterative reconstruction. Jpn J Radiol. 2021;39:598–604. https://doi.org/10.1007/s11604-021-01089-6.
33. Nakai H, Fujimoto K, Yamashita R, Sato T, Someya Y, Taura K, et al. Convolutional neural network for classifying primary liver cancer based on triple-phase CT and tumor marker information: a pilot study. Jpn J Radiol. 2021;39:690–702. https://doi.org/10.1007/s11604-021-01106-8.
34. Okuma T, Hamamoto S, Maebayashi T, Taniguchi A, Hirakawa K, Matsushita S, et al. Quantitative evaluation of COVID-19 pneumonia severity by CT pneumonia analysis algorithm using deep learning technology and blood test results. Jpn J Radiol. 2021;39:956–65. https://doi.org/10.1007/s11604-021-01134-4.

35. Kitahara H, Nagatani Y, Otani H, Nakayama R, Kida Y, Sonoda A, et al. A novel strategy to develop deep learning for image super-resolution using original ultra-high-resolution computed tomography images of lung as training dataset. Jpn J Radiol. 2022;40:38–47. https://doi.org/10.1007/s11604-021-01184-8.
36. Kaga T, Noda Y, Mori T, Kawai N, Miyoshi T, Hyodo F, et al. Unenhanced abdominal low-dose CT reconstructed with deep learning-based image reconstruction: Image quality and anatomical structure depiction. Jpn J Radiol. 2022;40:703–11. https://doi.org/10.1007/s11604-022-01259-0.
37. Ohno Y, Aoyagi K, Arakita K, Doi Y, Kondo M, Banno S, et al. Newly developed artificial intelligence algorithm for COVID-19 pneumonia: Utility of quantitative CT texture analysis for prediction of favipiravir treatment effect. Jpn J Radiol. 2022;40:800–13. https://doi.org/10.1007/s11604-022-01270-5.
38. Matsukiyo R, Ohno Y, Matsuyama T, Nagata H, Kimata H, Ito Y, et al. Deep learning-based and hybrid-type iterative reconstructions for CT: Comparison of capability for quantitative and qualitative image quality improvements and small vessel evaluation at dynamic CE-abdominal CT with ultra-high and standard resolutions. Jpn J Radiol. 2021;39:186–97. https://doi.org/10.1007/s11604-020-01045-w.
39. Koretsune Y, Sone M, Sugawara S, Wakatsuki Y, Ishihara T, Hattori C, et al. Validation of a convolutional neural network for the automated creation of curved planar reconstruction images along the main pancreatic duct. Jpn J Radiol. 2023;41:228–34. https://doi.org/10.1007/s11604-022-01339-1.
40. Anai K, Hayashida Y, Ueda I, Hozuki E, Yoshimatsu Y, Tsukamoto J, et al. The effect of CT texture-based analysis using machine learning approaches on radiologists' performance in differentiating focal-type autoimmune pancreatitis and pancreatic duct carcinoma. Jpn J Radiol. 2022;40:1156–65. https://doi.org/10.1007/s11604-022-01298-7.
41. Cay N, Mendi BAR, Batur H, Erdogan F. Discrimination of lipoma from atypical lipomatous tumor/well-differentiated liposarcoma using magnetic resonance imaging radiomics combined with machine learning. Jpn J Radiol. 2022;40:951–60. https://doi.org/10.1007/s11604-022-01278-x.
42. Wong LM, Ai QYH, Mo FKF, Poon DMC, King AD. Convolutional neural network in nasopharyngeal carcinoma: How good is automatic delineation for primary tumor on a non-contrast-enhanced fat-suppressed T2-weighted MRI? Jpn J Radiol. 2021;39:571–9. https://doi.org/10.1007/s11604-021-01092-x.
43. Yasaka K, Akai H, Sugawara H, Tajima T, Akahane M, Yoshioka N, et al. Impact of deep learning reconstruction on intracranial 1.5 T magnetic resonance angiography. Jpn J Radiol. 2022. https://doi.org/10.1007/s11604-021-01225-2.
44. Nomura Y, Hanaoka S, Nakao T, Hayashi N, Yoshikawa T, Miki S, et al. Performance changes due to differences in training data for cerebral aneurysm detection in head MR angiography images. Jpn J Radiol. 2021;39:1039–48. https://doi.org/10.1007/s11604-021-01153-1.
45. Ishihara M, Shiiba M, Maruno H, Kato M, Ohmoto-Sekine Y, Antoine C, et al. Detection of intracranial aneurysms using deep learning-based CAD system: usefulness of the scores of CNN's final layer for distinguishing between aneurysm and infundibular dilatation. Jpn J Radiol. 2023;41:131–41. https://doi.org/10.1007/s11604-022-01341-7.
46. Miki S, Nakao T, Nomura Y, Okimoto N, Nyunoya K, Nakamura Y, et al. Computer-aided detection of cerebral aneurysms with magnetic resonance angiography: usefulness of volume rendering to display lesion candidates. Jpn J Radiol. 2021;39:652–8. https://doi.org/10.1007/s11604-021-01099-4.
47. Nakaura T, Kobayashi N, Yoshida N, Shiraishi K, Uetani H, Nagayama Y, et al. Update on the use of artificial intelligence in hepatobiliary MR imaging. Magn Reson Med Sci. 2023;22:147–56. https://doi.org/10.2463/mrms.rev.2022-0102.
48. Naganawa S, Ito R, Kawai H, Kawamura M, Taoka T, Sakai M, et al. MR imaging of endolymphatic hydrops in five minutes. Magn Reson Med Sci. 2022;21:401–5. https://doi.org/10.2463/mrms.ici.2021-0022.
49. Kabasawa H, Kiryu S. Pulse sequences and reconstruction in Fast MR imaging of the liver. Magn Reson Med Sci. 2023;22:176–90. https://doi.org/10.2463/mrms.rev.2022-0114.
50. Iwamura M, Ide S, Sato K, Kakuta A, Tatsuo S, Nozaki A, et al. Thin-slice two-dimensional T2-weighted imaging with deep learning-based reconstruction: Improved lesion detection in the brain of patients with multiple sclerosis. Magn Reson Med Sci. 2023. https://doi.org/10.2463/mrms.mp.2022-0112.
51. Nai YH, Loi HY, O'Doherty S, Tan TH, Reilhac A. Comparison of the performances of machine learning and deep learning in improving the quality of low dose lung cancer PET images. Jpn J Radiol. 2022;40:1290–9. https://doi.org/10.1007/s11604-022-01311-z.
52. Nakao T, Hanaoka S, Nomura Y, Hayashi N, Abe O. Anomaly detection in chest 18F-FDG PET/CT by bayesian deep learning. Jpn J Radiol. 2022;40:730–9. https://doi.org/10.1007/s11604-022-01249-2.
53. Kumamaru KK, Machitori A, Koba R, Ijichi S, Nakajima Y, Aoki S. Global and Japanese regional variations in radiologist potential workload for computed tomography and magnetic resonance imaging examinations. Jpn J Radiol. 2018;36:273–81. https://doi.org/10.1007/s11604-018-0724-5.
54. Cozzi D, Cavigli E, Moroni C, Smorchkova O, Zantonelli G, Pradella S, et al. Ground-glass opacity (GGO): a review of the differential diagnosis in the era of COVID-19. Jpn J Radiol. 2021;39:721–32. https://doi.org/10.1007/s11604-021-01120-w.
55. Aoki R, Iwasawa T, Hagiwara E, Komatsu S, Utsunomiya D, Ogura T. Pulmonary vascular enlargement and lesion extent on computed tomography are correlated with COVID-19 disease severity. Jpn J Radiol. 2021;39:451–8. https://doi.org/10.1007/s11604-020-01085-2.
56. Zhu QQ, Gong T, Huang GQ, Niu ZF, Yue T, Xu FY, et al. Pulmonary artery trunk enlargement on admission as a predictor of mortality in in-hospital patients with COVID-19. Jpn J Radiol. 2021;39:589–97. https://doi.org/10.1007/s11604-021-01094-9.
57. Fukuda A, Yanagawa N, Sekiya N, Ohyama K, Yomota M, Inui T, et al. An analysis of the radiological factors associated with respiratory failure in COVID-19 pneumonia and the CT features among different age categories. Jpn J Radiol. 2021;39:783–90. https://doi.org/10.1007/s11604-021-01118-4.
58. Özer H, Kılınçer A, Uysal E, Yormaz B, Cebeci H, Durmaz MS, et al. Diagnostic performance of radiological society of North America structured reporting language for chest computed tomography findings in patients with COVID-19. Jpn J Radiol. 2021;39:877–88. https://doi.org/10.1007/s11604-021-01128-2.
59. Zhuang Y, Lin L, Xu X, Xia T, Yu H, Fu G, et al. Dynamic changes on chest CT of COVID-19 patients with solitary pulmonary lesion in initial CT. Jpn J Radiol. 2021;39:32–9. https://doi.org/10.1007/s11604-020-01037-w.
60. Kanayama A, Tsuchihashi Y, Otomi Y, Enomoto H, Arima Y, Takahashi T, et alAssociation of severe COVID-19 outcomes with radiological scoring and cardiomegaly: Findings from the COVID-19 inpatients database, Japan. Jpn J Radiol. 2022 https://doi.org/10.1007/s11604-022-01300-2
61. Inui S, Fujikawa A, Gonoi W, Kawano S, Sakurai K, Uchida Y, et al. Comparison of CT findings of coronavirus disease 2019 (COVID-19) pneumonia caused by different major variants. Jpn J Radiol. 2022;40:1246–56. https://doi.org/10.1007/s11604-022-01301-1.

62. Walston SL, Matsumoto T, Miki Y, Ueda D. Artificial intelligence-based model for COVID-19 prognosis incorporating chest radiographs and clinical data; a retrospective model development and validation study. Br J Radiol. 2022;95:20220058. https://doi.org/10.1259/bjr.20220058.
63. Matsumoto T, Walston SL, Walston M, Kabata D, Miki Y, Shiba M, et al. Deep learning-based time-to-death prediction model for COVID-19 patients using clinical data and chest radiographs. J Digit Imaging. 2023;36:178–88. https://doi.org/10.1007/s10278-022-00691-y.
64. Wynants L, Van Calster B, Collins GS, Riley RD, Heinze G, Schuit E, et al. Prediction models for diagnosis and prognosis of Covid-19: Systematic review and critical appraisal. BMJ. 2020;369:m1328. https://doi.org/10.1136/bmj.m1328.
65. Marmot M, Bell R. Fair society, healthy lives. Public Health. 2012;126(Suppl 1):S4–10. https://doi.org/10.1016/j.puhe.2012.05.014.
66. Ricci Lara MA, Echeveste R, Ferrante E. Addressing fairness in artificial intelligence for medical imaging. Nat Commun. 2022;13:4581. https://doi.org/10.1038/s41467-022-32186-3.
67. Gebru T, Morgenstern J, Vecchione B, Vaughan JW, Wallach H 3rd, Iii HD, et al. Datasheets for datasets. Commun ACM. 2021. New York: Association for Computing Machinery;64:86–92. https://doi.org/10.1145/3458723
68. Wenger NK. Women and coronary heart disease: A century after Herrick: Understudied, underdiagnosed, and undertreated. Circulation. 2012;126:604–11. https://doi.org/10.1161/CIRCULATIONAHA.111.086892.
69. Appelman Y, van Rijn BB, Ten Haaf ME, Boersma E, Peters SAE. Sex differences in cardiovascular risk factors and disease prevention. Atherosclerosis. 2015;241:211–8. https://doi.org/10.1016/j.atherosclerosis.2015.01.027.
70. Adamson AS, Smith A. Machine learning and health care disparities in dermatology. JAMA Dermatol. 2018;154:1247–8. https://doi.org/10.1001/jamadermatol.2018.2348.
71. Navarrete-Dechent C, Dusza SW, Liopyris K, Marghoob AA, Halpern AC, Marchetti MA. Automated dermatological diagnosis: hype or reality? J Invest Dermatol. 2018;138:2277–9. https://doi.org/10.1016/j.jid.2018.04.040.
72. Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann EK. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLOS Med. 2018. https://doi.org/10.1371/journal.pmed.1002683.
73. Wolff RF, Moons KGM, Riley RD, Whiting PF, Westwood M, Collins GS, et al. PROBAST: a tool to assess the risk of bias and applicability of prediction model studies. Ann Intern Med. 2019;170:51–8. https://doi.org/10.7326/M18-1376.
74. Park SH, Han K. Methodologic guide for evaluating clinical performance and effect of artificial intelligence technology for medical diagnosis and prediction. Radiology. 2018;286:800–9. https://doi.org/10.1148/radiol.2017171920.
75. Bluemke DA, Moy L, Bredella MA, Ertl-Wagner BB, Fowler KJ, Goh VJ, et al. Assessing radiology research on artificial intelligence: a brief guide for authors, reviewers, and readers-from the radiology editorial board. Radiology. 2020;294:487–9. https://doi.org/10.1148/radiol.2019192515.
76. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science. 2019;366:447–53. https://doi.org/10.1126/science.aax2342.
77. Meyer IH. Prejudice, social stress, and mental health in lesbian, gay, and bisexual populations: conceptual issues and research evidence. Psychol Bull. 2003;129:674–97. https://doi.org/10.1037/0033-2909.129.5.674.
78. Anderson M, Anderson SL. How should AI be developed, validated, and implemented in patient care? AMA J Ethics. Am Med Assoc. 2019;21:E125-130. https://doi.org/10.1001/amajethics.2019.125.
79. Dratsch T, Chen X, Rezazade Mehrizi M, Kloeckner R, Mähringer-Kunz A, Püsken M, et al. Automation bias in mammography: The impact of artificial intelligence BI-RADS suggestions on reader performance. Radiology. 2023. https://doi.org/10.1148/radiol.222176.
80. Walsh CG, Chaudhry B, Dua P, Goodman KW, Kaplan B, Kavuluru R, et al. Stigma, biomarkers, and algorithmic bias: recommendations for precision behavioral health with artificial intelligence. JAMIA Open. 2020;3:9–15. https://doi.org/10.1093/jamiaopen/ooz054.
81. Lehman CD, Wellman RD, Buist DSM, Kerlikowske K, Tosteson ANA, Miglioretti DL, et al. Diagnostic accuracy of digital screening mammography with and without computer-aided detection. JAMA Intern Med. 2015;175:1828–37. https://doi.org/10.1001/jamainternmed.2015.5231.
82. Phansalkar S, van der Sijs H, Tucker AD, Desai AA, Bell DS, Teich JM, et al. Drug—drug interactions that should be non-interruptive in order to reduce alert fatigue in electronic health records. J Am Med Inform Assoc Oxford. 2012. Academic Press;20:489–93
83. Fiscella K, Williams DR. Health disparities based on socioeconomic inequities: implications for urban health care. Acad Med. 2004;79:1139–47. https://doi.org/10.1097/00001888-200412000-00004.
84. Gamble VN. Under the shadow of Tuskegee: African Americans and health care. Am J Public Health. 1997;87:1773–8. https://doi.org/10.2105/ajph.87.11.1773.
85. Boulware LE, Cooper LA, Ratner LE, LaVeist TA, Powe NR. Race and trust in the health care system. Public Health Rep. 2003;118:358–65. https://doi.org/10.1093/phr/118.4.358.
86. Minkler M, Wallerstein N. Community-based participatory research for health: From process to outcomes. John Wiley & Sons; 2011
87. Haibe-Kains B, Adam GA, Hosny A, Khodakarami F, Massive Analysis Quality Control (MAQC) Society Board of Directors, Waldron L, et al. Transparency and reproducibility in artificial intelligence. Nature. 2020. p. E14–6
88. Buolamwini J, Gebru T. Gender shades: Intersectional accuracy disparities in commercial gender classification. In: Friedler SA, Wilson C, editors. Proceedings of the 1st conference on fairness, accountability and transparency. PMLR; 2018. p. 77–91
89. Finlayson SG, Subbaswamy A, Singh K, Bowers J, Kupke A, Zittrain J, et al. The clinician and dataset shift in artificial intelligence. N Engl J Med. 2021;385:283–6. https://doi.org/10.1056/NEJMc2104626.
90. Feng J, Phillips RV, Malenica I, Bishara A, Hubbard AE, Celi LA, et al. Clinical artificial intelligence quality improvement: Towards continual monitoring and updating of AI algorithms in healthcare. npj Digit Med. 2022. https://doi.org/10.1038/s41746-022-00611-y.
91. Cummings ML. Automation bias in intelligent time critical decision support systems. Decision making in aviation Routledge. 2004. https://doi.org/10.2514/6.2004-6313.
92. Gafni A, Charles C, McMaster University. Centre for health economics and policy analysis, Whelan T. Shared decision-making in the medical encounter: What does it mean?, or, it takes at least two to tango. Centre for Health Economics and Policy Analysis, McMaster University; 1994.
93. Coulter A, Collins A. Making shared decision-making a reality: No decision about me, without me. London: The King's Fund; 2011;621.

94. Vayena E, Blasimme A, Cohen IG. Machine learning in medicine: addressing ethical challenges. PLOS Med. 2018. https://doi.org/10.1371/journal.pmed.1002689.
95. Price WN 2nd, Cohen IG. Privacy in the age of medical big data. Nat Med. 2019;25:37–43. https://doi.org/10.1038/s41591-018-0272-7.
96. Grady C. Enduring and emerging challenges of informed consent. N Engl J Med. 2015;372:2172. https://doi.org/10.1056/NEJMc1503813.
97. Emanuel EJ, Wendler D, Grady C. What makes clinical research ethical? JAMA. 2000;283:2701–11. https://doi.org/10.1001/jama.283.20.2701.
98. Abouelmehdi K, Beni-Hessane A, Khaloufi H. Big healthcare data: Preserving security and privacy. J Big Data. 2018;5:1–18. https://doi.org/10.1186/s40537-017-0110-7.
99. Taylor L, Floridi L, van der Sloot B. Group privacy: New challenges of data technologies. Springer; 2016
100. Neri E, Coppola F, Miele V, Bibbolino C, Grassi R. Artificial intelligence: who is responsible for the diagnosis? Radiol Med. 2020;125:517–21. https://doi.org/10.1007/s11547-020-01135-9.
101. Price WN 2nd, Gerke S, Cohen IG. Potential liability for physicians using artificial intelligence. JAMA. 2019;322:1765–6. https://doi.org/10.1001/jama.2019.15064.
102. van der Velden BHM, Kuijf HJ, Gilhuijs KGA, Viergever MA. Explainable artificial intelligence (XAI) in deep learning-based medical image analysis. Med Image Anal. 2022. https://doi.org/10.1016/j.media.2022.102470.
103. Goebel R, Chander A, Holzinger K, Lecue F, Akata Z, Stumpf S, et al. Explainable AI: The new 42? Machine Learning and Knowledge Extraction. Springer International Publishing. 2018;295–303. https://doi.org/10.1007/978-3-319-99740-7_21
104. Gilpin LH, Bau D, Yuan BZ, Bajwa A, Specter M, Kagal L. Explaining explanations: An overview of interpretability of machine learning [Internet]; 2018. cs.AI. http://arxiv.org/abs/1806.00069. https://doi.org/10.1109/DSAA.2018.00018
105. Rudin C. Stop explaining Black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell. 2019;1:206–15. https://doi.org/10.1038/s42256-019-0048-x.
106. Miller T. Explanation in artificial intelligence: Insights from the social sciences. Artif Intell. 2019;267:1–38. https://doi.org/10.1016/j.artint.2018.07.007.
107. Lipton ZC. The mythos of model interpretability. In: machine learning, the concept of interpretability is both important and slippery. Queueing Syst. 2018. Association for Computing Machinery;16:31–57. https://doi.org/10.1145/3236386.3241340
108. Topol EJ. High-performance medicine: The convergence of human and artificial intelligence. Nat Med. 2019;25:44–56. https://doi.org/10.1038/s41591-018-0300-7.
109. Rajkomar A, Dean J, Kohane I. Machine learning in medicine. N Engl J Med. 2019;380:1347–58. https://doi.org/10.1056/NEJMra1814259.
110. Obermeyer Z, Emanuel EJ. Predicting the future - Big data, machine learning, and clinical medicine. N Engl J Med. 2016;375:1216–9. https://doi.org/10.1056/NEJMp1606181.
111. Yasaka K, Akai H, Kunimatsu A, Kiryu S, Abe O. Deep learning with convolutional neural network in radiology. Jpn J Radiol. 2018;36:257–72. https://doi.org/10.1007/s11604-018-0726-3.
112. Burrell J. How the machine "thinks": Understanding opacity in machine learning algorithms. Big Data Soc. 2016;3:2053951715622512. SAGE Publications Ltd;3. https://doi.org/10.1177/2053951715622512
113. Char DS, Shah NH, Magnus D. Implementing machine learning in health care - Addressing ethical challenges. N Engl J Med. 2018;378:981–3. https://doi.org/10.1056/NEJMp1714229.
114. Morley J, Machado CCV, Burr C, Cowls J, Joshi I, Taddeo M, et al. The ethics of AI in health care: a mapping review. Soc Sci Med. 2020. https://doi.org/10.1016/j.socscimed.2020.113172.
115. Epstein RM, Fiscella K, Lesser CS, Stange KC. Why the nation needs a policy push on patient-centered health care. Health Aff (Millwood). 2010;29:1489–95. https://doi.org/10.1377/hlthaff.2009.0888.
116. Barredo Arrieta A, Díaz-Rodríguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, et al. Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion. 2020;58:82–115. https://doi.org/10.1016/j.inffus.2019.12.012.
117. Vayena E, Dzenowagis J, Brownstein JS, Sheikh A. Policy implications of big data in the health sector. Bull World Health Organ. 2018;96:66–8. https://doi.org/10.2471/BLT.17.197426.
118. Mittelstadt BD, Floridi L. The ethics of big data: current and foreseeable issues in biomedical contexts. Sci Eng Ethics. 2016;22:303–41. https://doi.org/10.1007/s11948-015-9652-2.
119. Coulter A, Ellins J. Effectiveness of strategies for informing, educating, and involving patients. BMJ. 2007;335:24–7. https://doi.org/10.1136/bmj.39246.581169.80.
120. Open AI. GPT-4 technical report [Internet]. arXiv [cs.CL]; 2023. http://arxiv.org/abs/2303.08774
121. Eloundou T, Manning S, Mishkin P, Rock D. GPTs are GPTs: An early look at the labor market impact potential of large language models [Internet]. arXiv; 2023. econ.GN. http://arxiv.org/abs/2303.10130
122. Ueda D, Walston SL, Matsumoto T, Deguchi R, Tatekawa H, Miki Y. Evaluating GPT-4-based ChatGPT's clinical potential on the NEJM quiz [Internet]; 2023. medRxiv. https://www.medrxiv.org/content/10.1101/2023.05.04.23289493v1
123. Davis MA, Lim N, Jordan J, Yee J, Gichoya JW, Lee R. Imaging artificial intelligence: A framework for radiologists to address health equity, from the AJR special series on DEI. AJR Am J Roentgenol. 2023. American Roentgen Ray Society. https://doi.org/10.2214/AJR.22.28802
124. Shimazaki A, Ueda D, Choppin A, Yamamoto A, Honjo T, Shimahara Y, et al. Deep learning-based algorithm for lung cancer detection on chest radiographs using the segmentation method. Sci Rep. 2022;12:727. https://doi.org/10.1038/s41598-021-04667-w.
125. Ueda D, Yamamoto A, Nishimori M, Shimono T, Doishita S, Shimazaki A, et al. Deep learning for MR angiography: automated detection of cerebral aneurysms. Radiology. 2019;290:187–94. https://doi.org/10.1148/radiol.2018180901.
126. Ueda D, Yamamoto A, Shimazaki A, Walston SL, Matsumoto T, Izumi N, et al. Artificial intelligence-supported lung cancer detection by multi-institutional readers with multi-vendor chest radiographs: a retrospective clinical validation study. BMC Cancer. 2021;21:1120. https://doi.org/10.1186/s12885-021-08847-9.

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RESEARCH

A fairness-focused approach to recidivism prediction: implications for accuracy, trust, and equity

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Abstract
Fairness in AI has emerged as a critical priority, particularly in high-stakes domains like criminal justice, where algorithmic decisions profoundly affect individuals and demographic groups. This study addresses persistent challenges of bias in recidivism prediction by evaluating the integrated application of fairness-enhancing techniques across the AI pipeline. Specifically, we analyze the integrated use of Reweighing, Adversarial Learning, Disparate Impact Remover, Exponentiated Gradient Reduction, Reject Option Classification, and Equalized Odds optimization. These methods are assessed using key fairness metrics, Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), on two real-world recidivism datasets: COMPAS, a well-known publicly available dataset, and RisCanvi, a curated English dataset derived from a publicly available Spanish dataset. Our findings reveal trade-offs between fairness improvements and predictive accuracy, with fairness performance varying across integrated mitigation strategies. While individual techniques often fall short of effectively reducing bias, their integration across pre-processing, in-processing, and post-processing stages demonstrates a more comprehensive and robust capacity to address fairness concerns. In some instances, fairness mitigation strategies exacerbate disparities instead of reducing them, highlighting the complexity of achieving equitable AI outcomes. Some methods struggle with specific datasets, leading to worsened SPD, DI, EOD, or PED values, potentially reinforcing biases rather than eliminating them. Using multi-objective and bi-objective optimization frameworks, we identify optimal configurations that balance fairness and accuracy, contributing to a deeper understanding of fairness–accuracy trade-offs in AI systems. This study makes a novel contribution by systematically evaluating multi-phase fairness integration in recidivism prediction, bridging critical gaps in the literature. However, our results also underscore the risks of ineffective or counterproductive fairness interventions when not carefully tailored. Our findings highlight the necessity of tailoring bias mitigation strategies to specific datasets and contexts, providing actionable insights for developing equitable, diverse, and inclusive AI systems that are fair and reliable in the criminal justice system.

Keywords Fairness · Artificial intelligence · Criminal justice system · Recidivism · Trust · Multi-objective optimization · Equity

1 Introduction

Artificial intelligence (AI) is increasingly being integrated into the criminal justice system, particularly for predicting recidivism risk-the likelihood that an individual will re-offend upon release (Desmarais et al. 2016; Green 2020; Rodolfa et al. 2020; Farayola et al. 2023). This growing reliance on AI is driven by its potential to enhance judicial decision-making by improving efficiency, consistency, and objectivity, thereby mitigating human biases and promoting equitable treatment (Berk and Bleich 2014; Wang et al. 2022; Pelegrina et al. 2023). However, the application of AI in high-stakes legal decisions raises critical concerns

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regarding fairness, ethics, and public trust. A central issue is the risk of perpetuating historical and systemic biases, which disproportionately impact marginalized demographic groups, ultimately undermining the principles of fairness and equality that are foundational to the criminal justice system.

Fairness is recognized as an essential requirement for trustworthy AI, as outlined by the European Commission (AI 2019; Thiebes et al. 2021). Fairness is particularly crucial in the criminal justice domain, where AI-driven decisions carry profound and lasting consequences for individuals and society. Ensuring that AI systems treat all demographic groups equitably is imperative to prevent outcomes that unfairly disadvantage specific populations. Consequently, fairness in AI models predicting recidivism risk is not merely a technical challenge but a broader societal and ethical necessity that aligns with the fundamental tenets of justice and equity.

AI-driven recidivism models analyze diverse datasets, including criminal records, socioeconomic backgrounds, demographic attributes, and psychological factors, to generate risk scores that inform judicial decisions on bail, sentencing, parole, and rehabilitation. Ideally, these models promote consistency and evidence-based decision-making. However, in practice, they often replicate or amplify the existing social inequities. For example, studies have shown that tools such as the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) systematically overestimate recidivism risk for Black individuals while underestimating it for White individuals (Angwin et al. 2016). Such disparities can result in disproportionately punitive outcomes for marginalized populations, reinforcing cycles of poverty, incarceration, and social exclusion. These systemic biases highlight the urgent need for frameworks that enhance the equity and trustworthiness of AI-based recidivism prediction models.

Bias in AI systems originates from multiple sources, including historical data, algorithmic design, and deployment practices (Ferrara 2023; Farayola et al. 2023). Historical data often reflect entrenched societal inequalities, and AI models trained on such data may inadvertently learn and perpetuate these biases. Moreover, algorithmic design choices and deployment practices can further exacerbate disparities. In response, the literature has proposed various bias mitigation strategies categorized into three key phases: pre-processing, in-processing, and post-processing. Pre-processing techniques focus on correcting biases in datasets before training, ensuring a more equitable foundation. In-processing methods modify training algorithms to counteract biases during development, while post-processing approaches adjust model outputs to promote fairer outcomes (Pagano et al. 2023; Berk et al. 2023; Wan et al. 2023).

Despite these efforts, existing approaches often fall short in addressing fairness comprehensively. A key limitation is the tendency to apply fairness-enhancing techniques in isolation, targeting only one phase-pre-processing, in-processing, or post-processing—rather than considering their interconnected nature (Jain et al. 2019; Corbett-Davies et al. 2017; Jain et al. 2020; Biswas and Mukherjee 2021; Kobayashi and Nakao 2020; Miron et al. 2021). This fragmented approach overlooks the dynamic nature of bias, which can manifest and propagate across multiple stages of model development. Consequently, isolated mitigation strategies often fail to ensure fairness across the entire AI lifecycle. Moreover, this segmented approach risks undermining stakeholder trust, as addressing bias at one stage does not guarantee fairness throughout the system. To achieve equitable outcomes, rebuild trust, and effectively mitigate bias, an integrated approach is necessary—one that accounts for fairness considerations across all stages of AI system development.

Another key challenge is the trade-off between fairness and predictive accuracy. Fairness-enhancing techniques can sometimes reduce predictive accuracy, a critical concern in the criminal justice domain where erroneous predictions may have severe consequences, such as wrongful detention or premature release (Zhang and Sun 2022; Agarwal et al. 2018). Addressing this fairness–accuracy trade-off is essential to developing AI models that are not only equitable but also reliable and practical for real-world decision-making.

The deployment of biased AI models in the criminal justice system carries profound ethical implications, with far-reaching societal consequences (Angwin et al. 2016). Biased recidivism predictions can reinforce the existing disparities, shaping how individuals are perceived and treated within and beyond the judicial system. For example, being classified as "high risk for recidivism" may lead to stigmatization, limiting opportunities for employment, housing, and education (Jacobs and Gottlieb 2020; Graffam et al. 2014). This, in turn, can increase an individual's likelihood of re-offending, creating a self-fulfilling cycle of disadvantage.

Criminal justice stakeholders, including judges, parole officers, policymakers, and civil rights advocates, hold divergent views on AI's role in judicial decision-making. While some regard AI as a tool for enhancing consistency and efficiency, others raise concerns about its fairness and predictive reliability (Watamura et al. 2025; Farber 2024). These concerns underscore the need for AI models that balance fairness and accuracy. Achieving this balance is a complex challenge that requires integrating high predictive performance with equitable treatment across diverse populations. This study investigates a multi-stage approach that applies fairness interventions across all phases of the AI

modeling process, aiming to determine whether fairness can be improved without significantly compromising accuracy.

To validate this approach, we leverage two widely used recidivism datasets: the RisCanvi dataset and the COMPAS dataset. Validation across multiple datasets is crucial in this field, where the ethical and societal implications of AI predictions are significant. Demonstrating that fairness-enhancing techniques yield consistent and reliable outcomes across diverse datasets strengthens confidence in their real-world applicability. Specifically, we evaluate the integrated use of fairness-enhancing methods such as Reweighing (Krasanakis et al. 2018), Adversarial Learning (Wadsworth et al. 2018), Disparate Impact Remover (Feldman et al. 2015), Exponentiated Gradient Reduction (Agarwal et al. 2018), Reject Option Classification (Kamiran et al. 2012), and Equalized Odds optimization (Pleiss et al. 2017).

This study hypothesizes that integrating multiple fairness-enhancing techniques, each addressing biases at different stages of AI development, will provide a comprehensive and robust framework for mitigating bias throughout the AI lifecycle. By testing this approach on multiple datasets, we contribute to the broader discourse on AI fairness, exploring whether fairness-enhancing measures can maintain predictive stability while improving equity. This approach is motivated by limitations identified in prior research and the pressing need for a holistic strategy to enhance algorithmic fairness and rebuild public trust in AI-based recidivism risk assessment (Farayola et al. 2023; Caton and Haas 2023; Chakraborty et al. 2020; Raza et al. 2023; Chen et al. 2023; Zhang and Sun 2022; Farayola et al. 2024).

To the best of our knowledge, this paper is the first comprehensive study to systematically integrate and evaluate multiple fairness-enhancing techniques across all three phases of fairness intervention-pre-processing, in-processing, and post-processing-using two recidivism datasets, with a focus on fairness in recidivism prediction within the criminal justice system. We categorize these integrations into four distinct model types: PI, PP, IP, and PIP. The PI model combines pre-processing and in-processing techniques; the PP model incorporates pre-processing and post-processing methods; the IP model integrates in-processing and post-processing strategies; and the PIP model unifies methods across all three phases. This classification provides a structured framework for systematically assessing each model's impact on fairness and predictive accuracy.

Additionally, we employ a multi-objective optimization (MOO) methodology (Sharma and Kumar 2022) using the Pareto front concept to identify optimal models among the PI, PP, IP, and PIP configurations. This framework identifies non-dominated models, those that excel in at least one fairness metric, such as statistical parity, disparate impact, or equal opportunity, while maintaining strong predictive performance. By highlighting trade-offs between fairness and accuracy, the Pareto front provides a nuanced perspective on the effectiveness of each fairness-enhancing approach.

Our methodology offers criminal justice stakeholders actionable insights by presenting models that balance fairness and accuracy. By showcasing models along the Pareto front, stakeholders can select the most appropriate approach based on their priorities—whether emphasizing fairness, accuracy, or a balanced trade-off. Ultimately, this framework ensures that AI-driven recidivism risk assessments are optimized for equity and reliability, thereby enhancing the fairness and credibility of high-stakes judicial decision-making.

1.1 Research questions and hypothesis

This study contributes to the ongoing discourse on AI fairness by addressing the fairness-accuracy trade-off identified in Liu and Vicente (2022), Plecko and Bareinboim (2024), Mehrabi et al. (2021), Farayola et al. (2024). It aims to validate the applicability and generalizability of a multi-stage fairness-improving approach, integrating techniques across pre-processing, in-processing, and post-processing phases, on recidivism datasets, especially the RisCanvi dataset. This validation is essential to assess the robustness and applicability of fairness techniques in diverse, real-world contexts, particularly within high-stakes domains like criminal justice.

The central hypothesis of this study posits that combining fairness-improving techniques across different stages of the AI pipeline yields improved fairness without a significant loss of predictive accuracy, even when applied to a diverse dataset. To test this hypothesis, the study addresses the following research questions:

- RQ1: Can the integrated application of fairness-improving techniques across pre-processing, in-processing, and post-processing stages achieve comparable fairness and accuracy outcomes?
- RQ2: How does the effectiveness of various fairness metrics (e.g., Disparate Impact, Statistical Parity Difference, and Equal Opportunity Difference) vary when applying the integrated approach?
- RQ3: Which fairness-improving techniques or combinations are most effective in maintaining predictive accuracy?
- RQ4: Does the integration of fairness techniques across the AI pipeline enhance the predictive stability of the models?
- RQ5: What are the implications of validating fairness-enhanced models on new datasets for their real-world applicability in recidivism prediction?

2 Related works

The literature on fairness in machine learning underscores the urgent need to address biases in AI, especially in high-stakes domains like criminal justice system. Recent studies highlight the limitations of applying fairness-enhancing techniques at isolated stages of the AI development pipeline and advocate for a comprehensive, multi-stage approach to ensure fairness and maintain public trust in AI systems.

Farayola et al. (2023) provide a detailed examination of fairness-enhancing techniques applied within recidivism prediction models, noting the limitations of interventions focused on a single stage, whether pre-processing, in-processing, or post-processing. They argue that isolated interventions address only specific biases and do not comprehensively tackle fairness throughout the AI pipeline. Their findings support the need for multi-phase interventions, a key motivation for this study, to address fairness challenges at multiple stages of model development and build stakeholder trust in judicial AI applications.

Similarly, Caton and Haas (2024) provide a broad overview of fairness definitions, metrics, and mitigation strategies in their survey on machine learning fairness. They identify a persistent trade-off between fairness and accuracy, noting that isolated fairness interventions, whether at the pre-processing, in-processing, or post-processing stage, tend to yield suboptimal results by compromising either fairness or predictive accuracy. Caton et al. emphasize the importance of integrated fairness strategies to better balance these metrics, especially in high-stakes applications where both fairness and predictive performance are critical. Their survey informs this study's multi-stage approach, where combining techniques across all development stages aims to create a fairer and more accurate model for recidivism prediction.

Chen et al. (2023) present an empirical evaluation of various fairness-improving methods, demonstrating that single-phase interventions often struggle to balance multiple fairness metrics, such as disparate impact and statistical parity. By contrast, they show that integrated fairness approaches can enhance performance across multiple metrics without significantly compromising accuracy. Their findings underscore the practical advantages of adopting multi-phase interventions across the AI pipeline to achieve more consistent and equitable model performance; however, their work is limited to benchmarking 17 representative bias mitigation methods on eight widely adopted software decision tasks. The study systematically evaluates trade-offs between fairness and machine learning performance but does not explore fairness interventions specific to recidivism prediction which are the focus of our research.

Zhang and Sun (2022) propose a fairness framework that incorporates causal analysis to refine fairness interventions. They argue that causal relationships between features and outcomes, often overlooked in traditional fairness methods, are vital to mitigating bias effectively. By leveraging causal insights, their framework achieves a more nuanced mitigation of bias across multiple stages of development. This work reinforces the importance of a comprehensive fairness strategy, particularly in sensitive domains like criminal justice, where the integration of fairness techniques across all pipeline phases is essential for robust and equitable outcomes.

Kozodoi et al. (2022) examine fairness interventions in the financial sector, particularly credit scoring. Their analysis, grounded in the separation criterion, explores the trade-offs between fairness, accuracy, and profitability across various fairness-enhancing methods. Using multi-objective optimization, they identify non-dominated solutions that achieve balanced fairness and profitability, illustrating a practical application of the Pareto front in fairness optimization. While focused on profitability, their work contributes to a broader understanding of fairness trade-offs. In contrast, our study emphasizes a comprehensive fairness framework for recidivism prediction, a domain where diverse fairness metrics and stakeholder trust are crucial and profitability is not the primary consideration.

Raza et al. (2023) explore the intersection of fairness in AI and equity in healthcare, underscoring the need for multi-stage fairness interventions in sensitive domains. Through a case study on diabetic retinopathy prediction, they demonstrate that fairness techniques applied solely at one phase (e.g., pre-processing) are insufficient to address deep-seated inequities in healthcare data. By applying fairness interventions across pre-, in-, and post-processing stages, they achieve significant improvements in fairness and health equity. This finding supports the broader applicability of a multi-phase fairness strategy, validating its importance across critical domains beyond criminal justice.

Wang et al. (2024) extend the scope of fairness research by introducing a fairness-enhancing approach for large language models that integrates structural fairness mechanisms with active learning. A distinguishing aspect of their work lies in its exploration of the trade-offs between fairness and accuracy in large-scale models, offering valuable insights into how active learning can be employed to enhance structural fairness. While their study centers on language models, the integration of fairness-aware layers and iterative optimization resonates with the multi-phase approach of this study. Their method, which integrates fairness-aware loss functions with active learning for informed data selection, provides a scalable framework for ongoing fairness improvement, relevant to high-stakes AI applications where fairness is critical.

Chakraborty et al. (2020) take a two-stage approach, integrating pre-processing and in-processing fairness techniques to mitigate bias while maintaining model performance. By leveraging multi-objective optimization, their approach balances fairness and accuracy in fairness-sensitive datasets like COMPAS. However, their method does not include post-processing interventions, which are essential for refining predictions and addressing fairness issues post-deployment. Moreover, their approach lacks configurations that combine pre- and post-processing techniques (PP models) or in- and post-processing techniques (IP models), limiting its scope. In contrast, this study expands the methodology to integrate fairness-enhancing techniques across all three phases, pre-processing, in-processing, and post-processing, addressing bias in a more comprehensive manner and improving the fairness–accuracy trade-off in recidivism prediction.

Farayola et al. (2024) propose a fairness framework for recidivism prediction models by integrating fairness techniques across pre-processing, in-processing, and post-processing stages. Their study employs multi-objective optimization to balance fairness and accuracy. Key methods include Reweighing, Disparate Impact Remover, Adversarial Learning, Exponential Gradient Reduction, Reject Option-Based Classification, and Equalized Odds Optimization. The findings suggest that the combination of Disparate Impact Remover, Adversarial Learning, and Equalized Odds Optimization (`DIR+AL+EO`) yields notable fairness improvements with minimal accuracy loss. This multi-stage approach aligns with our work by demonstrating the effectiveness of integrated fairness interventions, particularly when guided by optimization frameworks to identify balanced, non-dominated models. However, a key limitation of their study is its reliance on a single dataset, the COMPAS dataset, for evaluating fairness-improving combinations. This constraint limits the generalizability of their approach, as datasets vary in attributes and inherent biases. In contrast, our study applies the integrated fairness-improving approach to the RisCanvi dataset, broadening the evaluation and enhancing both its applicability and potential generalizability across diverse recidivism prediction contexts.

These studies collectively highlight the importance of an integrated fairness approach across the AI development pipeline. Building on this foundation, our research aims to validate the robustness and generalizability of a multi-stage fairness framework by applying it to a new dataset, RisCanvi dataset. This approach provides a replicable model that seeks to enhance fairness while preserving accuracy, contributing to a broader body of knowledge focused on equitable outcomes in the criminal justice system when predicting recidivism risk, where fairness and predictive performance must be carefully balanced to maintain public trust.

3 Background

This section provides an overview of fairness in AI and the contemporary methodologies employed to improve fairness.

3.1 Fairness in AI

The issue of fairness and the methods to ensure its implementation in the design of AI systems, particularly for binary classification tasks, has garnered significant attention, since concerns about the equity of AI systems first arose. To address these challenges, researchers have proposed various metrics to measure and assess fairness in AI. These measures typically fall in two primary categories: individual fairness and group fairness.

Individual fairness emphasizes that individuals in similar situations should receive comparable treatment and outcomes from an algorithm, regardless of differences in protected attributes. This principle requires that individuals who are alike in relevant aspects are treated equally (Beutel et al. 2019). Notable examples of individual fairness include the Generalized Entropy Index and counterfactual fairness (Caton and Haas 2023.

Group fairness evaluates the outcomes of a classification algorithm across two or more protected groups, such as racial categories, to ensure parity based on statistical metrics. Unlike individual fairness, this paper focuses on group fairness and considers established metrics in the literature to evaluate the fairness of AI models used to predict recidivism risk.

3.2 Fairness metrics

This section outlines significant group fairness metrics, emphasizing their attributes and relevance to assessing fairness across demographic groups in AI models predicting recidivism risk (Chouldechova 2017; Mehrabi et al. 2021; Kleinberg et al. 2016; Pleiss et al. 2017). Many studies tend to focus on specific fairness metrics without considering the broader implications or effects of alternative metrics on different demographic groups. To address this limitation, our integrative approach incorporates multiple fairness metrics at various stages of AI system development. This comprehensive strategy enhances fairness assessments across multiple dimensions while addressing the trade-offs inherent in balancing different fairness metrics.

The literature has explored multiple fairness metrics during a single stage of AI development, often revealing conflicts between metrics (Chouldechova 2017; Mehrabi et al. 2021; Kleinberg et al. 2016; Pleiss et al. 2017). Our study builds upon this foundation by integrating several fairness-enhancing techniques and evaluating their impact

on key fairness metrics. Here, $G = g_i$ and $G = g_j$ denote the privileged and unprivileged groups, respectively.

Next, we will define one-by-one the fairness metrics of interest in the context of the study.

- **Statistical/demographic parity difference (SPD):** Statistical or demographic parity ensures that all groups defined by a protected attribute, such as race or gender, have an equal likelihood of receiving a positive classification from a model. Let $\hat{y} \in \{0, 1\}$ represent predictions and g_i and g_j represent groups based on a sensitive attribute G. Parity is achieved when the probability $P(\hat{y} = 1)$ is not influenced by G (Gao et al. 2022). This can be expressed as follows:

$$P(\hat{y} = 1 \mid G = g_i) - P(\hat{y} = 1 \mid G = g_j).$$

A value of zero indicates no disparity in positive outcomes across groups.
- **Disparate Impact (DI):** Disparate Impact measures the ratio of positive classification probabilities between privileged and unprivileged groups. It aligns with legal fairness principles, such as the 80% rule, which identifies discrimination when the ratio falls below 0.8. To ensure fairness, the ratio should ideally be close to 1 (Jo et al. 2023). The formula is as follows:

$$\frac{P(\hat{y} = 1 \mid G = g_i)}{P(\hat{y} = 1 \mid G = g_j)}.$$

- **Equal Opportunity Difference (EOD):** Equal Opportunity Difference evaluates fairness by comparing true positive rates (TPR) across groups. It ensures that qualified individuals in all groups have an equal chance of being correctly classified as positive (Beutel et al. 2019). The difference is calculated as follows:

$$P(\hat{y} = 1 \mid y = 1,\ G = g_i) - P(\hat{y} = 1 \mid y = 1,\ G = g_j).$$

A score of zero represents perfect fairness, while deviations indicate biases favoring or disadvantaging certain groups.
- **Predictive Equality Difference (PED):** Predictive Equality Difference measures the disparity in false positive rates (FPR) across groups, ensuring that no demographic group is disproportionately affected by incorrect positive predictions (Castelnovo et al. 2022). The metric is defined as follows:

$$P(\hat{y} = 1 \mid y = 0,\ G = g_i) - P(\hat{y} = 1 \mid y = 0,\ G = g_j).$$

A PED of zero indicates fairness, while deviations suggest unequal treatment among groups.

3.3 Fairness–metrics trade-off

Fairness metrics often conflict, requiring trade-offs when optimizing multiple objectives simultaneously Kleinberg et al. (2016); Chouldechova (2017); Chen et al. (2023). Key trade-offs include the following:

- **Statistical Parity vs. Equal Opportunity:** Statistical Parity ensures equal positive outcomes across groups but does not consider whether individuals are qualified. Equal opportunity ensures fairness for qualified individuals (true positives). Optimizing for Statistical Parity might inadvertently favor higher-risk individuals in some groups, violating Equal Opportunity.
- **Equal Opportunity vs. Predictive Equality:** Equal Opportunity seeks to equalize true-positive rates, while Predictive Equality focuses on equal false-positive rates. Improving one can worsen the other, as emphasizing true positives for qualified individuals may increase false positives.
- **Disparate Impact vs. Equal Opportunity:** Disparate Impact ensures proportional positive outcomes across groups, potentially conflicting with Equal Opportunity's focus on treating qualified individuals fairly. Balancing these may reduce fairness for those who clearly qualify or disqualify for certain outcomes.

These trade-offs highlight the complexity of achieving fairness in machine learning. In high-stakes contexts like criminal justice, stakeholders often prioritize metrics such as Predictive Equality to reduce unjust detentions, even at the expense of true-positive rate parity (Equal Opportunity).

3.4 Fairness-improving methods

Several fairness-enhancing techniques are designed to operate at specific phases of the AI development pipeline: pre-processing, in-processing, or post-processing (Farayola et al. 2023). In this study, we selected six fairness techniques: Reweighing, Disparate Impact Remover, Exponentiated Gradient Reduction, Adversarial Learning, Equalized Odds Optimization, and Reject Option-Based Classification. These techniques were chosen based on their empirical performance and comparative analyses, as identified in Zhang and Sun (2022).

Each method was selected for its capacity to improve at least one fairness metric widely used in the literature, including Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED). Moreover, as detailed by Bellamy et al. (2019), Reweighing and Disparate Impact Remover are especially effective at reducing SPD and improving DI, making them well suited for correcting disparities in dataset composition. Conversely, Exponentiated Gradient Reduction and Adversarial Learning focus on fairness during the training process and have demonstrated strong performance in mitigating classification error imbalances. Post-processing methods such as Equalized Odds Optimization and Reject Option-Based Classification are designed to mitigate bias at the evaluation stage of AI development outputs, thereby further aligning EOD and PED across privileged and unprivileged groups.

Importantly, no single method can comprehensively address all fairness concerns across the pipeline. Hence, the integration of these techniques is not arbitrary but reflects a deliberate design to leverage their complementary strengths.

- **Reweighing (Re):** A pre-processing method that assigns weights to correct biases in sensitive attributes while preserving the original data. Reweighing ensures balanced representation across groups and targets metrics like statistical parity and disparate impact (Bellamy et al. 2019; Krasanakis et al. 2018).
- **Disparate Impact Remover (DIR):** DIR modifies sensitive attribute values to achieve equitable decision distributions. It is a pre-processing method designed to address disparate impact while preserving relative rankings (Feldman et al. 2015).
- **Exponential Gradient Reduction (EGR):** An in-processing technique that creates a randomized classifier by solving fairness constraints as cost-sensitive classification tasks. EGR targets metrics like demographic parity and Equal Opportunity Difference (Agarwal et al. 2018).
- **Adversarial Learning (AL):** This in-processing technique trains models to minimize biases by introducing adversarial examples. AL improves fairness by reducing the influence of sensitive attributes on predictions (Wadsworth et al. 2018).
- **Reject Option-Based Classification (ROC):** A post-processing technique that adjusts predictions near decision boundaries to reduce discrimination. ROC targets metrics, such as statistical parity and Equal Opportunity Difference (Kamiran et al. 2012).
- **Equalized Odds Optimization (EO):** EO uses linear programming to modify labels, aligning predictions with fairness criteria. It focuses on metrics like Equal Opportunity Difference and Predictive Equality Difference (Pleiss et al. 2017).

By applying these methods across all stages of the pipeline, this study aims to mitigate the trade-offs between fairness metrics and accuracy, offering a more holistic approach to fairness in machine learning.

4 Methodology

This section outlines the empirical study designed to assess how integrating different fairness-enhancing methods influences fairness outcomes in AI systems. The study focuses on the criminal justice use case, more specifically, recidivism prediction, and uses two datasets described next.

4.1 Datasets

4.1.1 COMPAS dataset

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) dataset serves as the foundation for this study (Angwin et al. 2016). It comprises 6172 records from Broward County, Florida, capturing criminal history details across 11 features. The target variable, 2-year recidivism, indicates whether an individual reoffended within 2 years of release. For this analysis, a subset of 5278 records was used, focusing on individuals identified as African–American or Caucasian. Race was treated as the protected attribute, with Caucasians designated as the privileged group and African–Americans as the unprivileged group. Within the subset, 2795 records represented non-recidivists (label 0), and 2483 corresponded to recidivists (label 1), revealing a slight imbalance in class distribution.

Categorical variables were transformed into numeric formats to ensure compatibility with fairness models. Age was encoded as 0 (“Less than 25”), 1 (“25–45”), and 2 (“Greater than 45”). Recidivism risk text was mapped to 0 ("Low"), 1 (“Medium”), and 2 ("High"), while gender was encoded as 0 (“Female”) and 1 (“Male”). For consistency, race was standardized as 0 (African–American) and 1 (Caucasian), and charge severity was represented as 0 ("Misdemeanor") and 1 ("Felony"). Random sampling techniques were employed to address class imbalance, resulting in a more balanced dataset to facilitate fair evaluations across demographic groups.

4.1.2 RisCanvi dataset

The anonymized RisCanvi dataset used in this research is a carefully curated resource derived from the fifth installment in a longitudinal research series investigating recidivism among prison populations in Catalonia (CEJFE 2020).

Over 3 decades, this series has examined the post-release trajectories of incarcerated individuals, with the latest dataset focusing on those released in 2015 and followed for up to 5 years, ending in December 2019. In accordance with international recommendations, data collection concluded before March 2020 to prevent potential biases introduced by the COVID-19 pandemic.

This dataset was curated to ensure accuracy, completeness, and relevance for fairness-aware machine learning research. It includes N = 3,814 individuals, categorized into final releases (67.4%), conditional releases (30.7%), suspended sentences (1.9%), and expulsion cases (3.1%). Expulsion cases were excluded from recidivism monitoring, as these individuals do not reenter the penitentiary system. Recidivism is defined as penitentiary recidivism, involving new entries into the penitentiary system, either pre-trial or post-conviction. Additionally, the dataset supports focused studies on specific subpopulations, including individuals expelled from Spanish territory.

The RisCanvi dataset is geographically specific to Catalonia, an autonomous region of Spain, offering tailored insights into its penitentiary and judicial systems. This resource enables in-depth analysis of recidivism trends and the effectiveness of targeted interventions, providing critical support for advancing evidence-based policy and criminal justice practices within the region.

On average, inmates who undergo only one evaluation spend approximately 3 months in prison, whereas those with four evaluations remain incarcerated for an average of 2 years before being released on parole or regaining their freedom. Notably, evaluations cease once an inmate is released, ensuring that recidivism tracking accurately reflects post-release outcomes.

As part of the curation process, systematic data cleaning and pre-processing techniques were applied to address missing values. Missing data were imputed using corresponding values from either the previous or subsequent evaluations. For items with a yes/no/uncertain response format, missing values were replaced with "uncertain." Cases with irreparable missing data were excluded from the analysis, resulting in a final dataset of 2885 individuals.

The final class distribution consists of 2391 individuals who did not commit penitentiary recidivism and 464 individuals who did. To mitigate class imbalance, random sampling techniques were applied, creating a more balanced dataset for fair evaluations across demographic groups.

To facilitate accessibility for English-speaking researchers, the dataset and its documentation, originally in Spanish, were translated into English. Furthermore, the dataset has been structured to ensure ease of use for researchers across different disciplines, with clearly defined variable descriptions and standardized formats for seamless integration into various analytical workflows.

This curated dataset is publicly available for further research and reproducibility at the following repository: RisCanvi GitHub Repository.

4.2 Binary label dataset transformation

To assess fairness using AIF360's suite of fairness metrics (Bellamy et al. 2019), the datasets were converted into Binary Label Datasets via AIF360's 'BinaryLabelDataset' class. This conversion standardized the data, ensuring compatibility with fairness algorithms and enabling the computation of metrics, such as statistical parity difference, disparate impact, predictive equality difference, and equal opportunity difference. Once transformed, the Binary Label Datasets served as inputs for pre-processing, in-processing, and post-processing fairness interventions.

4.3 Experimental setup

The study explores the integration of fairness-enhancing methods at various stages of the AI pipeline: pre-processing, in-processing, and post-processing (see Fig. 1). Several configurations were implemented, including PI (Pre-Processing + In-Processing), PP (Pre-Processing + Post-Processing), IP (In-Processing + Post-Processing), and PIP (Pre-Processing + In-Processing + Post-Processing). These setups allowed for a comprehensive analysis of how the inclusion of fairness techniques at different pipeline stages impacted fairness outcomes while preserving predictive accuracy.

Pre-processing techniques, namely Reweighing (Re) and Disparate Impact Remover (DIR), modified the training data to address bias before model training. In-processing methods such as Adversarial Learning (AL) and Exponentiated Gradient Reduction (EGR) enforced fairness constraints during training. Post-processing techniques, including Equalized Odds Optimization (EO) and Reject Option-Based Classification (ROC), adjusted model predictions after training to ensure fairness in outcomes.

To enhance the reliability of the results, each experiment was repeated ten times using different random seeds.

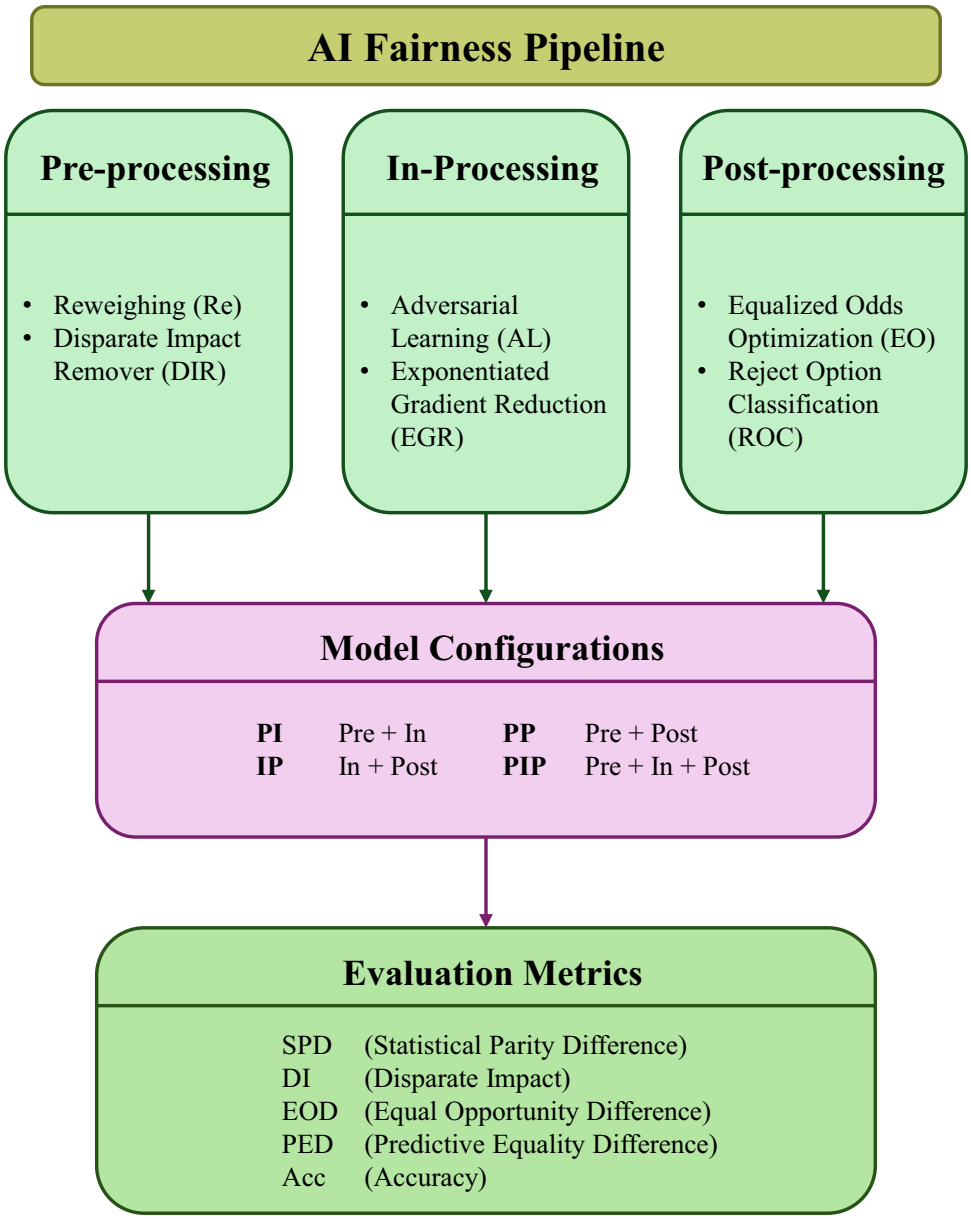


Fig. 1 Fairness-enhancing techniques integrated across the AI fairness pipeline

This repetition minimized the influence of randomness, providing consistent and reproducible insights into the effectiveness of fairness-enhancing techniques.

4.4 Handling of pre-processing techniques

Pre-processing methods aimed to reduce bias in the dataset before the training phase.

4.4.1 Reweighing (Re)

The reweighing method assigned weights to instances in the dataset, balancing the representation of privileged and unprivileged groups. By adjusting instance weights according to the protected attribute (race), it ensured equitable treatment of both groups during model training.

4.4.2 Disparate impact remover (DIR)

The Disparate Impact Remover adjusted feature values to reduce bias, promoting fair treatment of both demographic groups. A repair level of 1 was used, signifying complete bias correction based on race. The modified dataset was subsequently utilized for training models in the next stages.

4.5 Handling of in-processing techniques

In-processing methods integrated fairness constraints directly into the model training process.

4.5.1 Adversarial learning (AL)

Adversarial Learning incorporated a secondary adversarial network tasked with predicting the protected attribute (i.e., race), while the primary classifier focused on predicting recidivism. The adversarial network actively guided the primary classifier to avoid discriminatory patterns. The model was trained for 50 epochs using a multi-layer neural network with a batch size of 128. The weight parameter for adversary loss was set to 0.2, and the debias parameter was enabled to ensure fairness.

4.5.2 Exponentiated gradient reduction (EGR)

Exponentiated Gradient Reduction enforced fairness constraints by adjusting model parameters to optimize both fairness and accuracy. Logistic Regression served as the base classifier, and demographic parity was applied as the fairness constraint. The protected attribute (race) was retained in the dataset for evaluation, with training configured to a maximum of 20 iterations for consistency.

4.6 Handling of post-processing techniques

Post-processing methods adjusted predictions to ensure fairness in the final model outcomes.

4.6.1 Equalized odds optimization (EO)

Equalized Odds adjusted prediction labels to align true-positive and false-positive rates across privileged and unprivileged groups. This ensured balanced error rates and equitable outcomes for both groups.

4.6.2 Reject option-based classification (ROC)

Reject Option-Based Classification modified predictions for individuals near the decision boundary, prioritizing fairness for unprivileged individuals. A grid search optimized classification thresholds, using equal opportunity difference and average odds difference as guiding metrics to balance fairness and accuracy.

4.7 Model integration and metric computations

The fairness-enhancing techniques, pre-processing (Re, DIR), in-processing (AL, EGR), and post-processing (EO, ROC), were combined to develop fairness-enhanced models (PI, PP, IP, PIP). These models were evaluated using metrics such as Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), alongside accuracy metrics. This comprehensive evaluation captured the trade-off between fairness and predictive performance for each method.

4.8 Experimental evaluation

The datasets were split into a 70:30 train-test ratio, ensuring stratified sampling for proportional representation of the privileged and unprivileged groups. The same random state was used across all experiments to maintain consistency and comparability. Each model was trained and evaluated under identical conditions, with the Binary Label Dataset structure ensuring uniform fairness assessments. Predictions were analyzed using both fairness and accuracy metrics to assess the effectiveness of each configuration in promoting fairness while preserving predictive accuracy.

5 Multiple objective optimization framework

5.1 Balancing fairness and accuracy

Selecting the optimal model from the PI, IP, PP, and PIP configurations requires a careful equilibrium between fairness metrics and predictive accuracy. This process is challenging due to inherent trade-offs between these objectives and the absence of a universally accepted fairness metric. To address this complexity, a multi-objective optimization framework is adopted, enabling the identification of fairness-enhancing integrations that reconcile these competing priorities effectively.

5.2 Multi-objective approach and the pareto front

The multi-objective optimization framework resolves the fairness–accuracy trade-off by addressing diverse objectives without collapsing them into a singular metric. This study emphasizes identifying "best" approaches, which are characterized as non-dominated solutions within the set of PI, IP, PP, and PIP models. Non-dominated solutions excel on at least one fairness metric without being outperformed across all others. This approach leverages the Pareto front concept (Sharma and Kumar 2022), which systematically

evaluates trade-offs and identifies models that best balance competing objectives.

By employing this strategy, the study circumvents the limitations of single-objective optimization. Models situated on the Pareto front represent viable options that balance fairness and accuracy effectively, ensuring their applicability across diverse contexts.

5.3 Formulating the objectives

The multi-objective framework defines accuracy as a maximization objective to capture the predictive capability of models. Fairness metrics, on the other hand, require optimization toward specific target values. For Statistical Parity Difference (SPD), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), the optimal value is zero, signifying reduced disparities across demographic groups. For Disparate Impact (DI), the ideal value is one, indicating equitable treatment between privileged and unprivileged groups.

To unify these objectives within a single optimization framework, the study employs transformations. Absolute value transformations are applied to SPD, EOD, and PED, normalizing them around zero, while an inverse transformation adjusts DI values below one toward its target. This standardization ensures coherence and comparability across fairness objectives, enabling seamless integration with accuracy optimization.

5.4 Non-dominated solutions and pareto front analysis

The evaluation of non-dominated solutions involves assessing their performance across multiple objectives. Let $O = [O_1(x), \ldots, O_k(x)]$ represent the vector of k objective values (e.g., Accuracy, SPD, DI, EOD, PED) for each technique x in the set of techniques. A technique x_1 dominates another technique x_2, written as $x_1 \succ x_2$, if $\forall i \in \{1, \ldots, k\}, O_i(x_1) \geq O_i(x_2)$ and $\exists j \in \{1, \ldots, k\}$ such that $O_j(x_1) > O_j(x_2)$. A technique is non-dominated if no other technique outperforms it across all objectives. The set of all

Table 1 Many-objective optimization. Star (*) indicates a standardized value for minimization. Non-dominated approaches for both dataset and best metric result are in bold

Model	Riscanvi					COMPAS				
	SPD	DI	EOD	PED	Acc	SPD	DI	EOD	PED	Acc
NONE	0.059	1.224*	0.011	0.045	0.72	0.25	2.13	0.025	0.18	0.68
Re	0.046	1.155*	0.011	0.039	0.71	0.097	1.243	0.094	0.031	0.66
DIR	0.079	1.288*	0.056	0.067	0.71	0.284	2.092	0.281	0.211	0.67
EGR	0.082	1.274*	0.135	0.083	0.71	0.036	1.093	0.080	0.088	0.67
AL	0.064	1.237	0.059	0.055	0.71	0.053	1.174	0.115	0.085	0.66
EO	0.016	1.048*	0.033	0.004	0.70	0.040	1.125	0.014	0.015	0.64
ROC	0.079	1.288*	0.056	0.066	0.71	0.270	2.100	0.287	0.183	0.68
Re+EGR	0.058	1.189	0.068	0.067	0.71	0.033	1.086*	0.026	0.032	0.66
DIR+EGR	0.135	1.536*	0.169	0.142	0.73	0.055	1.142*	0.038	0.004	0.67
Re+AL	0.064	1.238*	0.059	0.055	0.71	0.061	1.168	0.052	0.004	0.67
DIR+AL	0.076	1.297	0.0820	0.0640	0.72	0.009	1.020*	0.028	0.067	0.67
Re+EO	0.010	1.035*	0.020	0.002	0.71	0.034	1.095	0.001	0.002	0.64
Re+ROC	0.086	1.340	0.092	0.075	0.71	0.081	1.230	0.091	0.002	0.67
DIR+EO	0.010	1.037*	0.007	0.0001	0.74	0.032	1.097	0.007	0.005	0.64
DIR+ROC	0.023	1.086	0.028	0.011	0.74	0.288	2.160	0.300	0.207	0.68
EGR+EO	0.009	1.024*	0.008	0.001	0.67	0.036	1.095	0.003	0.003	0.65
EGR+ROC	0.021	1.066	0.011	0.034	0.71	0.023	1.059	0.015	0.042	0.67
AL+EO	0.011	1.036*	0.015	0.0002	0.71	0.036	1.099	0.014	0.016	0.66
AL+ROC	0.024	1.092*	0.028	0.013	0.74	0.079	1.159	0.037	0.033	0.68
Re+EGR+EO	0.005	1.014*	0.033	0.010	0.69	0.034	1.088	0.001	0.0002	0.65
Re+EGR+ROC	0.021	1.066	0.011	0.034	0.71	0.023	1.059	0.015	0.042	0.67
Re+AL+ROC	0.024	1.092*	0.028	0.013	0.74	0.079	1.159	0.037	0.033	0.68
Re+AL+EO	0.011	1.036*	0.015	0.0002	0.71	0.036	1.099	0.014	0.016	0.66
DIR+EGR+ROC	0.080	1.314	0.076	0.093	0.74	0.055	1.142	0.038	0.004	0.67
DIR+EGR+EO	0.007	1.021*	0.011	0.0003	0.70	0.037	1.097	0.002	0.003	0.66
DIR+AL+ROC	0.031	1.081*	0.0002	0.026	0.66	0.030	1.056	0.006	0.032	0.68
DIR+AL+EO	0.010	1.034*	0.006	0.002	0.72	0.038	1.089	0.001	0.001	0.66

non-dominated techniques forms the Pareto front, wherein each solution represents an optimal balance that cannot be improved in one objective without compromising another.

Through Pareto front analysis, this framework identifies models that achieve an effective equilibrium between fairness and accuracy. These insights form a robust foundation for selecting models suitable for real-world deployment. The framework ensures that selected models not only address fairness comprehensively, but also maintain strong predictive performance, aligning ethical considerations with practical utility in fairness-enhanced AI systems.

Table 1 presents the results of a many-objective optimization evaluation, comparing the performance of various fairness-enhancing techniques, individually and in combination, across two datasets: *RisCanvi* and *COMPAS*. Each model is evaluated using four widely recognized group fairness metrics: Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), along with overall classification accuracy (Acc). For fairness metrics, lower values indicate better fairness performance, with Disparate Impact (DI) being standardized for minimization (as indicated by a star '*').

The table is organized to reflect parallel evaluations across the two datasets. Within each dataset column block, fairness metrics are presented in the same order for ease of comparison. Models are sorted by technique type, including baseline (NONE), single interventions (e.g., Re, DIR), and multi-intervention combinations (e.g., Re + EGR + EO). Bold values highlight non-dominated approaches, models that achieved optimal or near-optimal trade-offs across multiple metrics without being outperformed on all objectives. This allows for quick identification of models that maintain strong fairness across several dimensions while preserving predictive accuracy. The inclusion of multiple combinations emphasizes the importance of composite fairness strategies, as no single technique consistently performs best across all metrics or datasets.

5.5 Implementation of many-objective optimization

In our research, we simultaneously optimize five objectives—Accuracy (Acc), Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED)—using many-objective optimization along the Pareto front to achieve a balance across these diverse objectives within a unified framework. Table 1 presents our findings, identifying four optimal integrations of fairness-enhancing techniques across PI (pre-processing and in-processing), IP (in-processing and post-processing), PP (pre-processing and post-processing), and PIP (pre-processing, in-processing, and post-processing) models for the COMPAS dataset: `Re+EGR+EO`, `DIR+EGR+EO`, `DIR+AL+ROC`, and `DIR+AL+EO`. For the RisCanvi Dataset: `DIR+EO` and `DIR+AL+EO`. Significantly, the "NONE" model, which does not include any fairness interventions, does not appear on the Pareto front, emphasizing the benefits of integrating fairness-enhancing techniques to improve both fairness and accuracy.

5.6 Bi-objective optimization for targeted fairness metrics

Bi-objective optimization focuses on balancing two objectives, typically aiming to achieve a trade-off between a prioritized fairness metric and model accuracy. This approach is particularly useful in high-stakes domains, where decision-makers may emphasize specific fairness considerations alongside the need for predictive reliability.

When focusing on optimizing Statistical Parity Difference or Disparate Impact or equal opportunity difference alongside accuracy for RisCanvi dataset, models like `AL+ROC`, `DIR+EO`, `DIR+ROC`, `DIR+AL+EO`, and `Re+AL+ROC` stand out as optimal choices. For Predictive Equality and accuracy, four models emerge as optimal: `AL+ROC`, `DIR+EO`, `DIR+ROC`, and `Re+AL+ROC`.

When focusing on optimizing Statistical Parity Difference or Disparate Impact alongside accuracy for COMPAS dataset, models like `DIR+AL`, `EGR+ROC`, and `Re+EGR+ROC` stand out as optimal choices. Likewise, when prioritizing Equal Opportunity Difference in combination with accuracy, the non-dominated solutions include `DIR+EGR+EO`, `DIR+AL+EO`, and `DIR+AL+ROC`. For Predictive Equality and accuracy, five models emerge as optimal: "Re+AL," `DIR+EGR`, `DIR+EGR+ROC`, `DIR+AL+EO`, and `DIR+EGR+EO`.

6 Discussion

This section examines the integration of fairness-enhancing techniques to achieve a balance between fairness and predictive accuracy in recidivism prediction models. While single-phase fairness methods often have limitations, the integration of techniques across pre-processing, in-processing, and post-processing stages demonstrates promising improvements. A preliminary assessment of individual fairness techniques reveals their varying strengths and trade-offs, as summarized in Table 1.

6.1 Integrations enhancing fairness and accuracy

RQ1: Can the integrated application of fairness-improving techniques across pre-processing, in-processing, and post-processing stages achieve comparable fairness and accuracy outcomes?

The results demonstrate that integrated approaches consistently outperform individual fairness-enhancing techniques in achieving a balanced trade-off between fairness and predictive accuracy. Several combinations of methods achieved substantial improvements across multiple fairness metrics without significantly compromising classification accuracy.

For instance, on the RisCanvi dataset, the integration of `DIR+EGR+EO` resulted in an SPD of 0.007, a DI of 1.021, an EOD of 0.011, and a PED of 0.0003, with an accuracy of 0.70. This represents a substantial improvement over the baseline model (NONE), which had an SPD of 0.059, DI of 1.224, an EOD of 0.011, and a PED of 0.045 while achieving higher accuracy (0.72) but offering minimal fairness guarantees. Another competitive integration, Re+EGR+EO, yielded even lower SPD (0.005) and DI (1.014), with only a minor drop in accuracy to 0.69.

On the COMPAS dataset, similar patterns emerged. The model DIR+AL+EO achieved exceptional fairness outcomes with an SPD of 0.038, DI of 1.089, a near-perfect EOD of 0.001, and a PED of 0.001, while maintaining an accuracy of 0.66. Compared to the baseline, which recorded an SPD of 0.25, DI of 2.13, EOD of 0.025, PED of 0.18, and an Acc of 0.68, the integrated approaches reduced disparities by significant margins across all group fairness dimensions.

Top-performing models that combined fairness techniques across all three pipeline stages, such as `Re+EGR+EO` and `DIR+AL+EO`, consistently appeared as non-dominated solutions, achieving balanced improvements across all fairness metrics. In contrast, single techniques like EGR, while effective on specific metrics (e.g., DI = 1.093 on COMPAS), showed weaker performance on others (e.g., EOD = 0.080, PED = 0.088), highlighting the advantage of multi-stage integration.

These results confirm that no single technique is sufficient to address the complex and multidimensional nature of algorithmic bias. Instead, integrated models allow complementary strengths to emerge leading to broader fairness improvements. These results validate the hypothesis that integrated fairness techniques are effective in addressing bias comprehensively and can yield comparable outcomes across datasets. The findings underscore the value of a multi-phase approach in improving both fairness and accuracy, demonstrating its applicability to real-world scenarios.

6.2 Impact of dataset characteristics on metrics

RQ2: How does the effectiveness of various fairness metrics (e.g., Disparate Impact, Statistical Parity Difference, and Equal Opportunity Difference) vary when applying the integrated approach?

The dataset-specific impact of fairness techniques is evident in the performance variability of key metrics across RisCanvi and COMPAS datasets. Metrics, such as DI and SPD, showed greater variability, reflecting the influence of dataset characteristics on fairness outcomes. For instance, `Re+EO` achieved a DI of 1.035 on RisCanvi but a slightly higher DI of 1.095 on COMPAS, indicating that DI may be more sensitive to the underlying data distribution. Conversely, metrics like PED and EOD exhibited more consistent behavior across datasets, suggesting that they may be more reliable in evaluating fairness across different contexts. These observations highlight the importance of selecting fairness metrics that align with the specific characteristics and fairness objectives of the dataset. The findings also emphasize the need for practitioners to account for dataset variability when evaluating fairness-enhancing techniques, ensuring that models are tailored to the nuances of the data in real-world applications.

6.3 Effectiveness of techniques in maintaining accuracy

RQ3: Which fairness-improving techniques or combinations are most effective in maintaining predictive accuracy?

Integrated fairness techniques demonstrate better performance in maintaining accuracy compared to individual methods, particularly on new datasets. For example, on the RisCanvi dataset, DIR+AL+EO achieved an accuracy of 0.72 while significantly improving fairness metrics such as SPD (0.010) and DI (1.034). On the COMPAS dataset, combinations like EGR+EO and Re+EGR preserved accuracy levels of 0.65–0.67 while effectively reducing disparities across fairness metrics. In contrast, individual techniques often lead to greater trade-offs between fairness and accuracy. For instance, while EO alone achieved a DI of 1.048 on RisCanvi, its accuracy dropped to 0.70. The results suggest that combining techniques across the AI pipeline not only mitigates fairness disparities but also reduces the likelihood of substantial accuracy losses, making integrated

approaches more effective for maintaining performance on new datasets.

6.4 Predictive stability across datasets

RQ4: Does the integration of fairness techniques across the AI pipeline enhance the predictive stability of the models?

The integration of fairness techniques contributes to greater predictive stability across datasets, ensuring consistent performance even in diverse contexts. For instance, combinations like `DIR+AL+EO` and `Re+EGR+EO` exhibited similar trends in fairness and accuracy on both RisCanvi and COMPAS datasets, with only marginal differences in key metrics. This stability contrasts with the variability observed in individual techniques, where performance often fluctuated significantly between datasets. For example, DIR alone achieved an SPD of 0.079 on RisCanvi but a higher SPD of 0.284 on COMPAS, reflecting inconsistencies in its effectiveness. The robustness of integrated approaches stems from their ability to address fairness concerns comprehensively across the AI pipeline, mitigating the compounding effects of bias at different stages. These findings underscore the importance of adopting holistic strategies for fairness mitigation, particularly in high-stakes applications where predictive stability is critical for trust and reliability.

6.5 Real-world applicability and implications

RQ5: What are the implications of validating fairness-enhanced models on new datasets for their real-world applicability in recidivism prediction

The validation of fairness-enhanced models on diverse datasets such as RisCanvi and COMPAS reinforces their practical applicability in real-world contexts. The results demonstrate that integrated fairness techniques are adaptable to varying data distributions, making them more reliable for deployment in sensitive domains like criminal justice. For instance, combinations like `DIR+AL+EO` achieved strong fairness outcomes across both datasets, highlighting their generalizability. However, the dataset-specific variability observed in metrics like DI and SPD emphasizes the need for careful calibration of fairness techniques to align with the characteristics of the data and the fairness objectives of the application. These findings suggest that real-world deployment should prioritize integrated approaches that balance fairness and accuracy while being tailored to specific datasets. This ensures that fairness-enhanced models can address systemic biases effectively while maintaining stakeholder trust and confidence in their predictions.

7 Why do integrated fairness-enhancing techniques produce different results across datasets?

In the course of our analysis, a crucial question emerged repeatedly: why do different integrated fairness-enhancing techniques produce varying results? The variability in results arises from a complex interplay between dataset-specific characteristics, the design objectives of the fairness techniques, and the sensitivity of the fairness metrics employed. By examining the outcomes from both the COMPAS and RisCanvi datasets, we identify five key factors that can contribute to these differences.

7.1 Dataset-specific biases

Each dataset encodes distinct systemic and historical biases. COMPAS, for example, exhibits stronger correlations between demographic attributes (e.g., race) and the target variable (i.e., recidivism), leading to pronounced disparities in metrics like Disparate Impact (DI) and Statistical Parity Difference (SPD). Techniques like Disparate Impact Remover (DIR) are more effective on COMPAS due to its imbalanced feature distributions but show reduced impact on RisCanvi, where biases are subtler and less correlated with features. This highlights the importance of tailoring fairness techniques to the dataset's specific biases.

7.2 Mechanisms of fairness techniques

Different fairness techniques target different dimensions of bias, resulting in complementary, but dataset-dependent effects. For instance: pre-processing techniques (e.g., Reweighing, DIR) modify input data to balance group representation, leading to strong improvements in SPD and DI. However, their impact on metrics like Equal Opportunity Difference (EOD) may be limited by dataset-specific error distributions. In-processing techniques (e.g., Adversarial Learning, Exponential Gradient Reduction) enforce fairness constraints during training, excelling in metrics like Predictive Equality (PE) but being sensitive to the strength of correlations between features and sensitive attributes. Post-processing techniques (e.g., Equalized Odds Optimization) adjust decision thresholds to refine fairness in predictions, showing consistent improvements in PE and EOD but being less effective on datasets with residual feature imbalances.

7.3 Metric sensitivity to dataset properties

Fairness metrics capture different aspects of bias, and their sensitivity varies across datasets. For example: DI

and SPD, which focus on group-level disparities, are more sensitive to feature and demographic distribution, leading to variability between datasets. Likewise, EOD and PED, which evaluate error rate disparities, are influenced by both datasets bias and model generalization, making them more stable across datasets. Techniques like EGR+EO consistently improved EOD and PED on both datasets, highlighting their robustness.

7.4 Interactions in integrated techniques

Combining fairness techniques creates cumulative or competing effects depending on dataset characteristics. For instance: Re+EGR+EO showed strong synergy across both datasets, reducing DI, SPD, and EOD while maintaining accuracy. This is because reweighing balances the dataset by adding weights, EGR optimizes fairness constraints during training, and EO refines predictions. DIR+AL+EO, while effective on COMPAS, had diminished synergy on RisCanvi potentially due to weaker feature-label correlations, which limited the adversarial component's impact.

7.5 Trade-offs between fairness and accuracy

Integrated techniques must balance fairness improvements with accuracy preservation. Some combinations, such as Re+EGR+EO, achieved minimal accuracy loss while improving fairness across all metrics. Others, like DIR+AL+EO, demonstrated slightly reduced accuracy on RisCanvi, thus reflecting the cost of fairness and accuracy trade-offs when mitigating biases in the models.

8 Potential implications of the research findings

The findings of this study have far-reaching implications for society, the general public, and policymakers, particularly in deploying AI systems in high-stakes domains such as the criminal justice system for predicting the risk of recidivism. This research contributes to advancing fairness and trust in AI-driven decision-making by addressing systemic biases and improving fairness in recidivism prediction models.

For society, integrating fairness-enhancing techniques into AI systems is a crucial step toward promoting equity and mitigating systemic biases. Recidivism prediction models that fail to address fairness risk perpetuating cycles of inequality, disproportionately impacting marginalized communities. The findings of this study highlight that fairness-enhanced models can help disrupt these cycles by producing more equitable outcomes across demographic groups. In doing so, they contribute to fostering a society where AI-driven decisions align with democratic principles and ethical values, ensuring that individuals are treated fairly, regardless of their background.

Similarly, public trust in AI systems depends on their ability to deliver fair and unbiased outcomes, particularly in high-stakes applications such as predicting recidivism risk in the criminal justice system. By demonstrating the effectiveness of integrated fairness techniques across multiple datasets, this research provides evidence that AI systems can be designed to uphold fairness and equity while maintaining predictive reliability. This assurance has the potential to empower the public to trust and engage with AI technologies, fostering broader societal acceptance of their deployment in critical domains.

For policymakers, the findings underscore the importance of designing regulations and standards that prioritize fairness in AI systems. The variability in results across datasets highlights the need for policies that mandate fairness evaluations across diverse data contexts to ensure that AI models generalize effectively and do not perpetuate existing inequities. Additionally, this research provides a foundation for developing standardized frameworks for fairness metrics and mitigation techniques, enabling consistent and transparent evaluation of AI systems in practice. Policymakers can leverage these insights to establish guidelines for the responsible deployment of AI, ensuring that systems align with societal values and ethical principles.

Beyond the criminal justice system, the implications of this research extend to other high-stakes domains, such as healthcare, education, and employment, where algorithmic decisions significantly impact individuals and communities. The proposed fairness-enhancing framework offers a scalable and adaptable approach for addressing biases across various applications, providing a foundation for equitable and trustworthy AI systems in diverse contexts. Moreover, the emphasis on multi-dataset validation and fairness–accuracy trade-offs highlights the need for rigorous evaluation protocols to assess the generalizability and robustness of fairness-enhancing techniques.

Finally, this research demonstrates that fairness in AI systems is a technical challenge and an ethical imperative. Fairness-enhancing techniques play a pivotal role in advancing societal progress by addressing systemic inequities and fostering trust in AI-driven decision-making. The findings presented here provide actionable insights for researchers, practitioners, and policymakers, contributing to developing AI systems that align technological advancements with societal values of fairness, justice, and trust.

9 Potential challenges to the validity of the research

While our study demonstrates the effectiveness of integrated fairness-enhancing techniques in improving algorithmic fairness and balancing predictive accuracy, a few potential challenges to the validity of our research must be acknowledged to contextualize our findings and guide future research.

9.1 Limited dataset scope

This study focuses on two datasets, COMPAS and RisCanvi, both of which, while relevant and used in recidivism prediction research, may not capture the full range of demographic, legal, or geographic variability present in real-world criminal justice systems. The generalizability of our findings to other jurisdictions remains an open question. Future research should incorporate additional datasets with varying cultural, legal, and socio-political contexts to test the robustness and adaptability of integrated fairness approaches.

9.2 Absence of qualitative and stakeholder insights

This study adopts a quantitative approach centered on measurable fairness metrics and predictive accuracy. However, algorithmic fairness is inherently multidimensional, encompassing legal, ethical, and societal perspectives that may not be captured by metrics alone. The exclusion of insights from legal professionals, ethicists, and directly affected communities limits our ability to assess the practical implications and social acceptability of integrated fairness-enhancing interventions. Future work should incorporate participatory design methods and expert consultation to align fairness outcomes with stakeholder expectations and legal standards.

9.3 Static fairness evaluation

Our evaluation occurs at a single point in time based on historical data. We do not account for potential feedback loops that arise when AI systems influence future data (e.g., recidivism predictions affecting sentencing decisions, which then affect future data distributions). This static view overlooks the dynamic and evolving nature of fairness in deployed systems. Future work should explore longitudinal and causal analyses to understand how fairness interventions behave over time and under deployment.

9.4 Fairness techniques and metrics

This study focuses on six fairness-enhancing techniques and evaluates their impact using group fairness metrics, Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), which are commonly used to quantify disparities between demographic groups. While effective at capturing population-level bias, these metrics may obscure unfair treatment at the individual or subgroup level. Furthermore, fairness metrics can conflict with one another, making it difficult to achieve uniformly fair outcomes across all dimensions.

A further potential challenge to the validity of our research lies in the selection of a specific subset of fairness-enhancing techniques. Although the chosen methods were selected for their empirical performance and coverage across different stages of the AI pipeline, they do not represent the full range of available approaches. As the field of algorithmic fairness continues to evolve, future research should explore alternative fairness techniques to better account for context-sensitive and nuanced manifestations of bias.

10 Conclusion

The integration of fairness-enhancing techniques in AI systems holds significant promise for mitigating biases in recidivism prediction models within the criminal justice system. This study evaluates the performance of integrated techniques across pre-processing, in-processing, and post-processing stages using two recidivism datasets, COMPAS and RisCanvi, and employs a multi-objective optimization framework using the Pareto Front to select models that effectively balance fairness and predictive accuracy. The findings demonstrate that integrated approaches often outperform individual methods, yielding substantial improvements in fairness metrics, including Statistical Parity Difference (SPD), Disparate Impact (DI), Equal Opportunity Difference (EOD), and Predictive Equality Difference (PED), while maintaining minimal accuracy trade-offs. Integrated approaches, such as the combination of Reweighing, Exponential Gradient Reduction, and Equalized Odds Optimization (`Re+EGR+EO`) and as well as Disparate Impact Reduction and Adversarial Learning (`DIR+AL+EO`) exhibited robust performance across datasets, underscoring their potential for real-world deployment.

However, while the results indicate that integration can enhance fairness, performance varied significantly across datasets. The effectiveness of fairness-enhancing strategies is highly dependent on dataset characteristics, feature distributions, and demographic representation. For example, COMPAS benefited more from methods addressing

feature imbalances, whereas RisCanvi responded better to interventions targeting error distributions. These findings underscore the need for dataset-aware fairness strategies to improve generalizability and reliability across contexts.

Limitations and future work: Several limitations of this study also highlight promising directions for future research. First, the evaluation was conducted using only two datasets, COMPAS and RisCanvi, which may limit the generalizability of the results. Future work should include a broader range of datasets with diverse legal, cultural, and demographic characteristics to assess the robustness of integrated fairness-enhancing interventions. Second, this study adopted a purely quantitative approach, without incorporating the perspectives of key stakeholders, such as legal professionals, ethicists, and policymakers. Including these voices through participatory design and interdisciplinary consultation will help ensure that technical definitions of fairness align with real-world values and legal norms. Third, while this work evaluated a carefully selected subset of fairness-enhancing techniques, it did not explore the full spectrum of available methods. Future research should investigate alternative fairness strategies.

Finally, future studies should systematically investigate when and why integrated fairness approaches fail to produce expected improvements and explore the conditions under which integration may introduce new or unintended biases. Developing dynamic, data-aware fairness mechanisms and understanding their evolution across the AI pipeline will help advance more robust and adaptable solutions.

This work contributes to the field of AI fairness by providing a holistic framework, demonstrating the value of multi-dataset validation, and addressing fairness–accuracy trade-offs in a structured manner. Future research should extend this framework to other domains, incorporate real-time fairness adjustments, and assess the long-term societal impacts of fairness-aware AI. Achieving fairness in AI requires balancing technical innovation with ethical responsibility. By aligning AI systems with principles of inclusion, transparency, and justice, this work takes a step toward building more trustworthy and equitable technologies.

Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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References

Agarwal, A., Beygelzimer, A., Dudík, M., Langford, J., Wallach, H.: A reductions approach to fair classification. In: International Conference on Machine Learning, pp. 60–69 (2018). PMLR

AI, H.: High-level expert group on artificial intelligence. European Commission. Available at: https://ec.europa.eu/digital-single (2019)

Angwin, J., Larson, J., Mattu, S., Kirchner, L.: How We Analyzed the COMPAS Recidivism Algorithm (2016). https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm

Berk R, Bleich J (2014) Forecasts of violence to inform sentencing decisions. J Quant Criminol 30(1):79–96. https://doi.org/10.1007/s10940-013-9195-0

Beutel A, Chen J, Doshi T, Qian H, Woodruff A, Luu C, Kreitmann P, Bischof J, Chi EH (2019) Putting fairness principles into practice: Challenges, metrics, and improvements. In: Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society, pp. 453–459

Bellamy RK, Dey K, Hind M, Hoffman SC, Houde S, Kannan K, Lohia P, Martino J, Mehta S, Mojsilović A et al (2019) Ai fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. IBM J Res Dev 63(4/5):1–4

Berk RA, Kuchibhotla AK, Tchetgen Tchetgen E (2023) Fair risk algorithms. Annu Rev Stat Appl 10:165–187

Biswas A, Mukherjee S (2021) Ensuring fairness under prior probability shifts. In: Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society, pp. 414–424. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3461702.3462596

Castelnovo A, Crupi R, Greco G, Regoli D, Penco IG, Cosentini AC (2022) A clarification of the nuances in the fairness metrics landscape. Sci Rep 12(1):4209. https://doi.org/10.1038/s41598-022-07939-1

Corbett-Davies S, Pierson E, Feller A, Goel S, Huq A (2017) Algorithmic decision making and the cost of fairness. In: Proceedings of the 23rd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining, pp 797–806

Caton S, Haas C (2023) Fairness in machine learning: a survey. ACM Comput Surv 300–360 (2023) https://doi.org/10.1145/3616865

Caton S, Haas C (2024) Fairness in machine learning: A survey. ACM Comput Surv 56(7):1–38

Chouldechova A (2017) Fair prediction with disparate impact: a study of bias in recidivism prediction instruments. Big Data 5(2), 153–163. https://doi.org/10.48550/arXiv.1610.07524

Chakraborty J, Majumder S, Yu Z, Menzies T (2020) Fairway: a way to build fair ml software. In: Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, pp 654–665

Chen Z, Zhang JM, Sarro F, Harman M (2023) A comprehensive empirical study of bias mitigation methods for machine learning classifiers. ACM Trans Softw Eng Methodol 32(4):1–30

Formació Especialitzada (CEJFE), C.: Taxa de reincidència penitenciária 2020. Accessed: 2024-04-17 (2020). https://cejfe.gencat.cat/ca/recerca/opendata/presons/taxa-reincidencia-2020/index.html

Desmarais SL, Johnson KL, Singh JP (2016) Performance of recidivism risk assessment instruments in us correctional settings. Psychol Serv 13(3):206. https://doi.org/10.1037/ser0000075

Farber S (2024) Ai in terrorism sentencing: evaluating predictive accuracy and ethical implications. Crim Justice Stud 37(4):352–370. https://doi.org/10.1080/1478601X.2024.2415558

Ferrara E (2023) Fairness and bias in artificial intelligence: A brief survey of sources, impacts, and mitigation strategies. Science 6(1):3. https://doi.org/10.3390/sci6010003

Feldman, M., Friedler, S.A., Moeller, J., Scheidegger, C., Venkatasubramanian, S.: Certifying and removing disparate impact. In: Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 259–268 (2015)

Farayola, M.M., Malika, B., Saber, T., Connolly, R., Tal, I.: Enhancing algorithmic fairness: Integrative approaches and multi-objective optimization application in recidivism models. In: Proceedings of the 19th International Conference on Availability, Reliability and Security, pp. 1–10 (2024). https://doi.org/10.1145/3664476.3669978

Farayola, M.M., Tal, I., Malika, B., Saber, T., Connolly, R.: Fairness of ai in predicting the risk of recidivism: Review and phase mapping of ai fairness techniques. In: Proceedings of the 18th International Conference on Availability, Reliability and Security, pp. 1–10 (2023). https://doi.org/10.1145/3600160.3605033

Green, B.: The false promise of risk assessments: epistemic reform and the limits of fairness. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, pp. 594–606 (2020). https://doi.org/10.1145/3351095.3372869

Graffam J, Shinkfield AJ, Lavelle B (2014) Recidivism among participants of an employment assistance program for prisoners and offenders. Int J Offender Ther Comp Criminol 58(3):348–363. https://doi.org/10.1177/0306624X12470526

Gao, X., Zhai, J., Ma, S., Shen, C., Chen, Y., Wang, Q.: Fairneuron: improving deep neural network fairness with adversary games on selective neurons. In: Proceedings of the 44th International Conference on Software Engineering, pp. 921–933 (2022). https://doi.org/10.1145/3510003.3510087

Jo, N., Aghaei, S., Benson, J., Gomez, A., Vayanos, P.: Learning optimal fair decision trees: Trade-offs between interpretability, fairness, and accuracy. In: Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society, pp. 181–192 (2023)

Jacobs LA, Gottlieb A (2020) The effect of housing circumstances on recidivism: Evidence from a sample of people on probation in san francisco. Crim Justice Behav 47(9):1097–1115. https://doi.org/10.1177/0093854820942285

Jain B, Huber M, Elmasri R, Fegaras L (2020) Using bias parity score to find feature-rich models with least relative bias. Technologies 8(4):68. https://doi.org/10.3390/technologies8040068

Jain, B., Huber, M., Fegaras, L., Elmasri, R.A.: Singular race models: addressing bias and accuracy in predicting prisoner recidivism. In: Proceedings of the 12th ACM International Conference on PErvasive Technologies Related to Assistive Environments, pp. 599–607 (2019). https://doi.org/10.1145/3316782.3322787

Kozodoi N, Jacob J, Lessmann S (2022) Fairness in credit scoring: Assessment, implementation and profit implications. Eur J Oper Res 297(3):1083–1094

Kamiran, F., Karim, A., Zhang, X.: Decision theory for discrimination-aware classification. In: 2012 IEEE 12th International Conference on Data Mining, pp. 924–929 (2012). https://doi.org/10.1109/ICDM.2012.45 . IEEE

Kleinberg, J., Mullainathan, S., Raghavan, M.: Inherent trade-offs in the fair determination of risk scores. arXiv:1609.05807 (2016)

Kobayashi, K., Nakao, Y.: One-vs.-one mitigation of intersectional bias: A general method to extend fairness-aware binary classification. arXiv preprint arXiv:2010.13494 (2020)

Krasanakis, E., Spyromitros-Xioufis, E., Papadopoulos, S., Kompatsiaris, Y.: Adaptive sensitive reweighting to mitigate bias in fairness-aware classification. In: Proceedings of the 2018 World Wide Web Conference, pp. 853–862 (2018). https://doi.org/10.1145/3178876.3186133

Liu S, Vicente LN (2022) Accuracy and fairness trade-offs in machine learning: A stochastic multi-objective approach. CMS 19(3):513–537. https://doi.org/10.1007/s10287-022-00425-z

Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A (2021) A survey on bias and fairness in machine learning. ACM Computing Surveys (CSUR) 54(6):1–35

Miron M, Tolan S, Gómez E, Castillo C (2021) Evaluating causes of algorithmic bias in juvenile criminal recidivism. Artificial Intelligence and Law 29(2):111–147. https://doi.org/10.1007/s10506-020-09268-y

Plecko, D., Bareinboim, E.: Fairness-accuracy trade-offs: A causal perspective. arXiv preprint arXiv:2405.15443 (2024)

Pelegrina, G.D., Couceiro, M., Duarte, L.T.: A statistical approach to detect sensitive features in a group fairness setting. arXiv preprint arXiv:2305.06994 (2023)

Pagano TP, Loureiro RB, Lisboa FV, Peixoto RM, Guimarães GA, Cruz GO, Araujo MM, Santos LL, Cruz MA, Oliveira EL et al (2023) Bias and unfairness in machine learning models: a systematic review on datasets, tools, fairness metrics, and identification and mitigation methods. Big data and cognitive computing 7(1):15. https://doi.org/10.3390/bdcc7010015

Pleiss, G., Raghavan, M., Wu, F., Kleinberg, J., Weinberger, K.Q.: On fairness and calibration. Advances in neural information processing systems **30** (2017)

Raza, S., Pour, P.O., Bashir, S.R.: Fairness in machine learning meets with equity in healthcare. In: Proceedings of the AAAI Symposium Series, pp. 149–153 (2023)

Rodolfa, K.T., Salomon, E., Haynes, L., Mendieta, I.H., Larson, J., Ghani, R.: Case study: predictive fairness to reduce misdemeanor recidivism through social service interventions. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, pp. 142–153 (2020). https://doi.org/10.1145/3351095.3372863

Sharma S, Kumar V (2022) A comprehensive review on multi-objective optimization techniques: Past, present and future. Archives of Computational Methods in Engineering 29(7):5605–5633

Thiebes S, Lins S, Sunyaev A (2021) Trustworthy artificial intelligence. Electron Mark 31(2):447–464

Wang, C., Han, B., Patel, B., Rudin, C.: In pursuit of interpretable, fair and accurate machine learning for criminal recidivism prediction. Journal of Quantitative Criminology, 1–63 (2022) https://doi.org/10.1007/s10940-022-09545-w

Watamura, E., Liu, Y., Ioku, T.: Judges versus artificial intelligence in juror decision-making in criminal trials: Evidence from two preregistered experiments. PloS one 20(1), 0318486 (2025) https://doi.org/10.17605/OSF.IO/N6TM2

Wang, C., Li, J., Zhang, R.: A method to enhance structural fairness in large language models with active learning (2024)

Wadsworth, C., Vera, F., Piech, C.: Achieving fairness through adversarial learning: an application to recidivism prediction. arXiv preprint arXiv:1807.00199 (2018) https://doi.org/10.48550/arXiv.1807.00199

Wan M, Zha D, Liu N, Zou N (2023) In-processing modeling techniques for machine learning fairness: A survey. ACM Trans Knowl Discov Data 17(3):1–27

Zhang, M., Sun, J.: Adaptive fairness improvement based on causality analysis. In: Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, pp. 6–17 (2022). https://doi.org/10.1145/3540250.3549103

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Fairness and Bias in Algorithmic Hiring: A Multidisciplinary Survey

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Employers are adopting algorithmic hiring technology throughout the recruitment pipeline. Algorithmic fairness is especially applicable in this domain due to its high stakes and structural inequalities. Unfortunately, most work in this space provides partial treatment, often constrained by two competing narratives, optimistically focused on replacing biased recruiter decisions or pessimistically pointing to the automation of discrimination. Whether, and more importantly *what types of*, algorithmic hiring can be less biased and more beneficial to society than low-tech alternatives currently remains unanswered, to the detriment of trustworthiness. This multidisciplinary survey caters to practitioners and researchers with a balanced and integrated coverage of systems, biases, measures, mitigation strategies, datasets, and legal aspects of algorithmic hiring and fairness. Our work supports a contextualized understanding and governance of this technology by highlighting current opportunities and limitations, providing recommendations for future work to ensure shared benefits for all stakeholders.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Information systems** → *Data management systems*; • **Social and professional topics** → **Computing / technology policy**;

Additional Key Words and Phrases: Algorithmic hiring, Online recruitment, Algorithmic fairness, Bias, Anti-discrimination

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1 Introduction

New algorithms for **Human Resources (HR)** are developed and deployed every year. By one count, there are over 250 AI tools for HR on the market [99], with entire manuals for HR professionals available on the topic [80]. The average job posting yields more than 100 candidates [101, 206]. These mutually reinforcing factors were accelerated by the COVID-19 pandemic and recent advances in AI [170, 207]. As a result, prospective job applicants and their chances of success are increasingly influenced by algorithmic hiring technology, including automated job descriptions, resume parsers, and video interviews.

Workplaces and labor markets are fraught with biases, imbalances, and patterns of discrimination against vulnerable groups, including women, ethnic minorities, and people with disabilities [15, 23, 215]. While algorithmic hiring represents an opportunity to mitigate these biases, it also runs the risk of reinforcing and amplifying them, causing harm and hindering trustworthiness [168]. The debate on this topic is often polarized between techno-enthusiasm [139, 177] and pessimism [9, 132] due to partial perspectives on a field that is large and complex. In this article, we offer a multidisciplinary survey of fairness and bias in algorithmic hiring centered on computer science (focused on systems, algorithms, metrics, and datasets) and informed by related disciplines. We critically analyze available resources and methods, highlight common challenges, and identify opportunities to advance the field.

Related Work. Bogen and Rieke [30] presented a technical report that described algorithmic hiring tools available in 2018, together with selected sources of bias, and provided a US-centric review of relevant laws and policies. Köchling and Wehner [149] conducted a review of algorithms in HR from a business research perspective, focused on non-empirical articles. Their qualitative discussion raises awareness on discrimination in HR algorithms without delving into measures and mitigation strategies. The prior work most closely aligned to ours is the work of Rieskamp et al. [217], surveying nine articles on bias mitigation for algorithmic hiring. Our survey expands on this work by presenting more bias mitigation techniques, covering fairness measures, describing available datasets, and by situating them in the broader social and legal context characterizing algorithmic hiring. In concurrent work, Kumar et al. [154] survey the literature on fair recommender systems in the recruitment domain. While similar in spirit, their work focuses on ranking; our work presents a broader view of algorithmic hiring across different tasks and hiring stages, covering many technologies and **Bias Conducive Factors (BCFs)** throughout the algorithmic hiring pipeline.

Contributions and Audience. This work provides a contextualized treatment of fairness and bias in algorithmic hiring. It was carried out by a team with mixed backgrounds in computer science, law, and philosophy, including practitioners developing algorithmic hiring products. Our contributions, catering especially to practitioners and researchers, can roughly be divided as follows. Practitioners such as data scientists, engineers, and product managers will find (1) a detailed list and description of domain-specific factors that lead to biases in their systems (Section 3), (2) methods to mitigate these biases (Section 5), (3) guidance on their applicability in practice (Section 7), and (4) pointers to key legal references in the EU and the US (Section 8). Researchers will benefit from (4) an up-to-date description of hiring technology (Section 2), (5) a unified treatment of fair hiring measures (Section 4), and (6) a collection of datasets in this space (Section 6). Overall, an integrated coverage

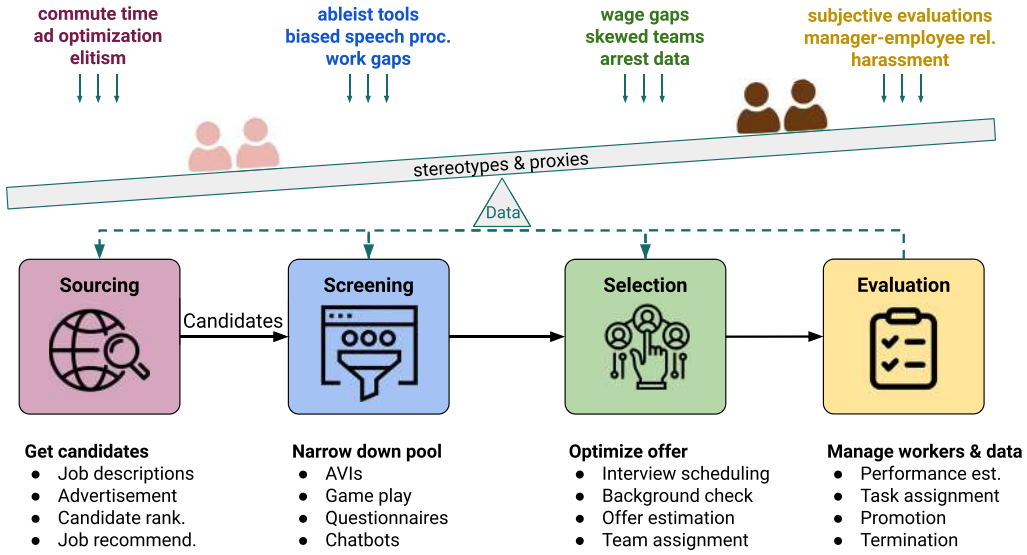


Fig. 1. Stages of algorithmic hiring with the main tools (below) and BCFs (above) tilting data against vulnerable groups, jointly with stereotypes and sensitive attribute proxies.

of these topics provides (7) a gentle primer for readers who are not experts in the field and (8) highlights important gaps and promising directions for future work in computer science at the intersection with law and policy (Section 10). Considering a broader audience, Sections 2, 3, 8, 10, and 11 cover a shared background which should be relevant for all readers, including HR professionals and legal scholars.

Structure. The remainder of this work is organized as follows. Section 2 introduces the main stages and systems in the algorithmic hiring pipeline. Section 3 focuses on bias, summarizing the most important factors in the labor market, the recruitment sector, and the tech industry that can lead to unfair recruitment systems. Sections 4–6 result from a systematic review of the literature with methods described in Appendix A. Sections 4 and 5 describe the main fairness measures and mitigation approaches, while Section 6 presents the datasets used in the algorithmic hiring literature. Section 7 discusses practical aspects of anti-discrimination in algorithmic hiring guiding the choice of fairness measures and mitigation strategies. Sections 8 and 9 widen the perspective beyond computer science by outlining the main legal frameworks and situating algorithmic hiring in its broader socio-technical context. Section 10 summarizes opportunities, limitations, and recommendations for future work; Section 11 provides concluding remarks.

2 The Algorithmic Hiring Pipeline

Algorithmic hiring comprises algorithms, tools, and systems to automate or assist HR decisions on candidate recruitment and evaluation. Elaborating on previous work from civil society [30], regulatory bodies [84], and academia [209], we distinguish four stages in the algorithmic hiring pipeline, reported in Figure 1.

Sourcing. The first stage of the hiring pipeline provides employers with a large pool of candidates for a given position. Historically, this is the hiring stage where algorithms are most prominent and well-researched. The main technological solutions and systems are summarized below.

— *Job advertisement* consists of tools to describe a vacancy and make it visible. Job descriptions [128], possibly written by language models [207], are optimized and shared through suitable channels, including employer websites [43], dedicated platforms [198], and ad delivery services [4, 158].
— *Search, ranking, and recommendation* algorithms favor matches between jobs and job seekers by ranking candidates for openings [103] and vice versa: recommending openings to job seekers [185], based on a combination of machine learning, Boolean matching, and sorting (e.g., skills, location, industry), supported by information extraction systems [47]. Fairness is a critical and well-studied property for these systems [154, 265].
— *Social networks*, especially those embedded in job platforms, play an important role in the visibility of candidates [239] and their awareness of opportunities [96], favoring professional connections [39], contributing to the dissemination of information, and facilitating referral [284].

It is worth noting that stage categorization is elastic and usage-dependent. For example, algorithms that extract information from CVs to score employees against job descriptions are central to the sourcing stage, but may also be used for screening [58].

Screening. After sourcing, employers must narrow down large pools of candidates into manageable subsets that can be more thoroughly evaluated by HR specialists and other employees.

— *Gameplay* is used for psychometric assessment and soft skills measurement. Companies develop proprietary suites of games and make them available to candidates, e.g., through a mobile app. Models analyze their gameplay to explicitly estimate competency scores [189] or implicitly look for similarities with desirable candidates such as current employees [271].
— ***Asynchronous Video Interviews (AVI)*** are recordings of candidates' answers to a specific set of questions in front of a camera. AVI models rely on different data modalities, including visual (e.g., facial expressions), verbal (e.g., length of sentences), and paraverbal features (e.g., tone) to infer a variety of traits related to personality and hireability [33, 120].
— *Questionnaires* in hiring cover a wide range of purposes, such as personality assessment [129], job performance inference [38, 95], or explicitly asking information about job requirements to filter candidates (e.g., Driver's license ownership) or prioritize them (e.g., years of experience).
— *Chatbots* can mediate interactions between employers and employees by asking basic screening questions for job seekers and scheduling interviews [30, 44]. Advancements in large language models [190] are likely to increase the influence of this technology on hiring processes, broadening its reach to other stages of recruitment, including sourcing and evaluation [170, 224].

Selection. After screening, candidates are interviewed by HR specialists or other employees. These interviews can include technical and cultural questions influenced by previous stages, e.g., by questionnaire answers. Several tools are available at this stage to help employers select the most desirable job seekers and extend them a suitable offer.

— *Compensation and benefit optimization* allows employers to target their offers to each candidate, thanks to tools that estimate the likelihood of acceptance based on salary, bonus, stock options, and other benefits [30, 176].
— *Background checks*, primarily focused on criminal records [59, 66, 157], are used by employers to obtain additional information on candidates before hiring them. Background checks can also target social media [127].

— *Team placement* concerns the assignment of a selected candidate to a specific role within a team or project [107, 202].

Evaluation. The hired employees are managed and evaluated on their career progression. While, strictly speaking, these are post-hiring processes, it is fundamental to consider them due to their feedback loops on algorithmic recruitment.

— *Performance and career management* tools facilitate employee development and monitoring. Technology in this area supports training assignment [36], work allocation [253], career development [188], along with monitoring and estimation through increasingly available tracking technology [64, 140, 232, 236].
— *Engagement analytics* measures employee satisfaction, commitment, and retention probability, often leveraging dedicated questionnaires and surveys [149, 277].
— *Career progression* deals with promotion and dismissal of employees. Succession plans and promotions can be partially automated [169, 188]. Turnover prediction is an active area of research, typically presented as part of analytics to favor retention [204]. It is worth noting that turnover data, frequently used to improve hiring systems [38], can also be repurposed toward employee termination. Indeed 98% of 300 HR leaders from US companies interviewed in a 2023 survey, report that software and algorithms will assist them with layoff decisions throughout the year [259].

The progression of candidates through the stages of the hiring pipeline is accompanied by a feedback loop in which the decisions at each stage generate data that influence the remaining stages in subsequent interactions. Evaluation, for example, can lead to job termination impacting tenure, which, in turn, represents a key desideratum and prediction target for sourcing and screening algorithms. These data are influenced by a diverse and complex set of factors, many of which can lead to undesirable discrimination, as described in the next section.

3 BCFs

Hiring data and algorithms can display undesirable groupwise patterns caused by BCFs affecting the employment domain. These patterns put some job seekers at a systematic disadvantage based on sensitive attributes such as age, disability, gender, religion or belief, racial or ethnic origin, or sexual orientation. Overall, disparities may exist in data sample composition and feature values across sensitive groups that reflect and amplify problematic structural differences in society. First, hiring datasets display measurement errors whose severity varies with protected groups, most critically in target variables related to employability, reflecting current biases in human ratings. Moreover, biased decisions at early stages result in downstream samples misrepresenting certain groups. Finally, there are some higher-order effects caused by technological blind spots and biases in external tools integrated into the hiring pipeline. Some of these biases are reported in Figure 1; at their root there are two overarching BCFs that are worth highlighting from the outset.

Stereotypes are widely held beliefs about groups of individuals with a common trait, including their propensity and ability to perform a given job [118]. Stereotypes are sustained by culture, socialization, and experience [50, 50]. They are often acquired at an early age [134] and can be activated unconsciously [181]. Even when outspokenly rejected, stereotypes affect the lives of individuals both descriptively and prescriptively based on coarse categories [76]. In turn, this ends up shaping expectations about the qualities, priorities, and needs that people have about themselves and others, including, and perhaps especially, about work [29, 50, 118, 248]. For example, *agency* (orientation toward leadership and goal attainment) is stereotypically associated with men, while communion (warmth and propensity for care) is frequently associated with women [72, 118], with far-reaching effects of gender roles and expectations in employment [118, 215].

Sensitive Attribute Proxies. Stereotype activation and entrenchment is not always direct; stereotypes about a group can be activated by proxies [211]. This is especially true for algorithms trained to make inferences from inputs that are strongly correlated with sensitive attributes. For instance, video interviews and resumes contain a wealth of information on gender, race, and other sensitive attributes [63, 69, 189, 218, 222], which can lead algorithms to learn stereotypical associations between sensitive and target variables encoded in the data. More specifically, voice timbre and physical appearance in videos may be used as proxies for gender, while names and spoken languages in CVs may correlate heavily with certain migration backgrounds. Sensitive attribute proxies allow models to learn and reflect the diverse BCFs described in the following section.

3.1 Institutional Biases

The first family of BCFs we introduce are institutional biases. These are practices, habits, and norms shared at institutions, such as companies and societies, which reflect negatively on the probability of positive algorithmic hiring outcomes for disadvantaged groups.

Direct discrimination takes place when disparate outcomes are explicitly caused by sensitive attributes. This type of bias can be difficult to prove, as it requires careful control over non-sensitive attributes to keep them as constant as possible, while only varying sensitive attributes. Famous correspondence studies have shown that black applicants are less likely to be contacted after applying for a job compared to otherwise identical white candidates [23, 147]. Similar results have been found in interview evaluations provided by US-based judges, showing a preference for standard American accents over international ones [67, 164, 205]. Numerous field experiments using matched pairs of applicants have highlighted discrimination against women and non-whites; experiments also point to a risk of direct discrimination against disabled and older applicants [208, 215]. This is the most obvious BCF; a variety of more subtle ones are listed below.

Horizontal Segregation. Job segregation (i.e., the world today) plays a fundamental role in hiring decisions (i.e., the world tomorrow). Prior experience is considered a fundamental predictor of suitability for a given position [101]. Horizontal segregation concerns differences in employment rate across industry sectors associated with sensitive attributes, such as gender and race [28, 243]. Strong gender patterns in diverse regions of the world are linked to persistent gender stereotypes about agency and communion that shape our perception and expectations [42, 83, 109]. Most field experiments, for example, demonstrate discrimination against women in men-dominated jobs and vice versa [215, 216].

Vertical segregation summarizes differences in career progression to leadership positions. Predominantly analyzed with respect to binary gender [53, 82], recent studies have also highlighted *glass ceiling effects* for non-binary workers [62], racial minorities [117], and intersectional identities [273]. When translated into data and dignified with a *ground truth* status, vertical segregation leads to models that reinforce wage gaps [222] and lack of diversity in high-status positions.

Cultural fit is often considered predictive of a candidate's ability to conform and adapt to the core values and collective behaviors of an organization. Evaluations of cultural fit by recruiters, subjective or objective, can contribute to maintaining a uniform workforce in the company, especially in more senior positions, reinforcing horizontal and vertical segregation [3, 219].

Elitism in recruiter evaluations favors candidates educated at prestigious institutions who can also list specific extracurricular accomplishments correlated with socioeconomic status [218], further entrenching family status and social class [51, 184].

Biased employee evaluation has different forms and derives from multiple causes. Two key drivers are stereotypes and employee–manager relationships. Gender and race stereotypes about competence drive people's perception on the workplace and potentially bias supervisor evaluations against female and black employees [116, 229, 233, 242]. Furthermore, a combination of vertical

job segregation and homophily entails that minority employees who are not well represented in management positions may receive lower ratings when manager–employee personal relationships have a positive influence on evaluations and promotions [26, 31, 152].

Stereotype violation has a disparate effect on the "transgressor," depending on their gender. Women are frequently penalized for gender norm violations that make them appear more agentic (e.g., competent and self-confident) and less communal (e.g., kind and warm) than stereotypically expected [115, 195, 247]. This is especially problematic in the hiring domain, where agency is deemed important for leadership positions and it is harder to demonstrate for women without giving an impression of low communion [160].

Workplace proximity and availability of reliable commutes can influence job satisfaction [201] and candidate–employer interactions [262], with amplifying effects introduced by technology [142]. Since discrimination has shaped residential patterns and influenced public transportation, this factor may have disproportionate impacts along racial and ethnic lines [144].

Wage gaps are a key part of power differences between genders and races [40, 81, 121, 269]. This BCF has several concurrent causes, including expectations reinforced by the *status quo* and lower success in salary negotiation [68, 110, 163]. Models that predict the likelihood that candidates will accept an offer can reinforce existing wage gaps.

Social networks topology influences job seekers' awareness of openings [96], and their likelihood of being successfully screened by recruiters due to a successful referral [136, 284]. Well-connected candidates have an inherent advantage over candidates with lower centrality; this advantage may be amplified by algorithms for link prediction [264]. The connection recommender system in a professional social network was found to underperform for women, recommending them less frequently for professional connections than men with similar success rates [39]. This effect may be exacerbated by homophily and biased preferential attachment [14].

3.2 Individual Preferences

Next, we consider BCFs that are an apparent consequence of individual preferences, but represent generalized patterns for protected groups. Listing a bias under this category does not make it an individual responsibility and a reasonable ground for discrimination. On the contrary, we aim to highlight the fact that some seemingly individual choices operated by candidates actually result from wider and recurrent patterns associated with protected attributes. Therefore, employers and providers of algorithmic hiring models should carefully consider these BCFs.

Job satisfaction influences job commitment [20, 228]. Historically disadvantaged groups such as transgender, non-binary, female, black, and disabled workers are more likely to experience discrimination and harassment on the job [225, 228, 261]. This fact may be reflected in datasets as a lower tenure for these groups, which can be penalized by algorithms trained to maximize tenure in order to reduce hiring costs and retain human capital.

Self-promotion gaps related to gender have been documented in the hiring domain [5] and beyond; men tend to self-evaluate higher than women even without intrinsic incentives [87]. This reflects on the visibility and perceived competence of candidates at the different stages of the hiring pipeline; it can be especially difficult to subvert for women due to the social and economic penalties incurred for violating female gender stereotypes [182].

Willingness to commute is another individual factor with gender-related patterns. Women tend to be more restrictive in their choice of job search area [78, 161]. This difference may be related to gender roles with respect to household and childcare responsibilities. Furthermore, people with a migration background are less likely to own a motor vehicle [146].

Salary negotiation differences between men and women are documented, including in propensity [163] and strategy [110]. Different interpretations have been advanced, including lower risk aversion

and the perceived chance of success [110, 122]. Although ostensibly a personal outcome of female candidates, unsuccessful salary negotiation is also based on an unfair *status quo* that influences group expectations [110].

Culture-based avoidance or attraction is a self-selection BCF that influences mainly the sourcing stage. Explicitly mentioning requirements such as community outreach can signal the posture of an employer and act as a pull factor for minority job seekers [192, 234]. On the contrary, job descriptions with unrealistic requirements discourage job seekers who lack one or more requirements, with a repulsive effect that can be stronger for candidates from vulnerable groups [101, 179]. Wording itself can also sustain inequality: gendered wording in job advertisements that makes use of stereotypically agentic (male) language, such as "leadership" or "delivery," can make a position less attractive for women [102]. The same alienating effect can occur by signaling an unwelcoming workplace culture at other stages of the hiring pipeline [274].

Work gaps, i.e., periods without formal employment, often reflect negatively on a candidate's probability of securing a job [101]. Gender asymmetries in caregiving responsibilities put women at a systematic disadvantage [93, 162].

3.3 Technology Blind Spots

Finally, we describe the biases introduced by biased components integrated into larger algorithmic hiring pipelines. This non-exhaustive list aims at demonstrating the need for proactive bias-preventing reasoning also (and perhaps especially) when using off-the-shelf tools.

Ad delivery optimization can skew the audiences reached by job advertisements. Multiple studies have shown that maximization of cost-effectiveness based on ad delivery metrics, such as impressions or clicks, makes delivery skewed in accordance with gender and race stereotypes in jobs [4, 10, 61, 158]. Ad text and images can further skew the audience [4]. This happens even though advertisers design neutral campaigns and is exacerbated if they target specific attributes [258], increasing the opportunity for bias and opacity at the sourcing stage. It should be noted that the platform(s) chosen to run a campaign can introduce a further bias in favor of its predominant demographics; campaigns run on platforms that cater to younger users, for example, are less likely to reach older segments of the population.

Accessibility issues and ableist norms can discourage disabled people from job applications [226] and tilt evaluations against them [245, 246]. AVIs can produce specific patterns from candidates with speech impairment (e.g., short answers) or mismeasure input from candidates with sight impairment (e.g., eye contact), which are judged unfavorably by algorithmic recruitment models [189].

Disparate performance of language processing and computer vision tools has been widely demonstrated with respect to gender, race, and other sensitive attributes [27, 37, 241]. Off-the-shelf algorithms from these domains integrated into hiring pipelines [141] are likely to underserve minority candidates for feature extraction and negatively affect algorithms that are based on these feature.

Differences in platform engagement broadly divide people into frequent and infrequent users. This causes an overrepresentation of the former in the training data, leading to rich-get-richer dynamics in job platforms [185]. Note that minority job seekers may suffer a lower quality of service, leading to disengagement and triggering a negative feedback loop by iteratively lowering training representation, quality of service, and engagement. Moreover, platforms are more likely to have rich profiles and metadata for their common users, while lacking information for less engaged job seekers, who are penalized due to missing data.

Biased psychological assessment performance for minorities can result in systematic disadvantages [212]. Systematic differences in the results of psychometric tests between subgroups [52, 130], caused

by low discriminant validity [133] or construct contamination [8], can be especially problematic for tests embedded in larger resource allocation processes, such as hiring systems.

Background check tools allow employers to obtain additional information on candidates from public records [244]. These databases contain a wealth of pre-conviction information, such as arrest data [157] whose validity as a proxy for crime is questionable [97]. Among other critical aspects, criminal background checks before employment run the risk of feeding the racial disparity of policing [203] and other areas of the criminal justice system [151] into hiring systems.

Interaction biases between employers and hiring technology can amplify tiny differences in algorithmic output [16]. During the sourcing stage, for instance, recruiters tend to focus on very specific credentials, certifications, and keywords that exclude atypical individuals from the initial pool of candidates, despite having the right skills [101]. More general examples include position bias that causes underexposure risks for candidates that are not at the top of a ranking [60] and automation bias leading to over-reliance on technology and reduced human oversight [106, 156].

3.4 Overlaps and Interactions

In the analysis above, we identified and described how the workings of key BCFs contribute to algorithmic discrimination within a digital and non-digital hiring context. It is important to note that these factors are interlocking and overlapping insofar as BCFs are mutually reinforced by the structure of institutions (Section 3.1), individual preferences (Section 3.2), and technological blind spots (Section 3.3).

Intersectional identities are particularly vulnerable as they are likely to be affected by multiple BCFs. Let us consider the case of JS, a woman with children and a migration background looking for a job in the hospitality industry. Her chances of securing suitable employment is hindered from the outset as (1) her connections to the local industries are weak and advertisements do not reach her, keeping her unaware of certain possibilities; (2) she finds out about certain openings but is discouraged from applying by a difficult commute based on public transport. Through public employment services, her data are entered into a shared database accessed by HRs companies. (3) Her profile is hastily filled in and lacks important metadata; this fact pushes her down in rankings every time her profile matches recruiters' queries, also because the time of last login influences job seekers' rankings. (4) Recruiters seek a catering and hospitality degree that JS does not have, making matches with her profile infrequent. (5) Due to a lack of leadership roles in her profile, the only matches she receives are at the apprentice level and (6) she is screened out from openings employing AVI models that do not recognize her accent. She is interviewed by few businesses; most turn her down because (7) they deem her a poor cultural fit and (8) are afraid that she will apply for parental leave. One business is finally willing to hire her through an HR company. (9) Despite a mid-level payment agreement between the employer and the HR company, the latter estimates that JS will accept the minimum wage (which she does) and pockets the difference.

This fictitious, yet plausible story sketches how multiple biases compound and reinforce each other. However, even if intersectional discrimination occurs, there are still reasons why the right approach to technological design can reduce bias. One upshot of understanding bias as an inherently intersectional process is that it also offers a way to reduce discrimination. Since the factors that create bias are interrelated and mutually reinforcing, by halting or ameliorating one BCF, we may introduce positive feedback loops on other BCFs. By removing the discriminatory effect of any one factor, we can hope to reduce its influence on the other factors that reinforce each other in a discriminatory way. For example, an increased representation of a minority group in a company can improve job satisfaction, increase their importance and visibility, and encourage other minority applicants in future hiring rounds. With time, this contributes to reducing vertical segregation,

Table 1. Main Notational Conventions Used in This Work

$x \in \mathcal{X}$	A vector of non-sensitive attributes
$s \in \mathcal{S}$	A vector of sensitive attributes
$s = g$	A sensitive attribute value
$s = g^{\mathsf{c}}$	Complement of a sensitive attribute value
$y \in \mathcal{Y}$	Target variable from domain $\mathcal{Y}$
$i = (x_i, s_i, y_i)$	A data point or item
$\hat{y} = f(x)$	A classifier $f : \mathcal{X} \to \mathcal{Y}$ issuing predictions in $\mathcal{Y}$ for data points in $\mathcal{X}$
$f_{\text{soft}}(x)$	A soft classifier, from which predictions $\hat{y} = f(x)$ can be derived through thresholding
$\tau = \text{argsort}(f_{\text{soft}}(x))$	Ranking of items, typically sorted by target estimates
$i = \tau(k)$	Item at rank k in τ
N	Number of items
N_g	Number of items in g
N_g^k	Number of items in g in the top k positions of a ranking
D_g	Desired representation for group g
$h(x)$	A classifier $h : \mathcal{X} \to \mathcal{S}$ issuing predictions in $\mathcal{S}$ for data points
$h_{\text{soft}}(x)$	A soft classifier for sensitive attributes

with minority employees in key positions shaping company culture, and attracting more candidates with a similar background. This optimistic view reflects positive spillover effects [155, 174].

4 Measures

Fairness measures for algorithmic hiring consider different types of systems, including AVI scoring, CV ranking, and advertising algorithms, operating at different stages and on different data modalities. In this section, we present them in a unified notation (Table 1) and discuss their key dimensions, summarized in Table 2. It is worth noting that the next three sections are based on a systematic review of the literature summarized in Appendix A.

4.1 Dimensions of Fairness Measures

We begin with a summary of dimensions that are important in the algorithmic fairness literature; some are established in the literature, such as *flavor* [178] and *conditionality* [275], others are highlighted as important in the hiring domain, such as *granularity* and *interpretability*. It should be noted that we focus on group fairness; we found a single article treating individual fairness [172].

Flavor. Fairness measures can target different properties of an algorithmic decision-making system. The main notions of fairness in hiring are the following.

— *Outcome fairness* [114] is the most common. It looks at predictions from the perspective of candidates, measuring differences in their preferred outcome, such as being screened in, typically corresponding to a positive prediction $\hat{y}$. Systems are deemed fair from an outcome perspective if quantities such as acceptance rates ($\Pr(\hat{y} = 1)$) or true-positive rates ($\Pr(\hat{y} = 1|y = 1)$) are similar between groups.
— *Accuracy fairness* [24] takes a closer perspective on the decision maker, requesting equalization of accuracy-related properties between groups, such as the average absolute error. Measures in this family lack the notion of preferable outcomes and candidate benefits.

Table 2. Main Measures of Fair Hiring and Their Properties

	Used in	Flavor	Cond.	Multi.	Gran.	Norm.	Interp.
skew@k	[103]	Outcome		No	No	✓	D
NDKL	[11, 103]	Outcome		✓	✓	✓	No
DI	[33, 38, 69, 148, 271]	Outcome		✓	No	✓	✓
DD	[222]	Outcome		No	No	✓	✓
RPP	[4, 132, 199]	Outcome		No	No	No	✓
DRD	[283]	Outcome		No	✓	✓	✓
SKL	[199]	Outcome		No	✓	✓	No
LRR	[48]	Outcome	x	No	✓	No	D
TPRD	[63, 119]	Outcome	y	No	No	✓	✓
FNRR	[148]	Outcome	y	No	No	✓	✓
MED	[231]	Outcome	y	✓	✓	✓	✓
RMS	[119]	Outcome	y	✓	No	No	No
xEO	[185]	Outcome	y	No	✓	No	No
MAE	[231, 276]	Accuracy	y	No	✓	✓	No
BCRD	[148]	Accuracy	y	No	No	✓	No
MID	[148]	Accuracy	y	No	No	✓	No
SD	[222]	Impact		No	No	✓	✓
GBS	[128]	Representational		No	n.a	No	✓
sAUC	[33, 119, 120, 194, 196, 222]	Process		No	n.a.	No	n.a.
GTR	[196]	Process	x	No	n.a.	No	n.a.
MIA	[276]	Process	y	✓	No	✓	n.a.

Columns report the fairness dimensions described in Section 4.1.
In the last column, "D" denotes direction-interpretability, for measures which clearly convey whether one group is at an advantage; ✓indicates both direction- and magnitude-interpretability, the latter being assigned to measures intuitively quantifying the advantage; "No" indicates none of the above; "n.a." in process and representational fairness stand for *not applicable*.

—*Impact fairness* relates algorithmic outcomes to downstream benefits or harms for individuals. These measures are rare in the literature, as they require a broader understanding and modeling of the sociotechnical system around the decision-making algorithm.
—*Process fairness* [111] is a notion that considers the equity of the procedure leading to a decision. Related to *procedural justice* [235], process fairness has been operationalized for algorithmic decision-making based on people's approval for the use of a given feature in a specific scenario. In hiring, this has been associated with the predictability of sensitive attributes from non-sensitive ones [33], implying that the absence of information on sensitive attributes in the data will lead to fair decision-making.
—*Representational fairness* [1] relates to stereotyping and biases in representations. In the context of algorithmic hiring, it is especially relevant for the wording of job descriptions which can skew the probability of application for different demographics [102].

Conditionality. Conditional fairness measures accept group differences, as long as they can be attributed to a set of variables deemed acceptable grounds for differentiation. This dimension is

strongly connected to world views [100, 123], and the extent to which differences in variables between sensitive groups are influenced by measurement errors and unjust social structures.

Granularity. The same model can lead to different results if used in different ways, including the use of different thresholds for classifiers or different cutoffs for rankers. Measures with fine granularity take this fact into account by measuring the fairness of a system across different operating conditions, as opposed to coarse-granularity measures which consider a single point of operation.

Normativity. Measures with explicit normative reasoning set a precise target for groupwise quantities, either in absolute terms or relative to each other. Implicit normative reasoning, on the other hand, is typical of measures introduced without a complete discussion of desiderata. Strong normative reasoning is, in general, preferable, but it is not a guarantee on its own. For example, while a measure providing a contextualized and accurate operationalization of a construct defined in the law displays normativity, the measure will inherit all the limitations of the underlying law.

Interpretability. We consider the interpretability of group fairness along two dimensions. Direction-interpretability allows us to immediately understand whether a group is at an advantage or disadvantage by comparing the measure against a threshold. Magnitude-interpretability lets us quantify the (dis)advantage by evaluating the distance from a threshold.

Native Multinarity. A measure is multinary if it accounts for more than two sensitive attribute values. Binary measures can sometimes be extended to the multinary case, whereas truly multinary measures natively account for this occurrence.

4.2 Notation

Let $x \in \mathcal{X}$ denote a vector containing non-sensitive features and let $y \in \mathcal{Y}$ be an unknown target variable, inferred at test time as $\hat{y} = f(x)$. Furthermore, let $f_{\text{soft}}(x)$ denote a scoring function supporting estimates $\hat{y}$ via thresholding and item ranking $\tau = \text{argsort}(f_{\text{soft}}(x))$ via sorting. Furthermore, let $s \in \mathcal{S}$ indicate a vector of sensitive attributes, so that $s = g$ defines a specific protected group, and $s = g^{\text{c}}$ represents its complement. Overall, a data point or item i is indicated as $i = (x_i, s_i, y_i)$ and $i = \tau(k)$ denotes the item at position k in ranking τ. For cardinality, let N denote the total number of data points in a sample of interest, let N_g indicate the number of items in group g, and let N_g^k be the number of items in g among the top k of a ranking τ. Additionally, let $D_g \in [0, 1]$ denote the desired representation for items in g among the ones receiving a favorable algorithmic outcome, e.g., the top-ranked items or the positively classified ones. Finally, let $h : \mathcal{X} \to \mathcal{S}$ denote a classifier issuing predictions in $\mathcal{S}$ and let $h_{\text{soft}}(x)$ indicate its soft scoring version (e.g., issuing posterior membership probabilities).

4.3 Measures

4.3.1 Outcome Fairness. These measures summarize the perspective of candidates by focusing on their preferred outcome; they are the most common.

Skew at k (skew@k) [103] evaluates a ranking at a specific rank k. It computes the logarithm of the ratio between the desired representation of a sensitive group D_g and the actual representation (N_g^k/k).

$$\text{skew@k}_g = \log\left(\frac{N_g^k/k}{D_g}\right). \tag{1}$$

This measure exhibits strong normativity, as it requires a precise definition of target representation D_g for each sensitive group. It is direction-interpretable, as values above zero indicate an advantage for group g, while the presence of the log function hinders its magnitude-interpretability. As a

summary for multiple groups, Geyik et al. [103] consider the maximum and minimum skew values, e.g., max-skew@k = $\max_{g\in\mathcal{S}}$ skew@k$_g$.

Normalized Discounted Cumulative Kullback-Leibler Divergence (NDKL) [11, 103] computes a divergence between the desired distribution of the representation ($D = \{D_g, \forall g \in \mathcal{S}\}$) and the actual one at rank k ($N^k = \{N^k_g/k, \forall g \in \mathcal{S}\}$). The measure consists of KL divergences, calculated at each rank, and aggregated with a logarithmic discount, making it granular. Notice that KL divergence is another way to summarize skew@k for multiple groups, weighted with a logarithmic discount. It is defined as

$$\text{NDKL} = \frac{1}{Z}\sum_{k=1}^{K}\frac{1}{\log_2(k+1)}d_{\text{KL}}(D, N^k), \tag{2}$$

where Z is a normalization constant. Note that several factors contribute to making NDKL difficult to interpret. First, advantages and disadvantages for a group g at some rank k can yield the same KL divergence value. Second, advantages and disadvantages at different ranks do not compensate each other; if a group is under-represented at low ranks, its overrepresentation at high ranks leads to even worse values of NDKL, despite being a form of compensation.

Acceptance Rate Ratio, better known as **Disparate Impact (DI)** [33, 38, 120, 148], is a measure of disparity in classification with a fixed threshold. Note that this measure is also used in rankings by translating a threshold (or percentile) into a cutoff rank k. This measure is related to US labor law, adverse impact, and the 80% rule (Section 8.2). Below, we report its popular min-over-max version, computing the selection rate ratio between the worst-off and best-off groups:

$$\begin{aligned}\text{DI} &= \frac{\min_{g\in\mathcal{S}} N^k_g/N_g}{\max_{g\in\mathcal{S}} N^k_g/N_g}\\ &= \frac{\min_{g\in\mathcal{S}} \Pr(\hat{y}=1|s=g)}{\max_{g\in\mathcal{S}} \Pr(\hat{y}=1|s=g)}.\end{aligned} \tag{3}$$

We consider this measure interpretable, so long as the best- and worst-off groups are reported.

Acceptance Rate Difference better known as **Demographic Disparity (DD)** [222] focuses on disparities in groupwise acceptance rates, similarly to DI, but computes the difference instead of the ratio. We define it below for classifiers and ranking algorithms, implicitly assuming a cutoff point at rank k for the latter:

$$\begin{aligned}\text{DD} &= N^k_g/N_g - N^k_{g^c}/N_{g^c}\\ &= \Pr(\hat{y}=1|s=g) - \Pr(\hat{y}=1|s=g^c).\end{aligned} \tag{4}$$

This approach may be easier to interpret [286], but fails to capture large disparities when acceptance rates are small. If, for example, $\Pr(\hat{y}=1|s=g) = 10\text{e}{-02}$ and $\Pr(\hat{y}=1|s=g^c) = 10\text{e}{-07}$, we measure a low value DD $\simeq 10\text{e}{-02}$, despite a difference of five orders of magnitude in acceptance rates.

Representation in Positive Predicted (RPP) [4, 132, 199] is a measure used in non-cooperative audits, where the overall population of interest for an algorithm is unknown. In online advertising, campaigners are informed about the size and composition of the audience reached by an ad (positive predicted class) but have no information on platform users and users who were active and thus candidates for an ad impression. This makes it impossible to compute groupwise acceptance rates ($\Pr(\hat{y}=1|s=g)$), but allows for estimates of groupwise representation in the positive predicted

class ($\Pr(s = g|\hat{y} = 1)$) and, trivially, its difference with respect to the complementary group:

$$\text{RPP}_g = \Pr(s = g|\hat{y} = 1) \tag{5}$$

$$\text{RPPD} = \text{RPP}_g - \text{RPP}_{g^c} = 2\,\text{RPP}_g - 1. \tag{6}$$

True-Positive Rate Difference (TPRD) [63, 119] measures disparities in true-positive rates (also known as *recall*) between different sensitive groups. This measure is closely related to equal opportunity [114] and separation [17] and presupposes the availability of a ground-truth variable y to condition on:

$$\begin{aligned} \text{TPR}_g &= \Pr(\hat{y} = 1|y = 1, s = g) \\ \text{TPRD} &= \text{TPR}_g - \text{TPR}_{g^c}. \end{aligned} \tag{7}$$

A closely related measure is the **False-Negative Rate Ratio (FNRR)** [148], defined as

$$\begin{aligned} \text{FNR}_g &= \Pr(\hat{y} = 0|y = 1, s = g) \\ \text{FNRR} &= \frac{\text{FNR}_g}{\text{FNR}_{g^c}}. \end{aligned} \tag{8}$$

As an aggregate measure for non-binary sensitive attributes, Hemamou and Coleman [119] propose the RMS of the vector containing all TPRD components:

$$\text{RMS} = \sqrt{\frac{1}{|\mathcal{S}|}\sum_{g\in\mathcal{S}} \text{TPRD}_g^2}. \tag{9}$$

Nandy et al. [185] propose an **eXtended Equality of Opportunity (xEO)** measure, which is a granular version of TPRD. They consider a soft classifier with a variable threshold and compute the Kolmogorov-Smirnov distance between the resulting score distributions for positives in different sensitive groups.

$$\text{xEO} = \max_x [|\Pr(f_{\text{soft}}(x) \leq t|y = 1, s = g) - \Pr(f_{\text{soft}}(x) \leq t|y = 1, s = g^c)|]. \tag{10}$$

Discounted Representation Difference (DRD) [283] is a granular measure of DD focused on rankings. It measures the difference in acceptance rates with a variable cutoff rank k by observing the sensitive attribute of the item $\tau(k)$ and applying a logarithmic rank-based discount before updating a counter. In other words, DRD measures groupwise representation as

$$\text{DRD} = \sum_{k=1}^{K} \frac{1}{\log_2(k+1)} [\mathbb{1}(s_{\tau(k)} = g) - \mathbb{1}(s_{\tau(k)} = g^c)]. \tag{11}$$

This measure is interpretable since it conveys the difference in candidates' exposure across groups under a recruiter browsing model with logarithmic decay. See Carterette [41] for an introduction to browsing models.

Score KL Divergence (SKL) [199] considers the distribution of scores ($f_{\text{soft}}(x_i)$) in different groups and measures unfairness by computing their KL divergence. Let D_g^f define the probability distribution of continuous scores $f_{\text{soft}}(x_i)$ for group g, then SKL is defined as

$$\text{SKL} = \text{KL}(D_g^f, D_{g^c}^f). \tag{12}$$

Log Rank Regression (LRR) [48] is a measure defined for search engines by fitting a linear model on the logarithm of rank $k = \tau^{-1}(i)$ at which candidates are presented on the result page. Independent variables comprise both sensitive s and non-sensitive attributes x, where the latter include skills and education. LRR reports the coefficient (and p-value) associated with the sensitive attribute. Fitting models to quantify the influence of a sensitive feature on an outcome variable

is typical of situations with limited access to the model(s) responsible for the outcomes. This is a common setting for external non-cooperative audits.

$$\log(\tau^{-1}(i)) = \beta_x x_i + \beta_s s_i + \mu + \epsilon$$
$$\text{LRR} = \hat{\beta}_s. \tag{13}$$

A similar approach is proposed in Lambrecht and Tucker [158] to estimate the influence of sensitive attributes such as age and gender in advertisement delivery optimization. This measure considers differences in outcomes acceptable, so long as they can be explained by non-sensitive attributes x.

Mean Error Difference (MED) [231] focuses on regression, measuring systematic groupwise biases. Following a more general paradigm, MED caters to the multinary case by considering the maximum and minimum (signed) bias.

$$\text{MED} = \max_{g \in \mathcal{S}} \frac{1}{N_g} \sum_{i \in g} (y_i - f_{\text{soft}}(x_i)) - \min_{g \in \mathcal{S}} \frac{1}{N_g} \sum_{i \in g} (y_i - f_{\text{soft}}(x_i)). \tag{14}$$

4.3.2 Accuracy Fairness. These measures study model accuracy across sensitive groups, aligning more closely with the perspective of decision makers. They are all conditional on the target variable y, inheriting the biases encoded in the ground truth.

Mean Absolute Error (MAE) [231, 276] targets regression problems similarly to MED (Equation (14)). It compares groupwise accuracy by measuring the absolute error for each individual and computing the average for items in the same sensitive group.

$$\text{MAE} = \frac{1}{N_g} \sum_{i \in g} |y_i - f_{\text{soft}}(x_i)| - \frac{1}{N_{g^c}} \sum_{i \in g^c} |y_i - f_{\text{soft}}(x_i)|. \tag{15}$$

Balanced Classification Rate Difference (BCRD) [148] is a measure of the disparity in classification accuracy between groups. It targets the balanced classification rate, defined as the average between the true-positive and the true-negative rate:

$$\text{BCR}_g = \frac{\text{TPR}_g + \text{TNR}_g}{2}$$
$$\text{BCRD} = \text{BCR}_g - \text{BCR}_{g^c}. \tag{16}$$

Mutual Information Difference (MID) [148] is another measure of classification accuracy fairness. It computes model accuracy within a group g as the **Mutual Information (MI)** between the target y and the prediction $\hat{y}$ for the items in g and compares it against the MI for the remaining items.

$$\text{MI}_g = \text{MI}_{\{i \in g\}}(\hat{y}_i, y_i)$$
$$\text{MID} = \text{MI}_g - \text{MI}_{g^c}. \tag{17}$$

4.3.3 Impact Fairness. This flavor of fairness models the impact of algorithmic outcomes on data subjects, measuring differences in harms and benefits between populations.

Salary Difference (SD) [222] is the only measure of this family proposed in the literature. Considering an embedding-based model, each candidate is matched to the closest vacancy in the embedding space. Based on the average wage for the role described in the vacancy, an average salary is calculated for male and female candidates in the same industry, and their difference quantifies

the gender impact of the model on earnings in that industry.

$$f(x_i) : \text{closest job match for candidate } i$$
$$W(f(x_i)) : \text{average wage for job} f(x_i)$$
$$\text{SD} = \frac{1}{N_g} \sum_{i \in g} W(f(x_i)) - \frac{1}{N_{g^c}} \sum_{i \in g^c} W(f(x_i)). \tag{18}$$

4.3.4 Representational Fairness. Measures to quantify stereotypes and representational harms are relatively understudied [1, 92], despite being a fundamental driver in the reinforcement of societal biases.

Gender Bias Score (GBS) [128] is a coarse measure of bias toward "masculine" or "feminine language" in job descriptions. It takes advantage of the word inventory from Konnikov et al. [150], which comprises traits associated with gender roles in labor, so that each term t in a job description can theoretically be coded as belonging to a set of stereotypically masculine, feminine, or neutral traits. The measure is defined as

$$\text{GBS} = \text{sign}(x_m - x_f) \cdot \max \left\{ \frac{x_m - x_f}{x_m}, \frac{x_f - x_m}{x_f} \right\}, \tag{19}$$

where x_m (x_f) represents the number of stereotypically male (female) words in a job description.

4.3.5 Process Fairness. These notions of fairness operationalize desiderata about decision-making beyond outcomes. Frequently, they derive from judgments of admissibility for specific variables x in a given scenario. In algorithmic hiring, the literature focuses on the amount of information on sensitive attributes s encoded by non-sensitive features x and the target variable y, with the goal of minimizing it.

Sensitive Area Under the Receiver Operating Characteristic Curve (sAUC) [33, 119, 120, 194, 196, 222] is a widely used measure of the information about sensitive attributes stored in (proxy) non-sensitive attributes. It is based on training a classifier $h(x)$ for the sensitive attribute s on non-sensitive features x; its accuracy is evaluated as the AUC. If little information on sensitive attributes can be recovered from the variables employed in the decisions, then the decision is deemed procedurally fair according to this definition.

$$\text{sAUC} = \text{AUC}(h_{\text{soft}}(x)). \tag{20}$$

Hemamou et al. [120] extend (sAUC) to the multi-class case by training one-vs-all classifiers and reporting their maximum AUC value.

Ground Truth Regression (GTR) [196] regresses (x, s) on y, and focuses on the latter coefficient ($\hat{\beta}_s$) similarly to LRR (Equation (13)), with the key difference that the target of the regression is the target variable. This measure estimates the importance of the sensitive attribute to predict the target, conditional on non-sensitive attributes. In other words, GTR audits the biases encoded in the target variable, recognizing it as a key factor for the decision-making process, regardless of the predictive model employed:

$$\log \left(\frac{y_i}{1 - y_i} \right) = \beta_x x_i + \beta_s s_i + \mu + \epsilon$$
$$\text{GTR} = \beta_s. \tag{21}$$

Both sAUC and GTR are *ex-ante* measures, i.e., they can be computed from the data before the final model is trained; therefore, the *granularity* dimension does not apply to them.

Mutual Information Amplification (MIA) [276] is another measure of process fairness, which considers the process fair if it does not reveal more information about sensitive attributes than the

ground truth does. This metric computes the MI between first $\hat{y}$ and s, and second between y and s; positive differences between the two are considered an undesirable amplification of information about sensitive attributes leaked through predictions.

$$\text{MIA} = \text{MI}(\hat{y}, s) - \text{MI}(y, s). \tag{22}$$

Differently from sAUC and GTR, MIA is an *ex-post*, model-dependent measure.

4.4 Discussion

In this section, we describe the main characteristics of the surveyed measures and how they inform fairness measurement choices.

Choosing a Measure. We introduce a made-up scenario to exemplify deliberation around fairness monitoring. *EquiHire*, a fictitious HR company, is choosing a fairness measure to monitor its candidate search systems for fairness with respect to ethnicity. For the sake of simplicity, they restrict the realm of possibilities to Table 2. Since ethnicity is a multinary sensitive attribute, they discard binary measures such as skew@k and DD. Moreover, EquiHire practitioners want to describe their equity strategy in white papers and external communication. Therefore, they prioritize an interpretable measure, excluding complex measures such as NDKL and SKL which would be difficult to explain. Although they have access to signals related to candidate fitness (y), they become available with great delay after hiring decisions are taken. To assess system fairness in a timely manner, they discard y-conditional measures such as TPR and RMS. Finally, since they have visibility into the downstream hiring decisions, they can use a low-granularity measure. These considerations lead EquiHire to choose DI as their primary fairness measure. For a complementary angle focused on outcome fairness, they choose to monitor wage differences for employees hired through their systems, adopting SD as an additional measure.

To reiterate, this is just one fictitious scenario and not a fixed recommendation for choosing the right measure. It demonstrates the importance of key characteristics of the measures summarized in Table 2 for practitioner decisions; in the paragraphs below, we expand on the factors driving these decisions.

Fairness Diagnostics vs. Fairness Optimization. Ease of interpretation is a desirable property, especially for fairness measures, as confirmed by the popularity of DI. Nevertheless, several fairness measures adopted in hiring are difficult to read in absolute terms, making it challenging to understand the severity of fairness violations based on their values. These values can only be interpreted relative to one another, i.e., it is clear that one value is more desirable than another, making them a possible target for optimization but less suited for diagnostics. This is especially true for measures of accuracy fairness such as BCRD and MID. Measures of outcome and impact fairness, such as DD and SD, tend to be more interpretable, since they are typically defined as differences between the probability of obtaining desirable outcomes for different groups. This holds especially for binary measures, while multinary measures, such as RMS, often sacrifice interpretability to summarize disparities across multiple groups.

Blurred Lines between Ranking and Classification. In the literature, classification and ranking are typically considered separate tasks with separate algorithmic fairness measures. In hiring, we find ranking systems evaluated according to classification measures such as DI [38], selection tasks solved with rankers [38], and target variables of similar nature encoded as binary, multinary, or continuous (Section 6). For this reason, the separation between classification and ranking is less rigid than in other domains, and the question of which measures are best suited to evaluate these systems arises. A fundamental part of the answer lies in the expected flexibility of use for these models. If the key aspects of their usage are fixed, including the cutoff thresholds for different outcomes, then a measure with coarse granularity, considering a single operating condition (typical of the

fair classification literature—e.g., DD, TPRD), is most suited for evaluation. On the contrary, for models whose operating conditions are more uncertain, requiring human interaction or threshold fine-tuning, it is preferable to adopt a measure with finer granularity (often derived from the fair ranking literature—e.g., NDKL, DRD), which averages outcomes across multiple realizations of a browsing model.

Diagnosing Biases. Fairness measures are especially useful when they provide actionable diagnostics by highlighting specific biases described in Section 3. Measures of process fairness such as sAUC and MIA detect the presence of sensitive attribute proxies. Dedicated proxy reduction approaches (Section 5) can mitigate the resulting risk of discrimination. Representational fairness (GBS) captures biases in representations; when applied to job descriptions, extreme values highlight stereotypically gendered language, signaling a risk of culture-based avoidance. Impact fairness can highlight nuanced aspects of hiring beyond selection rates; SD, for instance, captures wage gaps and salary negotiation differences prompting further analysis into remuneration packages. Measures of outcome (e.g., DI) and accuracy fairness (e.g., MAE) can serve as high-level metrics by highlighting the effects of multiple factors acting together. To exemplify, a combination of job segregation, work gaps, and disparate performance of language processing tools, even if relatively weak on their own, may trigger a very low (unfair) value of DI when they compound.

Assorted Fairness Flavors. No single measure provides a complete picture. Different fairness flavors have complementary strengths and weaknesses. Most of the fairness measures considered in hiring focus on outcome equity; the field is strongly influenced by the 80% rule (Section 8.2), which is appreciated as a quantitative rule of thumb, but also often criticized [74, 267]. Indeed, outcome fairness is based on a narrow view of algorithmic hiring as a single decision point abstracted away from its context, where equity is purely a function of algorithmic estimates and (sometimes) their similarity to target variables approximating a ground truth. Similar criticism can be moved to accuracy fairness, with the additional limitation that it is less interpretable and less aligned with the desiderata of job seekers. Departing from this approach, process fairness is operationalized with reference to the information on sensitive attributes encoded in the remaining variables. More work is required to understand process fairness in hiring more broadly and to suitably (if at all) operationalize it quantitatively. Impact fairness quantifies the downstream impacts of algorithmic outcomes on different populations; modeling important aspects of the surrounding socio-technical system, it goes beyond brittle algorithmic abstraction. Since these measures are exceedingly rare, we highlight the need for more context-specific research to understand and model the benefits and harms of decisions on job candidates at different stages of the hiring pipeline. Overall, practitioners developing equity monitoring protocols should consider multiple fairness angles to gain a nuanced and more complete picture.

Ignoring Privileged and Disadvantaged Groups. The definition of sensitive attributes in anti-discrimination law is informed by the historical (dis)advantage of specific groups [230]. Gender and race, for example, are considered sensitive attributes due to the recurrent structural disadvantages incurred by women and black people. This is especially true in hiring, where surveys and meta-analyses find consistent biases against women and ethnic minorities [23, 147, 208]. Although it is commonly held, especially in fairness research, that disparities against disadvantaged groups should be mitigated, there is less consensus on how to treat algorithms that happen to favor historically disadvantaged communities. In other words, should algorithmic fairness be symmetrical and reject *a priori* notions of privileged and disadvantaged groups? This is an important normative question, seldom acknowledged in the fairness literature, with immediate consequences on the choice of measures. For example, the version of DI reported in Equation (3) computes the ratio between the acceptance rates of the worse-off group (min) and the best-off group (max), ignoring prior knowledge of structural inequality and affected communities.

Table 3. Bias Mitigation Methods to Improve Fairness in Hiring

	In	Family	Measures	*s* availability	Mode	Approach	Summary
Rule-based Scraping or Substitution	[63, 194, 222]	PRE	sAUC, DD, TPRD, SD	None	Text	Proxy reduction	Remove or substitute gender identifiers based on hard-coded rules
Importance-Based Scraping	[33, 194]	PRE	sAUC, DI	Train		Proxy reduction	Iteratively remove proxy features most predictive of sensitive attribute
Balanced Sampling	[11, 276]	PRE	NDKL, MIA, MAE	Train		Re-balancing	Resample training set with same group cardinality in each class
Group Norming	[33]	PRE	sAUC, DI	Train		Groupwise feature transform	z-score normalization of each feature within each group
Subspace Projection	[194]	PRE	sAUC	None	Text	Proxy reduction	Remove gender information via embedding projection
Adversarial Inference	[120, 199, 222, 276]	IN	sAUC, DD, TPRD, SD, MIA, MAE, SKL	Train		Proxy reduction	Reduce sensitive info in latent representation through additional adv. loss
Face Decorrelation	[120]	IN	sAUC, DI	None	Image	Proxy reduction	Discourage intermediate representations correlated with face ID through adv. loss
Name Decorrelation	[119]	IN	sAUC, TPRD, RMS	None	Text	Proxy reduction	Discourage intermediate representation correlated with names by minimizing MI
Fair TF-IDF	[69]	IN	DI	Train	Text	Proxy reduction	Reduce feature weight according to its group-specificity
DetGreedy	[11, 103, 240]	POST	NDKL	Run-time		Output re-ranking	Re-rank items enforcing desired group representation in each prefix
CDF Rescoring	[185]	POST	xEO	Run-time		Output re-rescoring	Re-score items with groupwise CDF trick
Spatial Partitioning	[38]	POST	DI	Train		Output re-ranking	Promote candidates based on group membership probability

CDF, Cumulative Distribution Function.

5 Mitigation Strategies

Several algorithms have been proposed in the literature to improve model fairness; they are summarized in Table 3. We present them distinguishing between pre-processing, in-processing, and post-processing algorithms, depending on their applicability before, during, and after training.

5.1 Pre-Processing

Rule-based approaches perform a set of manipulations, typically defined by experts, on text data. They are related to process fairness (Section 4), as they focus on reducing the amount of information on sensitive attributes contained in non-sensitive features, i.e., to remove sensitive attribute proxies (Section 3). *Rule-based scraping* [63] is a heuristic for *text data* focused on gender, aimed at removing all words that explicitly refer to the gender of a person, including first names and titles. In general, it acts as a feature transformation $x' = \text{scrp}(x)$. Under bag-of-words representations, the scraping function is a feature selection mechanism that removes features in a censored vocabulary $\mathcal{V}$.

$$\text{scrp} : \mathcal{X} \to \mathcal{X}' \subseteq \mathcal{X}$$
$$\text{scrp}(x^j) = \begin{cases} x^j, & \text{if } x^j \notin \mathcal{V} \\ \emptyset, & \text{if } x^j \in \mathcal{V} \end{cases}.$$

De-Arteaga et al. [63] study the effectiveness of this approach in *job classification* from short biographies. They find it to be moderately effective in reducing TPRD. Parasurama and Sedoc [194]

take this approach one step further on a CV screening application. They define several rules for removing other strings related to gender, including e-mail addresses, intrinsically gendered words (e.g., "waitress"), and hobbies. *Rule-based substitution* [222] is a closely related approach, which replaces intrinsically gendered words with neutral words; for example, "his" is changed into "theirs." Due to redundant encoding of sensitive information in the data, rule-based approaches often represent a weak baseline with limited impact on fairness.

Importance-based scraping is an adversarial approach for feature removal based on a sensitive attribute classifier. Parasurama and Sedoc [194] consider a *resume screening* system and exploit contextualized word representations to train a gender classifier and iteratively scrape the words with the largest feature importance for gender classification. In other words, they train a sensitive attribute classifier

$$\hat{s} = h(x)$$

and assess the importance of features by selectively removing them from the prediction task. Proxy features $\mathcal{P}$ found to be most predictive of s (highest marginal contribution to $h(\cdot)$, e.g., measured via SHAP) are scraped:

$$\text{scrp}(x^j) = \begin{cases} x^j, & \text{if } x^j \notin \mathcal{P} \\ \emptyset, & \text{if } x^j \in \mathcal{P} \end{cases}$$

Parasurama and Sedoc [194] show that SHAP-based scraping can achieve a sizeable sAUC drop with a limited negative impact on performance. Booth et al. [33] develop an equivalent scheme for iterative feature removal in AVI analysis from multimodal data. They confirm the suitability of this method to improve process fairness (sAUC); outcome fairness (DI) also improves, but only for systems that were initially very unfair.

Subspace projection is a common approach to reduce gender bias in *text* representations based on word embeddings. Initially proposed by Bolukbasi et al. [32], this method is based on the observation that most intrinsically gendered information in word embeddings, such as the difference between the vectors for "mother" and "father," is contained within a small gender subspace. The algorithm is based on re-embedding each word by projecting it orthogonally to this space. Let $\vec{w}$ denote the embedding of a word and G the gender subspace. Each word is re-embedded to

$$\vec{w}' = \frac{\vec{w} - \text{proj}_G \vec{w}}{||\vec{w} - \text{proj}_G \vec{w}||}.$$

In the algorithmic hiring literature, subspace projection is used in Parasurama and Sedoc [194] to reduce gender biases in a *resume screening* application. Although in theory this approach is suitable to remove gender proxies and contrast the negative effects of stereotype violations, it proves ineffective in reducing the amount of gender information contained in resumes, as measured by sAUC, in alignment with prior art [108].

Balanced sampling is a broadly applicable scheme to reduce correlations between sensitive attributes and target variables in the training set. Arafan et al. [11] propose a down-sampling scheme to achieve an equal representation of sensitive groups among positive and negative points in the training set. For a binary sensitive attribute, this condition can be formulated as

$$\Pr_{\sigma}(s_i = g | y_i = \overline{y}) = \Pr_{\sigma}(s_i = g^c | y_i = \overline{y}), \forall \overline{y} \in \mathcal{Y},$$

where we let σ denote an algorithm's training set. Yan et al. [276] propose an upsampling approach to enforce the same condition before training a multimodal data fusion algorithm for the analysis of AVIs [141]. Overall, balanced sampling can reduce the influence of job segregation on hiring algorithms.

Group norming is a feature manipulation approach for *tabular data* enforcing a similar feature distribution in all sensitive groups, through normalization. Booth et al. [33] propose *Groupwise z-Normalization*, i.e., they divide data points based on their sensitive group membership, and standardize each feature by subtracting the groupwise mean and dividing by the groupwise standard deviation.

$$\mu_g = \frac{1}{N_g} \sum_{i \in g} x_i$$
$$\text{std}_g = \frac{1}{N_g} \sum_{i \in g} (x_i - \mu_g)^2$$
$$x'_i = \frac{x_i - \mu_g}{\text{std}_g} \text{ if } i \in g.$$

This approach represents an intermediate view on conditional discrimination: it prohibits *inter-group* discrimination based on a specific feature, while allowing it as a basis for *intra-group* discrimination. It is tested on multimodal data and found to decrease gender predictability (sAUC) for paraverbal and visual data, while increasing predictability for verbal features and only marginally improving DI. It is unclear how to generalize this method under multiple protected attributes (e.g., race and gender); a separate application to each intersectional group (e.g., black women) is the most straightforward extension, but increasingly smaller groups run the risk of unstable normalization. Considering its limited impact on both process and outcome fairness, in conjunction with its controversy in US employment law [22, 165], the use of group norming is not recommended.

5.2 In-Processing

Adversarial inference [120, 199, 222, 276] is an in-processing approach to remove sensitive information from latent representations that can be applied across tasks (e.g., classification, ranking) and data modalities (e.g., tabular, images). It leverages process fairness to reduce sensitive attribute proxies in pursuit of outcome fairness [73]. This approach directly models an adversary trying to infer an individual's sensitive attributes s_i from their latent representation $l(x_i)$. The latent representation is typically derived in one or more layers of a neural architecture whose goal is to predict the target variable y associated with employability. The adversarial loss for inference is defined as

$$L_{\text{INF}}^{\text{ADV}} = \frac{1}{N} \sum_i \text{dist}(s_i, d(l(x_i))),$$

where $d(\cdot)$ represents the layer(s) in the adversarial branch, so that $h(x) = d(l(x_i)) = \hat{s}_i$ is the inferred sensitive attribute value, and $\text{dist}(\cdot)$ computes its distance from the actual value s_i. Rus et al. [222] show the suitability of this method to improve selected process, outcome, and impact fairness indicators in a *job recommendation* scenario, while Peña et al. [199] demonstrate SKL improvements in a multimodal setting based on synthetic *resume ranking*.

Face decorrelation [120] was proposed for AVI systems and, more generally, algorithms that process *face images*. This approach is based on the key assumption that face data contain enough information on sensitive features such as gender and race, so that successful debiasing can be achieved without explicit sensitive attribute information. More in detail, let $l(x)$ denote a latent representation from a neural architecture used for employability prediction, and let $w(x)$ denote a representation of the candidates' face obtained from a state-of-the-art feature extraction method for face recognition [65]. Hemamou et al. [120] propose two schemes based on MSE and **Negative Sampling (NS)**. Under MSE, the adversary branch tries to leverage latent representations to

reconstruct face features by minimizing

$$L_{\text{MSE}}^{\text{ADV}} = \frac{1}{N}\sum_i [d(l(x_i)) - w(x_i)]^2,$$

where $d(\cdot)$ represents the final dense layer in the adversarial branch. Under NS, the adversary exploits the latent representation $l(x_i)$ extracted from an interview to discriminate the respective candidate from the remaining ones by maximizing the softmax

$$L_{\text{NS}}^{\text{ADV}}(x_i) = \frac{\exp(\text{sim}(l(x_i), w(x_i)))}{\sum_{j\neq i}\exp(\text{sim}(l(x_i), w(x_j)))},$$

where $\text{sim}(\cdot)$ represents a similarity function between the face recognition features $w(x_i)$ and the representations $l(x_i)$ learnt by the main branch of the network. Both variants of adversarial removal (NS, MSE) are shown reduce the sensitive information encoded in latent representations: notably, their sAUC is on par with adversarial methods explicitly trained to predict gender and ethnicity and may be suited for settings where sensitive attributes are unavailable during training. The effectiveness of face decorrelation for outcome fairness is more nuanced, showing positive effects on DI for very unfair models based on the video modality and limited effects on more equitable systems based on language and audio.

Name decorrelation [119] is a similar approach proposed for *text* data, based on sensitive information encoded in *names*. The goal of this method is to reduce the MI between the representations of individuals, such as document embeddings extracted from their resumes, and a word embedding of their name. To handle the complexity of MI estimation in high-dimensional continuous spaces, the method focuses on the MI between the latent representation of the input $l(x_i)$ and a low-dimensional projection of their name $\tilde{t}_i$. The adversarial loss function is defined as

$$L_{\text{name}}^{\text{ADV}} = \frac{1}{N}\sum_i \hat{\text{MI}}(\tilde{t}_i, l(x_i)).$$

This approach reduces the ability of adversaries to infer an individual's gender and ethnicity from their hidden representations as measured by sAUC in a *job classification* task. In terms of outcome fairness, name decorrelation is found to improve TPRD and RMS with respect to gender but not ethnicity. It is worth noting that the original disparities achieved by a vanilla model, measured by RMS and TPRD, were smaller for ethnicity than for gender. Similarly to Hemamou et al. [120], this result suggests that targeting this type of process fairness can improve outcome fairness in highly imbalanced situations, while only providing limited benefits in situations with lower inequity.

Feature weighting schemes learn to decrease the importance of a feature based on its likelihood of increasing the unfairness of a model. *Fair TF-IDF* [69] is an in-processing approach for *text* classification and ranking applications, such as *resume filtering*. As the name suggests, this algorithm is an extension of TF-IDF [237], one of the most popular and influential algorithms in text search engines. In response to a query describing a job, TF-IDF produces a score $f_{\text{soft}}(x_i)$ for each resume based on the count of query terms in item i and on the specificity of the terms in the entire resume collection. In other words, let t denote a term (word), potentially present in a resume i and in a query q describing a job posting; $\text{TF-IDF}(i,t)$ is the product of a term frequency $\text{tf}(i,t)$ and an inverse document frequency $\text{idf}(t)$. The $\text{tf}(i,t)$ factor summarizes the importance of t in resume i by counting the occurrences of t in i. The $\text{idf}(t)$ factor conveys the specificity of term t by counting how many resumes in the entire collection contain the term t and defining $\text{idf}(t)$ as its inverse; $\text{idf}(t)$ can be seen as a weight that increases (decreases) the importance of rare (common) words.

Deshpande et al. [69] propose a further weighting scheme based on penalizing terms that are highly specific to a given group.

$$\begin{aligned} p\text{-ratio}(t) &= \frac{\min_{g \in \mathcal{S}} \Pr(t \in i | i \in g)}{\max_{g \in \mathcal{S}} \Pr(t \in i | i \in g)} \\ \text{fair-tf-idf}(i,t) &= \text{tf(i,t)} \cdot \text{idf(t)} \cdot p\text{-ratio(t)} \\ f_{\text{soft}}(i,q) &= \sum_{t \in q} \text{fair-tf-idf}(i,t). \end{aligned}$$

The authors also propose more complex weighting schemes for the fairness factor p-ratio(t) and test their ability to improve system fairness as measured by DI.

5.3 Post-Processing

DetGreedy [103] and its variants are post-processing methods for *ranking* algorithms targeting NDKL. Given a desired representation of sensitive groups, expressed as a target distribution $D = \{D_g, \forall g \in \mathcal{S}\}$ (Equation (2)), Geyik et al. [103] seek to ensure this representation at different ranks. Starting from the target distribution D, DetGreedy populates the ranking progressively from the top to the bottom rank with the most relevant items from under-represented groups. To do so, it maintains a counter N_g^k for the number of items from group g that have already been placed in the top k positions of the ranking; this counter determines two priority sets from which items at rank $k+1$ can be drawn. The high priority set $\mathcal{G}_H$ consists of items from groups that are below the desired quota. The low-priority set $\mathcal{G}_L$ consists of groups that are above their desired quota, but only by a small margin.

$$\begin{aligned} \mathcal{G}_H &= \{i \in g : N_g^k < \lfloor D_g \cdot (k+1) \rfloor\} \\ \mathcal{G}_L &= \{i \in g : \lfloor D_g \cdot (k+1) \rfloor \leq N_g^k < \lceil D_g \cdot (k+1) \rceil\}. \end{aligned}$$

If $\mathcal{G}_H$ is not empty, DetGreedy chooses the most relevant item from the groups in $\mathcal{G}_H$; otherwise, it samples one from $\mathcal{G}_L$.

$$\tau(k+1) = \begin{cases} \text{argmax}_{i \in \mathcal{G}_H, i \notin \{\tau(1),\ldots,\tau(k)\}} f_{\text{soft}}(x_i), & \text{if } \mathcal{G}_H \setminus \{\tau(1),\ldots,\tau(k)\} \neq \emptyset \\ \text{argmax}_{i \in \mathcal{G}_L, i \notin \{\tau(1),\ldots,\tau(k)\}} f_{\text{soft}}(x_i), & \text{otherwise} \end{cases}$$

By prioritizing items with higher scores in $\mathcal{G}_L$, DetGreedy may end up violating some minimum representation constraints at some rank, i.e., $\exists g$ s.t. $N_g^k < \lfloor D_g \cdot k \rfloor$. This happens when more than one group is in $\mathcal{G}_H$. To mitigate this risk, Geyik et al. [103] propose two variants termed *DetCons* and *DetConsSort*. DetCons tries to avoid this occurrence by prioritizing groups that are more likely to enter $\mathcal{G}_H$ at the next iteration, while DetConsSort is a non-greedy algorithm that can re-order previous items dynamically to avoid constraint violations at the current rank. DetGreedy is implemented in *LinkedIn Recruiter* to ensure equitable gender representation in *candidate search*; the desired gender distribution D_g is made query-dependent and set to match the distribution of qualified candidates for the search criteria.

Cumulative Distribution Function (CDF) rescoring [185] is a post-processing method targeting xEO for *recommender systems*. xEO is a fine-granularity measure defined in Equation (10), studying the properties of the soft classifier $f_{\text{soft}}(x)$ at every possible threshold; it requires that the probability of positive items in a group $\{i \in g : y_i = 1\}$ achieving a score below a threshold t should be the same for every group, at every threshold t. This is achieved by re-mapping item scores to their groupwise CDF as

$$f'_{\text{soft}}(x_i) = \Pr(f_{\text{soft}}(x) \leq f_{\text{soft}}(x_i) | y = 1, s = s_i).$$

In other words, new soft scores are mapped to the $[0, 1]$ interval, according to the CDF of old scores for positive points in their group, ensuring xEO = 0. Nandy et al. [185] propose several extensions to this approach, to achieve xEO across all target class values, to account for position bias in the target, and to tradeoff fairness and accuracy. This method is tested on a *friendship recommendation* engine in a live proprietary system (most likely *LinkedIn*), where sensitive groups are defined based on the level of activity on the platform, mitigating potential differences in platform engagement.

Spatial partitioning [38] is a heuristic to select an optimal group of applicants σ from screening tests, with the additional difficulty that sensitive attribute values are unknown during testing. First, two estimators for the target ($f_{\text{soft}}(x)$) and protected variable ($h_{\text{soft}}(x)$) are developed on a training set. These estimators are applied to the test set, which is ranked customarily as $\tau = \text{argsort}(f_{\text{soft}}(x))$. The top candidates are selected from the ranking and put into σ. Given the systematic groupwise differences encoded in the data, the selected candidates tend to be mostly from the privileged group. Spatial partitioning mitigates this bias by replacing the candidates in σ who have the lowest values of $[f_{\text{soft}}(x) + h_{\text{soft}}(x)]$—or some other linear combination of the two—with the candidates who have the highest values of $[f_{\text{soft}}(x) + h_{\text{soft}}(x)]$ among the ones that were originally not chosen. This approach aims to rebalance the privileged and disadvantaged group in the final set σ while maintaining good accuracy in the selection of high-performance candidates.

5.4 Discussion

In this section, we present the main trends for fairness enhancement in the algorithmic hiring literature and describe important factors guiding these choices (summarized in Table 3). Section 7 will expand on this analysis presenting additional dimensions that guide fairness mitigation in practice.

Mitigating Biases. Some of the proposed approaches are suited to mitigate specific biases described in Section 3. Balanced sampling, for example, can counter the effect of under-representation in training sets caused by factors such as horizontal and vertical job segregation. Proxy reduction methods (e.g., decorrelation, adversarial inference) target sensitive attribute proxies, reducing the overreliance of models on sensitive information. Group norming seems applicable to counter systematic biases encoded in input features against protected groups, such as biased employee evaluations. However, it is certainly preferable to understand and improve the measurement system that provides these biased features rather than simply matching the distribution of input features across protected groups without a clear understanding of the underlying socio-technical system. Finally, output re-ranking can mitigate the joint effect of different BCFs. Its application comes with risks and opportunities described below.

Opportunities and Risks of Post-Processing. Post-processing approaches, such as DetGreedy, are the easiest to integrate into existing systems, as they can be modularly added to algorithmic pipelines [257]. Indeed, most post-processing methods in algorithmic hiring are proposed and deployed at *LinkedIn*, a large company with an established platform powered by interactions between complex data infrastructure and interdependent algorithmic modules [12, 104]. It is worth noting that post-processing explicitly takes into account sensitive attributes to change algorithmic outcomes for job candidates, which may be critical for disparate treatment and affirmative action (US) as well as direct discrimination and positive action (EU) [22, 112]. Additionally, post-processing requires runtime access to the sensitive attributes of all data subjects, as described in the next paragraph. These are likely two key factors explaining why post-processing is less popular and why hybrid approaches combining different types of fairness interventions, e.g., mitigating via pre- *and* post-processing, remain under-explored [11].

Sensitive Data. Different mitigation strategies have different requirements for sensitive attributes s. Post-processing methods, such as DetGreedy and CDF restoring, typically require knowledge

of sensitive attributes during runtime. Conversely, pre-processing and in-processing approaches, such as importance-based scraping and adversarial inference, require access to sensitive attributes only during training, not testing. Furthermore, specific data modalities, such as text and vision, facilitate specialized methods like subspace projection and face decorrelation, which function without sensitive attribute information about data subjects. Each of these methods has distinct data requirements, ranging from highly restrictive to more lenient. Having access to sensitive attributes at runtime enables precise and reliable interventions, whereas strategies that do not rely on this data must be continuously monitored to ensure they are meeting their intended goals.

Outcome Fairness through Proxy Reduction? A very clear trend in the literature is the popularity of proxy reduction methods, such as adversarial inference, that target process fairness (sAUC) in pursuit of outcome fairness (e.g., DI, TPRD). This approach probably gained popularity because it aligns with legislation against *disparate treatment* (US) and *direct discrimination* (EU), which prohibit basing hiring decisions on protected attributes [2, 158] and because it does not require runtime access to sensitive attributes. This approach is also adopted by vendors, such as *Pymetrics* and *Hirevue*, upon detecting violations of the 80% rule in pre-deployment audits [124, 209]. While this method can mitigate large outcome disparities, it does not produce convincing improvements when the inequity is less significant [33, 119, 120]. Moreover, it is worth noting that, in contrast to post-processing, this approach achieves mitigation on a held-out test set during development but provides no guarantee at deployment. More general studies are required to understand the effects of proxy removal on outcome fairness and ensure actual benefits for vulnerable candidates.

The Problem with Videos. Algorithmic screening can take advantage of data from multiple sources, including gameplay, psychological assessments, and video interviews. The latter provide three types of signals, namely visual, verbal, and paraverbal. Across multiple studies, systems trained on video signals are shown to yield the largest disparities and disadvantages for protected groups [33, 120, 276]. Visual signals have been removed from several products [175] because they inevitably encode sensitive information such as race and gender while lacking a solid foundation to justify their use in the hiring domain. Even if found to be accurate and fair in a specific evaluation, hiring algorithms based on face analysis are unlikely to predictably generalize and maintain accuracy or fairness under variable conditions, or at least they are less likely to do so than algorithms based on more established data modalities and better-understood correlations with job performance. Furthermore, it is worth noting regulation proposals against inference of emotions, states of mind, or intentions from face images in the workplace [84, 85]. Given this evidence, we invite particular caution against new proposals of hiring systems based on computer vision, advertised as capable of inferring the motivation [139] and personality traits [183] of candidates, even if accompanied by bias mitigation and fairness evaluation.

6 Data

Datasets used in algorithmic fairness research for the hiring domain are summarized in Table 4. The following sections present these resources divided into textual, visual, and tabular datasets. In line with recent literature, we find no graph dataset on algorithmic hiring [46, 71].

6.1 Text-Based Datasets

Textual datasets consist of job descriptions and job seekers' resumes or biographies, focusing on professional experience, training, and skills. Resumes and biographies contain strong proxies for sensitive attributes such as race and gender, including names and addresses.

Chinese bios is a textual dataset generated by Zhang et al. [283] to study gender bias in candidate rankings produced by BERT-based resume retrieval systems [70]. It consists of short resumes

Table 4. Datasets Used in Research on Fairness in Algorithmic Hiring

Dataset name	Used in	Type	Language	Geography	Target variable	Sensitive variable[a]	Hiring stage
Chinese Bios	[283]	Textual	Chinese	Synthetic	Mention of job-specific skills in bio	Gender (B, A)	Sourcing
Bias in Bios	[63, 119]	Textual	English	World	Candidate occupation	Gender (B, A)	Sourcing
Engineers and Scientists	[58]	Textual	English	Unknown	Candidate received an offer	Unknown	Screening
IT Resumes	[194, 196]	Textual	English	US	Candidate was called back	Gender (B, S)	Sourcing
CVs from Singapore	[69]	Textual	English	Singapore	CV match with job description	Ethnicity (A)	Sourcing
DPG Resumes	[222]	Textual	English, Dutch	The Netherlands	Candidate industry of interest	Gender (B, A)	Sourcing
ChaLearn First Impression	[120, 148, 276]	Visual	English	Unknown	Speaker hireability and personality annotated by AMT workers	Gender (B, A), ethnicity (A)	Screening
Student Interviews	[33]	Visual	English	Unknown	Speaker hireability annotated by research assistants	Gender (NB, S)	Screening
SHL Interviews	[231]	Visual	English	US, UK, India	Speaker communication skills rated by experts	Gender (B), age, race, country of residence	Screening
Oil Company Interviews	[139]	Visual	Dutch	The Netherlands	Candidate self-reported commitment	Gender, age	Screening
FairCVs	[119, 199]	Visual	English	Synthetic	Synthetic score	Gender (B, A), race (A)	Sourcing
Requirements and Candidates	[172]	Tabular	Synthetic	Synthetic	Synthetic score	Synthetic	Sourcing
Jobs and Candidates	[11]	Tabular	Unknown	The Netherlands	Candidate recruited or shortlisted	Gender (B, S)	Sourcing
Pymetrics Bias Group	[271]	Tabular	Unknown	Unknown	Similarity to current employees	Gender (S), race (S)	Screening
IBM HR Analytics	[105]	Tabular	Unknown	Unknown	Employee resignation	Gender (B), age, marital status	Evaluation
Resume Search Engines	[48]	Tabular	English	US	Unknown	Gender (B, S)	Sourcing
Walmart Employees	[38]	Tabular	English	US	Employee tenure and performance ratings	Synthetic	Screening
Web Developers Field Study	[13]	Tabular	English	US	Unknown	Gender (B,S), age (S), ethnicity (S)	Screening
Chinese Job Recommendations	[282]	Tabular	Chinese	China	Unknown	Gender (B,A), age (A)	Sourcing
Facebook Ads Audiences	[4]	Tabular	English	US	User clicks	Gender (B, S), race (B, A)	Sourcing

[a]Where reported, we indicate the provenance of sensitive attributes as annotated (A) or self-reported (S); we also indicate the gender encoding as binary (B) or non-binary (NB).

containing a binary gender indicator (he/she) and a two-sentence description of job skills for IT, finance, and administration.

Bias in bios [63] is composed of textual biographies written in English and extracted from the Common Crawl dataset. It was initially proposed to study fairness in occupation classification. Gender is automatically extracted based on the use of pronouns in the short biographies. Professions are the target variables; they are self-reported in the descriptions.

Engineers and scientists [58] results from a field study comparing human and algorithmic CV screening in an unspecified company. It is composed of applications, i.e., pairs of resumes and job postings for engineers (software and hardware) and technical scientists. The features were parsed from resumes, including candidate education (institutions, degrees, majors, awards), work experience (job titles and companies), skills, and relevant keywords. The target variable indicates whether the candidates were interviewed and extended an offer.

IT resumes [196] was used to study how men and women describe themselves on resumes and whether the difference impacts hiring outcomes. The dataset comprises approximately 900,000 resumes (without names, e-mails, and URLs) in the historical hiring records of eight IT firms based in the US, relevant to just over 6,000 job postings from the IT sector. The resumes were managed through an applicant tracking system, where the applicants self-reported their gender. The target variable encodes whether candidates received a callback after applying.

CVs from Singapore [69] was introduced to investigate ethnicity bias in automated resume filtering. It contains 135 resumes of candidates of Chinese, Malaysian, and Indian origin (the predominant ethnic groups in Singapore) who applied for vacancies in Singapore's accounting and finance sectors. The dataset curators annotated candidate ethnicity based on geographical information on their education and early employment. For example, candidates who completed their education in China were classified as ethnically Chinese. Nine job postings describing open positions in the financial sector are considered. Three annotators annotated each posting-resume pair with a binary variable indicating whether candidates appear qualified for a job based on their CV.

DPG resumes [222] contains over 10 million vacancies (including salary range, working hours, and job descriptions) and just under 1 million resumes augmented with job categories of interest and gender inferred from first names. Given the imbalance between vacancies and candidates, this dataset is suitable for studying recommender systems that propose jobs to candidates. The data are provided by DPG Recruitment in anonymized form, removing all names (including company names), dates, addresses, telephone numbers, e-mail addresses, and Web sites.

6.2 Visual Datasets

Several datasets in the hiring space are multimodal, with a strong focus on videos of candidates answering job interview questions in front of a camera [33, 139, 231], often called AVI. This data modality is relatively new in the hiring domain. Similarly to CVs, AVIs encode much information on sensitive attributes, including gender, race, age, and disability.

ChaLearn First Impression [79] contains 10,000 YouTube video clips of people facing a camera and speaking in English. Amazon Mechanical Turk workers were hired to annotate each clip with the personality traits of the speaker (openness, conscientiousness, extroversion, agreeableness, neuroticism) and a "variable indicating whether the subject should be invited to a job interview or not." The gender and ethnicity of the speakers were annotated by two dataset curators.

Student Interviews [33] consists of video interviews with 733 upper-level undergraduate students, who were asked to participate in a simulated interview answering six questions. Verbal (e.g., n-gram frequencies), paraverbal (e.g., loudness, jitter, shimmer), and visual (e.g., facial expressions, body motion) features were automatically extracted from videos using tools like OpenSmile [88], FACET [238], or Motion Tracker [270]. Data were annotated by three research assistants who assessed the candidates' "hireability" on a 5-point Likert scale. Gender is self-reported, including a non-binary option.

SHL Interviews [231] is a proprietary AVI dataset with more than 5,000 videos from 810 real job seekers from the US, UK, India, and Europe answering behavioral and domain knowledge questions. Videos were annotated by at least five assessors, who rated the presence of four social skills: engagement, positive emotion, calmness, and confidence, using a scale of 0 to 4. The sensitive attributes available with the dataset are country, age, gender, and race.

Oil Company Interviews [139] were curated to study the problem of predicting candidate motivation in job selection processes. It comprises AVIs with 154 students from Utrecht University carrying out a mock interview with a fictitious oil company. The participants self-reported their motivation ("To what degree are you motivated to work for the company") on a 10-point Likert scale, which represents the target variable. Software tools like OpenFace and EMFACS [75] were used to automatically create facial marker features and extract emotions from videos.

FairCVs [199] is a synthetic CV dataset combining short bios from *Bias in bios* [63], face images taken from the *DiveFace* database [180], and numerical features emulating desirable aspects such as availability or previous experience. Artificial target scores are generated for each CV, as a linear combination of numerical features; biased scores are derived from these with an ethnicity- and

gender-dependent additive penalty emulating biases in the data. The dataset has been employed to study the extent to which sensitive attribute proxies can contribute to discriminatory models when the target variables on which they are trained exhibit biases against certain protected groups.

6.3 Tabular Data

Tabular datasets encode structured data of a diverse nature, describing job seekers and employees at different stages of the algorithmic hiring pipeline.

Requirements and Candidates [172] is a synthetic dataset with numerical and Boolean values encoding both candidate skills and job requirements. Bias against certain candidates is deliberately introduced in the data through additive noise.

Jobs and Candidates [11] is a tabular dataset describing candidates and job postings with real-valued, categorical, and binary features. For candidates, their education, experience, preferences (minimum salary, preferred working hours, maximum travel distance), and self-reported gender are included. Regarding job postings, the dataset contains information about the industry, company size, and geographical location. Joint candidate-job features describing overlaps, such as the distance of candidates' residence from the place of work, are also present. The target variable indicates whether a candidate was recruited or short-listed for a job.

Pymetrics Bias Group [271] is a test set used for pre-deployment audits at Pymetrics, a company offering gameplay-based screening tools to clients. The gameplay of job seekers and current employees of the client company are compared to find candidates who resemble high-performing incumbent employees. The covariates consist of gamified psychological measurements; sensitive attributes, which can be self-reported on a voluntary basis, include gender and race.

IBM HR Analytics[1] is a synthetic dataset curated by IBM data scientists to study employee resignation, which is the target variable. Covariates include education, job satisfaction, income, years of service in the company, and commuting distance, along with sensitive attributes such as gender, age, and marital status.

Resume Search Engines [48] includes search results crawled from employment Web sites *Indeed*, *Monster*, and *CareerBuilder* for 35 job titles in 20 cities of the US, collecting data on 855,000 job candidates. Data were crawled in 2016 to study gender biases.

Walmart Employees was released as part of the Society for Industrial and Organizational Psychology machine learning competition[2] to study the problem of predicting employee retention and performance from pre-employment tests with questions on work history, personality, and behavioral scenarios. Target variables encode employee tenure and performance ratings. Each instance has a synthetic binary variable that mimics a protected attribute.

Web Developers Field Study [13] summarizes the results of an experiment on the impact of algorithmic hiring tools on gender diversity. The curators advertised a web developer position for US residents; upon applying, candidates provided information on their education, experience, and demographics, as well as free-form responses to selected questions. The responses were rated using a recruitment tool with a score of up to 100. Each applicant was then rated by a human assessor based on experience and education; assessors were divided into three groups based on whether they had access to algorithmic scores and candidate names.

Chinese Job Recommendations [282] consists of job recommendations from four Chinese boards to fictitious profiles that differ only by gender. Profiles are accessed programmatically every 2 weeks to record job recommendations, which are then compared between different genders.

[1] Available at https://github.com/IBM/employee-attrition-aif360
[2] https://eval.ai/web/challenges/challenge-page/527/overview

Several tabular datasets have been curated to measure discrimination in job ad delivery [4, 132, 158], using a common methodology. *Facebook Ads Audiences* [4] exemplifies this methodology by running a job ad campaign on Facebook and studying differential delivery along gendered and racial lines. They design advertising creatives (headline, text, and image) for different professions, using them in a campaign that optimizes clicks, which are then broken down into different demographics. Despite the fact that the Facebook Marketing API does not allow a breakdown by race, the curators perform this analysis with a careful design of target audiences leveraging phone numbers and racial information available from North Carolina voter records.

6.4 Discussion

Low Diversity. English is by far the dominant language; datasets with geographical information primarily represent US citizens (six datasets) and the Netherlands (three datasets). We offer two interpretations of these findings. On the one hand, they reflect the importance of both countries in the recruitment industry.[3] On the other hand, they are consistent with previous research reporting that most efforts to improve fairness in AI are influenced by the Global North [187, 221], particularly the US [135], and focus on English language resources [210].

Missing Stages. The vast majority of datasets describe the early stages of the hiring pipeline, i.e., sourcing (11 out of 20) and screening (8 out of 20). We found no datasets (and consequently no studies) for selection, and only one for evaluation (*IBM HR Analytics*). This is expected given the industry's tendency to use algorithms primarily for sourcing and screening [125, 166]. However, given the growing push to adopt this technology in later stages [259], there is a tangible risk that algorithms for selection and evaluation will be quietly deployed without a clear understanding of risks and limitations. Indeed, datasets that target employee tenure and performance, such as *Walmart Employees* and *IBM HR Analytics*, signal an interest from companies in understanding the factors that predict future productivity and loyalty.

Lacking Sensitive Attributes. Most importantly, attributes such as disability, religion, and sexual orientation are simply missing, despite the special legal status of these attributes and the evidence of workplace discrimination [7, 191, 193]. Gender is by far the most common sensitive attribute, overwhelmingly encoded as binary. Ethnicity and race are considered in 6 out of 20 datasets, making this the second most common attribute. They are frequently annotated, rather than self-reported, in textual and visual datasets, due to these data modalities carrying strong proxies for sensitive attributes.

Target Multiplicity. HR management and recruitment tasks allow multiple formulations. Many target variables can seem reasonable at face value; therefore, initial data curation and design choices have a prominent role. As candidates move through the hiring pipeline, their digital record goes through a data journey where they are marked as aware of a position, applying (or headhunted), proposed to a client, screened-in, interviewed, hired, retained, promoted, and so on. Companies interested in algorithmic hiring solutions pick one or more of these variables, balancing different priorities such as efficiency and quality of hire, with unpredictable effects on algorithmic fairness, a phenomenon called *multi-target multiplicity* [266]. Indeed, the target variables in Table 4 are very diverse. On the one hand, this reflects the length of hiring pipelines and the diversity of data journeys. On the other hand, it points to a lack of established best practices around target variables. Focusing on screening datasets, we find different constructs (e.g., communication skills vs. commitment) annotated by people with different competencies (AMT workers vs. experts) from disparate data sources (YouTube videos vs. mock interviews). Since the validity of these estimates

[3] https://www2.staffingindustry.com/eng/Editorial/Daily-News/World-World-s-largest-staffing-firms-post-224-billion-in-revenue-56012

and their connection with job performance has been called into question [19, 214], we call for caution in handling these variables and granting them the status of *ground truth*. This can be especially problematic when using conditional fairness measures whose very definition hinges on this so-called ground truth.

7 Measurement and Mitigation in Practice

In this section, we present practical considerations on bias mitigation and measurement that emerge from our review and from direct involvement in the industry. We focus on key technical factors guiding measurement and mitigation choices.

7.1 Data Modalities

First, different data modalities enable different bias mitigation strategies.

Textual data is a common data modality encountered in the sourcing stage, typically in the form of textual CV or job description data, or in the screening stage, e.g., transcripts of video interviews. Common bias mitigation methods that cater exclusively to textual data include rule-based scraping or dictionary-based methods. These methods, also offered by vendors such as *Textmetrics* and *Textio*, can be used for mitigating specific types of biases, e.g., age bias in job descriptions [98], or gender bias in biographies [63]. These mitigation strategies tend to be technically straightforward to implement, as they can be developed as standalone components and need not be integrated in complex algorithmic hiring pipelines. In addition, collecting the required data (e.g., constructing task-specific dictionaries) leans heavily on domain and context-specific knowledge, which is typically available to practitioners in the HR domain. A drawback of these dictionary-based methods is a different side of the same coin: due to reliance on domain, context, and task-specific data (dictionaries) they do not transfer well over different problems (e.g., scraping gendered words from resumes vs. substituting words associated with age discrimination in job descriptions require wholly different sets of terms), languages, or geographies (where cultural or regulatory differences may impose different requirements).

Additionally, when it comes to textual bias mitigation, the rapid uptake of LLMs has pulled into focus biases that arise from LLM-based natural language generation. For example, Salinas et al. [224] show how LLMs exhibit gender and nationality bias when generating job recommendations for job seekers of different genders and nationalities. They find that the types of jobs recommended follow common gender and nationality-based stereotypes. In a similar experiment, GPT-4 is found to more frequently use female pronouns in reference letters generated for female-dominated occupations (such as nannies), and male pronouns for male-dominated occupations (such as plumbers) [35]. A simple yet effective mitigation strategy here is prompt engineering: by appending the phrase "in an inclusive way" to the prompt, previously gendered pronouns are replaced with third person pronouns ("they/their"). While novel bias mitigation strategies for LLMs are actively studied [167], the previous examples show that being mindful of which information to include in a prompt (either explicit or implicit), and understanding how to effectively construct prompts are important first mitigation steps for leveraging LLMs in the context of algorithmic hiring.

Next, *visual data*, either through images or video, enable and require a different set of bias mitigation methods and strategies. As shown in Section 5, video-based systems tend to yield more disadvantageous effects for protected groups, also due to the strong encoding of sensitive information. In this light, *EASYRECRUE* present an adversarial method that removes sensitive information from latent representations of neural networks that were trained for predicting "hireability" given (features that represent) facial expressions of candidates in job interviews [120]. This type of adversarial bias mitigation can, however, be applied over a wider set of data modalities. In the end, due to video-based algorithms unreliability in the face of varying conditions, and incoming regulation

proposals that prohibit inference of emotions or intentions from facial data in workplace contexts (discussed in more detail in Section 5.4), it is advisable for practitioners to approach video-based hiring systems and tools with caution, irrespective of which mitigation strategy to choose.

Finally, *tabular data* is a common data modality in algorithmic hiring and machine learning more broadly. Mitigating bias in tabular data can be done through pre-processing approaches that directly intervene on the features, e.g., by removing features that are highly correlated with sensitive attributes via importance-based scraping, or increasing the representation of vulnerable populations in training sets with balanced sampling. These types of methods are widely available in open-source bias mitigation software packages such as the *Fairlearn* Python package [268] or the *AI Fairness 360* toolkit (available in Python and R) [21]. In addition, we see practitioners experiment with alternative pre-processing methods in algorithmic hiring, such as synthetic tabular data generation [197] for measuring [256] and mitigating bias [11].

7.2 Tasks

Next to different types of data modalities, different downstream tasks can enable and call for different measurement and bias mitigation strategies. The two most common tasks in algorithmic hiring are classification and ranking.

For classification tasks, getting started with bias mitigation is relatively straightforward, as there exist several open source or otherwise freely available software packages and libraries, as mentioned above, that provide different implementations of bias measurement methods and mitigation algorithms, specifically designed for classification tasks.

Ranking is a common task in the sourcing and screening stages of hiring, performed by search engines or recommender systems. However, the prevalence of classification-based bias metrics in algorithmic hiring (discussed in more detail in Section 4) is reflected in the aforementioned open source packages, which means that measuring and mitigating bias in ranking systems tends to follow the practice of re-purposing classification methods through, e.g., thresholding rankings (i.e., cutting off at k). We recommend using fair ranking measures for an evaluation of ranking systems that is more cognizant of user browsing behavior, although, of course, the latter needs to be suitably modeled. While several mitigation methods apply to both classification and ranking tasks (e.g., adversarial inference), practitioners should be aware of ranking-specific methods such as DetGreedy.

7.3 Scalability and Efficiency

Technical and infrastructural decisions further affect which bias mitigation methods to consider. For example, Geyik et al. [103] decouple their post-processing method from specific model choices and properties of input data, which means their method can naturally scale across the different (ranking) systems at *LinkedIn*. This also means that such a method can be developed and deployed as a standalone component or micro-service, which decreases development time, effort, and alignment, when compared to pre-processing or in-processing methods that may need to be designed and integrated in complex algorithmic hiring pipelines. In general, post-processing approaches offer an advantage in terms of system integration, but they should be applied with care due to anti-discrimination law (Section 8).

In terms of algorithmic efficiency, the re-ranking method by Geyik et al. [103] for bias mitigation can be considered computationally cheap as only a subset of a model's output needs to be processed (i.e., top-k items). At the same time, while low, the additional computational costs will be incurred for each ranker output. Bias mitigation through pre-processing, such as counterfactual data augmentation, and in-processing, such as feature weighting, can be comparatively more resource intensive as they involve (re-)training models. Adversarial inference methods, which have been applied to a

variety of data modalities and tasks, even require training multiple neural networks simultaneously, which means they can incur substantially higher costs compared to their non-mitigated counterparts. However, while these additional costs may be incurred at training time, the resulting models do not incur additional costs at inference time. In the end, algorithmic efficiency and computational costs will vary across bias mitigation methods in nature and magnitude, with simple rule-based scraping methods that involve dictionary look-ups on the cheap end of the spectrum, and bias mitigation methods that involve re-training LLMs at the resource-intensive end [167]. Aspects such as task, model complexity, architecture, training parameters, dataset size, and composition influence the scalability and efficiency of bias mitigation strategies.

7.4 Sensitive Attribute Data Availability and Usage

Different bias mitigation methods have different requirements around the availability of sensitive data, which can further steer mitigation method selection.

The availability of sensitive data is affected by several factors in practice. First, as we identify in Section 8, access to sensitive attributes may be at odds with privacy regulations such as the **General Data Protection Regulation (GDPR)**, which is an important real-world constraint. Next, availability of and access to sensitive attributes may require job seekers' explicit consent, which can be difficult to acquire at the scale needed for some bias mitigation methods. Finally, some sensitive attributes, such as age and gender, may have to be recorded in the hiring process for identification purposes and can hence be assumed to be available for all job seekers. Many other sensitive attributes (e.g., religious beliefs) will not be easily available and will have to be explicitly requested for bias measurement and mitigation purposes.

This lack of access to sensitive attributes means some bias mitigation methods may be less suited than others, as noted in Section 5.4; in particular, post-processing methods that require sensitive attributes for all candidates at inference time can be unrealistic. Here, pre-processing bias mitigation methods may prove more useful, as they (only) require access to sensitive attributes at training time and can operate on attributes even when available for only a subset of job seekers (e.g., those who provided consent). Furthermore, this allows these methods to be deployed and run in isolation from production systems in controlled batch scenarios, which can be another important practical benefit. Examples of these bias mitigation methods used by practitioners include adversarial inference [222], and re-balancing training data with synthetic data generation [11].

Finally, a third family of bias mitigation methods that can operate without requiring access to sensitive attributes at all are the rule-based approaches described above, which are commonly used for mitigating bias in textual data such as job descriptions or resumes. These methods rely solely on domain and task-specific gazetteers or dictionaries and hence can be used when access to sensitive attributes of individuals is not available or desirable.

7.5 Fairness Definitions and Intervention Targets

Once data modality, task, infrastructural and technical choices, and access to sensitive data are set, one important practical challenge is that of defining "fairness" and formulating the intervention target of a bias mitigation method—i.e., deciding "when to intervene" and "what to optimize for."

In the case of *LinkedIn*'s Talent Search [103], the DetGreedy algorithm was implemented to have the ranker's top-k output reflect the gender distribution of job seekers that meet the requirements of a recruiter-issued query, i.e., the desired distribution of the ranking is set to be equivalent to that of the underlying population of job seekers. Here, alignment with the company's goals and values, interpretability (or ability to explain), and collaboration across different stakeholders were mentioned as key factors in guiding the eventual target distribution. Defining fairness and formulating an intervention target is not a one-off task, as algorithmic hiring components will be

deployed in different, product-specific contexts and stages of the hiring pipeline. Candela et al. [39] present the (evolving) framework used at *LinkedIn* that guides definitions and operationalizations of AI fairness across their different (types of) products.

In general, the challenge in defining fairness and formulating intervention targets in algorithmic hiring is exacerbated by the multi-stakeholder nature of the hiring domain, where development teams may need to consult legal and compliance teams, HR professionals and recruiters, product managers, and executives, all of whom may need to be informed or provide input. Indeed, these choices should be guided by ethical, social, and legal dimensions. This challenge may explain in part the popularity of DI as a fairness metric, due to its ease of interpretation across a broad and diverse stakeholder group, or the popularity of complying with the **Equal Employment Opportunity Commission (EEOC)** 80% rule [77] as a fairness target (adopted, e.g., by *Pymetrics* [271]), as it appears, at least superficially, a legally grounded target.

7.6 Fairness vs. Utility Tradeoffs

In the sourcing or screening stages, depending on the intervention target, *outcome fairness* may come at the cost of utility; this can happen, for example, when optimizing for gender parity in heavily male- or female-dominated industries. However, perhaps surprisingly, different experiments by practitioners show that outcome fairness does not need to come at the cost of utility. First, in their online A/B-testing experiments, Geyik et al. [103] find their DetGreedy method improves their fairness metric, with no significant impact on utility. In addition, Arafan et al. [11] find that their pre-processing mitigation method of rebalancing training data may even improve utility over non-bias-mitigated methods, in an offline experiment using real data from an international HR company. Moreover, with respect to *impact fairness*, Peng et al. [200] show that "overcompensation" (i.e., artificially over-representing a gender in the output of a ranking system) as an intervention strategy for a hiring algorithm can in some cases mitigate human bias further down the hiring funnel.

Overall, this section described several practical considerations constraining bias mitigation interventions; desired utility levels are just one dimension, and often not the most restrictive.

8 Legal Landscape

In this section, we describe the main regulations and non-discrimination provisions concerning algorithmic hiring in the EU and the US. We list the main legal sources in both regions, emphasizing the former since it is less cited in the computer science literature on algorithmic hiring. We close this section with remarks on open challenges and practical concerns at the intersection of technology and policy.

8.1 European Law

Non-Discrimination Law. We focus on rules that apply in the whole European Union (27 Member States). In the absence of specific legal rules on non-discrimination in algorithmic hiring, general non-discrimination law applies. The right to non-discrimination is protected as a human right in Europe by the European Convention on Human Rights (1950) and the Charter of Fundamental Rights of the European Union (2020). The EU also adopted a number of legal acts called *directives*, prohibiting several types of discrimination in different contexts, which Member States implement (give effect to) by adopting national law [153]. The four most relevant non-discrimination directives are the following: the Racial Equality Directive (2000), the Employment Equality Directive (2000), the Gender Goods and Services Directive (2004), and the Recast Gender Equality Directive (2006) [54–56, 86]. Together, the directives offer protection against discrimination in hiring on the basis of six grounds, also called protected characteristics, or protected attributes: age; disability; gender;

religion or belief; racial or ethnic origin; and sexual orientation. EU law distinguishes two categories of prohibited discrimination: *direct* and *indirect*.

Direct discrimination means that the person is treated less favorably than another on the basis of a protected characteristic, such as ethnicity [54]. For example, if a company says it will not recruit people of a certain ethnicity, that is an example of direct discrimination [57]. Direct discrimination is always prohibited, with a few narrowly defined and specific examples. For instance, a women's clothing brand is allowed to hire only female models for its advertising pictures.

The second category of prohibited discrimination is *indirect discrimination*, occurring when a practice is neutral at first glance but ends up discriminating against people of a certain ethnicity (or another protected characteristic) [54]. For indirect discrimination, the law focuses on the effects of a practice; the intention of the alleged discriminator is not relevant. Hence, even if an organization can prove that it did not know that its algorithmic system discriminated unfairly, that will not help the organization.

There are three elements of indirect discrimination that can be summarized as follows [34, 54]. (1) The practice must be neutral. For example, rejecting job applications coming from a certain postal code would count as neutral. Rejecting applications from people with a certain ethnicity would not be neutral; it would be direct discrimination. (2) This neutral practice puts people with a certain ethnicity (or other sensitive attributes) at a "particular disadvantage compared with other persons." The word *disadvantage* must be interpreted in a wide way. (3) There is no objective justification for such practice. The apparently neutral practice is not prohibited if the "practice is objectively justified by a legitimate aim and the means of achieving that aim are appropriate and necessary."

As a hypothetical example, suppose that people with an immigrant background make more spelling errors in job application letters. A cleaning company never hires job applicants for cleaning jobs if the application contains spelling errors. This practice seems neutral at first glance, but results in the rejection of most applicants with an immigrant background. The cleaning company cannot justify this no-spelling-errors rule because people can be good cleaners, even if they make some spelling errors. Therefore, the cleaning company engages in illegal indirect discrimination. However, the situation would be different for a law firm. The main job of many lawyers is writing precise, and often official, documents. The law firm is allowed to reject applications with spelling errors, even if it results in most people with an immigrant background not being hired.

The organization is also responsible if it uses an algorithmic system provided by, for instance, a company. If the algorithmic system turns out to discriminate illegally, the organization using the system is responsible and the victim can sue it for damages, for instance. (Later, the organization could try to sue the AI developer, but the organization remains responsible toward the victim.) In sum, general non-discrimination law applies to new forms of algorithmic discrimination, also if that discrimination happens indirectly or accidentally.

Other Relevant Law. We briefly highlight some other relevant laws in the EU. The GDPR is, roughly summarized, a European-wide statute that aims to protect fairness and human rights when personal data are used. The GDPR is long and detailed. Among other things, GDPR bans the use of *special categories* of personal data (sometimes called sensitive data). These are data on, for example, someone's ethnicity, religion, trade union membership, health, or sexual orientation (article 9 GDPR). There are some exceptions to the ban. For instance, hospitals are allowed to use health data. Another exception is the individual's *explicit consent*. Generally speaking, consent is not freely given, and thus not valid, if an employer asks a job applicant or employee for their consent, because of the unequal power relation. This GDPR rule makes it difficult to use sensitive data to audit or train algorithmic systems [254].

There is a proposal for an AI Act in the EU, with many requirements for "high-risk" systems, including algorithmic systems for HR [84]. Developers of high-risk AI systems must, for instance, ensure that the training data are appropriate and do not lead to unlawful discrimination. At the time of writing, the EU did not officially adopt the AI Act yet, and did not publish the final text yet. The proposals in the AI Act also contain a new exception to the GDPR, to enable the use of sensitive data for debiasing algorithmic systems.

8.2 US Law

Federal Law. US non-discrimination law is similar to EU law in many respects, but also decidedly different in others. Like EU law, its sources are spread over different statutes, and case law plays a crucial role. For example, Title VII of the Civil Rights Act of 1964 constitutes a federal law prohibiting employment discrimination based on race, color, religion, sex, and national origin. Significantly, the EEOC issues non-binding, but practically important guidelines to interpret Title VII. Other important sources of federal law are the Age Discrimination in Employment Act of 1967, the Americans with Disabilities Act of 1990, and the Genetic Information Nondiscrimination Act of 2008.

For all these acts, US law distinguishes between two fundamental types of discrimination: disparate treatment and DI. They resemble, but do not perfectly mirror, the difference between direct and indirect discrimination in the EU. Importantly, disparate treatment requires not only an adverse action based on a protected attribute but also the proof of intent on the part of the discriminating individual—unlike the EU variety of direct discrimination [254]. To actually win in court, an injured person in the spelling mistake example would have to demonstrate that the cleaning agency introduced the spelling requirement with the purpose of treating unfavorably members of their protected group. In practice, this will often be difficult [251].

DI, in turn, closely resembles indirect discrimination. Intent is not required [249]. Rather, the DI doctrine prohibits actions that are seemingly neutral, but significantly disadvantage members of a protected group. In its guidelines, the EEOC suggested that such a disadvantage usually occurs if the chance of a member of the protected group being positively evaluated is 80% or less than that of a member of the privileged group (so-called 80% rule or 4/5 rule) [77]. Finally, DI can be justified if there is a legitimate reason for the practice [250], for example business necessity [18, 249]. Therefore, a law firm could arguably use a model evaluating orthography and grammar to rank candidates even if this disproportionately disadvantages members of one specific protected group.

Overall, unless intent can be shown, many cases of algorithmic discrimination will be argued under the DI prong. Therefore, many contributions to the literature on technical algorithmic fairness have provided tools to ensure that this rule is not violated at the statistical level [94, 278, 279]. However, courts will generally, both in the EU and in the US, look at factors beyond mere scores and numbers to determine whether a legally relevant disadvantage exists [112, 260].

Other Relevant Law. Apart from the acts mentioned, federal legislation specifically addressing discrimination in AI systems is unlikely to emerge anytime soon. The Algorithmic Accountability Act is stalled in a gridlocked Congress. Hence, several states and municipalities have taken the initiative and enacted AI hiring laws themselves. For example, the city of New York passed a law on automated employment decision tools [186]. From 5 July 2023, NYC Local Law 144 applies to employers using AI to substantially assist or replace discretionary decision-making in hiring. They are required to conduct and publish impartial bias evaluations by an independent auditor. Furthermore, the state of Illinois enacted the AI Video Interview Act [131]. In force since the year 2020, the law requires employers to put candidates on notice and obtain their consent, before

subjecting their video interview to AI analysis. Candidates may request deletion, and employers that rely solely on AI need to collect and report candidates' race and ethnicity.

Practitioners have to comply with these local rules if their model is applied in these jurisdictions. Finally, further constraints may arise from affirmative action law, particularly if the fairness intervention goes beyond what is necessary to remedy otherwise unjustified discrimination; this is a complex topic in both the US [22, 145, 252] and the EU legal framework [112, 126].

8.3 Operational Challenges

This outline of the EU and US legal landscapes provides some normative reference points for practitioners and offers an opportunity to discuss some practical challenges in assessing the legal compliance of algorithms.

In general terms, an algorithmic system does not cause indirect discrimination or DI if it pursues *legitimate aims* through *appropriate means*. In practice, several questions arise about both elements at the base of this legal principle. Is inferring a candidate's motivation, i.e., an internal combination of their emotions, states of mind, and intentions, a legitimate aim? Moreover, should estimates by algorithms trained on biased "ground truth" variables be considered appropriate means? In the absence of precise guidelines, these questions have to be assessed contextually to each algorithmic system, and algorithmic implementations should be compared, to the best extent possible, to (hypothetical) outcomes under non-algorithmic alternatives [112].

Direct discrimination, in turn, is generally much more difficult to legally justify [2]. A natural question arises about the compliance of bias mitigation approaches embedded in hiring algorithms. More specifically, are post-processing algorithms legitimate only if they enforce inter-group parity for equally qualified candidates, e.g., targeting TPR parity rather than DI? If so, what quantitative criteria should be applied to assess candidate qualifications? And to what extent is human intervention and a comprehensive assessment of any re-ranking required? Here, the answers depend on affirmative action (US) and positive action (EU) law, which are currently in flux after the US Supreme Court's decision against Harvard's and UNC's affirmative action programs. At a minimum, human oversight of post-processing operations, and an evaluation of its effects both on groups and on individuals particularly worthy of protection (single parents; chronically ill persons), seems advisable.

9 The Real World Is Messy

Our previous discussion has focused on how discrimination can be exacerbated by algorithms. Nevertheless, we have striven to point out the ways such discrimination can be mitigated and even how algorithms can be designed to actively discourage discrimination. It should now be clear that the design of technology plays a key role, either in creating discrimination or by reducing it. To overly value the role of technology design in discrimination or anti-discrimination would also be a mistake, however. This is because non-technological factors have been shown, on occasion, to strongly influence decisions and amplify technological bias, as we will explore in this section.

9.1 Algorithmic Uptake

During the COVID-19 pandemic, delivery companies such as Amazon, Deliveroo, and DoorDash rolled out algorithmic recruitment systems to avoid the danger of viral contagion in their HR teams, as well as experiment with a new technology that had the potential to save millions of dollars in HR bills [281]. One reason they were able to do this with ease was due to the regulatory environment, which was laxer than usual due to the ongoing emergency conditions. We have explored some of the ethical problems regarding discrimination in previous sections of this article, as well as some of the potential technological solutions to designing algorithms with anti-discrimination in mind.

Each of these solutions is a technological response, however, so we need to remain aware of the non-technological or reduced-technological options.

Although hiring technology can be more ethical than humans and reduce bias in decision-making [138], it often reinforces bias and results in distinctly non-ethical outcomes. This means we should at least consider the balance of ethical harms if we reintroduced a non-technological solution. The recent pandemic necessitated social distancing, but once this danger had passed, non-technological hiring procedures could be reintroduced. To understand why this has not happened, we must consider the institutional incentives of tech companies. These companies have invested heavily in their digital hiring technology as suppliers or customers, so are highly motivated to retain it, even once there is no longer a strong need for it.

9.2 Algorithmic Fairness

Recalling the example from Section 3.4, suppose that appropriate technical solutions have been deployed in hiring algorithms to counter the BCFs against intersectional minorities. Should she seek a new job, JS would have a sizeable probability of being sourced, screened-in, and recruited—equal to natives (male and non-male alike) with similar skills and experience in the hospitality industry. However, in the presence of an exogenous shock, such as the COVID-19 pandemic, workplaces are heavily affected, with large consequences on hiring and recruitment. The hospitality industry receives a major blow. Food services stop hiring waitstaff and dismiss most of the employees. Schools and childcare services become unavailable. All of a sudden, JS finds herself unemployed, urgently in need of a new job, which is now more difficult to obtain since demand for her skills has decreased and competition has surged. Sourcing algorithms place JS at the bottom positions of rankings, granting her low visibility. Childcare duties demand much time from her due to her gender [227]. She has fewer connections than native job seekers for recommendation and referral [284]. Overall, her probability of successfully reaching the end of a hiring pipeline has dropped substantially and more sharply than for privileged groups. In addition, her migration background makes JS less likely to gain support from social safety nets [45, 171], increasing the urgency of her need and the unfairness of the new *status quo*.

This section highlights two limitations of algorithmic fairness in hiring. On the one hand, by restricting their scope to a single system at a precise point in time, fairness measurements run the risk of missing the bigger picture, i.e., the broader socio-technical system in which hiring algorithms are embedded, which can change quickly and profoundly. A model deemed accurate and fair in the old context may perform poorly in the new one. On the other hand, these changes may be difficult to detect and quantify. Fairness evaluations by practitioners on pre-deployment test sets, such as *Pymetrics Bias Group* (Section 6.3), may quickly become obsolete. Strong shocks in the hiring domain are an issue for data representativeness more broadly. Fresh data become necessary. This is further complicated by changing incentive structures, by the complexity of handling sensitive data, and by the frequent delay between decisions and feedback in hiring.

10 Opportunities and Limitations

Algorithmic hiring benefits from, and contributes to, fairness and anti-discrimination work, as the previous sections have shown. In this section, we summarize the emerging opportunities and limitations, from which we derive a set of recommendations for researchers and practitioners, summarized in Table 5.

10.1 Bias and Validity

Opportunities. Algorithms for hiring can consider large pools of candidates, avoiding the preliminary exclusion of unusual profiles, as often done by human recruiters under time constraints.

Table 5. Opportunities, Limitations, and Recommendations for Algorithmic Hiring and Fairness Research

	Opportunities	Limitations	Recommendations
Bias and Validity	Consider large candidate pools, reduce human biases, and attract minority candidates	Risk of encoding individual biases along with inevitable societal biases; invalid target variables	Focus on vulnerable populations beyond acceptance rates; study individual fairness and exploratory policies; carefully scrutinize new technologies
Broader Context	Trigger positive feedback loops; consider tech-recruiter collaboration	Narrow focus on local outcomes can overlook fairness in entire hiring process; risk of repurposing for termination	Center on job seeker impacts; identify leaks in pipelines; design for recruiter-tech interaction; beware of performance and tenure prediction
Data	Support evaluations of diversity and inclusion	Reduced geographical, linguistic, and sensitive attribute coverage	Design data collection for diversity; develop IP-friendly audits
Law	Apply binding regulation to positively influence industry	Legal restrictions on fairness approaches: concerns about discrimination and data protection	Multidisciplinary research balancing fairness, privacy, and anti-discrimination; monitor EU AI Act

IP, intellectual property.

Under-representation and sampling biases can be mitigated as a result. Algorithmic hiring also has the potential to mitigate biases in imperfect human judgments. The simple fact of using algorithmic decision-making can reduce avoidance by vulnerable candidates [13]. Fair and trustworthy algorithms can lead to a positive form of automation bias and attract minority candidates.

Limitations. Data-driven algorithms tend to encode individual and societal biases. Some algorithms are explicitly trained and evaluated to "predict the competency scores candidates would have been given by trained human reviewers" [189], inheriting individual biases from recruiters. Previous experience is a preeminent feature for assessing candidates. In conjunction with current job segregation, this means that the most important features are inevitably skewed against historically disadvantaged groups. In addition to these biases, the epistemic validity of prediction targets such as candidates' *employability* and *motivation* is questionable. Job performance is famously difficult to define and measure, let alone predict [220, 263]. Algorithms cloaked in objectivity can promote bias while targeting and legitimizing ill-defined quantities.

Recommendations. Attention should be devoted not only to acceptance rates for vulnerable populations (DI) but also to their representation among applicants, as well as their progress downstream of the algorithm and post-hiring. Job descriptions and organizational communication can play an important role in attracting or repelling specific groups; automation attempts [207] should carefully include fairness evaluations. To mitigate biases against unusual candidates, individual fairness and exploratory policies should be studied. Individual fairness measures can surface problematic situations for individuals that may go unnoticed when studying group fairness. Exploratory policies based on partially stochastic mechanisms can provide new information in repeated decision-making scenarios. However, the social acceptability and procedural justice of such policies in the hiring domain remain to be studied. Finally, new technologies, including AVIs and personality prediction, deserve additional scrutiny, especially through the lens of validity theory [213, 214].

10.2 Broader Context

Opportunities. It is worth highlighting that improving fairness does not require completely removing bias. Algorithmic hiring can reduce certain disparities and trigger deflating feedback loops across

BCFs. These algorithms do not (and should not) operate autonomously. Effective and equitable hiring can result from a fruitful interaction between technology and recruiters, leveraging complementary strengths. For example, HR professionals are better suited to assess special cases and operate under changing conditions [166].

Limitations. Most fairness measures are focused on narrow algorithmic outcomes, neglecting the wider socio-technical context around these algorithms. Some of these measures are completely symmetric and consider advantages for vulnerable and privileged groups as equally problematic (which is, however, generally required by the law). Zooming out from single-algorithm evaluations, it is worth noting that outcome fairness at every stage does not guarantee fair outcomes for job seekers throughout the hiring pipeline. Furthermore, fairness for employers is currently missing; two-sided platforms would be well-positioned to study the performance of their algorithms across employers, devoting special attention to small businesses. Finally, it is worth noting that the discovery of patterns that predict job performance for hiring can open the way to models for termination decisions.

Recommendations. We call for contextualized and integrated evaluations of decision-making processes that go beyond the predictions of a single algorithm. This will help to address complex problems, such as identifying leaks in the hiring pipeline that are most critical for vulnerable groups and modeling impacts on job seekers, such as their efforts and benefits. To exemplify, rejected candidates may still benefit to some extent from a specific type of explanation. Moreover, it will be important to better understand the utility derived from these algorithms by different employers, considering recruitment workflows and developing new fairness measures. The prevalent *human vs. algorithm* evaluation framework is of limited utility; to overcome it, more research on recruiter-machine interaction is required [240], including candidate screening models [6] leading to more granular measures. Finally, we invite special caution in the development of predictive models for job performance and tenure, due to a risk of exploitation for termination decisions [285]; such an application of algorithms raises even stronger ethical and social concerns, which are only beginning to be discussed [223].

10.3 Data

Opportunities. Algorithmic fairness research is contributing additional analysis into hiring practices from a perspective of diversity and non-discrimination. More data entail more scrutiny and reflection, which can inform organizational frameworks, such as *Diversity, equity, and inclusion*, and scholarly fields, such as applied psychology and economics.

Limitations. Research on fairness in algorithmic hiring is based on data with reduced geographical and linguistic coverage. In addition, important sensitive attributes are missing from the data, making it difficult or impossible to evaluate algorithms for specific vulnerable groups. Data and research are constrained by a dual tension with the privacy of data subjects, on the one side, and the **Intellectual Property (IP)** of companies, on the other side.

Recommendations. Practitioners and researchers should seek more diverse data, with greater geographical and linguistic diversity, and better coverage of sensitive attributes that are relevant in hiring but are lacking, such as disability status and sexual orientation. Dedicated initiatives should be undertaken, including optional surveys for job applicants and broader data donation campaigns [25]. Innovative auditing protocols should be studied for employers and providers of algorithmic hiring solutions, including IP-friendly data disclosure procedures.

10.4 Law

Opportunities. In practice, binding regulation shapes algorithmic development more than ethical guidelines or self-regulation. Although clear guidance on algorithmic hiring is currently missing,

precise requirements set out in future regulation and case law, informed by research and practice on fair algorithmic hiring, have the potential to influence the industry positively and profoundly.

Limitations. Most of the fairness approaches developed so far are restricted to proxy reduction or removal, neglecting a wealth of solutions developed by the algorithmic fairness community. This is most likely due to concerns of infringing regulation on disparate treatment and direct discrimination. Furthermore, special categories of personal data are often lacking and difficult to process, particularly in the EU under the current data protection law. Therefore, it is difficult to assess hiring practices for certain vulnerable populations. Data protection law restricts these analyses and should consider exceptions for algorithmic fairness, as now suggested in the EU AI Act.

Recommendations. Algorithmic fairness can conflict with privacy and non-discrimination doctrine. The exact contours of legally compliant algorithmic fairness remain contested. The EU AI Act may offer a (limited) solution by allowing certain types of sensitive data processing to remove biases in high-risk scenarios. This guidance should be expanded to other areas and jurisdictions. We advocate for further multidisciplinary research on this topic, studying technical solutions and legal frameworks to reconcile these principles in light of their tradeoffs. Promising technological approaches include multiparty computation [143], sample-level estimators [89], and noise injection mechanisms [137].

11 Conclusions

The social, technological, and legal landscape around algorithmic hiring is rapidly evolving; algorithmic fairness has become a necessary component for both business-as-usual product development and frontier research. Practitioners and researchers in this field must understand BCFs, leveraging contextualized measures carried out on appropriate data to deploy suitable bias mitigation strategies. Multidisciplinary work at the intersection with legal scholarship is especially critical to implement and guide policy by defining technically achievable desiderata. Only a contextualized and balanced understanding of fair algorithmic hiring can guide research and practice to avoid the pitfalls of legitimizing questionable applications with misguided analyses and to reap truly shared benefits for society.

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References

[1] Mohsen Abbasi, Sorelle A. Friedler, Carlos Scheidegger, and Suresh Venkatasubramanian. 2019. Fairness in representation: Quantifying stereotyping as a representational harm. In *Proceedings of the 2019 SIAM International Conference on Data Mining (SDM '19)*. Tanya Y. Berger-Wolf and Nitesh V. Chawla (Eds.), SIAM, 801–809. DOI: https://doi.org/10.1137/1.9781611975673.90

[2] Jeremias Adams-Prassl, Reuben Binns, and Aislinn Kelly-Lyth. 2023. Directly discriminatory algorithms. *The Modern Law Review* 86, 1 (2023), 144–175.

[3] Ifeoma Ajunwa. 2019. The paradox of automation as anti-bias intervention. *Cardozo Law Review* 41 (2019), 1671.

[4] Muhammad Ali, Piotr Sapiezynski, Miranda Bogen, Aleksandra Korolova, Alan Mislove, and Aaron Rieke. 2019. Discrimination through optimization: How Facebook's ad delivery can lead to biased outcomes. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 199:1–199:30. DOI: https://doi.org/10.1145/3359301

[5] Kristen M. Altenburger, Rajlakshmi De, Kaylyn Frazier, Nikolai Avteniev, and Jim Hamilton. 2017. Are there gender differences in professional self-promotion? An empirical case study of LinkedIn profiles among recent MBA graduates. In *Proceedings of the 11th International Conference on Web and Social Media (ICWSM '17)*. AAAI Press, 460–463. Retrieved from https://aaai.org/ocs/index.php/ICWSM/ICWSM17/paper/view/15615

[6] Jose M. Alvarez and Salvatore Ruggieri. 2023. The initial screening order problem. arXiv:2307.15398. Retrieved from https://arxiv.org/abs/2307.15398

[7] Mason Ameri, Lisa Schur, Meera Adya, F. Scott Bentley, Patrick McKay, and Douglas Kruse. 2018. The disability employment puzzle: A field experiment on employer hiring behavior. *ILR Review* 71, 2 (2018), 329–364.

[8] American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. 2014. The standards for educational and psychological testing.

[9] Lori Andrews and Hannah Bucher. 2022. Automating discrimination: AI hiring practices and gender inequality. *Cardozo Law Review* 44 (2022), 145.

[10] Julia Angwin, Noam Scheiber, and Ariana Tobin. 2017. *Dozens of companies are using facebook to exclude older workers from job ads*. Machine Bias. ProPublica, New York, NY. Retrieved from https://www.propublica.org/article/facebook-ads-age-discrimination-targeting

[11] Adam Mehdi Arafan, David Graus, Fernando P. Santos, and Emma Beauxis-Aussalet. 2022. End-to-end bias mitigation in candidate recommender systems with fairness gates. In *Proceedings of the 2nd Workshop on Recommender Systems for Human Resources (RecSys-in-HR '22)*. CEUR-WS, 1–8.

[12] Aditya Auradkar, Chavdar Botev, Shirshanka Das, Dave De Maagd, Alex Feinberg, Phanindra Ganti, Lei Gao, Bhaskar Ghosh, Kishore Gopalakrishna, Brendan Harris, Joel Koshy, Kevin Krawez, Jay Kreps, Shi Lu, Sunil Nagaraj, Neha Narkhede, Sasha Pachev, Igor Perisic, Lin Qiao, Tom Quiggle, Jun Rao, Bob Schulman, Abraham Sebastian, Oliver Seeliger, Adam Silberstein, Boris Shkolnik, Chinmay Soman, Roshan Sumbaly, Kapil Surlaker, Sajid Topiwala, Cuong Tran, Balaji Varadarajan, Jemiah Westerman, Zach White, David Zhang, and Jason Zhang. 2012. Data infrastructure at LinkedIn. In *Proceedings of the IEEE 28th International Conference on Data Engineering (ICDE '12)*. Anastasios Kementsietsidis and Marcos Antonio Vaz Salles (Eds.), IEEE Computer Society, 1370–1381. DOI: https://doi.org/10.1109/ICDE.2012.147

[13] Mallory Avery, Andreas Leibbrandt, and Joseph Vecci. 2023. Does artificial intelligence help or hurt gender diversity? Evidence from two field experiments on recruitment in tech. Retrieved from http://monash-econ-wps.s3.amazonaws.com/RePEc/mos/moswps/2023-09.pdf

[14] Chen Avin, Barbara Keller, Zvi Lotker, Claire Mathieu, David Peleg, and Yvonne-Anne Pignolet. 2015. Homophily and the glass ceiling effect in social networks. In *Proceedings of the 2015 Conference on Innovations in Theoretical Computer Science (ITCS '15)*. Tim Roughgarden (Ed.), ACM, 41–50. DOI: https://doi.org/10.1145/2688073.2688097

[15] Ghazala Azmat and Barbara Petrongolo. 2014. Gender and the labor market: What have we learned from field and lab experiments? *Labour Economics* 30 (2014), 32–40.

[16] Ricardo Baeza-Yates. 2018. Bias on the web. *Communications of the ACM* 61, 6 (2018), 54–61. DOI: https://doi.org/10.1145/3209581

[17] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2019. *Fairness and Machine Learning: Limitations and Opportunities*. Retrieved from http://www.fairmlbook.org

[18] Solon Barocas and Andrew D. Selbst. 2016. Big data's disparate impact. *California Law Review* 104 (2016), 671–732.

[19] Murray R. Barrick and Michael K. Mount. 1991. The big five personality dimensions and job performance: A meta-analysis. *Personnel Psychology* 44, 1 (1991), 1–26.

[20] Arthur G. Bedeian, Gerald R. Ferris, and K. Michele Kacmar. 1992. Age, tenure, and job satisfaction: A tale of two perspectives. *Journal of Vocational Behavior* 40, 1 (1992), 33–48.

[21] Rachel K. E. Bellamy, Kuntal Dey, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilovic, Seema Nagar, Karthikeyan Natesan Ramamurthy, John Richards, Diptikalyan Saha, Prasanna Sattigeri, Moninder Singh, Kush R. Varshney, and Yunfeng Zhang. 2018. AI fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted algorithmic bias. Retrieved from https://arxiv.org/abs/1810.01943

[22] Jason R. Bent. 2019. Is algorithmic affirmative action legal. *The Georgetown Law Journal* 108 (2019), 803.

[23] Marianne Bertrand and Sendhil Mullainathan. 2004. Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. *American Economic Review* 94, 4 (2004), 991–1013.

[24] Alex Beutel, Jilin Chen, Tulsee Doshi, Hai Qian, Li Wei, Yi Wu, Lukasz Heldt, Zhe Zhao, Lichan Hong, Ed H. Chi, and Cristos Goodrow. 2019. Fairness in recommendation ranking through pairwise comparisons. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2212–2220.

[25] Matthew Bietz, Kevin Patrick, and Cinnamon Bloss. 2019. Data donation as a model for citizen science health research. *Citizen Science: Theory and Practice* 4, 1 (2019).

[26] Lee E. Biggerstaff, Joanna T. Campbell, and Bradley A. Goldie. 2023. Hitting the "Grass Ceiling": Golfing CEOs, exclusionary schema, and career outcomes for female executives. *Journal of Management* 50 (2023), 1502–1535.

[27] Su Lin Blodgett, Lisa Green, and Brendan T. O'Connor. 2016. Demographic dialectal variation in social media: A case study of African-American English. In *Proceedings of the 2016 Conference on Empirical Methods in Natural*

Language Processing (EMNLP '16). Jian Su, Xavier Carreras, and Kevin Duh (Eds.), The Association for Computational Linguistics, 1119–1130. DOI: https://doi.org/10.18653/v1/d16-1120

[28] Lotte Bloksgaard. 2011. Masculinities, femininities and work–the horizontal gender segregation in the Danish Labour market. *Nordic Journal of Working Life Studies* 1, 2 (2011), 5–21.

[29] Donna Bobbitt-Zeher. 2011. Gender discrimination at work: Connecting gender stereotypes, institutional policies, and gender composition of workplace. *Gender & Society* 25, 6 (2011), 764–786.

[30] Miranda Bogen and Aaron Rieke. 2018. *Help Wanted: An Examination of Hiring Algorithms, Equity, and Bias*. Technical Report. Upturn.

[31] Jasmijn C. Bol. 2011. The determinants and performance effects of managers' performance evaluation biases. *The Accounting Review* 86, 5 (2011), 1549–1575.

[32] Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam Tauman Kalai. 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. In *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016*. Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett (Eds.), 4349–4357. Retrieved from https://proceedings.neurips.cc/paper/2016/hash/a486cd07e4ac3d270571622f4f316ec5-Abstract.html

[33] Brandon M. Booth, Louis Hickman, Shree Krishna Subburaj, Louis Tay, Sang Eun Woo, and Sidney K. D'Mello. 2021. Bias and fairness in multimodal machine learning: A case study of automated video interviews. In *Proceedings of the International Conference on Multimodal Interaction (ICMI '21)*. Zakia Hammal, Carlos Busso, Catherine Pelachaud, Sharon L. Oviatt, Albert Ali Salah, and Guoying Zhao (Eds.), ACM, 268–277. DOI: https://doi.org/10.1145/3462244.3479897

[34] Frederik Zuiderveen Borgesius. 2020. Price discrimination, algorithmic decision-making, and European non-discrimination law. *European Business Law Review* 31, 3 (2020), 401–422.

[35] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nori, Hamid Palangi, Marco Tulio Ribeiro, and Yi Zhang. 2023. Sparks of artificial general intelligence: Early experiments with GPT-4. arXiv:2303.12712.

[36] Pawan Budhwar, Ashish Malik, M. T. Thedushika De Silva, and Praveena Thevisuthan. 2022. Artificial intelligence–challenges and opportunities for international HRM: A review and research agenda. *The International Journal of Human Resource Management* 33, 6 (2022), 1065–1097.

[37] Joy Buolamwini and Timnit Gebru. 2018. Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Proceedings of the Conference on Fairness, Accountability and Transparency (FAT '18)*. Sorelle A. Friedler and Christo Wilson (Eds.), Vol. 81, PMLR, 77–91. Retrieved from http://proceedings.mlr.press/v81/buolamwini18a.html

[38] Ian Burke, Robin Burke, and Goran Kuljanin. 2021. Fair candidate ranking with spatial partitioning: Lessons from the SIOP ML competition. In *Proceedings of the 1st Workshop on Recommender Systems for Human Resources (RecSys in HR '21) Co-located with the 15th ACM Conference on Recommender Systems (RecSys '21)*, Vol. 2967.

[39] Joaquin Quiñonero Candela, Yuwen Wu, Brian Hsu, Sakshi Jain, Jennifer Ramos, Jon Adams, Robert Hallman, and Kinjal Basu. 2023. Disentangling and operationalizing AI fairness at LinkedIn. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*. ACM, 1213–1228. DOI: https://doi.org/10.1145/3593013.3594075

[40] David Card, Ana Rute Cardoso, and Patrick Kline. 2016. Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women. *The Quarterly Journal of Economics* 131, 2 (2016), 633–686.

[41] Ben Carterette. 2011. System effectiveness, user models, and user utility: A conceptual framework for investigation. In *Proceeding of the 34th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '11)*. Wei-Ying Ma, Jian-Yun Nie, Ricardo Baeza-Yates, Tat-Seng Chua, and W. Bruce Croft (Eds.), ACM, 903–912. DOI: https://doi.org/10.1145/2009916.2010037

[42] Census Bureau. 2023. Current population survey. Retrieved from https://stats.bls.gov/news.release/empsit.toc.htm

[43] Maura Cerioli, Maurizio Leotta, and Filippo Ricca. 2020. What 5 million job advertisements tell us about testing: A preliminary empirical investigation. In *Proceedings of the 35th ACM/SIGAPP Symposium on Applied Computing*. Chih-Cheng Hung, Tomás Cerný, Dongwan Shin, and Alessio Bechini (Eds.), ACM, 1586–1594. DOI: https://doi.org/10.1145/3341105.3373961

[44] Simon Chandler. 2018. The AI ChatBot will hire you now. Retrieved from https://www.wired.com/story/the-ai-chatbot-will-hire-you-now/

[45] Lei Che, Haifeng Du, and Kam Wing Chan. 2020. Unequal pain: A sketch of the impact of the Covid-19 pandemic on migrants' employment in China. *Eurasian Geography and Economics* 61, 4–5 (2020), 448–463.

[46] April Chen, Ryan A. Rossi, Namyong Park, Puja Trivedi, Yu Wang, Tong Yu, Sungchul Kim, Franck Dernoncourt, and Nesreen K Ahmed. 2023. Fairness-aware graph neural networks: A survey. *ACM Transactions on Knowledge Discovery from Data* 18 (2023), 1–23.

[47] Jie Chen, Chunxia Zhang, and Zhendong Niu. 2018. A two-step resume information extraction algorithm. *Mathematical Problems in Engineering* 2018 (2018), 5761287.

[48] Le Chen, Ruijun Ma, Anikó Hannák, and Christo Wilson. 2018. Investigating the impact of gender on rank in resume search engines. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Regan L. Mandryk, Mark Hancock, Mark Perry, and Anna L. Cox (Eds.), ACM, 651. DOI: https://doi.org/10.1145/3173574.3174225

[49] Ruey-Cheng Chen, Qingyao Ai, Gaya Jayasinghe, and W. Bruce Croft. 2019. Correcting for recency bias in job recommendation. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM '19)*. Wenwu Zhu, Dacheng Tao, Xueqi Cheng, Peng Cui, Elke A. Rundensteiner, David Carmel, Qi He, and Jeffrey Xu Yu (Eds.), ACM, 2185–2188. DOI: https://doi.org/10.1145/3357384.3358131

[50] Sapna Cheryan, Allison Master, and Andrew N. Meltzoff. 2015. Cultural stereotypes as gatekeepers: Increasing girls' interest in computer science and engineering by diversifying stereotypes. *Frontiers in Psychology* 6 (2015), 49.

[51] Raj Chetty, David J. Deming, and John N. Friedman. 2023. Diversifying society's leaders? The causal effects of admission to highly selective private colleges. Working Paper 31492. National Bureau of Economic Research. DOI: https://doi.org/10.3386/w31492

[52] T. Anne Cleary. 1968. Test bias: Prediction of grades of Negro and white students in integrated colleges. *Journal of Educational Measurement* 5, 2 (1968), 115–124.

[53] David A. Cotter, Joan M. Hermsen, Seth Ovadia, and Reeve Vanneman. 2001. The glass ceiling effect. *Social Forces* 80, 2 (2001), 655–681.

[54] Council of the European Union. 2000. Council Directive 2000/43/EC implementing the principle of equal treatment between persons irrespective of racial or ethnic origin. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32000L0043

[55] Council of the European Union. 2000. Council Directive 2000/78/EC of 27 November 2000 establishing a general framework for equal treatment in employment and occupation. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32000L0078

[56] Council of the European Union. 2004. Council Directive 2004/113/EC of 13 December 2004 implementing the principle of equal treatment between men and women in the access to and supply of goods and services. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32004L0113

[57] Court of Justice of the European Union. 2008. Centrum voor gelijkheid van kansen en voor racismebestrijding v Firma Feryn NV. Retrieved from https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX%3A62007CJ0054

[58] Bo Cowgill. 2018. *Bias and Productivity in Humans and Algorithms: Theory and Evidence from Resume Screening*. Columbia Business School, Columbia University, Vol. 29.

[59] Terry-Ann Craigie. 2020. Ban the box, convictions, and public employment. *Economic Inquiry* 58, 1 (2020), 425–445.

[60] Nick Craswell, Onno Zoeter, Michael J. Taylor, and Bill Ramsey. 2008. An experimental comparison of click position-bias models. In *Proceedings of the International Conference on Web Search and Web Data Mining (WSDM '08)*. Marc Najork, Andrei Z. Broder, and Soumen Chakrabarti (Eds.), ACM, 87–94. DOI: https://doi.org/10.1145/1341531.1341545

[61] Amit Datta, Michael Carl Tschantz, and Anupam Datta. 2015. Automated experiments on ad privacy settings. *Proceedings on Privacy Enhancing Technologies* 2015, 1 (2015), 92–112. DOI: https://doi.org/10.1515/popets-2015-0007

[62] Skylar Davidson. 2016. Gender inequality: Nonbinary transgender people in the workplace. *Cogent Social Sciences* 2, 1 (2016), 1236511.

[63] Maria De-Arteaga, Alexey Romanov, Hanna M. Wallach, Jennifer T. Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Cem Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. 2019. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. danah boyd and Jamie H. Morgenstern (Eds.), ACM, 120–128. DOI: https://doi.org/10.1145/3287560.3287572

[64] Jenny Yang Deirdre Mulligan. 2023. Hearing from the American people: How are automated tools being used to surveil, monitor, and manage workers? Retrieved from https://www.whitehouse.gov/ostp/news-updates/2023/05/01/hearing-from-the-american-people-how-are-automated-tools-being-used-to-surveil-monitor-and-manage-workers/

[65] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. ArcFace: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR '19)*. Computer Vision Foundation/IEEE, 4690–4699. DOI: https://doi.org/10.1109/CVPR.2019.00482

[66] Megan Denver. 2020. Criminal records, positive credentials and recidivism: Incorporating evidence of rehabilitation into criminal background check employment decisions. *Crime & Delinquency* 66, 2 (2020), 194–218.

[67] Anne-Sophie Deprez-Sims and Scott B. Morris. 2010. Accents in the workplace: Their effects during a job interview. *International Journal of Psychology* 45, 6 (2010), 417–426.

[68] Ellora Derenoncourt, Chi Hyun Kim, Moritz Kuhn, and Moritz Schularick. 2022. *Wealth of Two Nations: The US Racial Wealth Gap, 1860-2020*. Technical Report. National Bureau of Economic Research.

[69] Ketki V. Deshpande, Shimei Pan, and James R. Foulds. 2020. Mitigating demographic bias in AI-based resume filtering. In *Adjunct Publication of the 28th ACM Conference on User Modeling, Adaptation and Personalization (UMAP '20)*. Tsvi Kuflik, Ilaria Torre, Robin Burke, and Cristina Gena (Eds.), ACM, 268–275. DOI: https://doi.org/10.1145/3386392.3399569

[70] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805. Retrieved from https://arxiv.org/abs/1810.04805

[71] Yushun Dong, Jing Ma, Song Wang, Chen Chen, and Jundong Li. 2023. Fairness in graph mining: A survey. *IEEE Transactions on Knowledge and Data Engineering* 35, 10 (2023), 10583–10602. DOI: https://doi.org/10.1109/TKDE.2023.3265598

[72] Alice H. Eagly, Christa Nater, David I. Miller, Michèle Kaufmann, and Sabine Sczesny. 2019. Gender stereotypes have changed: A cross-temporal meta-analysis of US public opinion polls from 1946 to 2018. *American Psychologist* 75, 3 (2019), 301–315.

[73] Harrison Edwards and Amos J. Storkey. 2016. Censoring representations with an adversary. In *Proceedings of the 4th International Conference on Learning Representations (ICLR '16)*. Yoshua Bengio and Yann LeCun (Eds.). arXiv:1511.05897. Retrieved from http://arxiv.org/abs/1511.05897

[74] EEOC - US Equal Employment Opportunity Commission. 2023. Select issues: Assessing adverse impact in software, algorithms, and artificial intelligence used in employment selection procedures under title VII of the Civil Rights Act of 1964. Retrieved from https://www.eeoc.gov/select-issues-assessing-adverse-impact-software-algorithms-and-artificial-intelligence-used

[75] Paul Ekman and Wallace V. Friesen. 2003. Unmasking the Face: A Guide to Recognizing Emotions from Facial Clues. Vol. 10. ISHK.

[76] Naomi Ellemers. 2018. Gender Stereotypes. *Annual Review of Psychology* 69, 1 (2018), 275–298.

[77] Equal Employment Opportunity Commission. 2015. Uniform guidelines on employment selection procedures.

[78] Stefan Eriksson and Jonas Lagerström. 2012. The labor market consequences of gender differences in job search. *Journal of Labor Research* 33 (2012), 303–327.

[79] Hugo Jair Escalante, Heysem Kaya, Albert Ali Salah, Sergio Escalera, Yağmur Güçlütürk, Umut Güçlü, Xavier Baró, Isabelle Guyon, Julio C. S. Jacques Junior, Meysam Madadi, Evelyne Viegas, Furkan Gürpinar, Achmadnoer Sukma Wicaksana, Cynthia C. S. Liem, Marcel A. J. van Gerven, and Rob van Lier. 2020. Modeling, recognizing, and explaining apparent personality from videos. *IEEE Transactions on Affective Computing* 13, 2 (2020), 894–911.

[80] Ben Eubanks. 2022. *Artificial Intelligence for HR: Use AI to Support and Develop a Successful Workforce*. Kogan Page Publishers.

[81] European Commission. 2020. The gender pay gap situation in the EU. Retrieved from https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/gender-equality/equal-pay/gender-pay-gap-situation-eu_en

[82] European Institute for Gender Equality. 2020. Gender Equality Index 2020. Retrieved from https://eige.europa.eu/publications/gender-equality-index-2020-key-findings-eu

[83] European Institute for Gender Equality. 2023. Gender Equality Index. Retrieved from https://eige.europa.eu/gender-equality-index/2022/domain/work

[84] European Parliament. 2021. Proposal for a regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (artificial intelligence act) and amending certain union legislative acts. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A52021PC0206

[85] European Parliament. 2023. Artificial Intelligence Act: Amendments adopted by the European Parliament on 14 June 2023 on the proposal for a regulation of the European Parliament and of the Council on laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts. Retrieved from https://www.europarl.europa.eu/doceo/document/TA-9-2023-0236_EN.pdf

[86] European Parliament and Council of the European Union. 2004. Directive 2006/54/EC of the European Parliament and of the Council of 5 July 2006 on the implementation of the principle of equal opportunities and equal treatment of men and women in matters of employment and occupation (recast). Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32006L0054

[87] Christine L. Exley and Judd B. Kessler. 2022. The gender gap in self-promotion. *The Quarterly Journal of Economics* 137, 3 (2022), 1345–1381.

[88] Florian Eyben, Martin Wöllmer, and Björn Schuller. 2010. Opensmile: The munich versatile and fast open-source audio feature extractor. In *Proceedings of the 18th ACM International Conference on Multimedia*, 1459–1462.

[89] Alessandro Fabris, Andrea Esuli, Alejandro Moreo, and Fabrizio Sebastiani. 2023. Measuring fairness under unawareness of sensitive attributes: A quantification-based approach. *Journal of Artificial Intelligence Research* 76 (2023), 1117–1180. DOI: https://doi.org/10.1613/jair.1.14033

[90] Alessandro Fabris, Stefano Messina, Gianmaria Silvello, and Gian Antonio Susto. 2022. Algorithmic fairness datasets: The story so far. *Data Mining and Knowledge Discovery* 36, 6 (2022), 2074–2152. DOI: https://doi.org/10.1007/s10618-022-00854-z

[91] Alessandro Fabris, Stefano Messina, Gianmaria Silvello, and Gian Antonio Susto. 2022. Tackling documentation debt: A survey on algorithmic fairness datasets. In *Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '22)*. ACM, 2:1–2:13. DOI: https://doi.org/10.1145/3551624.3555286

[92] Alessandro Fabris, Alberto Purpura, Gianmaria Silvello, and Gian Antonio Susto. 2020. Gender stereotype reinforcement: Measuring the gender bias conveyed by ranking algorithms. *Information Processing and Management* 57, 6 (2020), 102377. DOI: https://doi.org/10.1016/j.ipm.2020.102377

[93] Lídia Farré. 2016. Parental leave policies and gender equality: A survey of the literature. *Studies of Applied Economics* 34, 1 (2016), 45–60.

[94] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and removing disparate impact. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Longbing Cao, Chengqi Zhang, Thorsten Joachims, Geoffrey I. Webb, Dragos D. Margineantu, and Graham Williams (Eds.), ACM, 259–268. DOI: https://doi.org/10.1145/2783258.2783311

[95] Elena Fernández-del Río, Linda Koopmans, Pedro J. Ramos-Villagrasa, and Juan R. Barrada. 2019. Assessing job performance using brief self-report scales: The case of the individual work performance questionnaire. *Revista de Psicología del Trabajo y de las Organizaciones* 35, 3 (2019), 195–205.

[96] Benjamin Fish, Ashkan Bashardoust, danah boyd, Sorelle A. Friedler, Carlos Scheidegger, and Suresh Venkatasubramanian. 2019. Gaps in information access in social networks? In *The World Wide Web Conference (WWW '19)*. Ling Liu, Ryen W. White, Amin Mantrach, Fabrizio Silvestri, Julian J. McAuley, Ricardo Baeza-Yates, and Leila Zia (Eds.), ACM, 480–490. DOI: https://doi.org/10.1145/3308558.3313680

[97] Riccardo Fogliato, Alice Xiang, Zachary C. Lipton, Daniel Nagin, and Alexandra Chouldechova. 2021. On the validity of arrest as a proxy for offense: Race and the likelihood of arrest for violent crimes. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES '21)*. Marion Fourcade, Benjamin Kuipers, Seth Lazar, and Deirdre K. Mulligan (Eds.), ACM, 100–111. DOI: https://doi.org/10.1145/3461702.3462538

[98] A. S. Fokkens, C. J. Beukeboom, and E. Maks. 2018. *Leeftijdsdiscriminatie in vacatureteksten: Een geautomatiseerde inhoudsanalyse naar verboden leeftijd-gerelateerd taalgebruik in vacatureteksten: Rapport in opdracht van het College voor de Rechten van de Mens.*

[99] World Economic Forum. 2021. Human-centred artificial intelligence for human resources: A toolkit for human resources professionals. Retrieved from https://www3.weforum.org/docs/WEF_Human_Centred_Artificial_Intelligence_for_Human_Resources_2021.pdf

[100] Sorelle A. Friedler, Carlos Scheidegger, and Suresh Venkatasubramanian. 2016. On the (im) possibility of fairness. arXiv:1609.07236. Retrieved from https://arxiv.org/abs/1609.07236

[101] Joseph Fuller, Manjari Raman, Eva Sage-Gavin, and Kristen Hines. 2021. *Hidden Workers: Untapped Talent.* Technical Report. Harvard Business School.

[102] Danielle Gaucher, Justin Friesen, and Aaron C. Kay. 2011. Evidence that gendered wording in job advertisements exists and sustains gender inequality. *Journal of Personality and Social Psychology* 101, 1 (2011), 109.

[103] Sahin Cem Geyik, Stuart Ambler, and Krishnaram Kenthapadi. 2019. Fairness-aware ranking in search & recommendation systems with application to LinkedIn talent search. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '19)*. Ankur Teredesai, Vipin Kumar, Ying Li, Rómer Rosales, Evimaria Terzi, and George Karypis (Eds.), ACM, 2221–2231. DOI: https://doi.org/10.1145/3292500.3330691

[104] Sahin Cem Geyik, Qi Guo, Bo Hu, Cagri Ozcaglar, Ketan Thakkar, Xianren Wu, and Krishnaram Kenthapadi. 2018. Talent search and recommendation systems at LinkedIn: Practical challenges and lessons learned. In *Proceedings of the 41st International ACM SIGIR Conference on Research & Development in Information Retrieval (SIGIR '18)*. Kevyn Collins-Thompson, Qiaozhu Mei, Brian D. Davison, Yiqun Liu, and Emine Yilmaz (Eds.), ACM, 1353–1354. DOI: https://doi.org/10.1145/3209978.3210205

[105] Azin Ghazimatin, Matthäus Kleindessner, Chris Russell, Ziawasch Abedjan, and Jacek Golebiowski. 2022. Measuring fairness of rankings under noisy sensitive information. In *Proceedings of the 2022 ACM Conf. on Fairness, Accountability, and Transparency (FAccT '22)*. ACM, 2263–2279. DOI: https://doi.org/10.1145/3531146.3534641

[106] Kate Goddard, Abdul V. Roudsari, and Jeremy C. Wyatt. 2012. Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association* 19, 1 (2012), 121–127. DOI: https://doi.org/10.1136/amiajnl-2011-000089

[107] Diego Gómez-Zará, Leslie A. DeChurch, and Noshir S. Contractor. 2020. A taxonomy of team-assembly systems: Understanding how people use technologies to form teams. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2 (2020), 181:1–181:36. DOI: https://doi.org/10.1145/3415252

[108] Hila Gonen and Yoav Goldberg. 2019. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. In *Proceedings of the 2019 Conference of the North American Chapter of the*

Association for Computational Linguistics: Human Language Technologies (NAACL-HLT '19) (Long and Short Papers). Jill Burstein, Christy Doran, and Thamar Solorio (Eds.), Vol. 1, Association for Computational Linguistics, 609–614. DOI: https://doi.org/10.18653/v1/n19-1061

[109] Carlos Gradín. 2021. Occupational gender segregation in post-apartheid South Africa. *Feminist Economics* 27, 3 (2021), 102–133.

[110] Kelsey Gray, Angela Neville, Amy H. Kaji, Mary Wolfe, Kristine Calhoun, Farin Amersi, Timothy Donahue, Tracy Arnell, Benjamin Jarman, Kenji Inaba, Marc Melcher, Jon B. Morris, Brian Smith, Mark Reeves, Jeffrey Gauvin, Edgardo S. Salcedo, Richard Sidwell, Kenric Murayama, Richard Damewood, V. Prasad Poola, Daniel Dent, and Christian de Virgilio. 2019. Career goals, salary expectations, and salary negotiation among male and female general surgery residents. *JAMA Surgery* 154, 11 (2019), 1023–1029.

[111] Nina Grgic-Hlaca, Muhammad Bilal Zafar, Krishna P. Gummadi, and Adrian Weller. 2016. The case for process fairness in learning: Feature selection for fair decision making. In *NIPS Symposium on Machine Learning and the Law*, Vol. 1, 11.

[112] Philipp Hacker. 2018. Teaching fairness to artificial intelligence: Existing and novel strategies against algorithmic discrimination under EU law. *Common Market Law Review* 55, 4 (2018), 1143–1185.

[113] Aniko Hannak, Claudia Wagner, David García, Alan Mislove, Markus Strohmaier, and Christo Wilson. 2017. Bias in online freelance marketplaces: Evidence from TaskRabbit and Fiverr. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW '17)*. Charlotte P. Lee, Steven E. Poltrock, Louise Barkhuus, Marcos Borges, and Wendy A. Kellogg (Eds.), ACM, 1914–1933. DOI: https://doi.org/10.1145/2998181.2998327

[114] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, 29 (2016), 3315–3323.

[115] Joyce C. He and Sonia K. Kang. 2021. Covering in cover letters: Gender and self-presentation in job applications. *Academy of Management Journal* 64, 4 (2021), 1097–1126.

[116] Joyce C. He, Sonia K Kang, Kaylie Tse, and Soo Min Toh. 2019. Stereotypes at work: Occupational stereotypes predict race and gender segregation in the workforce. *Journal of Vocational Behavior* 115 (2019), 103318.

[117] Deepak Hegde, Alexander Ljungqvist, and Manav Raj. 2022. Race, glass ceilings, and lower pay for equal work. *Swedish House of Finance Research Paper* 21, 09.

[118] Madeline E. Heilman. 2012. Gender stereotypes and workplace bias. *Research in Organizational Behavior* 32 (2012), 113–135.

[119] Léo Hemamou and William Coleman. 2022. Delivering fairness in human resources AI: Mutual information to the rescue. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (AACL/IJCNLP '22)*. Yulan He, Heng Ji, Yang Liu, Sujian Li, Chia-Hui Chang, Soujanya Poria, Chenghua Lin, Wray L. Buntine, Maria Liakata, Hanqi Yan, Zonghan Yan, Sebastian Ruder, Xiaojun Wan, Miguel Arana-Catania, Zhongyu Wei, Hen-Hsen Huang, Jheng-Long Wu, Min-Yuh Day, Pengfei Liu, and Ruifeng Xu (Eds.), Long Papers, Vol. 1. Association for Computational Linguistics, 867–882. Retrieved from https://aclanthology.org/2022.aacl-main.64

[120] Léo Hemamou, Arthur Guillon, Jean-Claude Martin, and Chloé Clavel. 2021. Don't judge me by my face: An indirect adversarial approach to remove sensitive information from multimodal neural representation in asynchronous job video interviews. In *Proceedings of the 9th International Conference on Affective Computing and Intelligent Interaction (ACII '21)*. IEEE, 1–8. DOI: https://doi.org/10.1109/ACII52823.2021.9597443

[121] Morela Hernandez, Derek R. Avery, Sabrina D. Volpone, and Cheryl R. Kaiser. 2019. Bargaining while black: The role of race in salary negotiations. *Journal of Applied Psychology* 104, 4 (2019), 581.

[122] Iñigo Hernandez-Arenaz and Nagore Iriberri. 2019. A review of gender differences in negotiation. In *Oxford Research Encyclopedia of Economics and Finance*.

[123] Corinna Hertweck, Christoph Heitz, and Michele Loi. 2021. On the moral justification of statistical parity. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT '21 '21)*. Madeleine Clare Elish, William Isaac, and Richard S. Zemel (Eds.), ACM, 747–757. DOI: https://doi.org/10.1145/3442188.3445936

[124] HireVue. 2022. Explainability statement. Retrieved from https://hirevue-api.dev-directory.com/wp-content/uploads/2022/04/HV_AI_Short-Form_Explainability_1pager.pdf

[125] Bilal Hmoud and Varallyai Laszlo. 2019. Will artificial intelligence take over human resources recruitment and selection. *Network Intelligence Studies* 7, 13 (2019), 21–30.

[126] Holly Hoch, Corinna Hertweck, Michele Loi, and Aurelia Tamò. 2021. Discrimination for the sake of fairness: Fairness by design and its legal framework. Available at SSRN 3773766.

[127] Md Sajjad Hosain, Ping Liu, and Mohitul Ameen Ahmed Mustafi. 2021. Social networking information and pre-employment background check: Mediating effects of perceived benefit and organizational branding. *International Journal of Manpower* 42 (2021).

[128] Shenggang Hu, Jabir Alshehabi Al-Ani, Karen D. Hughes, Nicole Denier, Alla Konnikov, Lei Ding, Jinhan Xie, Yang Hu, Monideepa Tarafdar, Bei Jiang, Linglong Kong, and Hongsheng Dai. 2022. Balancing gender bias in job advertisements with text-level bias mitigation. *Frontiers Big Data* 5 (2022), 805713. DOI: https://doi.org/10.3389/fdata.2022.805713

[129] David J. Hughes and Mark Batey. 2017. Using personality questionnaires for selection. In *The Wiley Blackwell Handbook of the Psychology of Recruitment, Selection and Employee Retention*, 151–181.

[130] Ben Hutchinson and Margaret Mitchell. 2019. 50 years of test (un)fairness: Lessons for machine learning. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. danah boyd and Jamie H. Morgenstern (Eds.), ACM, 49–58. DOI: https://doi.org/10.1145/3287560.3287600

[131] Illinois General Assembly. 2020. Artificial Intelligence Video Interview Act, 820 ILCS 42. Retrieved from https://www.ilga.gov/legislation/ilcs/ilcs3.asp?ActID=4015 & ChapterID=68

[132] Basileal Imana, Aleksandra Korolova, and John S. Heidemann. 2021. Auditing for discrimination in algorithms delivering job ads. In *The World Wide Web Conference 2021 (WWW '21)*. Jure Leskovec, Marko Grobelnik, Marc Najork, Jie Tang, and Leila Zia (Eds.), ACM / IW3C2, 3767–3778. DOI: https://doi.org/10.1145/3442381.3450077

[133] Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. ACM, New York, NY, 375–385. DOI: https://doi.org/10.1145/3442188.3445901

[134] Jilana Jaxon, Ryan F. Lei, Reut Shachnai, Eleanor K. Chestnut, and Andrei Cimpian. 2019. The acquisition of gender stereotypes about intellectual ability: Intersections with race. *Journal of Social Issues* 75, 4 (2019), 1192–1215.

[135] Anna Jobin, Marcello Ienca, and Effy Vayena. 2019. The global landscape of AI ethics guidelines. *Nature Machine Intelligence* 1, 9 (01 Sep. 2019), 389–399. DOI: https://doi.org/10.1038/s42256-019-0088-2

[136] Jobvite. 2021. *2021 Recruiter Nation Report*. Technical Report. Retrieved from https://www.jobvite.com/lp/2021-recruiter-nation-report/

[137] Marc Juarez and Aleksandra Korolova. 2023. "You can't fix what you can't measure": Privately measuring demographic performance disparities in federated learning. In *Workshop on Algorithmic Fairness through the Lens of Causality and Privacy*. PMLR, 67–85.

[138] Daniel Kahneman, Olivier Sibony, and Cass R. Sunstein. 2021. *Noise: A Flaw in Human Judgment*. Hachette UK.

[139] Mitchel Kappen and Marnix Naber. 2021. Objective and bias-free measures of candidate motivation during job applications. *Scientific Reports* 11, 1 (2021), 21254.

[140] Navroop Kaur and Sandeep K. Sood. 2017. A game theoretic approach for an IoT-based automated employee performance evaluation. *IEEE Systems Journal* 11, 3 (2017), 1385–1394. DOI: https://doi.org/10.1109/JSYST.2015.2469102

[141] Heysem Kaya, Furkan Gürpinar, and Albert Ali Salah. 2017. Multi-modal score fusion and decision trees for explainable automatic job candidate screening from video CVs. In *Proceedings of the 2017 IEEE Conf. on Computer Vision and Pattern Recognition Workshops, CVPR Workshops 2017*. IEEE Computer Society, 1651–1659. DOI: https://doi.org/10.1109/CVPRW.2017.210

[142] Nicolas Kayser-Bril. 2023. *LinkedIn Automatically Rates "Out-of-Country" Candidates as "Not Fit" in Job Applications*. Technical Report. AlgorithmWatch.

[143] Niki Kilbertus, Adrià Gascón, Matt J. Kusner, Michael Veale, Krishna P. Gummadi, and Adrian Weller. 2018. Blind justice: Fairness with encrypted sensitive attributes. In *Proceedings of the 35th International Conference on Machine Learning (ICML '18)*. Jennifer G. Dy and Andreas Krause (Eds.), Vol. 80, PMLR, 2635–2644. Retrieved from http://proceedings.mlr.press/v80/kilbertus18a.html

[144] Pauline T. Kim. 2016. Data-driven discrimination at work. *William & Mary Law Review* 58 (2016), 857.

[145] Pauline T. Kim. 2022. Race-aware algorithms: Fairness, nondiscrimination and affirmative action. *California Law Review* 110 (2022), 1539.

[146] Nicholas J. Klein and Michael J. Smart. 2017. Car today, gone tomorrow: The ephemeral car in low-income, immigrant and minority families. *Transportation* 44 (2017), 495–510.

[147] Patrick Kline, Evan K. Rose, and Christopher R. Walters. 2022. Systemic discrimination among large US employers. *The Quarterly Journal of Economics* 137, 4 (2022), 1963–2036.

[148] Alina Köchling, Shirin Riazy, Marius Claus Wehner, and Katharina Simbeck. 2021. Highly accurate, but still discriminatory: A fairness evaluation of algorithmic video analysis in the recruitment context. *Business & Information Systems Engineering* 63 (2021), 39–54.

[149] Alina Köchling and Marius Claus Wehner. 2020. Discriminated by an algorithm: A systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development. *Business Research* 13, 3 (2020), 795–848.

[150] Alla Konnikov, Nicole Denier, Yang Hu, Karen D. Hughes, Jabir Alshehabi Al-Ani, Lei Ding, Irina Rets, and Monideepa Tarafdar. 2022. BIAS Word inventory for work and employment diversity,(in) equality and inclusivity (Version 1.0). SocArXiv (2022). Retrieved from https://ideas.repec.org/p/osf/socarx/t9v3a.html

[151] Margaret Bull Kovera. 2019. Racial disparities in the criminal justice system: Prevalence, causes, and a search for solutions. *Journal of Social Issues* 75, 4 (2019), 1139–1164.
[152] Kurt Kraiger and J. Kevin Ford. 1985. A meta-analysis of ratee race effects in performance ratings. *Journal of Applied Psychology* 70, 1 (1985), 56.
[153] Jasper Krommendijk and Frederik Zuiderveen Borgesius. 2023. EU law analysis 'How to read EU legislation?' Retrieved from http://eulawanalysis.blogspot.com/p/how-to-read-eu-legislation.html
[154] Deepak Kumar, Tessa Grosz, Navid Rekabsaz, Elisabeth Greif, and Markus Schedl. 2023. Fairness of recommender systems in the recruitment domain: An analysis from technical and legal perspectives. *Frontiers in Big Data* 6 (2023).
[155] Astrid Kunze and Amalia R. Miller. 2017. Women helping women? Evidence from private sector data on workplace hierarchies. *Review of Economics and Statistics* 99, 5 (2017), 769–775.
[156] Cordula Kupfer, Rita Prassl, Jürgen Fleiß, Christine Malin, Stefan Thalmann, and Bettina Kubicek. 2023. Check the box! How to deal with automation bias in AI-based personnel selection. *Frontiers in Psychology* 14 (2023), 1118723.
[157] Sarah E. Lageson, Elizabeth Webster, and Juan R. Sandoval. 2021. Digitizing and disclosing personal data: The proliferation of state criminal records on the internet. *Law & Social Inquiry* 46, 3 (2021), 635–665.
[158] Anja Lambrecht and Catherine Tucker. 2019. Algorithmic bias? An empirical study of apparent gender-based discrimination in the display of STEM career ads. *Management Science* 65, 7 (2019), 2966–2981. DOI: https://doi.org/10.1287/mnsc.2018.3093
[159] Maude Lavanchy, Patrick Reichert, Jayanth Narayanan, and Krishna Savani. 2023. Applicants' fairness perceptions of algorithm-driven hiring procedures. *Journal of Business Ethics* 188 (2023), 1–26.
[160] M. Asher Lawson, Ashley E. Martin, Imrul Huda, and Sandra C. Matz. 2022. Hiring women into senior leadership positions is associated with a reduction in gender stereotypes in organizational language. *Proceedings of the National Academy of Sciences* 119, 9 (2022), e2026443119.
[161] Thomas Le Barbanchon, Roland Rathelot, and Alexandra Roulet. 2021. Gender differences in job search: Trading off commute against wage. *The Quarterly Journal of Economics* 136, 1 (2021), 381–426.
[162] Yeonjung Lee and Fengyan Tang. 2015. More caregiving, less working: Caregiving roles and gender difference. *Journal of Applied Gerontology* 34, 4 (2015), 465–483.
[163] Andreas Leibbrandt and John A. List. 2015. Do women avoid salary negotiations? Evidence from a large-scale natural field experiment. *Management Science* 61, 9 (2015), 2016–2024.
[164] Chee Wee Leong, Katrina Roohr, Vikram Ramanarayanan, Michelle P. Martin-Raugh, Harrison Kell, Rutuja Ubale, Yao Qian, Zydrune Mladineo, and Laura McCulla. 2019. Are humans biased in assessment of video interviews? In *Adjunct of the 2019 International Conference on Multimodal Interaction (ICMI '19)*. ACM, 9:1–9:5. DOI: https://doi.org/10.1145/3351529.3360653
[165] Eve A. Levin. 2018. Gender-normed physical-ability tests under Title VII. *Columbia Law Review* 118, 2 (2018), 567–604.
[166] Lan Li, Tina Lassiter, Joohee Oh, and Min Kyung Lee. 2021. Algorithmic hiring in practice: Recruiter and HR Professional's perspectives on AI use in hiring. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, 166–176.
[167] Yingji Li, Mengnan Du, Rui Song, Xin Wang, and Ying Wang. 2023. A survey on fairness in large language models. arXiv:2308.10149.
[168] Haochen Liu, Yiqi Wang, Wenqi Fan, Xiaorui Liu, Yaxin Li, Shaili Jain, Yunhao Liu, Anil K. Jain, and Jiliang Tang. 2023. Trustworthy AI: A computational perspective. *ACM Transactions on Intelligent Systems and Technology* 14, 1 (2023), 4:1–4:59. DOI: https://doi.org/10.1145/3546872
[169] Yuxi Long, Jiamin Liu, Ming Fang, Tao Wang, and Wei Jiang. 2018. Prediction of employee promotion based on personal basic features and post features. In *Proceedings of the International Conference on Data Processing and Applications (ICDPA '18)*. ACM, 5–10. DOI: https://doi.org/10.1145/3224207.3224210
[170] Ashish Malik, Pawan Budhwar, Charmi Patel, and N. R. Srikanth. 2022. May the bots be with you! Delivering HR cost-effectiveness and individualised employee experiences in an MNE. *The International Journal of Human Resource Management* 33, 6 (2022), 1148–1178.
[171] Don Mar, Paul Ong, Tom Larson, and James Peoples. 2022. Racial and ethnic disparities in who receives unemployment benefits during COVID-19. *SN Business & Economics* 2, 8 (2022), 102.
[172] Karla Markert, Afrae Ahouzi, and Pascal Debus. 2022. Fairness in regression – Analysing a job candidates ranking system. In *INFORMATIK 2022.* Daniel Demmler, Daniel Krupka, and Hannes Federrath (Eds.), Gesellschaft für Informatik, 1275–1285. DOI: https://doi.org/10.18420/inf2022_109
[173] Yoosof Mashayekhi, Nan Li, Bo Kang, Jefrey Lijffijt, and Tijl De Bie. 2022. A challenge-based survey of e-recruitment recommendation systems. arXiv:2209.05112. Retrieved from https://arxiv.org/abs/2209.05112
[174] David A. Matsa and Amalia R. Miller. 2011. Chipping away at the glass ceiling: Gender spillovers in corporate leadership. *American Economic Review* 101, 3 (2011), 635–639.

[175] Roy Maurer. 2021. HireVue discontinues facial analysis screening. Retrieved from https://www.shrm.org/resourcesandtools/hr-topics/talent-acquisition/pages/hirevue-discontinues-facial-analysis-screening.aspx

[176] Qingxin Meng, Keli Xiao, Dazhong Shen, Hengshu Zhu, and Hui Xiong. 2022. Fine-grained job salary benchmarking with a nonparametric Dirichlet process-based latent factor model. *INFORMS Journal on Computing* 34, 5 (2022), 2443–2463. DOI: https://doi.org/10.1287/ijoc.2022.1182

[177] Alex Miller. 2018. Want less-biased decisions? Use algorithms. Retrieved from https://hbr.org/2018/07/want-less-biased-decisions-use-algorithms

[178] Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, and Kristian Lum. 2021. Algorithmic fairness: Choices, assumptions, and definitions. *Annual Review of Statistics and Its Application* 8 (2021), 141–163.

[179] Tara Sophia Mohr. 2014. Why women don't apply for jobs unless they're 100% qualified. Retrieved from https://hbr.org/2014/08/why-women-dont-apply-for-jobs-unless-theyre-100-qualified

[180] Aythami Morales, Julian Fierrez, Ruben Vera-Rodriguez, and Ruben Tolosana. 2020. SensitiveNets: Learning agnostic representations with application to face images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43, 6 (2020), 2158–2164.

[181] Gordon B. Moskowitz, Jeff Stone, and Amanda Childs. 2012. Implicit stereotyping and medical decisions: Unconscious stereotype activation in practitioners' thoughts about African Americans. *American Journal of Public Health* 102, 5 (2012), 996–1001.

[182] Corinne A. Moss-Racusin and Laurie A. Rudman. 2010. Disruptions in women's self-promotion: The backlash avoidance model. *Psychology of Women Quarterly* 34, 2 (2010), 186–202.

[183] Dena F. Mujtaba and Nihar R. Mahapatra. 2021. Multi-task deep neural networks for multimodal personality trait prediction. In *Proceedings of the 2021 International Conference on Computational Science and Computational Intelligence (CSCI '21)*. IEEE, 85–91.

[184] Ann L. Mullen. 2009. Elite destinations: Pathways to attending an Ivy League university. *British Journal of Sociology of Education* 30, 1 (2009), 15–27.

[185] Preetam Nandy, Cyrus DiCiccio, Divya Venugopalan, Heloise Logan, Kinjal Basu, and Noureddine El Karoui. 2022. Achieving fairness via post-processing in web-scale recommender systems. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*. ACM, 715–725. DOI: https://doi.org/10.1145/3531146.3533136

[186] New York City Council. 2021. Automated employment decision tools, 144. Retrieved from https://legistar.council.nyc.gov/LegislationDetail.aspx?ID=4344524&GUID=B051915D-A9AC-451E-81F8-6596032FA3F9&Options=Advanced&Search

[187] Chinasa T. Okolo, Nicola Dell, and Aditya Vashistha. 2022. Making AI explainable in the global south: A systematic review. In *ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '22)*, 439–452.

[188] Oracle. 2023. Welcome to Oracle AI Apps for Talent Management. Retrieved from https://docs.oracle.com/en/cloud/saas/talent-management/22d/faimh/welcome-to-ai-apps-for-talent-management.html#u30010414

[189] ORCAA. 2020. *Description of Algorithmic Audit: Pre-built Assessments*. Technical Report. Retrieved from https://techinquiry.org/HireVue-ORCAA.pdf

[190] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, Vol. 35, 27730–27744.

[191] Emir Ozeren. 2014. Sexual orientation discrimination in the workplace: A systematic review of literature. *Procedia-Social and Behavioral Sciences* 109 (2014), 1203–1215.

[192] KerryAnn O'Meara, Dawn Culpepper, and Lindsey L. Templeton. 2020. Nudging toward diversity: Applying behavioral design to faculty hiring. *Review of Educational Research* 90, 3 (2020), 311–348.

[193] Aasim I. Padela, Huda Adam, Maha Ahmad, Zahra Hosseinian, and Farr Curlin. 2016. Religious identity and workplace discrimination: A national survey of American Muslim physicians. *AJOB Empirical Bioethics* 7, 3 (2016), 149–159.

[194] Prasanna Parasurama and João Sedoc. 2021. Degendering resumes for fair algorithmic resume screening. arXiv:2112.08910. Retrieved from https://arxiv.org/abs/2112.08910

[195] Prasanna Parasurama and João Sedoc. 2022. Gendered information in resumes and its role in algorithmic and human hiring bias. In *Academy of Management Proceedings*, Vol. 2022. Academy of Management Briarcliff, Manor, NY, 17133.

[196] Prasanna Parasurama, João Sedoc, and Anindya Ghose. 2022. Gendered information in resumes and hiring bias: A predictive modeling approach. Available at SSRN 4074976.

[197] Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. 2016. The synthetic data vault. In *IEEE International Conference on Data Science and Advanced Analytics (DSAA '16)*, 399–410. DOI: https://doi.org/10.1109/DSAA.2016.49

[198] Mirjana Pejic-Bach, Tine Bertoncel, Maja Mesko, and Zivko Krstic. 2020. Text mining of industry 4.0 job advertisements. *International Journal of Information Management* 50 (2020), 416–431. DOI: https://doi.org/10.1016/j.ijinfomgt.2019.07.014

[199] Alejandro Peña, Ignacio Serna, Aythami Morales, and Julian Fiérrez. 2020. Bias in multimodal AI: Testbed for fair automatic recruitment. In *Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR Workshops 2020*. Computer Vision Foundation/IEEE, 129–137. DOI: https://doi.org/10.1109/CVPRW50498.2020.00022

[200] Andi Peng, Besmira Nushi, Emre Kiciman, Kori Inkpen, Siddharth Suri, and Ece Kamar. 2019. What you see is what you get? The impact of representation criteria on human bias in hiring. In *Proceedings of the 7th AAAI Conference on Human Computation and Crowdsourcing (HCOMP '19)*. Edith Law and Jennifer Wortman Vaughan (Eds.). AAAI Press, 125–134. Retrieved from https://ojs.aaai.org/index.php/HCOMP/article/view/5281

[201] Anders Persson. 2016. Implicit bias in predictive data profiling within recruitments. In *Privacy and Identity Management. Facing up to Next Steps - 11th IFIP WG* 9.2, 9.5, 9.6/11.7, 11.4, 11.6/SIG 9.2.2 International Summer School. Anja Lehmann, Diane Whitehouse, Simone Fischer-Hübner, Lothar Fritsch, and Charles D. Raab (Eds.), IFIP Advances in Information and Communication Technology, Vol. 498, 212–230. DOI: https://doi.org/10.1007/978-3-319-55783-0_15

[202] Dana Pessach, Gonen Singer, Dan Avrahami, Hila Chalutz Ben-Gal, Erez Shmueli, and Irad Ben-Gal. 2020. Employees recruitment: A prescriptive analytics approach via machine learning and mathematical programming. *Decision Support System* 134 (2020), 113290. DOI: https://doi.org/10.1016/j.dss.2020.113290

[203] Emma Pierson, Camelia Simoiu, Jan Overgoor, Sam Corbett-Davies, Daniel Jenson, Amy Shoemaker, Vignesh Ramachandran, Phoebe Barghouty, Cheryl Phillips, Ravi Shroff, and Sharad Goel. 2020. A large-scale analysis of racial disparities in police stops across the United States. *Nature Human Behaviour* 4, 7 (2020), 736–745.

[204] Rohit Punnoose and Pankaj Ajit. 2016. Prediction of employee turnover in organizations using machine learning algorithms. *International Journal of Advanced Research in Artificial Intelligence* 5, 9 (2016). DOI: https://doi.org/10.14569/IJARAI.2016.050904

[205] Sharon L, Segrest Purkiss, Pamela L. Perrewé, Treena L. Gillespie, Bronston T. Mayes, and Gerald R. Ferris. 2006. Implicit sources of bias in employment interview judgments and decisions. *Organizational Behavior and Human Decision Processes* 101, 2 (2006), 152–167.

[206] PwC. 2017. Artificial Intelligence in HR: A no-brainer. Retrieved from https://www.pwc.nl/nl/assets/documents/artificial-intelligence-in-hr-a-no-brainer.pdf

[207] Chuan Qin, Kaichun Yao, Hengshu Zhu, Tong Xu, Dazhong Shen, Enhong Chen, and Hui Xiong. 2023. Towards automatic job description generation with capability-aware neural networks. *IEEE Transactions on Knowledge and Data Engineering* 35, 5 (2023), 5341–5355. DOI: https://doi.org/10.1109/TKDE.2022.3145396

[208] Lincoln Quillian, Devah Pager, Ole Hexel, and Arnfinn H. Midtbøen. 2017. Meta-analysis of field experiments shows no change in racial discrimination in hiring over time. *Proceedings of the National Academy of Sciences* 114, 41 (2017), 10870–10875.

[209] Manish Raghavan, Solon Barocas, Jon M. Kleinberg, and Karen Levy. 2020. Mitigating bias in algorithmic hiring: evaluating claims and practices. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '20)*. Mireille Hildebrandt, Carlos Castillo, L. Elisa Celis, Salvatore Ruggieri, Linnet Taylor, and Gabriela Zanfir-Fortuna (Eds.), ACM, 469–481. DOI: https://doi.org/10.1145/3351095.3372828

[210] Krithika Ramesh, Sunayana Sitaram, and Monojit Choudhury. 2023. Fairness in language models beyond English: Gaps and challenges. In *Findings of the Association for Computational Linguistics (EACL '23)*. Association for Computational Linguistics, Dubrovnik, Croatia, 2106–2119. Retrieved from https://aclanthology.org/2023.findings-eacl.157

[211] Christine Reyna, Mark Brandt, and G. Tendayi Viki. 2009. Blame it on hip-hop: Anti-rap attitudes as a proxy for prejudice. *Group Processes & Intergroup Relations* 12, 3 (2009), 361–380.

[212] Cecil R. Reynolds and Lisa A. Suzuki. 2012. Bias in psychological assessment: An empirical review and recommendations. In *Handbook of Psychology (2nd. ed.)*, Vol. 10.

[213] Alene K. Rhea, Kelsey Markey, Lauren D'Arinzo, Hilke Schellmann, Mona Sloane, Paul Squires, and Julia Stoyanovich. 2022. Resume format, LinkedIn URLs and other unexpected influences on AI personality prediction in hiring: Results of an audit. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES '22)*. Vincent Conitzer, John Tasioulas, Matthias Scheutz, Ryan Calo, Martina Mara, and Annette Zimmermann (Eds.), ACM, 572–587. DOI: https://doi.org/10.1145/3514094.3534189

[214] Alene K. Rhea, Kelsey Markey, Lauren D'Arinzo, Hilke Schellmann, Mona Sloane, Paul Squires, Falaah Arif Khan, and Julia Stoyanovich. 2022. An external stability audit framework to test the validity of personality prediction in AI hiring. *Data Mining and Knowledge Discovery* 36, 6 (2022), 2153–2193.

[215] Peter A Riach and Judith Rich. 2002. Field experiments of discrimination in the market place. *The Economic Journal* 112, 483 (2002), F480–F518.

[216] Judith Rich. 2014. What do field experiments of discrimination in markets tell us? A meta analysis of studies conducted since 2000. Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2517887

[217] Jonas Rieskamp, Lennart Hofeditz, Milad Mirbabaie, and Stefan Stieglitz. 2023. Approaches to improve fairness when deploying AI-based algorithms in hiring - using a systematic literature review to guide future research. In *Proceedings*

of the 56th Hawaii International Conference on System Sciences (HICSS '23). Tung X. Bui (Ed.), ScholarSpace, 216–225. Retrieved from https://hdl.handle.net/10125/102654

[218] Lauren A. Rivera. 2011. Ivies, extracurriculars, and exclusion: Elite employers' use of educational credentials. *Research in Social Stratification and Mobility* 29, 1 (2011), 71–90.

[219] Lauren A. Rivera. 2012. Hiring as cultural matching: The case of elite professional service firms. *American Sociological Review* 77, 6 (2012), 999–1022.

[220] David Robotham and Richard Jubb. 1996. Competences: Measuring the unmeasurable. *Management Development Review* 9, 5 (1996), 25–29.

[221] Cathy Roche, Dave Lewis, and P. J. Wall. 2021. Artificial intelligence ethics: An inclusive global discourse? arXiv:2108.09959. Retrieved from https://arxiv.org/abs/2108.09959

[222] Clara Rus, Jeffrey Luppes, Harrie Oosterhuis, and Gido H. Schoenmacker. 2022. Closing the gender wage gap: Adversarial fairness in job recommendation. In *Proceedings of the 2nd Workshop on Recommender Systems for Human Resources (RecSys-in-HR 2022) co-located with the 16th ACM Conference on Recommender Systems (RecSys 2022)*, Seattle, Washington, 1–10.

[223] Mary-Ann Russon. 2020. Uber sued by drivers over 'automated robo-firing'. *BBC News* 26 (2020).

[224] Abel Salinas, Parth Shah, Yuzhong Huang, Robert McCormack, and Fred Morstatter. 2023. The unequal opportunities of large language models: Examining demographic biases in job recommendations by ChatGPT and LLaMA. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '23)*. ACM. DOI: https://doi.org/10.1145/3617694.3623257

[225] Kimberly T. Schneider, Suzanne Swan, and Louise F. Fitzgerald. 1997. Job-related and psychological effects of sexual harassment in the workplace: Empirical evidence from two organizations. *Journal of Applied Psychology* 82, 3 (1997), 401.

[226] Frederike Scholz. 2020. Taken for granted: Ableist norms embedded in the design of online recruitment practices. In *The Palgrave Handbook of Disability at Work*, 451–469.

[227] Almudena Sevilla and Sarah Smith. 2020. Baby steps: The gender division of childcare during the COVID-19 pandemic. *Oxford Review of Economic Policy* 36, Supplement_1 (2020), S169–S186.

[228] Michael A. Shields and Stephen Wheatley Price. 2002. Racial harassment, job satisfaction and intentions to quit: Evidence from the British nursing profession. *Economica* 69, 274 (2002), 295–326.

[229] Jim Sidanius and Marie Crane. 1989. Job evaluation and gender: The case of university faculty. *Journal of Applied Social Psychology* 19, 2 (1989), 174–197.

[230] Jan Simson, Alessandro Fabris, and Christoph Kern. 2024. Lazy data practices harm fairness research. In *Proceeding of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24)*. ACM, 642–659. DOI: https://doi.org/10.1145/3630106.3658931

[231] Abhishek Singhania, Abhishek Unnam, and Varun Aggarwal. 2020. Grading video interviews with fairness considerations. arXiv:2007.05461. Retrieved from https://arxiv.org/abs/2007.05461

[232] Clea Skopeliti. 2023. 'I feel constantly watched': The employees working under surveillance. Retrieved from https://www.theguardian.com/money/2023/may/30/i-feel-constantly-watched-employees-working-under-surveillance-monitorig-software-productivity

[233] David G. Smith, Judith E. Rosenstein, Margaret C. Nikolov, and Darby A. Chaney. 2019. The power of language: Gender, status, and agency in performance evaluations. *Sex Roles* 80 (2019), 159–171.

[234] Daryl G. Smith, Caroline S. Turner, Nana Osei-Kofi, and Sandra Richards. 2004. Interrupting the usual: Successful strategies for hiring diverse faculty. *The Journal of Higher Education* 75, 2 (2004), 133–160.

[235] Lawrence B. Solum. 2004. Procedural justice. *Southern California Law Review* 78 (2004), 181.

[236] Keith E. Sonderling, Bradford J. Kelley, and Lance Casimir. 2022. The promise and the peril: Artificial intelligence and employment discrimination. *University of Miami Law Review* 77 (2022), 1.

[237] Karen Sparck Jones. 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation* 28, 1 (1972), 11–21.

[238] Sabrina Stöckli, Michael Schulte-Mecklenbeck, Stefan Borer, and Andrea C. Samson. 2018. Facial expression analysis with AFFDEX and FACET: A validation study. *Behavior Research Methods* 50 (2018), 1446–1460.

[239] Ana-Andreea Stoica, Christopher J. Riederer, and Augustin Chaintreau. 2018. Algorithmic glass ceiling in social networks: The effects of social recommendations on network diversity. In *Proceedings of the 2018 World Wide Web Conference on World Wide Web (WWW '18)*. Pierre-Antoine Champin, Fabien Gandon, Mounia Lalmas, and Panagiotis G. Ipeirotis (Eds.). ACM, 923–932. DOI: https://doi.org/10.1145/3178876.3186140

[240] Tom Sühr, Sophie Hilgard, and Himabindu Lakkaraju. 2021. Does fair ranking improve minority outcomes? Understanding the interplay of human and algorithmic biases in online hiring. In *AAAI/ACM Conference on AI, Ethics, and Society, Virtual Event (AIES '21)*. Marion Fourcade, Benjamin Kuipers, Seth Lazar, and Deirdre K. Mulligan (Eds.). ACM, 989–999. DOI: https://doi.org/10.1145/3461702.3462602

[241] Rachael Tatman. 2017. Gender and dialect bias in YouTube's automatic captions. In *Proceedings of the 1st ACL Workshop on Ethics in Natural Language Processing (EthNLP@EACL '17)*. Dirk Hovy, Shannon L. Spruit, Margaret Mitchell, Emily M. Bender, Michael Strube, and Hanna M. Wallach (Eds.), Association for Computational Linguistics, 53–59. DOI: https://doi.org/10.18653/v1/w17-1606

[242] Josh Terrell, Andrew Kofink, Justin Middleton, Clarissa Rainear, Emerson R. Murphy-Hill, Chris Parnin, and Jon Stallings. 2017. Gender differences and bias in open source: Pull request acceptance of women versus men. *PeerJ Computer Science* 3 (2017), e111. DOI: https://doi.org/10.7717/peerj-cs.111

[243] Rebbeca Tesfai and Kevin J. A. Thomas. 2020. Dimensions of inequality: Black immigrants' occupational segregation in the United States. *Sociology of Race and Ethnicity* 6, 1 (2020), 1–21.

[244] Kerri A. Thompson. 2020. Countenancing employment discrimination: Facial recognition in background checks. *Texas A & M Law Review* 8 (2020), 63.

[245] Nicholas Tilmes. 2022. Disability, fairness, and algorithmic bias in AI recruitment. *Ethics and Information Technology* 24, 2 (2022), 21. DOI: https://doi.org/10.1007/s10676-022-09633-2

[246] Shari Trewin, Sara H. Basson, Michael J. Muller, Stacy M. Branham, Jutta Treviranus, Daniel M. Gruen, Daniel Hebert, Natalia Lyckowski, and Erich Manser. 2019. Considerations for AI fairness for people with disabilities. *AI Matters* 5, 3 (2019), 40–63. DOI: https://doi.org/10.1145/3362077.3362086

[247] James M. Tyler and Jennifer Dane McCullough. 2009. Violating prescriptive stereotypes on job resumes: A self-presentational perspective. *Management Communication Quarterly* 23, 2 (2009), 272–287.

[248] UNDP - United Nations Development Programme. 2023. Breaking down gender biases: Shifting social norms towards gender equality. Retrieved from https://hdr.undp.org/system/files/documents/hdp-document/gsni202302pdf.pdf

[249] U.S. Supreme Court. 1971. Griggs v. Duke Power Co., 401 U.S. 424. Retrieved from https://supreme.justia.com/cases/federal/us/401/424/

[250] U.S. Supreme Court. 1973. McDonnell Douglas Corp. v. Green, 411 U.S. 792. Retrieved from https://supreme.justia.com/cases/federal/us/411/792/

[251] U.S. Supreme Court. 1989. Price Waterhouse v. Hopkins, 490 U.S. 228. Retrieved from https://supreme.justia.com/cases/federal/us/490/228/

[252] U.S. Supreme Court. 2009. Ricci v. DeStefano, 557 U.S. 557. Retrieved from https://supreme.justia.com/cases/federal/us/557/557/

[253] Chris Vallance. 2023. TUC: Government failing to protect workers from AI. Retrieved from https://www.bbc.com/news/technology-65301630

[254] Marvin Van Bekkum and Frederik Zuiderveen Borgesius. 2023. Using sensitive data to prevent discrimination by artificial intelligence: Does the GDPR need a new exception? *Computer Law & Security Review* 48 (2023), 105770.

[255] Elmira van den Broek, Anastasia V. Sergeeva, and Marleen Huysman. 2019. Hiring algorithms: An ethnography of fairness in practice. In *Proceedings of the 40th International Conference on Information Systems (ICIS '19)*. Helmut Krcmar, Jane Fedorowicz, Wai Fong Boh, Jan Marco Leimeister, and Sunil Wattal (Eds.), Association for Information Systems. Retrieved from https://aisel.aisnet.org/icis2019/future_of_work/future_work/6

[256] Sarah-Jane van Els, David Graus, and Emma Beauxis-Aussalet. 2022. Improving fairness assessments with synthetic data: A practical use case with a recommender system for human resources. In *Proceedings of the 1st International Workshop on Computational Jobs Marketplace* (CompJobs '22), 5 pages.

[257] Sriram Vasudevan and Krishnaram Kenthapadi. 2020. LiFT: A scalable framework for measuring fairness in ML applications. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM '20)*. Mathieu d'Aquin, Stefan Dietze, Claudia Hauff, Edward Curry, and Philippe Cudré-Mauroux (Eds.), ACM, 2773–2780. DOI: https://doi.org/10.1145/3340531.3412705

[258] Giridhari Venkatadri and Alan Mislove. 2020. On the potential for discrimination via composition. In *ACM Internet Measurement Conference (IMC '20)*. ACM, 333–344. DOI: https://doi.org/10.1145/3419394.3423641

[259] Pranshu Verma. 2023. AI is starting to pick who gets laid off. Retrieved from https://www.washingtonpost.com/technology/2023/02/20/layoff-algorithms/

[260] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2021. Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Computer Law & Security Review* 41 (2021), 105567.

[261] Sean Waite. 2021. Should I stay or should I go? Employment discrimination and workplace harassment against transgender and other minority employees in Canada's federal public service. *Journal of Homosexuality* 68, 11 (2021), 1833–1859.

[262] Joseph Walker. 2012. Meet the new boss: Big Data. Retrieved from https://www.wsj.com/articles/SB10000872396390443890304578006252019616768

[263] Angelina Wang, Sayash Kapoor, Solon Barocas, and Arvind Narayanan. 2022. Against predictive optimization: On the legitimacy of decision-making algorithms that optimize predictive accuracy. *Available at SSRN*.

[264] Yu Wang and Tyler Derr. 2022. Degree-related bias in link prediction. In *IEEE International Conference on Data Mining Workshops (ICDM '22)*. K. Selçuk Candan, Thang N. Dinh, My T. Thai, and Takashi Washio (Eds.), IEEE, 757–758. DOI: https://doi.org/10.1109/ICDMW58026.2022.00103

[265] Yifan Wang, Weizhi Ma, Min Zhang, Yiqun Liu, and Shaoping Ma. 2023. A survey on the fairness of recommender systems. *ACM Transactions on Information Systems* 41, 3 (2023), 52:1–52:43. DOI: https://doi.org/10.1145/3547333

[266] Jamelle Watson-Daniels, Solon Barocas, Jake M. Hofman, and Alexandra Chouldechova. 2023. Multi-target multiplicity: Flexibility and fairness in target specification under resource constraints. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*. ACM, 297–311. DOI: https://doi.org/10.1145/3593013.3593998

[267] Amy L. Wax. 2011. Disparate impact realism. *William & Mary Law Review* 53 (2011), 621.

[268] Hilde Weerts, Miroslav Dudík, Richard Edgar, Adrin Jalali, Roman Lutz, and Michael Madaio. 2023. Fairlearn: Assessing and improving fairness of AI systems, 8 pages. Retrieved from http://jmlr.org/papers/v24/23-0389.html

[269] Doris Weichselbaumer and Rudolf Winter-Ebmer. 2005. A meta-analysis of the international gender wage gap. *Journal of Economic Surveys* 19, 3 (2005), 479–511.

[270] Jacqueline Kory Westlund, Sidney K. D'Mello, and Andrew M. Olney. 2015. Motion tracker: Camera-based monitoring of bodily movements using motion silhouettes. *PloS One* 10, 6 (2015), e0130293.

[271] Christo Wilson, Avijit Ghosh, Shan Jiang, Alan Mislove, Lewis Baker, Janelle Szary, Kelly Trindel, and Frida Polli. 2021. Building and auditing fair algorithms: A case study in candidate screening. In *Proccedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. Madeleine Clare Elish, William Isaac, and Richard S. Zemel (Eds.). ACM, 666–677. DOI: https://doi.org/10.1145/3442188.3445928

[272] Claes Wohlin. 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering*, 1–10.

[273] Carol Woodhams, Ben Lupton, and Marc Cowling. 2015. The presence of ethnic minority and disabled men in feminised work: Intersectionality, vertical segregation and the glass escalator. *Sex Roles* 72 (2015), 277–293.

[274] Alison T. Wynn and Shelley J. Correll. 2018. Puncturing the pipeline: Do technology companies alienate women in recruiting sessions? *Social Studies of Science* 48, 1 (2018), 149–164.

[275] Renzhe Xu, Peng Cui, Kun Kuang, Bo Li, Linjun Zhou, Zheyan Shen, and Wei Cui. 2020. Algorithmic decision making with conditional fairness. In *Proccedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '20)*. ACM, 2125–2135. DOI: https://doi.org/10.1145/3394486.3403263

[276] Shen Yan, Di Huang, and Mohammad Soleymani. 2020. Mitigating biases in multimodal personality assessment. In *International Conference on Multimodal Interaction (ICMI '20)*. ACM, 361–369. DOI: https://doi.org/10.1145/3382507.3418889

[277] Maya Yaneva. 2018. Employee satisfaction vs. employee engagement vs. employee NPS. *European Journal of Economics and Business Studies* 4, 1 (2018), 221–227.

[278] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. 2017. Fairness constraints: Mechanisms for fair classification. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS '17)*, , Vol. 54. PMLR, 962–970. Retrieved from http://proceedings.mlr.press/v54/zafar17a.html

[279] Meike Zehlike, Philipp Hacker, and Emil Wiedemann. 2020. Matching code and law: Achieving algorithmic fairness with optimal transport. *Data Mining and Knowledge Discovery* 34, 1 (2020), 163–200. DOI: https://doi.org/10.1007/s10618-019-00658-8

[280] Meike Zehlike, Ke Yang, and Julia Stoyanovich. 2023. Fairness in ranking, part I: Score-based ranking. *ACM Computing Surveys* 55, 6 (2023), 118:1–118:36. DOI: https://doi.org/10.1145/3533379

[281] Lixuan Zhang and Christopher Yencha. 2022. Examining perceptions towards hiring algorithms. *Technology in Society* 68 (2022), 101848.

[282] Shuo Zhang and Peter Kuhn. 2022. Understanding algorithmic bias in job recommender systems: An audit study approach. (2022).

[283] Sijing Zhang, Ping Li, and Ziyan Cai. 2022. Are male candidates better than females? Debiasing BERT resume retrieval system. In *IEEE International Conference on Systems, Man, and Cybernetics (SMC '22)*. IEEE, 616–621. DOI: https://doi.org/10.1109/SMC53654.2022.9945184

[284] Yiguang Zhang and Augustin Chaintreau. 2021. Unequal opportunities in multi-hop referral programs. arXiv:2112.00269. Retrieved from https://arxiv.org/abs/2112.00269

[285] Dave Zielinski. 2023. Should algorithms make layoff decisions? Retrieved from https://www.shrm.org/hr-today/news/hr-magazine/summer-2023/pages/should-algorithms-make-layoff-decisions-.aspx

[286] Indre Zliobaite. 2015. A survey on measuring indirect discrimination in machine learning. arXiv:1511.00148. Retrieved from https://arxiv.org/abs/1511.00148

Appendix

A Systematic Review Methodology

This article is an interdisciplinary survey aimed at informing researchers and practitioners interested in fairness and bias in algorithmic hiring. We focused on a Computer Science perspective while summarizing key topics from HR Management, Industrial and Organizational Psychology, Philosophy and Law with mixed methods, including the analysis of influential technical reports, industry white papers and legal literature. Three sections of this work, summarizing measures, mitigation strategies, and data (Sections 4–6), result from a systematic literature review summarized below.

(1) To ensure a broad coverage of the scientific literature centered on computer Science, we leverage three scholarly search engines: IEEE Xplore, ACM Digital Library, and Google Scholar.
(2) We use the query `algorithmic` ∧ `hiring` ∧ `fairness` on article titles, where each term is expanded as follows:
   — `algorithmic`: algorithm* ∨ AI ∨ search* ∨ recommend* ∨ rank* ∨ screen* ∨ retriev*
   — `hiring`: hir* ∨ recruit* ∨ candidate* ∨ job* ∨ work* ∨ resum* ∨ CV ∨ interview* ∨ eval* ∨ appraisal. The last two terms target the evaluation stage.
   — `fairness`: *bias* ∨ *ethic* ∨ *fair* ∨ discriminat* ∨ *equit* ∨ *equal* ∨ *parit* ∨ *symmetr* ∨ gap
(3) To ensure high recall, we consider the top 100 results. To guarantee precision, we manually analyze each article and only select the ones that treat fairness in algorithmic hiring from a quantitative perspective, i.e., performing a fairness audit or introducing a novel method. To exemplify, we discard articles on related yet different topics, such as freelancing [113], focused on human perceptions [159], on qualitative aspects [255], or mitigating biases to improve accuracy without any fairness consideration [49].
(4) We take additional steps to further improve recall. For each included article, we perform forward and backward snowballing [272], pre-filtering article titles with a "`hiring` ∧ `fairness`" query and applying the inclusion criteria in (3). Finally, we consider articles and datasets presented in related surveys [90, 91, 173, 280] or published at dedicated venues,[4] finding one additional dataset (IBM HR Analytics) and no additional article.

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[4]Workshop on Recommender Systems for Human Resources https://recsyshr.aau.dk/

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Review

Surveying Racial Bias in Facial Recognition: Balancing Datasets and Algorithmic Enhancements

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Abstract: Facial recognition systems frequently exhibit high accuracies when evaluated on standard test datasets. However, their performance tends to degrade significantly when confronted with more challenging tests, particularly involving specific racial categories. To measure this inconsistency, many have created racially aware datasets to evaluate facial recognition algorithms. This paper analyzes facial recognition datasets, categorizing them as racially balanced or unbalanced while limiting racially balanced datasets to have each race be represented within five percentage points of all other represented races. We investigate methods to address concerns about racial bias due to uneven datasets by using generative adversarial networks and latent diffusion models to balance the data, and we also assess the impact of these techniques. In an effort to mitigate accuracy discrepancies across different racial groups, we investigate a range of network enhancements in facial recognition performance across human races. These improvements encompass architectural improvements, loss functions, training methods, data modifications, and incorporating additional data. Additionally, we discuss the interrelation of racial and gender bias. Lastly, we outline avenues for future research in this domain.

Keywords: biometrics; deep learning; deep learning bias; facial recognition; race bias

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1. Introduction

Although current facial recognition systems achieve high average accuracy, numerous current improvements are directed towards addressing the disproportionate accuracies across different categories, including race [1–4]. Many attribute the majority of these issues to unbalanced datasets [5–8]. Suresh and Guttag [9] from MIT explore the factors contributing to data bias. They categorize biases into five subcategories: historical, representational, measurement, evaluation, and aggregation. These biases frequently arise unintentionally, often remaining unnoticed by researchers and consequently impact results [9].

In an attempt to quantify and mitigate varying accuracy across race in facial recognition systems, many researchers have encouraged the use of racially balanced datasets over the use of those that are racially unbalanced [10–13]. When a racially balanced dataset is not available for training, a racially balanced evaluation is encouraged [14–17]. The use of racially balanced facial recognition systems is critical to applications that require high security, regardless of race. As such, we examine publicly available datasets and provide information on whether they are racially balanced or unbalanced.

Recently, large improvements have been made in the field of image generation [18]. The effects of these generated images on facial recognition tasks are now emerging. We provide a discussion on how image generation has impacted deep learning systems corresponding to the face in cross-race transformation. Many use this cross-race transformation to generate or balance through augmentation datasets that were originally racially unbalanced to be racially balanced. This is accomplished by transforming the race of individuals in the images through the use of generative adversarial networks (GANs) or latent diffusion models.

Various tradeoffs come from balancing datasets and point to the need for other improvements besides only balancing the dataset. In response to this need, various network improvements were developed that reduce variability in performance across different races. These improvements vary from loss functions and architectural changes to data modification and the use of additional data. We compile and analyze these contributions to demonstrate the field's current state of overcoming racial bias in facial recognition.

In summary, our contributions are as follows:

1. An analysis of both racially balanced and unbalanced datasets reveals significant imbalance across race in numerous widely used facial recognition datasets. Additionally, a list of the datasets that are racially balanced is included.

2. An examination of dataset balancing techniques through data generation, accompanied by an exploration of the implications of these methods.

3. Discussion on various network enhancements that effectively narrow the error gaps between different racial groups in facial recognition tasks.

Additionally, as gender bias is greatly interrelated with racial bias, we discuss gender bias in the context of these three contributions.

The rest of this paper is structured as follows: In Section 2, we present our methods. In Section 3, we discuss the differences between various facial recognition datasets comparing those that are racially unbalanced with those that are racially balanced. In Section 4, we discuss balancing datasets through image generation and the impact of data generation. In Section 5, we discuss various network improvements to decrease the skew of accuracies of facial recognition across race. We recommend future research directions in Section 6 and conclude in Section 7.

2. Methods

This section describes the approach for our survey paper for the dataset comparison in Section 2.1, balancing datasets through data generation in Section 2.2, the network improvements across the human race in Section 2.3, and our decision on terminology in Section 2.4. We consulted the PRISMA review methodology [19] and incorporated the majority of its checklist items to enhance the organization and clearly demonstrate the contributions of this work.

2.1. Datasets Methods

In this work, we selected 22 datasets for comparison in Section 3 based on Google Scholar searches for facial recognition datasets and selected the most popular datasets as defined by Serna et al. [20]. We analyzed the 22 datasets by reviewing the title, the abstract, and the dataset collection description. Each paper was initially reviewed by a single researcher. If inconsistencies or questions arose, a second researcher conducted an additional review. To maintain consistency and minimize bias, the primary researcher performed all initial reviews. Although this does not eliminate all potential biases, it ensures consistent biases throughout the analysis. Any papers that contained grayscale image datasets, such as that from the work of Samaria and Harter [21], were removed from the analysis.

After the datasets were selected, we compiled the racial and gender distributions for each dataset using the original publication of the dataset. When these distributions were not provided by the original publication, we referenced Serna et al. [20] to report the available distributions. When neither source provided the required data, we extracted the racial and gender labels directly from the dataset to obtain the distributions. These three methods allowed us to identify each dataset's racial and gender distributions without the need for hand labeling. We acknowledge that the varying methods may result in inconsistencies in race classification.

We then determined which racially balanced datasets contained overlap with racially unbalanced datasets, as discussed in Section 3.3. Our definition of a racially balanced

dataset is when the proportionate difference between the most and least represented races in the dataset does not exceed 5 percentage points. As described by the following equation:

$$\left| \frac{n_{\max} - n_{\min}}{N} \right| \leq 0.05 \quad (1)$$

where $n_{\max}$ is the number of samples of the most represented race, $n_{\min}$ is the number of samples of the least represented race, and N is the total number of samples in the dataset. This definition is based on the principle of an alpha value of 0.05 in statistics. Our decision on the 0.05 alpha value is discussed more in depth in Section 3. The papers for the racially balanced datasets were further analyzed to determine the details of creation, benefits, tradeoffs, and what overlap exists with other popular datasets.

2.2. Dataset Balancing through Generation Methods

One of the first papers to propose generating a balanced dataset through generation was the work by Yucer et al. [22] as discussed more fully in Section 4.2. For this work, we examined all papers on Google Scholar that cited the work of Yucer et al. [22] and additionally searched for works that balance datasets through generation for facial recognition. The titles and abstracts were used to identify papers of interest. Papers were removed that did not focus on generating faces for creating a racially balanced dataset for facial recognition. Each paper's contributions were retrieved and reported in this work.

As the impact of generating images is a growing research field with the growth of additional image generation platforms and networks, we chose to include a discussion on the impacts of data generation. We acknowledge that entire survey papers are made to discuss the impact of data generation. This paper provides a small sample from the literature to highlight the use and benefits of data generation to reduce racial bias. Not including this small sample of data generation impacts would do the reader a disservice by not demonstrating the tradeoffs of the given approach.

2.3. Network Improvements across Human Race Methods

Upon reviewing various network improvements using Google Scholar we chose to focus our review of network improvements across the human race on loss/training, architecture, dataset modification, and the use of additional data. To compare works that made network improvements, we focus on works that were trained on the BUPT-Balancedface dataset while using the ArcFace loss and evaluated on the RFW dataset. The only exceptions are when the improvement is an adjustment to the loss function. This limitation allows for accurate comparison on the RFW dataset.

As we reviewed the articles surveyed in this paper, we discovered many approaches benefit from race classification. As such, we discuss the field of racial classification with articles found through Google Scholar. These articles were analyzed and the split between the two-class and multiple-class race classification categories.

2.4. Terminology

Across the facial recognition literature, the terms "race" and "ethnicity" are used interchangeably. For example, the work by Wang et al. [15] uses the term "race" while the work by Li et al. [16] uses the term "ethnicity" to define the same concept—varying physical appearance that comes from ancestral heritage. In our work, we use the similar, but distinct definitions of race and ethnicity provided by the Merriam-Webster Dictionary. Race is defined as "any one of the groups that humans are often divided into based on physical traits regarded as common among people of shared ancestry" [23], and ethnicity is defined as "ethnic quality or affiliation" [24] with ethnic being defined as "of or relating to large groups of people classed according to common racial, national, tribal, religious, linguistic, or cultural origin or background" [25]. Thus, we refer to "race" as the physical characteristics of the face and "ethnicity" as the cultural aspects of the individual. This

clarification is provided to shed light on the use of these terms throughout the paper, acknowledging that others may use different definitions for these terms.

Notably, this survey paper's referenced works refer to 2–7 racial classifications. We acknowledge that there are many more racial classifications than these 2–7 categories and that many individuals identify as multicultural. As evident by the publication of a statistical policy directive defining what classifications and definitions of race ought to be used throughout the United States by the Office of Information and Regulatory Affairs and the Office of Management and Budget, racial classification best practices are still evolving [26]. This lengthy report discusses the various intricacies of racial identification, and includes multiple recent changes that demonstrate the complexity of classifying racial groups.

While we acknowledge the complexities of racial classification, in this work, we maintain the 2–7 categories used in previous racial facial recognition scholarships for consistency's sake. In line with one of the first racially balanced datasets, RFW, which created the standard across the field of racially aware facial recognition tasks [15], we primarily use the following ordering and groupings for racial classification: Caucasian, African, Indian, and Asian. In addition, throughout this paper, various approaches refer to "White" and "Non-White" classifications. We acknowledge that this definition may vary depending on where the authors of the various approaches reside. However, the use in this paper is primarily to emphasize the differing accuracies between different races.

Overall, our method is centered around using Google Scholar's search to find relevant papers. Titles and abstracts were manually checked to identify articles of interest. Further analysis of the articles was then performed to remove any articles that did not meet the focus of this work. Finally, for a paper to be included, a complete analysis of the paper was performed to identify contributions, results, and any pertinent background information. We used the PRISMA checklist [19] as a guide in the process of the creation of this paper with many of the checklist items included throughout the content of this paper.

3. Facial Recognition Dataset Comparison

Early facial recognition datasets often implemented constraints for what variation was acceptable in lighting, head pose, background, and facial expression. One early dataset, the AT&T Database of Faces (originally known as the ORL Database of Faces), incorporated several of these constraints [21]. Although this was a novel task in 1994, after only two years researchers shifted towards performing facial recognition in an unconstrained fashion seeking a model to increase generalization beyond the specified constraints [27]. In 2007, the Labeled Faces in the Wild (LFW) dataset was released with images containing a variety of head poses, lightings, camera parameters, and resolutions moving toward generalization across more scenarios [28]. However, in the last 5 years, the racial imbalance of these web-scraped datasets and the overfitting to the distribution of these datasets were demonstrated [15]. To overcome these datasets' inaccurate distributions, recent datasets synthetically generate images to enforce the prior of proper data distributions [29].

A recent dataset, Digi-Face 1M, released in 2023, is generated with synthetic images with a focus placed on previously overlooked biases such as race, lighting, and even make-up [29]. Although synthetic image generation has advanced significantly, authentic images remain essential in certain aspects of deep learning training pipelines [30]. Although approaches have identified the need for proper racial distributions across training datasets [15,29], there is no standard criteria to measure racial equality in deep learning systems [31].

To address the absence of criteria for measuring racially balanced datasets, we define a racially balanced dataset as a dataset with a 5 (or less) percentage point difference in the proportion of images from the highest to the lowest represented racial categories. This 5 percentage point difference is chosen based on statistics containing a p-value of less than 0.05 considered statistically significant for an alpha of 0.05. The 0.05 alpha value was originally proposed by Fisher [32] and was adopted as the default alpha value across many statistical approaches [33]. Following this trend, we set our alpha value to 0.05 resulting in

the 5 percentage point maximum difference between race distributions to be considered a racially balanced dataset. A graph demonstrating the racial distribution across prevalent datasets is given in Figure 1.

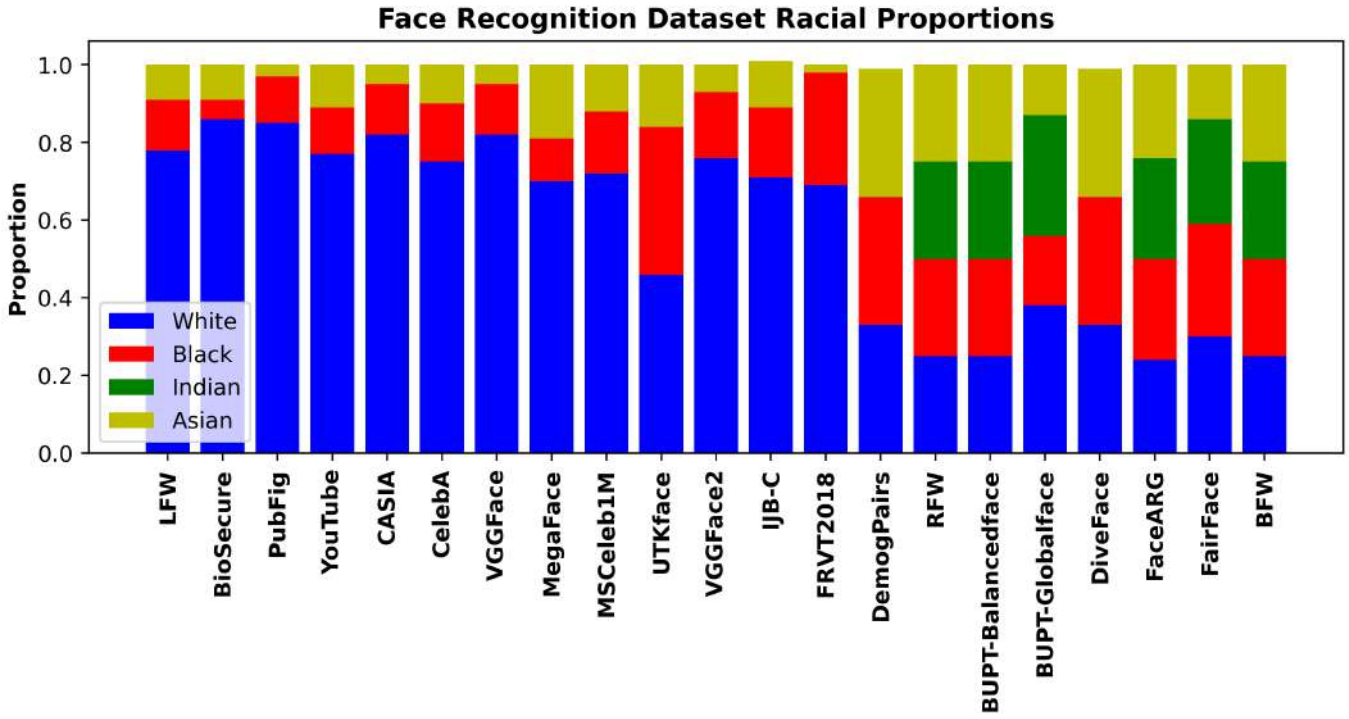


Figure 1. Dataset Racial Distributions: Visualization of racial distributions across prevalent facial recognition datasets. Consistent with the literature, we report only the races included in the RFW dataset [15].

3.1. Unbalanced Datasets

The pursuit to improve facial recognition led to the creation of many datasets. To create the largest datasets possible, web scraping Yahoo, Flickr, and other internet platforms are popular methods for compiling datasets. Some of the prevalent datasets include LFW [28], BioSecure [34], PubFig [35], YouTube Faces [36], CASIA [37], CelebA [38], VGGFace [39], MegaFace [40], MSCeleb1M [41], UTKFace [42], VGGFace2 [43], IJB-C [44], and FRVT2018 [45]. Each of these datasets was collected using different methods. However, each of these datasets was collected with a focus on obtaining the most images in the most cost-effective and efficient method available. Convenience sampling is defined as a method of focusing on ease of access when selecting participants from a target population [46]. When the population of interest is the entire human race, these datasets can be considered a convenience sample. The racial distribution of each of these datasets is provided in Table 1. We define a racially unbalanced dataset as a dataset that contains larger than a five percentage point difference from the most represented race to the least represented. As such, we define each of these datasets as racially unbalanced.

Despite being unbalanced, some racially unbalanced datasets remain valuable for racially focused facial recognition systems. These racially aware datasets employ targeted data collection methodologies. One of these datasets is the BUPT-Globalface dataset. Rather than basing the racial distribution on maintaining equal proportions across the dataset, they base their distribution on the worldwide distribution of race [10].

Table 1. Dataset Distribution Information: Descriptions of various racially balanced and unbalanced datasets. The datasets are sorted chronologically by release year, then alphabetically by dataset name. For those papers that did not originally report the race distribution, the distribution was taken from the work of Serna et al. [20].

Dataset	Year	Number of Images (In Thousands)	Number of Identities (In Thousands)	Average Images Per Identity	Caucasian	African	Indian	Asian	Racially Balanced/ Unbalanced
LFW [28]	2008	13	5.7	2	77.6%	12.9%	** Combined In African	9.4%	Unbalanced
BioSecure [34]	2009	2.7	0.667	4	86.1%	5.2%	** Combined In African	8.8%	Unbalanced
PubFig [35]	2011	58	0.2	294	85.0%	12.0%	** Combined In African	3.0%	Unbalanced
YouTube Faces [36]	2011	621	1.6	390	77.2%	11.7%	** Combined In African	10.9%	Unbalanced
CASIA [37]	2014	500	10.5	48	82.0%	12.9%	** Combined In African	5.2%	Unbalanced
CelebA [38]	2015	203	10.2	20	75.4%	14.6%	** Combined In African	9.9%	Unbalanced
VGGFace [39]	2015	2600	2.6	1000	82.3%	12.7%	** Combined In African	5.0%	Unbalanced
MegaFace [40]	2016	4700	660	7	70.3%	10.9%	** Combined In African	18.7%	Unbalanced
MSCeleb1M [41]	2016	10,000	100	100	71.6%	16%	** Combined In African	12.2%	Unbalanced
UTKFace [42]	2017	24	-	-	46.2%	37.8%	** Combined In African	16.0%	Unbalanced
IJB-C [44]	2018	21	3.5	6	70.5%	17.8%	** Combined In African	11.6%	Unbalanced
VGGFace2 [43]	2018	3300	9	370	76.1%	16.8%	** Combined In African	7%	Unbalanced
DemogPairs [14]	2019	10.8	0.6	18	33.3%	33.3%	** Combined In African	33.3%	Balanced
FRVT2018 [45]	2019	2700	1200	2	64.9%	27.3%	** Combined In African	1.6%	Unbalanced
RFW [15]	2019	40	12	3.3	25%	25%	25%	25%	Balanced
BUPT-Balancedface [10]	2020	1300	28	46.4	25%	25%	25%	25%	Balanced
BUPT-Globalface [10]	2020	2000	38	52.6	38%	13%	18%	31%	Unbalanced
CeFA [16]	2020	23.5	1.6	14.7	N/A	33.3%	33.3%	33.3%	Balanced
DiveFace [11]	2020	125	24	5.2	33.3%	33.3%	** Combined In African	33.3%	Balanced
FaceARG [12]	2021	175	-	-	24.42%	24.02%	25.94%	25.60%	Balanced
FairFace [13]	2021	108	-	-	* 29.7%	* 29.5%	* 26.6%	* 14.2%	* Unbalanced
BFW [17]	2023	20	0.8	25	25%	25%	25%	25%	Balanced

* The FairFace dataset [13] distributions are 14.1% Black 19.0% Caucasian, 14.1% East Asian, 14.2% Indian, 15.3% Latino, 10.7% Middle Eastern, and 12.5% Southeast Asian. Despite its emphasis on capturing diverse racial distributions, we classify this dataset as unbalanced due to the maximum difference of 8.3 percentage points.
** The Indian column is classified along with the African column.

In contrast to the majority of racially aware datasets that collected their images by subsampling existing facial recognition datasets, the FairFace dataset [13] collected new images for facial recognition. FairFace used a large public dataset not originally designed for facial recognition purposes, Yahoo YFCC100M [47], and detected faces in the images. They note that another dataset, Diversity in Faces (DIF) [48], also created a dataset in this

manner. However, DIF does not focus on racial distribution. Additionally, FairFace varies from other racially aware datasets as they used seven race classifications based on the accepted race classification from the U.S. Census Bureau: Black, East Asian, Indian, Latino, Middle-East, South East Asian, and White. While we acknowledge that the U.S. Census Bureau's definition of race is not based on entirely visual aspects, the FairFace dataset is based on visual aspects as the race, gender, and age groups were manually labeled using images that were labeled on Amazon's Mechanical Turk. The racial distribution proportions for the FairFace dataset found in Table 1 were derived from the dataset labels as they were not specified in the original work. Despite the FairFace dataset's clear emphasis on racial equality, it is classified as unbalanced according to our criteria due to a maximum racial disparity of 8.3 percentage points, which exceeds the limit of 5 percentage points.

3.2. Balanced Datasets

While there are many scenarios in which convenience sampling is sufficient to represent a population of interest, strata sampling is a more structured approach to ensure proportionate representation. Strata sampling is defined as taking proportionately equal samples from each category or stratum from the population of interest [49]. Many assert that strata sampling is optimal for facial recognition datasets, as it employs proportional sampling from each race [10–15,48].

One of the first datasets presented with relatively proportionally equal distributions across race was the Racial Faces in the Wild (RFW) dataset. The RFW dataset was created by subsampling the MS-Celeb-1M dataset [50] with an equal proportion of images across the Caucasian, African, Indian, and Asian races. To identify the race, the researchers used the nationality attribute in FreeBase celebrities [51] to select individuals of Asian or Indian race. Then, they used the Face++ API [52] to estimate race for Caucasians and Africans. Afterward, the dataset was manually and thoroughly cleaned. This resulted in a dataset of four racial subsets (African, Asian, Caucasian, and Indian), each with 10K images of 3K individuals [15]. We acknowledge that their race labels came from the "nationality" label, and nationality is defined as "a people having a common origin, tradition, and language and capable of forming or actually constituting a nation-state" [53]. The nationality label is not equivalent to a race label but is more in line with the ethnicity of an individual. While the nationality label is not directly correlated to race, we acknowledge their manual cleaning to ensure the racial labels' accuracy. The racial distribution of the RFW dataset is compared with other unbalanced and balanced datasets in Table 1.

Released in the same year as the RFW dataset, the DemogPairs dataset followed a similar pattern of taking a subsample of unbalanced datasets [14]. However, instead of taking the samples all from one dataset, samples were taken from the CWF, VGGFace, and VGGFace2 datasets. The DemogPairs dataset is designed to be a validation dataset that can be used to evaluate model performance across the full spectrum of human racial diversity. If a model is trained on CWF, VGGFace, or VGGFace2, and validated on DemogPairs, the overlapping images must be removed from the training dataset to prevent polluting the test set. The DemogPairs dataset contains 0.13% of VGGFace2, 0.02% of VGGFace, and 1.27% of CWF [14].

In a trend to increase the size of racially balanced datasets, DiveFace [11] was assembled as a subset of the Megaface datasets MF2 [11]. This sampling brought variation in head pose, lighting, age, facial expression, and quality from the Flickr scraped images. However, they limit their racial categories to three races: combining the Indian and African categories. In Table 1, we have placed the African and Indian categories into the African category as African falls before Indian in alphabetical order.

Among the largest racially balanced datasets to date are the BUPT-Globalface and BUPT-Balancedface datasets. BUPT-Balancedface follows our racially balanced definition while the BUPT-Globalface represents each race according to the global proportions of that race and is racially unbalanced by our definition [10]. Both are subsets of the MS-Celeb-1M dataset and are extended using the one-million FreeBase celebrity list [50] and used the nationality labels in the FreeBase celebrities [51] along with the Face++ API [52], similar to RFW [15]. In addition,

they removed any overlapping images with RFW, allowing for models that are trained on BUPT-Balancedface and BUPT-Globalface to be evaluated on the RFW dataset.

Another racially balanced dataset is the CASIA-SURF Cross-ethnicity Face Anti-spoofing (CeFA) dataset. The CeFA dataset was designed as an anti-spoofing dataset with 3 races, 1607 subjects, 3 modalities, and 2D plus 3D spoofing attack types [16]. This dataset was designed to counteract spoofing attacks by providing balanced races and depth information collected by infrared sensors.

The FaceARG dataset was collected by scraping the web for pictures of celebrities and resulted in over 175,000 facial images. The individuals' race was assigned labels corresponding to four main races, following the RFW split [12]. This dataset is referred to as an "in the wild" dataset due to the variety of head poses and orientations of the individuals' faces. More information about the racial distribution of the FaceARG dataset is given in Table 1.

The Balanced Faces in the Wild (BFW) dataset [17] was released recently in 2023 as a subset of the VGG2 dataset [43]. The BFW dataset was balanced across identity, gender, and race. The BFW dataset demonstrates the ongoing research on racially aware facial recognition systems along with the need to balance across multiple attributes in addition to balancing across race. However, to achieve the desired balance, the BFW dataset is less than 1% the size of the VGGFace2 dataset. Balancing datasets across race often requires significantly reducing the size of large datasets, resulting in a severe tradeoff.

In this work, we focus on datasets with images of individuals taken from monocular and stationary cameras. It is important to acknowledge other datasets exist with different emphases, such as those stated in past review articles [54]. Since the introduction of the RFW and DemogPairs datasets in 2019, there has been an ongoing effort to create racially balanced datasets. We emphasize that although using an authentic racially balanced dataset for training is ideal, the tradeoff tends to decrease the overall size of the dataset. To overcome this tradeoff, dataset balancing through data generation and various network improvements have been developed; these are discussed in Sections 4 and 5.

3.3. Evaluating Networks Trained on Unbalanced Datasets

The impact of training on a racially unbalanced dataset is best seen when evaluated on a racially balanced dataset. To properly evaluate network performance on facial recognition across race, it is crucial to have no overlap between the training and testing datasets. As many of the racially balanced datasets contain overlap with frequently used training datasets, we analyze which data should be partially excluded to ensure valid evaluation, as detailed for each of the racially conscious datasets in Table 2.

Table 2. Dataset Overlap: To properly use the racially conscious datasets as a test dataset, there needs to be no overlap between the training dataset and the test dataset. This table lists what racially balanced datasets contain overlap with popular racially unbalanced facial recognition datasets. We note that BUPT-Balancedface and BUPT-Globalface do not supply what percentage came from MS-Celeb-1M.

Dataset	Overlaps with Other Datasets	Percentage Overlap
DemogPairs [14]	VGGFace2, VGGFace, and CWF	0.13%, 0.02% and 1.27%
RFW [15]	MS-Celeb-1M	0.4%
BUPT-Balancedface [10]	MS-Celeb-1M	Not Available
BUPT-Globalface [10]	MS-Celeb-1M	Not Available
DiveFace [11]	Megaface dataset MF2	2.7%

In addition to publishing the RFW dataset, Wang et al. [15] also evaluated many state-of-the-art facial recognition methods on the RFW dataset. They demonstrated that certain races had lower individual accuracies, as seen in Table 3. For example, African individuals had nearly twice the average error rate than their Caucasian counterparts. To simplify comparison, we also report the skewed error ratio (SER) as defined by $SER = \frac{MaxError_r}{MinError_r}$ with r being the races: Caucasian, Indian, Asian, and African.

Table 3. Commercial API comparison: This table compares commercial APIs and popular algorithms trained on racially unbalanced datasets and evaluated on racially unbalanced (Labeled Faces in the Wild (LFW)) and racially balanced (Racial Faces in the Wild (RFW)) datasets as reported by Wang et al. [15]. The algorithms that are trained on racially unbalanced datasets have high accuracies on unbalanced datasets. Their generalization to balanced datasets such as RFW is limited. These accuracies and skewed error ratio (SER) values can be compared to the results of networks trained on racially balanced approaches discussed in Section 5. Reprinted/adapted with permission from Ref. Wang et al. [15]. 2019, IEEE.

	Model	**LFW**	**RFW**				
		Verification Accuracy	**Caucasian**	**Indian**	**Asian**	**African**	**SER (↓)**
Commercial API	Microsoft [55]	-	87.60%	82.83%	79.67%	75.83%	1.95
	Face++ [52,56]	99.5%	93.90%	88.55%	92.47%	87.50%	2.05
	Baidu [57]	-	89.13%	86.53%	90.27%	77.97%	2.26
	Amazon [58]	-	90.45%	87.20%	84.87%	86.27%	1.58
	mean	-	90.27%	86.28%	86.82%	81.89%	1.96
Algorithms trained on racially unbalanced datasets	Center-loss [59]	99.0%	87.18%	81.92%	79.32%	78.00%	1.72
	Sphereface [60]	99.42%	90.80	87.02	82.95	82.28	1.93
	Arcface [61]	99.40%	92.15%	88.00%	83.98%	84.93	2.04
	VGGface2 [43]	99.0%	89.90%	86.13%	84.93%	83.38	1.64
	mean	99.21%	90.01%	85.77%	82.80%	82.15%	1.83

3.4. Intersection of Racially Balanced Datasets and Gender Bias

As our survey focuses primarily on racial bias, we do not conduct an in depth analysis of gender bias across the datasets. However, as gender and race bias intersection is well known [62], we provide an analysis of the gender bias that remains in multiple racially balanced and racially unbalanced datasets.

In Table 4, we provide the distribution across race of both racially balanced and racially unbalanced datasets. We note that the BUPT-Balancedface and BUPT-Globalface datasets are not presented in the table as the initial work, nor subsequent works, have published labels for gender across these two datasets. In addition, the CeFA dataset only provides the distribution across gender and not the gender across the different races. We obtained the gender distributions for the FaceARG, DiveFace, and FairFace datasets by manually analyzing the datasets' provided gender labels. The RFW dataset distribution was obtained from Sarridis et al. [63]. We note that the BFW dataset is the most gender-balanced network that is also racially balanced. In contrast, the RFW dataset's gender distribution contains the greatest unbalance across gender for racially balanced datasets. In the discussion of racially balanced datasets, it is critical to understand which datasets are also balanced across gender.

Table 4. Dataset Gender Distributions: In the discussion of racial bias, gender bias is closely related. The gender distribution across multiple datasets is presented in this table. The table is sorted initially by the difference in percentage points between the overall Male and Female distributions (labeled in the table as Δ) and then by publication year. Although certain datasets are racially balanced, many remain unbalanced across gender.

Dataset	**Racially Balanced**	**Caucasian**		**African**		**Indian**		**Asian**		**Overall**		**Δ**
		Female	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	$\lvert F - M \rvert$
LFW [28]	✗	18.7%	58.9%	3.3%	9.6%	–	–	2.2%	7.2%	24.2%	75.7%	51.5%
RFW [15]	✓	7.9%	17.2%	1.3%	24.4%	9.1%	14.7%	7.7%	17.6%	26.1%	73.9%	47.8%
YouTube Faces [36]	✗	20.3%	56.9%	4.0%	7.7%	–	–	3.0%	7.9%	27.3%	72.5%	45.2%
FRVT2018 [45]	✗	16.5%	48.4%	7.4%	19.9%	–	–	0.4%	1.2%	24.3%	69.5%	45.2%
MSCeleb1M [41]	✗	19.2%	52.4%	3.9%	12.1%	–	–	4.5%	7.7%	27.6%	72.2%	44.6%
VGGFace2 [43]	✗	30.2%	45.9%	6.3%	10.5%	–	–	3.6%	3.4%	40.1%	59.8%	19.7%
CASIA [37]	✗	33.2%	48.8%	5.7%	7.2%	–	–	2.6%	2.6%	41.5%	58.6%	17.1%

Table 4. *Cont.*

Dataset	Racially Balanced	Caucasian		African		Indian		Asian		Overall		Δ
		Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	$\lvert F - M \rvert$
PubFig [35]	✗	35.5%	49.5%	5.5%	6.5%	–	–	1.0%	2.0%	42.0%	58.0%	16.0%
IJB-C [44]	✗	30.2%	40.3%	6.0%	11.8%	–	–	6.2%	5.4%	42.4%	57.5%	15.1%
BioSecure [34]	✗	36.0%	50.1%	2.1%	3.1%	–	–	4.5%	4.3%	42.6%	57.5%	14.9%
MegaFace [40]	✗	30.3%	40.0%	4.7%	6.2%	–	–	8.1%	10.6%	43.1%	56.8%	13.7%
FaceARG [12]	✓	14.5%	9.8%	10.3%	13.8%	15.7%	10.3%	16.2%	9.5%	56.7%	43.3%	13.4%
CeFA [16]	✓	–	–	–	–	–	–	–	–	43.9%	56.1%	12.2%
CelebA [38]	✗	41.5%	33.9%	8.2%	6.4%	–	–	5.5%	4.4%	55.2%	44.7%	10.5%
UTKFace [42]	✗	20.0%	26.2%	16.3%	21.5%	–	–	8.9%	7.1%	45.2%	54.8%	9.6%
FairFace [13]	✗	12.3%	17.4%	14.8%	14.7%	6.8%	7.3%	13.1%	13.6%	47.0%	53.0%	6%
VGGFace [39]	✗	38.6%	43.7%	6.9%	5.8%	–	–	2.9%	2.1%	48.4%	51.6%	3.2%
DiveFace [11]	✓	19.7%	20.2%	14.0%	15.0%	–	–	16.3%	14.8%	50.5%	49.5%	1.0%
DemogPairs [14]	✓	16.7%	16.7%	16.7%	16.7%%	–	–	16.7%	16.7%	50%	50%	0%
BFW [17]	✓	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	50.0%	50.0%	0%

3.5. Discussion on Racially Balanced Datasets

As the field of facial recognition continues to grow, certain biases, such as bias across race begin to be more understood. Many point to the unbalance of training and validation datasets as the cause for this bias. We provide a definition for racially balanced and racially unbalanced datasets based on statistical principles. We define racially balanced datasets as those datasets that contain less than a 5 percentage point difference between the least represented race and the most represented race. In contrast, a racially unbalanced dataset contains a greater than five percentage point difference.

The first racially balanced dataset was released in 2019, with continual releases of additional racially balanced datasets. We provide a list of racially balanced datasets in Table 1 and describe the overlap between racially balanced and racially unbalanced datasets in Table 3.

As racially balanced datasets are becoming more available, the intersection of race and gender is highlighted. We identify that although many racially balanced datasets are created to decrease the racial bias, the gender bias often remains. The most recently released racially balanced dataset, BFW, is not only balanced across race but also gender.

4. Balancing Datasets through Data Generation

While there are various datasets that are balanced [10–12,14–17], many of the largest datasets remain unbalanced [39–41,43,45]. One method for obtaining racially balanced datasets from a racially unbalanced dataset is to augment the dataset through race transformations. The concept is centered around taking all images of people in one race and transferring the images to also be in all other races, creating an equally balanced dataset. Facial recognition focused on racially balancing datasets through race transformation frequently uses generative adversarial networks (GANs) [64] and diffusion models [65,66] to perform this augmentation.

Since the original publication of the GAN by Goodfellow et al. [64] in 2014, GANs have continued to increase in complexity and ability. Some of the top research papers on GANs are written on topics including adding details to segmentations or outlines of items such as handbags or cars [67], style-based networks that generate human faces [68] and even creating anime characters simply off of a list of attributes given in text [69]. In this paper, we focus on the effect that GANs have on increasing and decreasing racial equality as well as cross-race transformation.

Although stable diffusion was created in 2015 [65], it significantly grew in popularity and performance in 2022 with the release of stable diffusion in the latent space [66]. Latent stable diffusion is currently the leading method for image generation in various applications [70]. A recent variant, the diffusion transformer [71], has shown promising results in further enhancing this technology's capabilities.

4.1. Understanding the Impact of Data Generation

As GANs and latent stable diffusion continuously improve, many users remain unaware of the unintended consequences of data generation. One team of researchers demonstrated unintended consequences by using headshots of engineers to train a deep donvolutional GAN [72], resulting in the majority of GAN-produced images having white skin tones and masculine facial features [73]. They point out that GANs can increase data biases that are already in the dataset if GANs are used to generate facial data.

Before the publication of the work by Jain et al. [73], another team used a GAN to attempt to lower racial inequality with their focus on quantifying attractiveness to simulate equality of opportunity [74]. To accomplish this, they trained a network to take a subset of images, such as one gender, and be able to distinguish whether they are attractive or not. They then compared across the different subsets using publicly available datasets: CelebA [38], Soccer (many analysts, one dataset) [75], and Quick Draw! [76]. They designed their generator and discriminator network structures based on previous works [77–79] and used conditional batch normalization [80]. Overall, they improved on the simulated equality of opportunity with their GAN-based approach [74].

A larger example of a GAN demonstrating racial inequality is seen with a team that created a face-generating GAN [81]. The GAN was used to generate faces that would pass on multiple individuals when compared using top-performing face authentication systems such as DLib [82], FaceNet [83], and SphereFace [60]. They termed these faces "master faces" [81]. To generate face images, they use Dlib's embedding representation as the input to their network. Their network has a linear layer and four de-convolutional layers, based on the DC-GAN architecture [72]. They trained on DLib's embedding representations of the FFHQ dataset [68]. Upon inspection of the faces generated that are displayed in their paper, the highest proportion of any race of generated images is Caucasian [81]. Examples of generated images are shown in Figure 2.

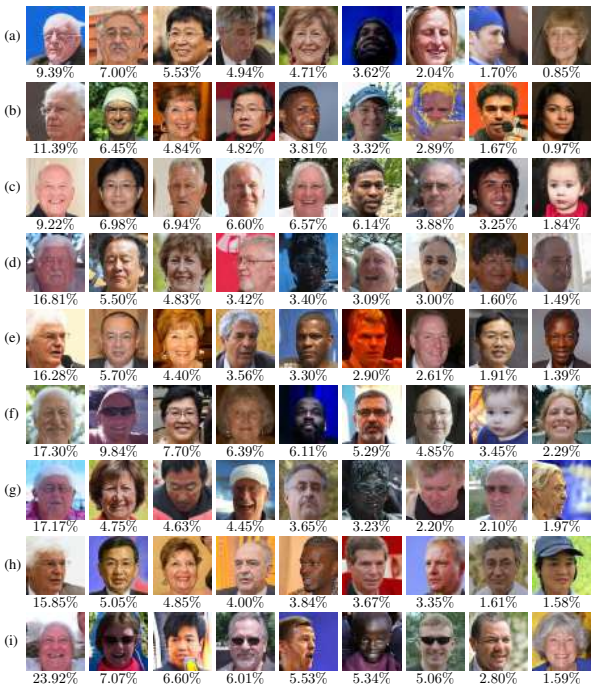


Figure 2. Example Images of MasterFace: MasterFace is an adversarial approach to the facial recognition task where a high proportion of generated images pass for the majority of faces on many facial recognition models [81]. The (**a**–**i**) rows demonstrate 9 different sets of Masterfaces. We include this figure to demonstrate the highest proportion of any race generated is Caucasian faces, demonstrating the skew across race in many facial recognition models. Reprinted/adapted with permission from Ref. [81]. 2021, IEEE.

While there are various examples of GANs having large racial biases, there are also GANs that display the ability to decrease racial biases. One such work used a classical GAN-based approach to lower the racial bias in machine learning. Their approach consisted of training a network to identify race using California Census data, followed by an adversarial network to train against the identifying network. Their method increased race identification accuracy from 39% to 76% on their dataset [84]. This demonstrates the ability of GANs to greatly increase the ability of classification networks particularly with race. It also demonstrates that improvements continue to be made with GANs to decrease racial inequality.

4.2. Cross-Race Transformation

While using a racially balanced dataset to train a facial recognition model is beneficial, it often results in a tradeoff with decreasing the size of the datasets. To continue the use of large-scale facial recognition datasets, a beneficial method is race transformation. This method augments datasets to be racially balanced by transforming the race of images to other races.

Yucer et al. [22] overcomes the issues of racially unbalanced datasets using a racial transformation data augmentation approach. They used a four-race dataset: African, Asian, Caucasian, and Indian to compare more easily with the RFW (Racial Faces in the Wild) dataset. For the racial transformation, they trained six CycleGAN models [85] on the BUPT-Transferface dataset [15]. For facial recognition, they trained a common DCNN, Resnet [86], and used a ResNet100 [61] to obtain the final 512-D feature space representation. They experimented with various loss functions: Softmax [60], CosFace [87], and ArcFace [61]. They trained on a subset of 1200 images from VGGFace2 [43] with 300 images for each of their four races. Each of these 300 images was then transferred to the other 3 races using the CycleGANs for the augmented dataset. Their results not only improve accuracy on the RFW dataset but also the overall accuracy on the LFW (Labeled Faces in the Wild) dataset [22]. Example images of the transferred race images are available in Figure 3.

A similar approach was taken by Ge et al. [88] as they use a Fan-Shaped GAN (FGAN) to also transfer the race of an individual to another race. Their network structure is a combination of previous architectures [85,89] with the addition of spectral normalization [78]. For the dataset, they note that RFW is collected from MS-Celeb-1M. So, they collect their images from MS-Celeb-1M with 5000 images in each racial subset for their dataset. They compare their network against the StarGAN [89] and CycleGAN [85]. In doing so, they displayed that their approach was more accurate in all races for FID and received a better inception score on all races besides African [88].

One recent work in racial image generation transforms gender and age in the same system [90]. Their pipeline takes the input image, detects the faces and eyes, crops, normalizes, generates the face image by GRA-GAN, and estimates gender/race/age using a CNN. This pipeline creates a method to control for extra variation that is outside the scope of the GRA-GAN. They demonstrate that their GRA-GAN outperforms or has comparable results to top image-to-image GANs such as Pix2Pix [67], CycleGAN [85], Modified CycleGAN [91], and Enhanced CycleGAN [92]. They state that their network could be used to augment data to prevent training from overfitting to certain genders, races, or ages [90].

While many methods for balancing datasets through data generation have focused on training a new model for transferring race, the work by Jain et al. [93] uses an existing Style GAN to perform the generation. Previous methods identified the need to scale their method up with additional subjects and computational power, the work by Jain et al. [93] provides a method for transferring style to an image to transfer the race. Using this method, Jain et al. [93] pre-trained a facial recognition network on the augmented dataset, which outperforms corresponding models not pre-trained with their augmented dataset. Their approach demonstrates a solution to previous methods' need to scale up the transformation approaches.

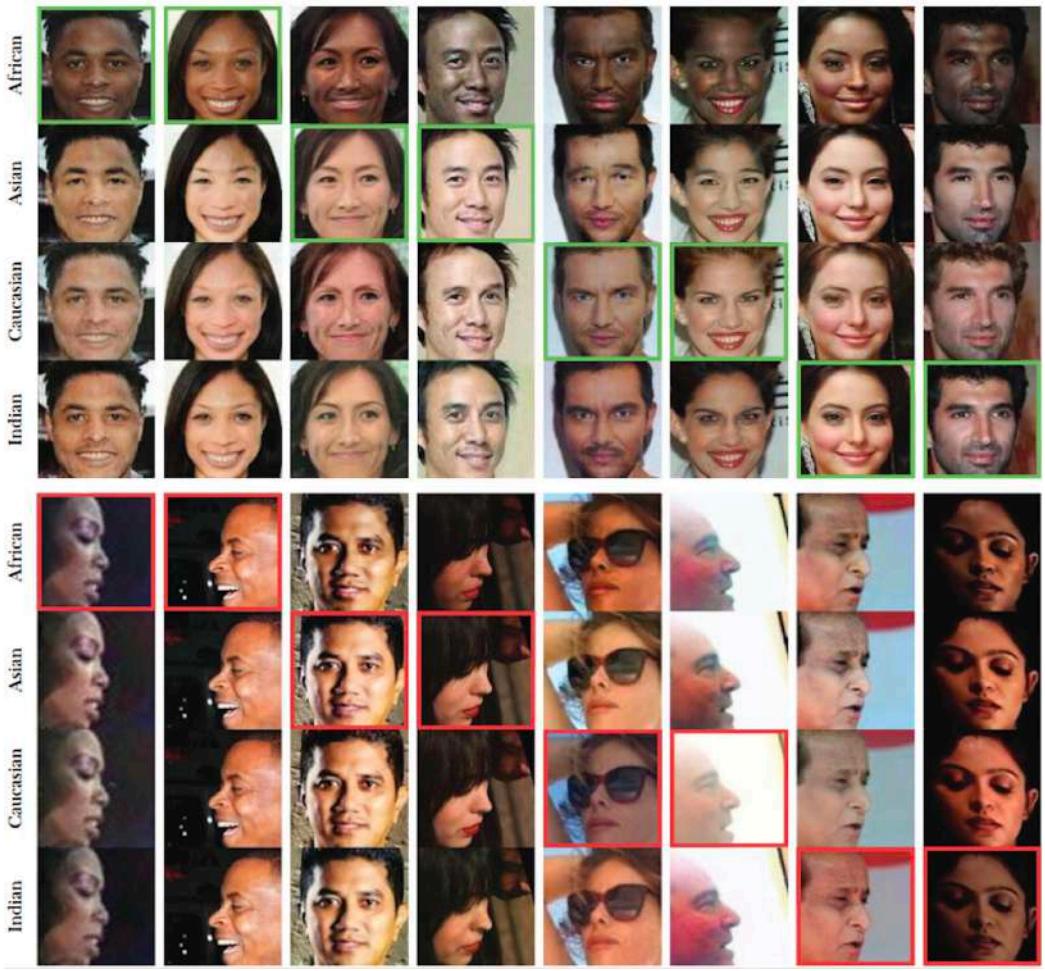


Figure 3. Example Images of Transferring Race: The top images demonstrate successful race transfers while the bottom images demonstrate failures. The green and red bounding boxes demonstrate the original image. As observed in the bottom images, the GAN struggles with extreme head poses and varying lighting [22].

While racial transformation sounds promising to balance a dataset, there remain many problems. Some of these problems include whether or not the images of a person that has been racially transformed to a new race should be considered as positive, negative, or not even considered to the original image or other images of that same person [94]. In contrast, the approach of limiting the dataset to only use an equal number of images from each race is a more defined approach to the solution. However, it does limit the size of the dataset instead of increasing the size. The benefits and the problems of each of these approaches are an open research question.

One of the most recent image generation methods has generated an entire dataset for the use of overcoming the skew across race in facial recognition. DigiFace 1M provides 1 million synthetically generated faces that are balanced across race and adds various accessories to make the datasets encompass a larger spread of data. Using only their dataset, they were able to achieve a 95.82% on the LFW dataset without using any of the LFW training images. As opposed to other methods of using GANs or stable diffusion, DigiFace 1M was created using a computer graphics pipeline [29]. Training on this racially balanced dataset resulted in a high accuracy on the LFW dataset, demonstrating promise for racially aware facial recognition systems. An example from the DigiFace 1M dataset is provided in Figure 4.



Figure 4. DigiFace 1M Example Image: DigiFace 1M [29], one of the most recent image generation datasets, generates the face and then changes lighting, background, and pose.

4.3. Intersection of Racial Dataset Balancing and Gender

As we discuss racially balancing datasets through data generation, we note that similar responses have also been proposed for balancing across gender. Previously, we discussed the GRA-GAN approach that uses the same network to generate gender, race, and age [90]. While other works had transferred across race and age [92], the GRA-GAN was the first approach to also transfer across gender. Previous related approaches classified gender for tasks such as face aging [95].

The results of the GRA-GAN demonstrate the improvement over multiple biases when training to overcome the biases in the same network. Their work shows the large interrelation between racial and gender bias.

4.4. Discussion on Balancing Datasets through Data Generation

Balancing datasets through data generation to minimize racial bias is relatively new to the field of image generation. Preliminary methods published in 2020 demonstrated the success on samples of larger datasets [22], while recent publications have scaled to full datasets by using pre-existing networks [93]. In addition, the benefit of training a network to not only transfer across race but also gender and age demonstrates an improvement over only transferring across race [90].

The field of racially balancing datasets through data generation has followed the progression of GANs using both CycleGANs and FanGANS. Other image networks mitigating racial bias in related tasks [96] have used diffusion to generate images while balancing datasets through data generation for facial recognition has turned to using computer graphic pipelines [29]. One of the next steps for balancing datasets through data generation for facial recognition is to use latent diffusion models as various other fields of image generation have.

5. Network Improvements across Human Race

Efforts to mitigate the performance disparities across different races in facial recognition systems continue to evolve with various innovative approaches. This work focuses on approaches that include refining loss functions, optimizing training methodologies, modifying datasets, and novel network architectures. In addition, as various methods rely on race recognition within the overarching solution, we include a discussion on identifying race. As gender bias is interrelated with racial bias, we discuss network improvements that decrease both racial and gender bias. We provide a brief discussion on related tasks' network improvements and an overall discussion on network improvements in facial recognition across human races.

5.1. Network Improvements on the RFW Dataset

This section focuses on improvements measured on the RFW dataset [15], a popular racially balanced test dataset. Table 5 compares various improvements discussed in this section. The SER (as defined in Section 3) is also reported in this table to facilitate straightforward comparison to methods trained on unbalanced datasets shown in Table 3.

Table 5. RFW Accuracies Across Network Improvements: This table presents the accuracies, in percentages, achieved by state-of-the-art (SOTA) methods on the RFW dataset. Each method is trained on the BUPT-Balancedface dataset and uses the ArcFace loss, except for IMAN and CosFace approaches. The highest accuracy in each column is bolded and enclosed in brackets. "STD" denotes the standard deviation and "SER" represents the skewed error ratio. "↓" describes that a smaller value is an improvement while "↑" demonstrates that a larger value is an improvement.

Method	Loss-ArcFace	White	Black	East Asian	South Asian	Average (↑)	STD (↓)	SER (↓)
IMAN [15]	✓	93.92	92.98	90.60	90.98	92.12	1.38	1.55
CosFace [87]	X	95.12	93.93	92.98	92.93	93.74	0.89	1.45
ArcFace [61]	✓	96.18	94.67	93.72	93.98	94.67	0.96	1.64
ACNN [97]	✓	96.12	94.00	93.67	94.55	94.59	0.94	1.63
PFE [98]	✓	96.38	95.17	94.27	94.60	95.10	0.80	1.58
DebFace [99]	✓	95.95	93.67	94.33	94.78	94.68	0.83	1.56
GAC [100]	✓	96.20	94.77	94.87	94.98	95.21	0.58	1.37
RL-RBN [10]	✓	96.27	94.68	94.82	95.00	95.19	0.63	1.43
MBN [101]	✓	96.25	95.38	95.32	94.85	95.45	0.51	1.37
Rethinking [102]	✓	89.1	85.5	71.8	75.8	80.55	7.01	2.59
PCT [103]	✓	97.00	**[96.22]**	95.73	96.38	96.33	0.52	1.42
Sensitive Loss [20]	X	**[97.23]**	95.82	**[96.50]**	**[96.95]**	**[96.63]**	0.53	1.51
GABN [104]	✓	95.78	94.71	94.51	95.21	95.05	**[0.49]**	**[1.30]**

Accompanying the RFW dataset, Wang et al. [15] provides an initial baseline method for the dataset. This baseline, the Information Maximization Adaptation Network (IMAN), has two main stages: pseudo-adaptation and a custom mutual information loss. The pseudo-adaptation focuses on pre-clustering to obtain initial improvement in the target domain. Then, the mutual information loss uses the distribution of the target classifier to obtain larger decision margins.

The CosFace solution presents a loss function that incorporates a cosine margin to the distances between intra-class and inter-class face representations [87]. This margin increases inter-class pair distances and decreases intra-class pair distances. Although CosFace was released prior to the RFW dataset, results are included from Gong et al. [100] in Table 5 that evaluated the CosFace loss with a ResNet34 architecture.

A pivotal improvement on the RFW task and facial recognition in general is the ArcFace (Additive Angular Margin) loss [61]. Instead of relying on direct distance measurements, the innovative ArcFace loss uses the angle of the arc between the two facial representations. This puts the facial representations along the exterior of the arc in multidimensional space. Inserting this defined embedding representation prior allows for a more robust facial representation. This has become widely adopted across facial recognition with the majority of the following methods within this section using the ArcFace loss as illustrated by the Loss-ArcFace column in Table 5. It is important to note that ArcFace is now a popular loss function used in various applications [105–108].

Along with updating loss functions, there is an additional focus on incorporating conventional methods with deep learning techniques, specifically hand-crafted features. Some of these hand-crafted features are crowd counting, camera angle, camera height, appearance, and scale of the individuals. The ACNN method applies these hand-crafted features to the neural network facial recognition approach [97]. This extra information outperforms a similar-sized CNN without the hand-crafted features. This approach was also released before the RFW dataset, Gong et al. [100] evaluated the ACNN method on the RFW dataset, and we include these results in Table 5.

The use of conventional methods to overcome racial bias in facial recognition shifted towards using probabilistic approaches. The Probabilistic Face Embeddings (PFE) approach learns a Gaussian latent space representation of the face [98]. While most facial recognition approaches use neural networks to separate the multi-dimensional representation of the face, the PFE uses a variational auto-encoder, similar to image generation methods such as latent diffusion models [66], to encode the face into a latent space representation with a mean and a variance. The latent space Gaussian representations are compared to determine whether the facial representations correspond to the same individual. Although this approach is not widely adopted, it remains competitive with other racially aware methods, as seen in Table 5.

Architectural advancements have been developed to further improve facial recognition across race. One such improvement, the debiasing adversarial network (DebFace), uses an image-to-feature encoder, four attribute classifiers, a distribution classifier, and a feature aggregation network [100]. The four classifiers—gender, age, race, and identity—turn potential biases into informed features, thus improving performance across underrepresented groups.

Continuing in architectural improvements, the Group Adaptive Classifier (GAC) method uses an adaptive classification within the network structure to obtain higher accuracies [100]. They use a group of adaptive classifiers that assign the individual to a demographic group. A demographic-specific kernel then performs the final classification. These demographic-specific kernels learn unique racial characteristics, thus improving accuracy.

Recent studies have explored significant enhancements in training methodologies. One notable approach, the Reinforcement Learning-based Race Balance Network (RL-RBN) [10], uses reinforcement learning as a novel method for training facial recognition. They deliberately pick the action, reward, and objective functions to ensure balanced learning across race. The action function permits three outcomes: keeping the margin between classes the same, shifting the margin to a larger value, and shifting the margin to a smaller value. The reward model aims to standardize distances and mitigate skew across racial groups. To achieve this the reward is calculated using the distance between the Caucasians and the race of the individual [10]. The objective function is a deep Q-Learning function [109,110] which learns the optimal policy for the agent [10]. While reinforcement learning is not frequently used in facial recognition tasks, this study underscores the importance of enhancing training methods beyond merely refining datasets, losses, and architectures.

Wang et al. [101] proposes another significant improvement in training methodology with the introduction of the Meta Balanced Network (MBN), a meta-learning method that improves fairness across skin tones. The MBN uses two loss functions: training loss and meta loss. The meta loss weights the training loss to ensure equal weight adjustment across all skin tones during backpropagation. While other studies emphasize the necessity of balanced training data, this approach is able to use skewed datasets without skewing the final result.

Coe and Atay [1] investigates the optimal network architecture to minimize racial bias in facial recognition systems. To demonstrate that each model learns different races in different proportions, they compare AlexNet [111], VGG16 [112], and ResNet50 [113]. Their results demonstrate that VGG16 outperforms both AlexNet and ReseNet50 in terms of accuracy across different races [1]. This indicates that the performance across race varies based on network architecture, despite being trained on the same data. As their approach did not report their results on the RFW dataset, the results are not included in Table 5.

A not-expected approach is seen as the researchers claim that training only on one race is not inherently disadvantageous [102]. They demonstrate that by training only using African faces, they achieved less skew across race than by training with a balanced dataset. They also found that they obtained higher accuracy across race by using additional images for an individual instead of adding more identities. To come to their findings they performed their testing on four recent top facial recognition models: VGGFace2 [43], CenterLoss [59], SphereFace [60], and ArcFace [61]. They note that training on one race

decreases the overall size of the dataset and emphasize that having a larger dataset would result in even higher metrics [102]. Their experiments and findings result in a new and unique method for obtaining higher accuracies across race.

Serna et al. [20] proposed Sensitive Loss, a modification of a sensitive triplet generator and the triplet loss function [114]. Their approach emphasized how transfer learning a racially skewed facial recognition model can decrease the skew across race. To demonstrate the performance of the sensitive loss, they evaluated their loss function across three baseline models: VGG-Face [39], ResNet-50 [113], and ArcFace [61]. Their loss greatly improved the Equal Error Rate and decreased the standard deviation across the DiveFace dataset, RFW dataset, and the BUPT-Balancedface dataset.

An additional method returned to the architectural improvements in the Adversarial Information Network (AIN) approach. Here, they minimize the target classifier and maximize the feature extractor. Then, they use a graph neural network to find the likelihood of the target data [115]. As the AIN approach reported the RFW results in a non-conventional method, we do not include the AIN results in Table 5.

Research has shown that networks tend to focus on varying facial regions based on the individual's race. To address this issue, Li et al. [103] introduced the Progressive Cross Transformer (PCT). This innovative approach employs a dual transformer arrecognition process.

Introduced in 2023, the Gradient Attention Balance Network (GABN) is a facial recognition model designed to analyze the same facial regions regardless of the race of the individual [104]. They use attention masks to enforce structured regions of interest, effectively minimizing racial disparities, as illustrated by the results in Table 5.Introduced in 2023, the Gradient Attention Balance Network (GABN) is a facial recognition model designed to analyze the same facial regions regardless of the race of the individual [104]. They use attention masks to enforce structured regions of interest, effectively minimizing racial disparities, as illustrated by the results in Table 5.

5.2. Racial Classification

Various facial recognition approaches require racial classification methods, either directly [10,20,99,101,102,104,115] or indirectly [100,103]. Thus, discussing racial classification methods and their improvements is crucial. In 2014, a review article [54] cited the highest accuracy for identifying race as nearly 99% [116], misleading some to consider race classification as a solved problem. However, this impressive accuracy used 3D imaging techniques to identify race [117], whereas this work focuses exclusively on race identification with monocular images—a significantly more challenging task.

5.2.1. Two-Class Race Classification

Instead of focusing on classifying across many races, initial research focused on distinguishing between two races. One such group focused on distinguishing Black and White individuals in one network and in another network distinguishing Chinese and Non-Chinese [118]. Their network was based on the CIFAR-10 network [119]. For their datasets, they collected images from various sources. From the MORPH-II dataset [120], they used 43,130 face images of Black and White individuals, from the Casia-Webface dataset [37], they used 101,771 images of Black and White individuals, from the CASIA-PEAL dataset [121] they chose 5429 images of Asian individuals, from the FERET dataset [122] they used 3407 images of Black and White individuals [118]. The reported overall accuracy was 100% for identifying Black individuals and 99.4% for White individuals [118]. On the Chinese and non-Chinese classification task, they achieved 91.6% accuracy for identifying Chinese individuals, and 93.5% for non-Chinese individuals. These high accuracies demonstrate a strong ability to identify race in a narrow scope of Chinese and non-Chinese individuals or Black and White individuals.

Some may argue that identifying just a few races is oversimplifying a complex task. Within one race or demographic category, there may be significant racial variation. To ex-

plore intra-racial performance, one group of researchers developed a network to identify a face as either from northern India or from southern India. Upon collecting and labeling a custom dataset, the method achieved human-level accuracy but with different error patterns [123]. Their findings demonstrate the ability of neural networks to classify race at a more precise level, even within a specific race.

Vo et al. [124] based their racial recognition model on the VGG architecture [112]. They scraped individuals' profile pictures and race from Facebook. This was done to obtain images at different poses, lighting, accessories, and imaging conditions. They then used a Haar Cascade classifier [125] to crop, and label 6100 faces with 2892 Vietnamese and 3208 images in other races, creating their VNFaces dataset. They obtained an 88.87% overall accuracy using this model [124]. This dataset contains increased variation across lighting, accessories, and image conditions, resulting in a more difficult classification task.

5.2.2. Multiple-Class Race Classification

While various methods focus on differentiating between two races for race classification, many facial recognition models require multiple races for their approach. One method that identifies race across the four labeled races in the UTKFace dataset [42] based their network architecture on ResNet V1 [126] followed by L2 normalization and fully connected layers achieved an average race classification of 90.1% [127].

One set of researchers focused their neural network on identifying the race of individuals that are Chinese, Japanese, or Korean [128]. To collect a dataset to train on, they scrapped Twitter, specifically the followers of Asian celebrities. They used the username and the primary language of the profile description to label the race. Then, their approach predicted the correct race of 78.21% Chinese, 72.80% Japanese, and 73.80% Korean [128], demonstrating great success in race classification while also demonstrating that race classification remains unsolved.

As researchers published the FaceARG dataset, they also trained and compared four unique SOTA convolutional neural networks on the task of recognizing race [12]. They chose to include VGG19 [112], Inception ResNet v2 [126], SE-ResNet24 [129], and MobileNetV3 [130]. The results of each of these four networks are shown in Table 6. Their MobileNetV3 achieved their highest overall F1 score at 96.64 on the four-class human race classification [12]. To test the robustness of their system, they altered the images with experiments including eye blur, eye occlusion, nose blur, nose occlusion, mouth blur, mouth occlusion, grayscale, increasing brightness, decreasing brightness, and image blur. For overall accuracy in identifying race, all robustness experiments maintained over 90% accuracy, except for eye occlusion, which dropped to 80.74% accuracy [12]. All of this demonstrates the ability of CNNs to classify race at a fairly high accuracy, even with some important parts of the face occluded.

Table 6. Race Classification Results: This table presents the results of human race classification obtained by four networks on the FaceARG dataset [12]. The maximum value of each metric within its respective row is highlighted in bold and enclosed in brackets.

Race	Metric	VGG-19	Inception ResNet-v2	SeNet	MobileNetV3
Afro-American	P_r	96.68	96.88	95.27	**[96.97]**
Afro-American	R_e	97.96	98.20	98.32	**[98.44]**
Afro-American	F_1	97.32	97.54	96.77	**[97.70]**
Asian	P_r	98.27	98.43	97.94	**[98.52]**
Asian	R_e	97.76	97.92	96.96	**[98.20]**
Asian	F_1	97.32	98.18	97.45	**[98.36]**
Caucasian	P_r	94.91	95.61	95.43	**[96.20]**
Caucasian	R_e	94.00	**[95.00]**	94.28	94.12
Caucasian	F_1	94.45	**[95.30]**	94.85	95.15

Table 6. *Cont.*

Race	Metric	VGG-19	Inception ResNet-v2	SeNet	MobileNetV3
Indian	P_r	94.37	94.51	**[94.95]**	94.89
Indian	R_e	94.52	94.32	94.00	**[95.80]**
Indian	F_1	94.44	94.41	94.47	**[95.34]**
Overall	P_r	94.06	96.36	95.90	**[96.64]**
Overall	R_e	96.06	96.36	95.89	**[96.64]**
Overall	F_1	96.06	96.36	95.89	**[96.64]**

5.3. Related Tasks

While there are various network improvements that are being accomplished in facial recognition to decrease the variance across race, other tasks also benefit from focusing on race. One similar task is race, gender, and age classification. In the release of the FairFace dataset, Karkkainen and Joo [13] trained four different classification networks all with the same setup: DLib's face detector [131], a ResNet-34 [113], and the ADAM optimizer [132]. The difference between the four networks was what dataset they were trained on: UTKFace [42], LFWA+, CelebA [133], and their new racially aware dataset, FairFace. They took these four trained networks and evaluated them. The cross-dataset accuracy findings on the White race vs. Not White race are shown in Table 7a for race classification, Table 7b for gender classification, and Table 7c for age classification. They note that the model trained on the FairFace dataset outperforms all others. The results emphasize the benefit that comes from focusing on race while training not only facial recognition or race classification tasks but also gender and age classification. Their findings demonstrate the need for generalization across race on more tasks than facial recognition. We acknowledge that the White vs. Not White definitions are used although the definition of White or Not White will vary across which country the labeler resides in or cultural backgrounds that the labeler has.

Table 7. (**a**) Race Classification: Race classification accuracies are presented as percentages for White and Not White categories, emphasizing the differences across racial groups [13]. We note that the balanced dataset achieves the highest accuracies for most test datasets. The highest accuracy in each column is highlighted in bold and enclosed in brackets. (**b**) Gender Classification Across Race: Gender classification accuracies are presented as percentages for White and Not White categories, emphasizing that race has lower accuracies across various tasks [13]. We note that the balanced dataset achieves the highest accuracies for most datasets tested. The highest accuracy in each column is highlighted in bold and enclosed in brackets. (**c**) Age Classification Across Race: Age classification accuracies are presented as percentages for White and Not White categories, emphasizing that race has lower accuracies across various tasks [13]. The highest accuracy in each column is highlighted in bold and enclosed in brackets. Reprinted/adapted with permission from Ref. [13]. 2021, IEEE.

(**a**)

		Tested on (White Accuracy—Not White Accuracy):		
		FairFace (Balanced)	UTKFace (Unbalanced)	LFWA+ (Unbalanced)
Trained on:	FairFace (Balanced)	**[93.7%]–[75.4%]**	93.6–80.1%	**[97.0%]–[96.0%]**
	UTKFace (Unbalanced)	80.0–69.3%	91.8%–**[83.9%]**	92.5–88.7%
	LFWA+ (Unbalanced)	87.9–54.1%	**[94.7%]**–38.0%	96.1–86.6%

Table 7. *Cont.*

(**b**)

		Tested on (White Accuracy—Not White Accuracy):			
		FairFace (Balanced)	UTKFace (Unbalanced)	LFWA+ (Unbalanced)	CelebA (Unbalanced)
Trained on:	FairFace (Balanced)	**[94.2%]–[94.4%]**	**[94.0%]–[93.9%]**	92.0%–**[93.0%]**	**[98.1%]–[98.1%]**
	UTKFace (Unbalanced)	86.0–82.3%	93.5–92.5%	91.6–90.8%	96.2–96.2%
	LFWA+ (Unbalanced)	76.1–73.8%	84.2–83.3%	**[93.0%]**–89.4%	94.0%–94.0%
	CelebA (Unbalanced)	81.2–78.1%	88.0–88.6%	90.5–90.1%	97.1–92.1%

(**c**)

		Tested on (White Accuracy—Not White Accuracy):	
		FairFace (Balanced)	UTKFace (Unbalanced)
Trained on:	FairFace (Balanced)	**[59.7%]–[60.7%]**	56.5–61.6%
	UTKFace (Unbalanced)	41.3–41.8%	**[57.6%]–[61.7%]**

Another example of benefiting performance based on race is seen with the Inclusive-FaceNet solution. Their model is used to detect face attributes. It does this by using the race and gender representations that are transfer-taught to the model. To transfer-learn race and gender, they train a FaceNet model [83] and extract features from the avgpool layer. This layer learns race and gender. Then, to further learn race, they obtained over 100,000 images from famous individuals [134–136]. Then, after combining these transfer learnings, they obtained higher results in face attribute detection across race and gender than the network without the transfer learnings [137].

Pastaltzidis et al. [138] explored the bias in the RWF-2000 dataset [139], a law enforcement project created for violent activity recognition. They note that certain demographics are over-represented in the training dataset. To counteract this bias, they implemented a unique data augmentation method that force-balances the dataset to be more representative of the population. They achieved this by modifying videos that over-represent minority groups, using body movement tracking to replace the individual with others from a different race. This resulted in a more balanced dataset, and their results show promise for using synthetically generative models to balance various datasets. These examples of network improvements across related tasks demonstrate the improvements that come from focusing on race during trainings.

These three studies demonstrate the benefit of a racially balanced dataset, race classification pre-training, and force balancing a dataset through augmentation are beneficial to more than only facial recognition. These improvements also transfer to other classification and detection tasks by decreasing the disproportionate accuracies between races. This improvement is critical for the integration of systems in real-world applications.

5.4. Intersection of Network Improvements across Race and Gender

Many of the network improvements across human races could also be applied to improving the bias across gender. Our work separates network improvements across race into four categories: loss/training, architecture, dataset modification, and the use of additional data. Each of these four categories not only decreases racial bias but also has been seen to decrease gender bias. The improvement in loss/training can be seen with Conti et al. [140] presenting the use of a von Mises–Fisher mixture model, which takes a trained facial recognition model and trains a shallow network with the fair von

Mises–Fisher loss. The improvement across architecture can be seen with the work of Gwyn and Roy [141], which focuses on analyzing different architectures to identify the optimal architecture for minimizing gender bias. They find that VGG-16 is one of the architectures with the least amount of gender bias. This is the same conclusion that Coe and Atay [1] concluded was optimal for reducing racial bias. From these two studies, we see the VGG-16 network minimizing not only racial bias but also gender bias.

Just as dataset modifications decrease the racial bias in facial recognition, Tian et al. [142] demonstrates that data augmentation affects gender bias as well. Similar results are seen with the use of additional data. The BFW dataset [17], demonstrating that having additional data to balance datasets benefits not only race bias but also gender bias. As the discussion around minimizing racial bias through network improvements begins to grow, the overlap between minimizing gender bias also grows. The decrease in racial bias and gender bias are connected, allowing improvements from one to be modified to benefit the other.

5.5. Discussion on Network Improvements

General trends emerge in the analysis of the varying network improvements for decreasing the skew across race in facial recognition. In Table 8, we classify each improvement into one or more of the following categories: loss/training improvements, architectural changes, dataset modification, and using additional data.

Table 8. Network Improvements Discussion: While there are various improvements that are made in overcoming racial bias in facial recognition, we classify these improvements into four categories: loss/training, architecture, dataset modification, and additional data. We also define which methods are improved from further race classification research. The table is sorted by publication year and then alphabetically by method. The checkmark (✓) defines when it is categorized into each column.

Method	Year	Loss/Training	Architecture	Dataset Modification	Additional Data	Benefits from Race Classification
ACNN [97]	2017				✓	
InclusiveFaceNet [137]	2017		✓		✓	✓
CosFace [87]	2018	✓				
ArcFace [61]	2019	✓				
IMAN [15]	2019	✓	✓			✓
PFE [98]	2019		✓			
DebFace [99]	2020		✓		✓	✓
RL-RBN [10]	2020	✓				✓
GAC [100]	2021		✓			✓
MBN [101]	2021	✓	✓			✓
PCT [103]	2021		✓			✓
Rethinking [102]	2021	✓		✓		✓
Using VGG16 [1]	2021		✓			
AIN [115]	2022		✓			✓
Fairness aware augmentation [138]	2022			✓		✓
Sensitive Loss [20]	2022	✓				✓
GABN [104]	2023	✓	✓			✓

The loss and training improvements began with the CosFace loss [87] and then was built on by the IMAN approach with its custom mutual information loss [15]. But the

most commonly used loss for facial recognition, particularly for racially balanced facial recognition, is the ArcFace loss [61]. Other losses have been developed [104]; however, none have been as widely accepted as the ArcFace loss [61]. Along with adapting new loss functions, there was also a focus on improving training approaches such as new attempts to use reinforcement learning [10] or losses that force the network to focus on the same parts of the face regardless of race [104].

Most publications on improvements for mitigating the skew across race for facial recognition focus on improving the architecture of the neural network. Initial methods were built to learn the race and gender initially and then learn the facial recognition task [137]. These methods were followed by having multiple stages of the network to learn the race and the face separately [15] or within the same architecture [99]. Other methods learned a probabilistic latent space representation of the face [98] or compared different popular architectures on the task [1]. Most recently, the contributions have focused on architectures that can compare skin tone [101] and approaches that focus on maintaining the networks focus on the same parts of the face, regardless of race [104,115].

There have also been improvements in modifying the dataset with novel ideas such as only training on one race that improves cross-race results [102] or more common ideas of force balancing datasets [138]. Other approaches have attempted to include previous methodologies, such as using hand-picked features [97] and incorporating different tasks (gender, age, race, and identity) [99].

Overall, many of the methods are centered around inserting prior information within loss functions [15,61,87,104], modifying architectures to learn additional tasks [15,99,101,104,115], and modifying the dataset or incorporating additional data [97,99,102]. Each method contributes to mitigating the skew across race in facial recognition.

6. Future Work

The field of facial recognition focusing on substrata, such as the human race, is a growing field. There are many methods where future work could be created and added upon. As discussed previously, much work has been carried out to create racially balanced datasets. This approach offers multiple benefits. It primarily enhances the ability of a system to achieve high accuracy in realistic, generalized scenarios. This area of research is still lacking as the number and size of racially unbalanced datasets greatly dwarf those of racially balanced datasets. Future work could focus on making larger datasets with the intent to have equal sampling from the stratum of the human race.

Another future work was discussed previously in race transformation. This would be used to take an image of an individual in one race and transfer it to all other races that we are interested in. Then, doing that with all the images for that individual would create a new individual that could be used as an additional training sample. With high quality race transformation, one could take any racially unbalanced dataset and augment the dataset to have an equal distribution across all racial groups. This could be carried out by replacing GAN racial transformation networks with diffusion racial transformation networks. While GANs were originally used as the SOTA generative deep learning approach, diffusion has surpassed GANs in many categories and shows promise for the ability to create a race transformation network [143].

To better aid racially balanced performance across human races, race recognition using monocular images must be improved. With certain strata outperforming other strata in classification, it is difficult to scrap images for races with lower classification accuracies. To create the optimal racially balanced dataset, the race classification of human faces must also be improved.

As this work focuses on comparing racially unbalanced/balanced datasets, balancing datasets through data generation, and network improvements across the human race using monocular cameras there are limitations to our study. Some of these limitations include that bias across datasets is not limited to race, but rather also is connected or even interconnected with other biases such as gender or age. We recognize that gender [62,144,145] and age [146]

are not covered in great detail within this survey. Future survey papers could center around the conjugation of race, gender, and age in facial recognition.

7. Conclusions

Understanding bias in deep learning is a widespread and evolving field, presenting complex challenges. We provide an analysis of race imbalance across many popular facial recognition datasets and the increasing trend of new facial recognition datasets to be racially balanced. We discuss the promising attempts to balance datasets through data generation accompanied by a discussion on the impact of data generation. We discuss various network improvements that aim to reduce racial bias in facial recognition systems. These include loss modifications, training methods, architecture improvements, data modification, and incorporating additional data. Finally, we provide a list of future work that researchers can follow. Overall, the skew across race in facial recognition is decreasing but requires further research to mitigate the problem fully.

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References

1. Coe, J.; Atay, M. Evaluating impact of race in facial recognition across machine learning and deep learning algorithms. *Computers* **2021**, *10*, 113. [CrossRef]
2. Cavazos, J.G.; Phillips, P.J.; Castillo, C.D.; O'Toole, A.J. Accuracy comparison across face recognition algorithms: Where are we on measuring race bias? *IEEE Trans. Biom. Behav. Identity Sci.* **2020**, *3*, 101–111. [CrossRef] [PubMed]
3. Krishnapriya, K.; Albiero, V.; Vangara, K.; King, M.C.; Bowyer, K.W. Issues related to face recognition accuracy varying based on race and skin tone. *IEEE Trans. Technol. Soc.* **2020**, *1*, 8–20. [CrossRef]
4. Pagano, T.P.; Loureiro, R.B.; Araujo, M.M.; Lisboa, F.V.N.; Peixoto, R.M.; Guimaraes, G.A.d.S.; Santos, L.L.d.; Cruz, G.O.R.; de Oliveira, E.L.S.; Cruz, M.; et al. Bias and unfairness in machine learning models: A systematic literature review. *arXiv* **2022**, arXiv:2202.08176.
5. Parikh, R.B.; Teeple, S.; Navathe, A.S. Addressing bias in artificial intelligence in health care. *JAMA* **2019**, *322*, 2377–2378. [CrossRef] [PubMed]
6. Ntoutsi, E.; Fafalios, P.; Gadiraju, U.; Iosifidis, V.; Nejdl, W.; Vidal, M.E.; Ruggieri, S.; Turini, F.; Papadopoulos, S.; Krasanakis, E.; et al. Bias in data-driven artificial intelligence systems—An introductory survey. *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.* **2020**, *10*, e1356. [CrossRef]
7. Fountain, J.E. The moon, the ghetto and artificial intelligence: Reducing systemic racism in computational algorithms. *Gov. Inf. Q.* **2022**, *39*, 101645. [CrossRef]
8. Abdurrahim, S.H.; Samad, S.A.; Huddin, A.B. Review on the effects of age, gender, and race demographics on automatic face recognition. *Vis. Comput.* **2018**, *34*, 1617–1630. [CrossRef]
9. Suresh, H.; Guttag, J.V. A framework for understanding unintended consequences of machine learning. *arXiv* **2019**, arXiv:1901.10002
10. Wang, M.; Deng, W. Mitigating bias in face recognition using skewness-aware reinforcement learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Virtual, 14–19 June 2020 ; pp. 9322–9331.
11. Morales, A.; Fierrez, J.; Vera-Rodriguez, R.; Tolosana, R. Sensitivenets: Learning agnostic representations with application to face images. *IEEE Trans. Pattern Anal. Mach. Intell.* **2020**, *43*, 2158–2164. [CrossRef]

12. Darabant, A.S.; Borza, D.; Danescu, R. Recognizing Human Races through Machine Learning—A Multi-Network, Multi-Features Study. *Mathematics* **2021**, *9*, 195. [CrossRef]
13. Karkkainen, K.; Joo, J. Fairface: Face attribute dataset for balanced race, gender, and age for bias measurement and mitigation. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, Virtual, 5–9 January 2021; pp. 1548–1558.
14. Hupont, I.; Fernández, C. Demogpairs: Quantifying the impact of demographic imbalance in deep face recognition. In Proceedings of the 2019 14th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2019), Lille, France, 14–18 May 2019; pp. 1–7.
15. Wang, M.; Deng, W.; Hu, J.; Tao, X.; Huang, Y. Racial faces in the wild: Reducing racial bias by information maximization adaptation network. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Seoul, Republic of Korea, 27 October–2 November 2019; pp. 692–702.
16. Li, A.; Tan, Z.; Li, X.; Wan, J.; Escalera, S.; Guo, G.; Li, S.Z. CASIA-SURF CeFA: A Benchmark for Multi-modal Cross-ethnicity Face Anti-spoofing. *arXiv* **2020**, arXiv:2003.05136.
17. Robinson, J.P.; Qin, C.; Henon, Y.; Timoner, S.; Fu, Y. Balancing biases and preserving privacy on balanced faces in the wild. *IEEE Trans. Image Process.* **2023**, *32*, 4365–4377. [CrossRef] [PubMed]
18. Ning, X.; Nan, F.; Xu, S.; Yu, L.; Zhang, L. Multi-view frontal face image generation: A survey. *Concurr. Comput. Pract. Exp.* **2023**, *35*, e6147. [CrossRef]
19. Page, M.J.; Moher, D.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *bmj* **2021**, *372*, n160. [CrossRef] [PubMed]
20. Serna, I.; Morales, A.; Fierrez, J.; Obradovich, N. Sensitive loss: Improving accuracy and fairness of face representations with discrimination-aware deep learning. *Artif. Intell.* **2022**, *305*, 103682. [CrossRef]
21. Samaria, F.S.; Harter, A.C. Parameterisation of a stochastic model for human face identification. In Proceedings of the 1994 IEEE Workshop on Applications of Computer Vision, Sarasota, FL, USA, 5–7 December 1994; pp. 138–142.
22. Yucer, S.; Akçay, S.; Al-Moubayed, N.; Breckon, T.P. Exploring racial bias within face recognition via per-subject adversarially-enabled data augmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, Virtual, 14–19 June 2020; pp. 18–19.
23. Merriam-Webster. Race. Available online: https://www.merriam-webster.com/dictionary/race (accessed on 24 May 2024).
24. Merriam-Webster. Ethnicity. Available online: https://www.merriam-webster.com/dictionary/ethnicity (accessed on 24 May 2024).
25. Merriam-Webster. Ethnic. Available online: https://www.merriam-webster.com/dictionary/ethnic (accessed on 24 May 2024).
26. Office of Management and Budget. *Revisions to OMB's Statistical Policy Directive No. 15: Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity*; Office of Information and Regulatory Affairs, Office of Management and Budget, Executive Office of the President: Washington, DC, USA, 2024.
27. Howell, A.J.; Buxton, H. Towards unconstrained face recognition from image sequences. In Proceedings of the Second International Conference on Automatic Face and Gesture Recognition, Killington, VT, USA, 14–16 October 1996; pp. 224–229.
28. Huang, G.B.; Mattar, M.; Berg, T.; Learned-Miller, E. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. In Proceedings of the Workshop on faces in 'Real-Life' Images: Detection, Alignment, and Recognition, Marseille, France, 17–20 October 2008.
29. Bae, G.; de La Gorce, M.; Baltrušaitis, T.; Hewitt, C.; Chen, D.; Valentin, J.; Cipolla, R.; Shen, J. Digiface-1m: 1 million digital face images for face recognition. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, Waikoloa, HI, USA, 3–7 January 2023; pp. 3526–3535.
30. Shorten, C.; Khoshgoftaar, T.M. A survey on image data augmentation for deep learning. *J. Big Data* **2019**, *6*, 60. [CrossRef]
31. Mehrabi, N.; Morstatter, F.; Saxena, N.; Lerman, K.; Galstyan, A. A survey on bias and fairness in machine learning. *ACM Comput. Surv. (CSUR)* **2021**, *54*, 1–35. [CrossRef]
32. Fisher, R.A. Statistical methods for research workers. In *Breakthroughs in Statistics: Methodology and Distribution*; Springer: Berlin/Heidelberg, Germany; 1970; pp. 66–70.
33. Cowles, M.; Davis, C. On the origins of the. 05 level of statistical significance. *Am. Psychol.* **1982**, *37*, 553. [CrossRef]
34. Ortega-Garcia, J.; Fierrez, J.; Alonso-Fernandez, F.; Galbally, J.; Freire, M.R.; Gonzalez-Rodriguez, J.; Garcia-Mateo, C.; Alba-Castro, J.L.; Gonzalez-Agulla, E.; Otero-Muras, E.; et al. The multiscenario multienvironment biosecure multimodal database (bmdb). *IEEE Trans. Pattern Anal. Mach. Intell.* **2009**, *32*, 1097–1111. [CrossRef]
35. Kumar, N.; Berg, A.; Belhumeur, P.N.; Nayar, S. Describable visual attributes for face verification and image search. *IEEE Trans. Pattern Anal. Mach. Intell.* **2011**, *33*, 1962–1977. [CrossRef] [PubMed]
36. Wolf, L.; Hassner, T.; Maoz, I. Face recognition in unconstrained videos with matched background similarity. In Proceedings of the CVPR 2011, Colorado Springs, CO, USA, 21–25 June 2011; pp. 529–534. [CrossRef]
37. Yi, D.; Lei, Z.; Liao, S.; Li, S.Z. Learning face representation from scratch. *arXiv* **2014**, arXiv:1411.7923.
38. Yang, S.; Luo, P.; Loy, C.C.; Tang, X. From facial parts responses to face detection: A deep learning approach. In Proceedings of the IEEE International Conference on Computer Vision, Santiago, Chile, 13–16 December 2015; pp. 3676–3684.
39. Parkhi, O.M.; Vedaldi, A.; Zisserman, A. Deep Face Recognition. In Proceedings of the BMVC 2015—British Machine Vision Conference 2015, British Machine Vision Association, Swansea, UK, 7–10 September 2015.

40. Kemelmacher-Shlizerman, I.; Seitz, S.M.; Miller, D.; Brossard, E. The megaface benchmark: 1 million faces for recognition at scale. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 26 June–1 July 2016; pp. 4873–4882.
41. Guo, Y.; Zhang, L.; Hu, Y.; He, X.; Gao, J. Ms-celeb-1m: A dataset and benchmark for large-scale face recognition. In Proceedings of the European Conference on Computer Vision, 2016, Amsterdam, The Netherlands, 11–14 October 2016; Springer: Berlin/Heidelberg, Germany 2016; pp. 87–102.
42. Zhang, Z.; Song, Y.; Qi, H. Age progression/regression by conditional adversarial autoencoder. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, HI, USA, 21–26 July 2017; pp. 5810–5818.
43. Cao, Q.; Shen, L.; Xie, W.; Parkhi, O.M.; Zisserman, A. Vggface2: A dataset for recognising faces across pose and age. In Proceedings of the 2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018), Xi'an, China, 15–19 May 2018; pp. 67–74.
44. Maze, B.; Adams, J.; Duncan, J.A.; Kalka, N.; Miller, T.; Otto, C.; Jain, A.K.; Niggel, W.T.; Anderson, J.; Cheney, J.; et al. Iarpa janus benchmark-c: Face dataset and protocol. In Proceedings of the 2018 International Conference on Biometrics (ICB), Gold Coast, QLD, Australia, 20–23 February 2018; pp. 158–165.
45. Grother, P.; Ngan, M.; Hanaoka, K. *Face Recognition Vendor Test (FVRT): Part 3, Demographic Effects*; National Institute of Standards and Technology: Gaithersburg, MD, USA, 2019.
46. Golzar, J.; Tajik, O.; Noor, S. Convenience Sampling. *IJELS* **2022**, *1*, 72–77. [CrossRef]
47. Thomee, B.; Shamma, D.A.; Friedland, G.; Elizalde, B.; Ni, K.; Poland, D.; Borth, D.; Li, L.J. YFCC100M: The new data in multimedia research. *Commun. ACM* **2016**, *59*, 64–73. [CrossRef]
48. Merler, M.; Ratha, N.; Feris, R.S.; Smith, J.R. Diversity in faces. *arXiv* **2019**, arXiv:1901.10436.
49. Birmajer, D.; Blount, B.; Boyd, S.; Einsohn, M.; Helmreich, J.; Kenyon, L.; Lee, S.; Taub, J. Introductory Statistics. 2021. Available online: https://touroscholar.touro.edu/oto/26/ (accessed on 5 April 2024).
50. s-celeb-1m Challenge 3: Face Feature Test/Trillion Pairs. Originally. Available online: http://trillionpairs.deepglint.com/overview (accessed on 14 June 2020).
51. Google. Freebase Data Dumps. Originally. Available online: https://developers.google.com/freebase/data (accessed on 24 May 2024).
52. Face++ Research Toolkit. Available online: https://www.faceplusplus.com/(accessed on 24 May 2024).
53. Merriam-Webster. Nationality. Available online: https://www.merriam-webster.com/dictionary/nationality (accessed on 24 May 2024).
54. Fu, S.; He, H.; Hou, Z.G. Learning race from face: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **2014**, *36*, 2483–2509. [CrossRef] [PubMed]
55. Microsoft Azure. Available online: https://www.azure.cn. (accessed on 24 May 2024).
56. Zhou, E.; Cao, Z.; Yin, Q. Naive-deep face recognition: Touching the limit of LFW benchmark or not? *arXiv* **2015**, arXiv:1501.04690.
57. Baidu Cloud Vision Api. Available online: http://ai.baidu.com/ (accessed on 24 May 2024).
58. Amazon's Rekognition Tool. Available online: https://aws.amazon.com/rekognition/ (accessed on 24 May 2024).
59. Wen, Y.; Zhang, K.; Li, Z.; Qiao, Y. A discriminative feature learning approach for deep face recognition. In Proceedings of the European Conference on Computer Vision, Amsterdam, The Netherlands, 11–14 October 2016; Springer: Berlin/Heidelberg, Germany, 2016; pp. 499–515.
60. Liu, W.; Wen, Y.; Yu, Z.; Li, M.; Raj, B.; Song, L. Sphereface: Deep hypersphere embedding for face recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, HI, USA, 21–26 July 2017; pp. 212–220.
61. Deng, J.; Guo, J.; Xue, N.; Zafeiriou, S. Arcface: Additive angular margin loss for deep face recognition. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Long Beach, CA, USA, 16–20 July 2019; pp. 4690–4699.
62. Acien, A.; Morales, A.; Vera-Rodriguez, R.; Bartolome, I.; Fierrez, J. Measuring the gender and ethnicity bias in deep models for face recognition. In Proceedings of the Progress in Pattern Recognition, Image Analysis, Computer Vision, and Applications: 23rd Iberoamerican Congress, CIARP 2018, Madrid, Spain, 19–22 November 2018; Proceedings 23; Springer: Berlin/Heidelberg, Germany, 2019; pp. 584–593.
63. Sarridis, I.; Koutlis, C.; Papadopoulos, S.; Diou, C. Towards Fair Face Verification: An In-depth Analysis of Demographic Biases. *arXiv* **2023**, arXiv:2307.10011.
64. Goodfellow, I.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; Bengio, Y. Generative adversarial nets. In Proceedings of the Advances in Neural Information Processing Systems 27 (NIPS 2014), Montreal, QC, Canada, 8–13 December 2014.
65. Sohl-Dickstein, J.; Weiss, E.; Maheswaranathan, N.; Ganguli, S. Deep unsupervised learning using nonequilibrium thermodynamics. In Proceedings of the International Conference on Machine Learning, PMLR, Lille, France, 7–9 July 2015; pp. 2256–2265.
66. Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; Ommer, B. High-resolution image synthesis with latent diffusion models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, New Orleans, LA, USA, 19–24 June 2022; pp. 10684–10695.
67. Isola, P.; Zhu, J.Y.; Zhou, T.; Efros, A.A. Image-to-image translation with conditional adversarial networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, HI, USA, 21–26 July 2017; pp. 1125–1134.

68. Karras, T.; Laine, S.; Aila, T. A Style-Based Generator Architecture for Generative Adversarial Networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 16–20 June 2019.
69. Jin, Y.; Zhang, J.; Li, M.; Tian, Y.; Zhu, H.; Fang, Z. Towards the automatic anime characters creation with generative adversarial networks. *arXiv* **2017**, arXiv:1708.05509.
70. Croitoru, F.A.; Hondru, V.; Ionescu, R.T.; Shah, M. Diffusion models in vision: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **2023**, *45*, 10850–10869. [CrossRef]
71. Peebles, W.; Xie, S. Scalable diffusion models with transformers. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Paris, France, 2–6 October 2023; pp. 4195–4205.
72. Radford, A.; Metz, L.; Chintala, S. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv* **2015**, arXiv:1511.06434.
73. Jain, N.; Manikonda, L.; Hernandez, A.O.; Sengupta, S.; Kambhampati, S. Imagining an engineer: On GAN-based data augmentation perpetuating biases. *arXiv* **2018**, arXiv:1811.03751.
74. Ting, D.S.W.; Cheung, C.Y.L.; Lim, G.; Tan, G.S.W.; Quang, N.D.; Gan, A.; Hamzah, H.; Garcia-Franco, R.; San Yeo, I.Y.; Lee, S.Y.; et al. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA* **2017**, *318*, 2211–2223. [CrossRef] [PubMed]
75. Silberzahn, R.; Uhlmann, E.L.; Martin, D.P.; Anselmi, P.; Aust, F.; Awtrey, E.; Bahník, Š.; Bai, F.; Bannard, C.; Bonnier, E.; et al. Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Adv. Methods Pract. Psychol. Sci.* **2018**, *1*, 337–356. [CrossRef]
76. Wheatley, G. *Quick Draw*; Mathematics Learning: Bethany Beach, DE, USA, 2007
77. Miyato, T.; Koyama, M. cGANs with projection discriminator. *arXiv* **2018**, arXiv:1802.05637.
78. Miyato, T.; Kataoka, T.; Koyama, M.; Yoshida, Y. Spectral normalization for generative adversarial networks. *arXiv* **2018**, arXiv:1802.05957.
79. Gulrajani, I.; Ahmed, F.; Arjovsky, M.; Dumoulin, V.; Courville, A.C. Improved training of wasserstein gans. In Proceedings of the Advances in Neural Information Processing Systems 30 (NIPS 2017), Long Beach, CA, USA, 4–9 December 2017.
80. Dumoulin, V.; Shlens, J.; Kudlur, M. A learned representation for artistic style. *arXiv* **2016**, arXiv:1610.07629.
81. Shmelkin, R.; Wolf, L.; Friedlander, T. Generating master faces for dictionary attacks with a network-assisted latent space evolution. In Proceedings of the 2021 16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021), Jodhpur, India, 15–18 December 2021; pp. 1–8.
82. King, D.E. Dlib-ml: A machine learning toolkit. *J. Mach. Learn. Res.* **2009**, *10*, 1755–1758.
83. Schroff, F.; Kalenichenko, D.; Philbin, J. Facenet: A unified embedding for face recognition and clustering. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Boston, MA, USA, 7–12 June 2015; pp. 815–823.
84. Mandis, I.S. Reducing Racial and Gender Bias in Machine Learning and Natural Language Processing Tasks Using a GAN Approach. *Int. J. High Sch. Res.* **2021**, *3*, 17–24 [CrossRef]
85. Zhu, J.Y.; Park, T.; Isola, P.; Efros, A.A. Unpaired image-to-image translation using cycle-consistent adversarial networks. In Proceedings of the IEEE International Conference on Computer Vision, Venice, Italy, 22–29 October 2017; pp. 2223–2232.
86. He, L.; Wang, Z.; Li, Y.; Wang, S. Softmax dissection: Towards understanding intra-and inter-class objective for embedding learning. In Proceedings of the AAAI Conference on Artificial Intelligence, New York, NY, USA, 7–12 February 2020; Volume 34, pp. 10957–10964.
87. Wang, H.; Wang, Y.; Zhou, Z.; Ji, X.; Gong, D.; Zhou, J.; Li, Z.; Liu, W. Cosface: Large margin cosine loss for deep face recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–22 June 2018; pp. 5265–5274.
88. Ge, J.; Deng, W.; Wang, M.; Hu, J. FGAN: Fan-Shaped GAN for Racial Transformation. In Proceedings of the 2020 IEEE International Joint Conference on Biometrics (IJCB), Houston, TX, USA, 28 September–1 October 2020; pp. 1–7. [CrossRef]
89. Choi, Y.; Choi, M.; Kim, M.; Ha, J.W.; Kim, S.; Choo, J. Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–22 June 2018; pp. 8789–8797.
90. Kim, Y.H.; Nam, S.H.; Hong, S.B.; Park, K.R. GRA-GAN: Generative adversarial network for image style transfer of Gender, Race, and age. *Expert Syst. Appl.* **2022**, *198*, 116792. [CrossRef]
91. Kim, Y.H.; Lee, M.B.; Nam, S.H.; Park, K.R. Enhancing the accuracies of age estimation with heterogeneous databases using modified CycleGAN. *IEEE Access* **2019**, *7*, 163461–163477. [CrossRef]
92. Kim, Y.H.; Nam, S.H.; Park, K.R. Enhanced cycle generative adversarial network for generating face images of untrained races and ages for age estimation. *IEEE Access* **2020**, *9*, 6087–6112. [CrossRef]
93. Jain, A.; Memon, N.; Togelius, J. Zero-shot racially balanced dataset generation using an existing biased stylegan2. In Proceedings of the 2023 IEEE International Joint Conference on Biometrics (IJCB), Ljubljana, Slovenia, 25–28 September 2023; pp. 1–18.
94. Sumsion, A.; Torrie, S.; Sun, Z.; Lee, D.J. Overcoming deep learning subclass imbalances: Comparing the transfer of identity across a racial transformation. *Electron. Imaging* **2023**, *35*, 1–6. [CrossRef]
95. Yang, C.; Lv, Z. Gender based face aging with cycle-consistent adversarial networks. *Image Vis. Comput.* **2020**, *100*, 103945. [CrossRef]

96. Lui, N.; Chia, B.; Berrios, W.; Ross, C.; Kiela, D. Leveraging Diffusion Perturbations for Measuring Fairness in Computer Vision. In Proceedings of the AAAI Conference on Artificial Intelligence, Vancouver, BC, Canada, 20–27 February 2024; Volume 38, pp. 14220–14228.
97. Kang, D.; Dhar, D.; Chan, A. Incorporating side information by adaptive convolution. In Proceedings of the Advances in Neural Information Processing Systems 30 (NIPS 2017), Long Beach, CA, USA, 4–9 December 2017.
98. Shi, Y.; Jain, A.K. Probabilistic face embeddings. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Seoul, Republic of Korea, 27 October–2 November 2019; pp. 6902–6911.
99. Gong, S.; Liu, X.; Jain, A.K. Jointly de-biasing face recognition and demographic attribute estimation. In Proceedings of the Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, 23–28 August 2020; Proceedings, Part XXIX 16; Springer: Berlin/Heidelberg, Germany, 2020; pp. 330–347.
100. Gong, S.; Liu, X.; Jain, A.K. Mitigating face recognition bias via group adaptive classifier. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Virtual, 19–25 June 2021; pp. 3414–3424.
101. Wang, M.; Zhang, Y.; Deng, W. Meta balanced network for fair face recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* **2021**, *44*, 8433–8448. [CrossRef] [PubMed]
102. Gwilliam, M.; Hegde, S.; Tinubu, L.; Hanson, A. Rethinking common assumptions to mitigate racial bias in face recognition datasets. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Virtual, 11–17 October 2021; pp. 4123–4132.
103. Li, Y.; Sun, Y.; Cui, Z.; Shan, S.; Yang, J. Learning fair face representation with progressive cross transformer. *arXiv* **2021**, arXiv:2108.04983.
104. Huang, L.; Wang, M.; Liang, J.; Deng, W.; Shi, H.; Wen, D.; Zhang, Y.; Zhao, J. Gradient attention balance network: Mitigating face recognition racial bias via gradient attention. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Vancouver, BC, Canada, 18–22 June 2023; pp. 38–47.
105. Li, Z.; Liu, F.; Yang, W.; Peng, S.; Zhou, J. A survey of convolutional neural networks: Analysis, applications, and prospects. *IEEE Trans. Neural Netw. Learn. Syst.* **2021**, *33*, 6999–7019. [CrossRef] [PubMed]
106. Chan, E.R.; Lin, C.Z.; Chan, M.A.; Nagano, K.; Pan, B.; De Mello, S.; Gallo, O.; Guibas, L.J.; Tremblay, J.; Khamis, S.; et al. Efficient geometry-aware 3d generative adversarial networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, New Orleans, LA, USA, 19–24 June 2022; pp. 16123–16133.
107. Chen, S.; Wang, C.; Chen, Z.; Wu, Y.; Liu, S.; Chen, Z.; Li, J.; Kanda, N.; Yoshioka, T.; Xiao, X.; et al. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE J. Sel. Top. Signal Process.* **2022**, *16*, 1505–1518. [CrossRef]
108. Wang, M.; Deng, W. Deep face recognition: A survey. *Neurocomputing* **2021**, *429*, 215–244. [CrossRef]
109. Mnih, V.; Kavukcuoglu, K.; Silver, D.; Rusu, A.A.; Veness, J.; Bellemare, M.G.; Graves, A.; Riedmiller, M.; Fidjeland, A.K.; Ostrovski, G.; et al. Human-level control through deep reinforcement learning. *Nature* **2015**, *518*, 529–533. [CrossRef]
110. Watkins, C.J.; Dayan, P. Q-learning. *Mach. Learn.* **1992**, *8*, 279–292. [CrossRef]
111. Krizhevsky, A.; Sutskever, I.; Hinton, G.E. Imagenet classification with deep convolutional neural networks. In Proceedings of the Advances in Neural Information Processing Systems 25 (NIPS 2012), Lake Tahoe, NV, USA, 3–6 December 2012.
112. Simonyan, K.; Zisserman, A. Very deep convolutional networks for large-scale image recognition. *arXiv* **2014**, arXiv:1409.1556.
113. He, K.; Zhang, X.; Ren, S.; Sun, J. Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 26 June–1 July 2016; pp. 770–778.
114. Hoffer, E.; Ailon, N. Deep metric learning using triplet network. In Proceedings of the Similarity-Based Pattern Recognition: Third International Workshop, SIMBAD 2015, Copenhagen, Denmark, 12–14 October 2015; Proceedings 3; Springer: Berlin/Heidelberg, Germany, 2015; pp. 84–92.
115. Wang, M.; Deng, W. Adaptive Face Recognition Using Adversarial Information Network. *IEEE Trans. Image Process.* **2022**, *31*, 4909–4921. [CrossRef]
116. Manesh, F.S.; Ghahramani, M.; Tan, Y.P. Facial part displacement effect on template-based gender and ethnicity classification. In Proceedings of the 2010 11th International Conference on Control Automation Robotics & Vision, Singapore, 7–10 December 2010; pp. 1644–1649.
117. Toderici, G.; O'malley, S.M.; Passalis, G.; Theoharis, T.; Kakadiaris, I.A. Ethnicity-and gender-based subject retrieval using 3-D face-recognition techniques. *Int. J. Comput. Vis.* **2010**, *89*, 382–391. [CrossRef]
118. Wang, W.; He, F.; Zhao, Q. Facial ethnicity classification with deep convolutional neural networks. In Proceedings of the Chinese Conference on Biometric Recognition, Chengdu, China, 14–16 October 2016; pp. 176–185.
119. Krizhevsky, A.; Hinton, G. Convolutional deep belief networks on cifar-10. *Unpubl. Manuscr.* **2010**, *40*, 1–9.
120. Ricanek, K.; Tesafaye, T. Morph: A longitudinal image database of normal adult age-progression. In Proceedings of the 7th International Conference on Automatic Face and Gesture Recognition (FGR06), Southamtpon, UK, 10–12 April 2006; pp. 341–345.
121. Gao, W.; Cao, B.; Shan, S.; Chen, X.; Zhou, D.; Zhang, X.; Zhao, D. The CAS-PEAL large-scale Chinese face database and baseline evaluations. *IEEE Trans. Syst. Man Cybern.-Part A Syst. Hum.* **2007**, *38*, 149–161.
122. Phillips, P.J.; Moon, H.; Rizvi, S.A.; Rauss, P.J. The FERET evaluation methodology for face-recognition algorithms. *IEEE Trans. Pattern Anal. Mach. Intell.* **2000**, *22*, 1090–1104. [CrossRef]
123. Katti, H.; Arun, S. Can you tell where in India I am from? Comparing humans and computers on fine-grained race face classification. *arXiv* **2017**, arXiv:1703.07595.

124. Vo, T.; Nguyen, T.; Le, C. Race recognition using deep convolutional neural networks. *Symmetry* **2018**, *10*, 564. [CrossRef]
125. Lyons, M.; Akamatsu, S.; Kamachi, M.; Gyoba, J. Coding facial expressions with gabor wavelets. In Proceedings of the Proceedings Third IEEE International Conference on Automatic Face and Gesture Recognition, Nara, Japan, 14–16 April 1998; pp. 200–205.
126. Szegedy, C.; Ioffe, S.; Vanhoucke, V.; Alemi, A.A. Inception-v4, inception-resnet and the impact of residual connections on learning. In Proceedings of the Thirty-first AAAI Conference on Artificial Intelligence, San Francisco, CA, USA, 4–9 February 2017.
127. Das, A.; Dantcheva, A.; Bremond, F. Mitigating bias in gender, age and ethnicity classification: A multi-task convolution neural network approach. In Proceedings of the European Conference on Computer Vision (ECCV) Workshops, Munich, Germany, 8–14 September 2018.
128. Wang, Y.; Feng, Y.; Liao, H.; Luo, J.; Xu, X. Do they all look the same? Deciphering chinese, japanese and koreans by fine-grained deep learning. In Proceedings of the 2018 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), Miami, FL, USA, 10–12 April 2018; pp. 39–44.
129. Hu, J.; Shen, L.; Sun, G. Squeeze-and-excitation networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–22 June 2018; pp. 7132–7141.
130. Howard, A.; Sandler, M.; Chu, G.; Chen, L.C.; Chen, B.; Tan, M.; Wang, W.; Zhu, Y.; Pang, R.; Vasudevan, V.; et al. Searching for mobilenetv3. In Proceedings of the IEEE/CVF International Conference on Computer Vision, Seoul, Republic of Korea, 27 October–2 November 2019; pp. 1314–1324.
131. King, D.E. Max-Margin Object Detection. *arXiv* **2015**, arXiv:1502.00046.
132. Kingma, D.P.; Ba, J. Adam: A method for stochastic optimization. *arXiv* **2014**, arXiv:1412.6980.
133. Liu, Z.; Luo, P.; Wang, X.; Tang, X. Deep learning face attributes in the wild. In Proceedings of the IEEE International Conference on Computer Vision, Santiago, Chile, 13–16 December 2015; pp. 3730–3738.
134. Vrandečić, D.; Krötzsch, M. Wikidata: A free collaborative knowledgebase. *Commun. ACM* **2014**, *57*, 78–85. [CrossRef]
135. Rothe, R.; Timofte, R.; Van Gool, L. Deep expectation of real and apparent age from a single image without facial landmarks. *Int. J. Comput. Vis.* **2018**, *126*, 144–157. [CrossRef]
136. Somandepalli, K. *Prediction Race from Face for Movie Data*; University of Southern California: Los Angeles, CA, USA, 2017.
137. Ryu, H.J.; Adam, H.; Mitchell, M. Inclusivefacenet: Improving face attribute detection with race and gender diversity. *arXiv* **2017**, arXiv:1712.00193.
138. Pastaltzidis, I.; Dimitriou, N.; Quezada-Tavarez, K.; Aidinlis, S.; Marquenie, T.; Gurzawska, A.; Tzovaras, D. Data augmentation for fairness-aware machine learning: Preventing algorithmic bias in law enforcement systems. In Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency, Seoul, Republic of Korea, 21–24 June 2022; pp. 2302–2314.
139. Cheng, M.; Cai, K.; Li, M. RWF-2000: An open large scale video database for violence detection. In Proceedings of the 2020 25th International Conference on Pattern Recognition (ICPR), Milan, Italy, 10–15 January 2021; pp. 4183–4190.
140. Conti, J.R.; Noiry, N.; Clemencon, S.; Despiegel, V.; Gentric, S. Mitigating gender bias in face recognition using the von mises-fisher mixture model. In Proceedings of the International Conference on Machine Learning, PMLR, Baltimore, MD, USA, 17–23 July 2022; pp. 4344–4369.
141. Gwyn, T.; Roy, K. Examining Gender Bias of Convolutional Neural Networks via Facial Recognition. *Future Internet* **2022**, *14*, 375. [CrossRef]
142. Tian, F.; Liu, W.; Zhao, S.; Liu, J. Face Recognition Fairness Assessment based on Data Augmentation: An Empirical Study. In Proceedings of the 2022 IEEE 22nd International Conference on Software Quality, Reliability, and Security Companion (QRS-C), Guangzhou, China, 5–9 December 2022; pp. 315–318.
143. Avrahami, O.; Fried, O.; Lischinski, D. Blended latent diffusion. *ACM Trans. Graph. (TOG)* **2023**, *42*, 1–11. [CrossRef]
144. Palmer, M.A.; Brewer, N.; Horry, R. Understanding gender bias in face recognition: Effects of divided attention at encoding. *Acta Psychol.* **2013**, *142*, 362–369. [CrossRef] [PubMed]
145. Ng, C.B.; Tay, Y.H.; Goi, B.M. A review of facial gender recognition. *Pattern Anal. Appl.* **2015**, *18*, 739–755. [CrossRef]
146. Terhörst, P.; Kolf, J.N.; Huber, M.; Kirchbuchner, F.; Damer, N.; Moreno, A.M.; Fierrez, J.; Kuijper, A. A comprehensive study on face recognition biases beyond demographics. *IEEE Trans. Technol. Soc.* **2021**, *3*, 16–30. [CrossRef]

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